AN13297

Embedded Wi-Fi Subsystem API Specification v17 - Host driver firmware interface

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Application Note

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Document information

Information	Content
Keywords	Driver, firmware, host, data path, Access Point (AP), client, host commands, MAC events, micro access point, MACEvents, WPA, QoS, RSSI, TLV
Abstract	Includes the specifications for the interface between the host driver and the firmware that powers NXP Wi-Fi System-on-Chip (SoC) products for client (station) applications

1 Revision History

Table 1: Revision history

Version	Date	Modifications
v.4	20230501	**Section - Host commands** Section 6.1.8, CMD_STACMD_SYS_CONFIGURE: updated the CmdCode description of the request Section 6.2.5, CMD_802_11_REMAIN_ON_CHANNEL: corrected the CmdCode for the REQUEST Section *CMD_802_11_HEART_BEAT*: removed Section 6.18.1, CMD_11N_CFG: updated bit [0] description for HTTxCapability field (REQUEST and RESPONSE) Section 6.29.1, CMD_ED_CTRL: • updated the description of the command • Updated the fields ed_ctrl_2g, ed_offset_2g, ed_ctrl_5g, and ed_offset_5g for the REQUEST and RESPONSE • removed the fields ed_on_period_2g, ed_off_preriod_2g, ed_bitmap_2g, ed_on_period_5g, ed_off_preriod_5g, ed_bitmap_5g • Added the field ed_ctrl_txq_lock (REQUEST and RESPONSE) **Section - Command and Event List** Table 89, Host Command List: updated the value of CMD_STACMD_SYS_CONFIGURE
v.3	20220629	**Section - Data Path** Section 4.1, Receive Packet Descriptor: • rx_rate: added *HE rate (rate_info Bits[1:0] = 11)* • rate_info: updated bit [7] and bits [1:0] of HT information **Section - Host commands** Section 6.1.10, CMD_ADD_NEW_STATION: added the last TLV Section 6.5.1, CMD_GET_HW_SPEC response: added the TLVs Section 6.8.1, CMD_802_11_ASSOCIATE: added a sentence about 802.11ax support and added HE Capabilities IE field to the request table and to the response (successful association) table Section 6.10.1, CMD_TX_RATE_QUERY: • Added HeRateInfo field to the request and response tables. • Updated TxPacket HT information definition (Table 23) • Added HE Rate Value to Table 25 Section 6.10.2, CMD_TX_RATE_CFG: updated *TxRateCfg* TLV in the query and response Section 6.40, 802.11ax: • Section 6.40.2, CMD_TWT: added • Section 6.40.3, CMD_802_11AX_CMD: added Section 6.40.1, CMD_802_11AX_CFG: • Updated CmdCode in Request and Response tables • Updated result description in the response table -- Continued -->

Table 1: Revision history (Continued)

Version	Date	Modifications
v.3 (continued)	20220629	-- Continued -- **Section - TLV Usage** Section 10.4, NXP Proprietary IE Formats: • Added *MRVL_DOT11AX_OBSS_PD_OFFSET_TLV_ID* • Added *MRVL_DOT11AX_ENABLE_SR_TLV_ID* Section 10.4.56, MrvlIETypes_ChanTRPCConfig_t: • ModulationGroup1: added the description for 0x1C to 0x23 Section 10.4.106, MrvlIETypes_ChanRuPwrCfg _t: added Section 10.4.107, MRVL_DOT11AX_OBSS_PD_OFFSET_TLV_ID: added Section 10.4.108, MRVL_DOT11AX_ENABLE_SR_TLV_ID: added Table 81, NXP Micro AP TLVs: added *MrvlIETypes_Api_Ver_Info_t*, *MrvlIETypes_Max_Supported_STA_Count_t*, and *MrvlIETypes_FW_Cap_Info_t* Table 89, Host Command List: updated the list for 802.11ax Section 10.5, Micro AP TLV • Section 10.5.46, MrvlIETypes_Api_Ver_Info_t: added • Section 10.5.47, MrvlIETypes_Max_Supported_STA_Count_t: added • Section 10.5.48, MrvlIETypes_FW_Cap_Info_t: added
v.2	20210615	Applied NXP branding and version numbering scheme. Replaced WLAN occurrences with Wi-Fi.
v.1	20190611	Initial release

2 Introduction

This document contains the specifications for the interface between the host driver and the firmware that powers NXP Wi-Fi System-on-Chip (SoC) products for client (station) applications.

> ***Note:*** This specification uses little-endian byte order.

2.1 Theory of Operation

2.1.1 Establishing Link with AP

When the host driver wants to establish a link with a particular Access Point (AP), it issues a command to the firmware to initiate the joining process.

When joining an AP, the station must pass 2 processes:

- Authentication process
- Association process

2.1.2 Terminating Link with AP

When the host driver wants to terminate the link with a particular AP, it issues a command to the firmware to initiate the de-authentication process.

2.1.3 Joining Existing Ad-Hoc Network

When the host driver wants to join an existing Ad-Hoc network, it issues a command to the firmware to initiate the joining process. When joining an Ad-Hoc network, the flow would be the same as joining an Infrastructure except that it does not initiate the Authentication and Association processes.

2.1.4 Starting Non-Existing Ad-Hoc Network

When the host driver wants to create a new Ad-Hoc network, it issues a command to the firmware to initiate that start process. This command should only be sent if no other network exist that meet the driver criteria to join with.

2.1.5 Terminating Membership in Ad-Hoc Network

When the host driver wants to terminate its membership in an Ad-Hoc network, it issues a command to the firmware to initiate the termination process.

2.1.6 Scanning for Existing Networks Within the Proximity

When the host driver wants to discover wireless networks within its proximity, it issues a command to the firmware to initiate the scanning process. The host driver can specify if the scan is that of an active or passive scan. The flow of the scan process is the same except that there are no probe requests sent out during a passive scan. Also the driver can specify specific filters bases on Service Set Identifier (SSID), Basic Service Set Identifier (BSSID), or Type.

3 Driver and Firmware

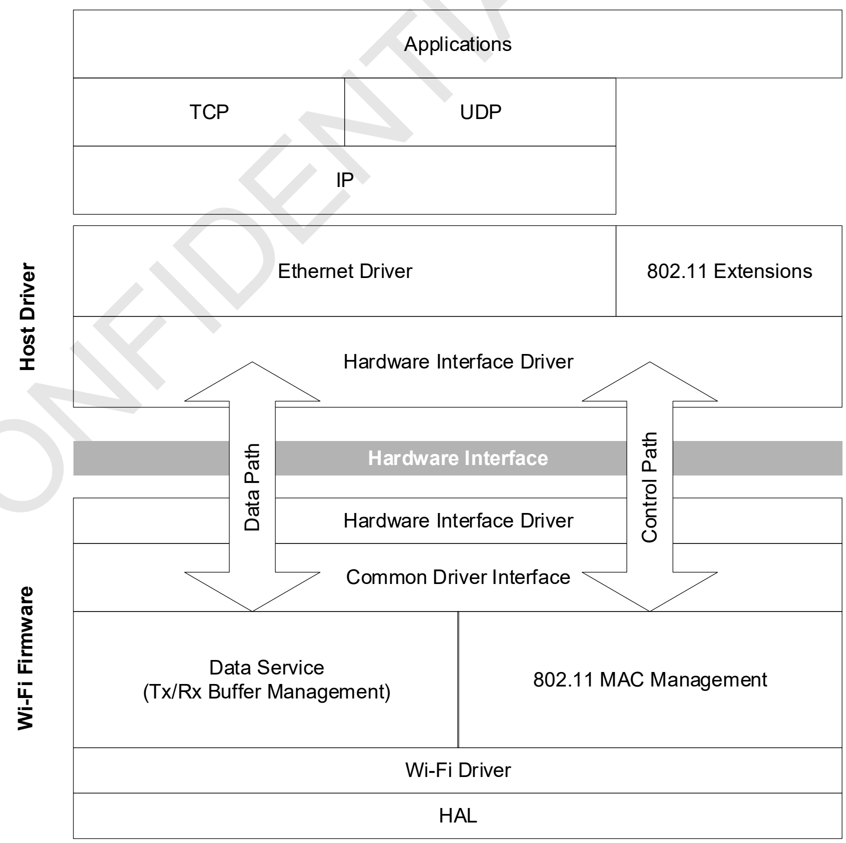
3.1 Driver and Firmware Architecture

A simplified view of the overall architecture of the typical host driver and basic architecture of the device firmware is shown in Figure 1. The Wi-Fi firmware architecture is modular and allows the porting of some modules (that is, 802.11 MAC Sublayer Management Entity (MLME)) to reside in the host driver.

This basic architecture is typical of a thick firmware architecture, where the Wi-Fi firmware handles all 802.11 MAC Management tasks.

- The host driver downloads standard 802.3 frames to the firmware to transmit over the wireless link as 802.11 frames
- The Wi-Fi firmware processes the received 802.11 frames and converts them into 802.3 frames before forwarding them to the host driver.

Figure 1: Standard Host Driver and Wi-Fi Firmware Architecture



3.1.1 Host Driver Modules

The host driver modules are:

- **Ethernet Driver:** Standard Ethernet driver.
- **802.11 Extensions:** This module extends the standard Ethernet driver in order to view and control the state of the Wi-Fi adapter.
- **Hardware Interface Driver:** This module controls the hardware interface on the host side.

3.1.2 Host Driver Control Pipes into Firmware

The host driver control pipes into the firmware are:

- **Command Requests:** The host driver can download command requests to the Wi-Fi firmware using the control path, (see Section 5, Control Path, on page 16).
- **Embed Control Directives:** The host driver can embed control directives in each data packet it downloads to the Wi-Fi firmware for transmission over the air. Similarly, the Wi-Fi firmware embeds Wi-Fi receive information into each data packet uploaded to the host driver, (see Section 4, Data Path, on page 8).

3.1.3 Firmware Modules

The available firmware modules are:

- **Hardware Interface Driver:** Controls the hardware interface on the Wi-Fi system on the device (SoC) side.
- **Common Driver Interface:** Supports the common driver interface.
- **Data Service:** Manages data buffers on transmit and receive paths and their flow. This module also handles any frame type conversion, such as 802.11 to 802.3.
- **802.11 MLME:** Handles 802.11 Medium Access Control (MAC) layer management tasks.
- **Wi-Fi Driver:** Manages the Wi-Fi MAC interrupts and controls the hardware MAC.
- **HAL:** This module is the hardware abstraction layer.

3.2 Driver and Firmware After Card Insertion

This section describes the interaction between the host driver and firmware after a Wi-Fi card is inserted. After a card has been inserted, the host driver:

1. Downloads firmware into the Wi-Fi module.
2. Waits until the firmware completes its boot sequence and indicates that it is ready.
   If the device does not contain EEPROM, then it needs to download EEPROM information at this time. This can be accomplished by calling CMD_802_11_CAL_DATA_EXT.
3. Sends the scan command to the firmware to prepare the firmware for association with the specified AP.
4. Sends the Authenticate command to the firmware to prepare the firmware for association with the specified AP.
5. Sends the Associate command to the firmware to start the association process with the specified AP.

   Note: See Driver Porting Guide for firmware download procedures.

4 Data Path

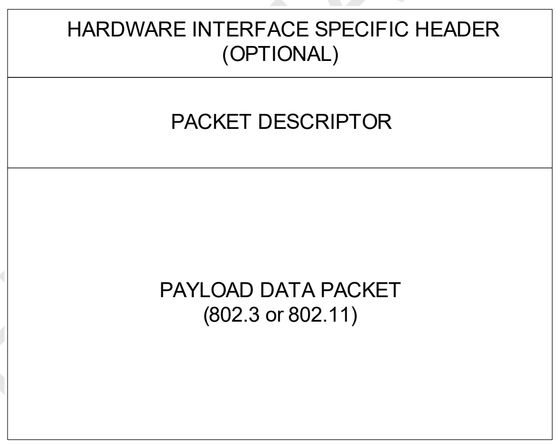
Each data packet (either 802.3 or 802.11) transferred across the hardware interface begins with a hardware specific header (optional) followed by the packet descriptor (required).

The hardware interface specific header is optional when the hardware interface supports data and command channels. Otherwise, it is mandatory. The header is needed to distinguish command packets from data packets on hardware interfaces that do not support channels with commands separate from the data channel (SDIO interface).

> ***Note:*** The fragmentation threshold for all data or management type frames must be set to a multiple of 4 bytes.

Figure 2 shows the general packet structure. The structure of the packet descriptor for packets sent from the firmware to the host is different from the structure of the packet descriptor sent from the host to the firmware.

Figure 2: General Packet Structure



For 802.11n, if firmware performs Aggregated MAC Service Data Unit (AMSDU) aggregation, Traffic Stream Identifier (TID) is sent from driver in the TID field of the Quality of Service (QoS) Control field in TxInfo. Event DOT11N_RX_REORDERING_WINDOW_ADJUST will need to be added if driver performs receive reordering. A new TxPD and RxPD will be used for all packets including aggregated and non-aggregated packets.

4.1 Receive Packet Descriptor

Table 2 shows the structure of the Receive Packet Descriptor (RxPD).

Table 2: Fields in Receive Packet Descriptor

Field Name	Type	Description
bss_type	UINT8	BSS type 0x00 = STA 0x01 = uAP 0x02 = Point-to-Point (P2P) 0x04 = NAN 0x05 = 802.11p Others = reserved
bss_num	UINT8	BSS number 0 to 15
rx_pkt_length	UINT16	Number of bytes in the payload
rx_pkt_offset	UINT16	Offset from the start of the packet to the beginning of the payload data packet
rx_pkt_type	UINT16	Rx packet type 0x00 = IEEE 802.3 default 0x01 = IEEE 802.3 with LLC 0x02 = Ethernet version 2 frame 0x03 = IEEE 802.3 with LLC + SNAP frame 0x05 = IEEE 802.11 frame 0x07 = ETCP socket data 0x08 = raw data 0x09 = mesh 0xD0 = mac80211 0xDD = timesync 0xDE = TDLS 0xDF = EAPOL message 0xE0 = NXP Bluetooth AMP 0xE2 = forward management frame 0xE5 = management frame 0xE6 = AMSDU 0xE7 = BAR 0xE8 = loopback 0xE9 = data more 0xEA = data last 0xEB = data null 0xED = send to dataswitch 0xEE = send to host 0xEF = debug
seq_num	UINT16	Sequence number
priority	UINT8	Specifies the user priority of received packet

Table 2: Fields in Receive Packet Descriptor (Continued)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>rx_rate</td><td>UINT8</td><td>Rate at which this packet is received
Legacy and HT rates (rate_info Bits[1:0] = 00 or 01):
<table>
<tr><th rowspan="2">RxRate[7:0]</th><th colspan="2">Data Rate</th></tr>
<tr><th>HTInfo Bit[0] = 0</th><th>HTInfo Bit[0] = 1</th></tr>
<tr><td>0x0</td><td>1 Mbps</td><td>MCS0</td></tr>
<tr><td>0x1</td><td>2 Mbps</td><td>MCS1</td></tr>
<tr><td>0x2</td><td>5.5 Mbps</td><td>MCS2</td></tr>
<tr><td>0x3</td><td>11 Mbps</td><td>MCS3</td></tr>
<tr><td>0x4</td><td>Reserved</td><td>MCS4</td></tr>
<tr><td>0x5</td><td>6 Mbps</td><td>MCS5</td></tr>
<tr><td>0x6</td><td>9 Mbps</td><td>MCS6</td></tr>
<tr><td>0x7</td><td>12 Mbps</td><td>MCS7</td></tr>
<tr><td>0x8</td><td>18 Mbps</td><td>MCS8</td></tr>
<tr><td>0x9</td><td>24 Mbps</td><td>MCS9</td></tr>
<tr><td>0xA</td><td>36 Mbps</td><td>MCS10</td></tr>
<tr><td>0xB</td><td>48 Mbps</td><td>MCS11</td></tr>
<tr><td>0xC</td><td>54 Mbps</td><td>MCS12</td></tr>
<tr><td>0xD and up</td><td>Reserved</td><td>MCS13 ... MCSn</td></tr>
</table>
VHT rate (rate_info Bits[1:0] = 10):
<table>
<tr><th>RxRate[7:4]</th><th>Number of Spatial Stream</th><th>RxRate[3:0]</th><th>MCS Index</th></tr>
<tr><td>0x0</td><td>Nss=1</td><td>0x0</td><td>MCS0</td></tr>
<tr><td>0x1</td><td>Nss=2</td><td>0x1</td><td>MCS1</td></tr>
<tr><td>0x2</td><td>Reserved</td><td>0x2</td><td>MCS2</td></tr>
<tr><td>0x3</td><td>Reserved</td><td>0x3</td><td>MCS3</td></tr>
<tr><td>0x4</td><td>Reserved</td><td>0x4</td><td>MCS4</td></tr>
<tr><td>0x5</td><td>Reserved</td><td>0x5</td><td>MCS5</td></tr>
<tr><td>0x6</td><td>Reserved</td><td>0x6</td><td>MCS6</td></tr>
<tr><td>0x7</td><td>Reserved</td><td>0x7</td><td>MCS7</td></tr>
<tr><td>0x8</td><td>Reserved</td><td>0x8</td><td>MCS8</td></tr>
<tr><td>0x9</td><td>Reserved</td><td>0x9</td><td>MCS9</td></tr>
<tr><td>0xA and up</td><td>Reserved</td><td>0xA and up</td><td>Reserved</td></tr>
</table>
Continues --></td></tr>
</table>

Table 2: Fields in Receive Packet Descriptor (Continued)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>rx_rate
(continued)</td><td>UINT8</td><td>HE rate (rate_info Bites[1:0] = 11)
<table>
<tr><th>RxRate[7:4]</th><th>Number of Spatial Stream</th><th>RxRate[3:0]</th><th>MCS Index</th></tr>
<tr><td>0x0</td><td>Nss=1</td><td>0x0</td><td>MCS0</td></tr>
<tr><td>0x1</td><td>Nss=2</td><td>0x1</td><td>MCS1</td></tr>
<tr><td>0x2</td><td>Reserved</td><td>0x2</td><td>MCS2</td></tr>
<tr><td>0x3</td><td>Reserved</td><td>0x3</td><td>MCS3</td></tr>
<tr><td>0x4</td><td>Reserved</td><td>0x4</td><td>MCS4</td></tr>
<tr><td>0x5</td><td>Reserved</td><td>0x5</td><td>MCS5</td></tr>
<tr><td>0x6</td><td>Reserved</td><td>0x6</td><td>MCS6</td></tr>
<tr><td>0x7</td><td>Reserved</td><td>0x7</td><td>MCS7</td></tr>
<tr><td>0x8</td><td>Reserved</td><td>0x8</td><td>MCS8</td></tr>
<tr><td>0x9</td><td>Reserved</td><td>0x9</td><td>MCS9</td></tr>
<tr><td>0xA</td><td>Reserved</td><td>0xA</td><td>MCS10</td></tr>
<tr><td>0xB</td><td>Reserved</td><td>0xB</td><td>MCS11</td></tr>
<tr><td>0xC and up</td><td>Reserved</td><td>0xC and up</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>snr</td><td>SINT8</td><td>Signal to noise ratio for this packet (dB)</td></tr>
<tr><td>nf</td><td>SINT8</td><td>Noise floor for this packet (dBm)
Noise floor is always negative. The absolute value is passed.</td></tr>
<tr><td>rate_info</td><td>UINT8</td><td>HT/VHT/HE information
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7</td><td>[Bit 7] [Bit 4] 11ax GI
00 = 01xHELTF + GI0.8us.8us
01 = 2xHELTF + GI0.8us
10 = 2xHELTF + GI1.6us
11 = 4xHELTF + GI0.8us, if DCM=0 and STBC=0
11 = 4xHELTF + GI3.2us, otherwise</td></tr>
<tr><td>6</td><td>LDPC</td></tr>
<tr><td>5</td><td>STBC</td></tr>
<tr><td>4</td><td>HT/VHT guard interval
0 = long guard interval
1 = short guard interval</td></tr>
<tr><td>3:2</td><td>HT/VHT bandwidth
00 = BW 20
01 = BW 40
10 = BW 80
11 = BW 160</td></tr>
<tr><td>1:0</td><td>Rx Rate format
00 = legacy rates
01 = HT rates
10 = VHT rates
11 = HE rates</td></tr>
</table></td></tr>
</table>

Table 2: Fields in Receive Packet Descriptor (Continued)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>reserved</td><td>UINT8[3]</td><td>Reserved for specific customer projects</td></tr>
<tr><td>flags</td><td>UINT8</td><td>Receive packet flags
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1</td><td>extraHeader</td></tr>
<tr><td>0</td><td>TDLS packet
0 = no
1 = yes</td></tr>
</table></td></tr>
<tr><td>antenna</td><td>UINT8</td><td>For 1x1 device:
0x00 = antenna A
0x01 = antenna B
0x03 = antenna C
For 2x2 device:
0x01 = antenna A
0x02 = antenna B
0x03 = antenna AB</td></tr>
<tr><td>ToA_ToD_
tstamps</td><td>UINT64</td><td>Time of Arrival (ToA) and Time of Departure (ToD) timestamps (ns)
Bits[63:32]: ToD of ACK for Rx packet
Bits[31:0]: ToA of Rx packet</td></tr>
<tr><td>rx_info</td><td>UINT32</td><td>Rx information
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>31:14</td><td>Reserved</td></tr>
<tr><td>13:5</td><td>Rx channel
Non-0 channel number on which this packet is received.</td></tr>
<tr><td>4</td><td>Valid firmware bandwidth
0 = driver ignores Bits[3:0]
1 = bandwidth in Bits[3:0] is valid</td></tr>
<tr><td>3:0</td><td>Channel bandwidth which this packet is received
0x0 = 5 MHz
0x1 = 10 MHz
0x2 = 20 MHz
0x3 = 40 MHz
0x4 = 80 MHz
0x5 to 0xF = reserved</td></tr>
</table></td></tr>
</table>

4.2 Transmit Packet Descriptor

Table 3 shows the fields of the Transmit Packet Descriptor (TxPD).

Table 3: Fields in Transmit Packet Descriptor

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>bss_type</td><td>UINT8</td><td>BSS type
0x00 = STA
0x01 = uAP
0x02 = P2P
0x04 = NAN
0x05 = 802.11p
Others = reserved</td></tr>
<tr><td>bss_num</td><td>UINT8</td><td>BSS number
0 to 15</td></tr>
<tr><td>tx_pkt_length</td><td>UINT16</td><td>Number of bytes in the payload data frame</td></tr>
<tr><td>ty_pkt_offset</td><td>UINT16</td><td>Offset of the beginning of the payload data packet (802.11n frames) from the beginning of the packet (byte)</td></tr>
<tr><td>tx_pkt_type</td><td>UINT16</td><td>Packet type
0x00 = normal 802.3
0x05 = 802.11
0xE5 = management frame
0xE6 = AMSDU</td></tr>
<tr><td>tx_control</td><td>UINT32</td><td>Tx control
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>31:24</td><td>Reserved</td></tr>
<tr><td>23:16</td><td>RateID as in Section 3.2</td></tr>
<tr><td>15</td><td>Host rate control
0 = firmware rate setting
1 = specified in Bits[23:16]</td></tr>
<tr><td>14:13</td><td>Ack policy
0x = firmware ACK setting
10 = ACK_IMMD
11 = NO_ACK</td></tr>
<tr><td>12</td><td>Host retry control
0 = firmware retry setting
1 = specified in Bits[11:8]</td></tr>
<tr><td>11:8</td><td>Retry limit</td></tr>
<tr><td>7</td><td>Host Tx power control
0 = firmware setting for Tx power
1 = specified in Bit[6] and Bits[5:0] are valid</td></tr>
<tr><td>6</td><td>Sign of power specified in Bits[5:0]
0 = positive
1 = negative</td></tr>
<tr><td>5:0</td><td>Power used for transmission (dBm)</td></tr>
</table></td></tr>
<tr><td>priority</td><td>UINT8</td><td>Specifies the user priority of transmit packet</td></tr>
</table>

Table 3: Fields in Transmit Packet Descriptor (Continued)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>flags</td><td>UINT8</td><td>Transmit packet flags
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>7:6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5</td><td>Tx packet status report
0 = disable
1 = enable</td></tr>
<tr><td>4</td><td>TDLS packet
0 = no
1 = yes</td></tr>
<tr><td>3</td><td>Last packet indicator (used to indicate if the attached packet is the last packet in the service period)</td></tr>
<tr><td>2</td><td>pmVal</td></tr>
<tr><td>1</td><td>overRideFwPM</td></tr>
<tr><td>0</td><td>When set to 1:
UAPSD/MRVL PPS mode: this bit is the last packet indicator when there is no data to transmit
UAPSD mode only: firmware transmit a null data packet
Bit[3] is ignored when this bit is set.</td></tr>
</table></td></tr>
<tr><td>pkt_delay_
2ms</td><td>UINT8</td><td>Amount of time the packet has been queued in the driver layer for WMM implementations
Ignored when WMM is not enabled and should be set to 0 when not used.
Units are 2 ms.</td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>Reserved</td></tr>
</table>

Table 3: Fields in Transmit Packet Descriptor (Continued)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>tx_control_1</td><td>UINT32</td><td>Tx control1
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>31:30</td><td>Reserved</td></tr>
<tr><td>29:21</td><td>Tx channel
0 = firmware setting
Others = channel number to transmit this packet</td></tr>
<tr><td>20</td><td>Host bandwidth control
0 = firmware setting
1 = bandwidth setting specified in Bits[3:0] for transmit of this packet</td></tr>
<tr><td>19:16</td><td>Bandwidth for transmit of this packet
0x0 = 5 MHz
0x1 = 10 MHz
0x2 = 20 MHz
0x3 = 40 MHz
0x4 = 80 MHz
0x5 to 0xF = reserved</td></tr>
<tr><td>15:8</td><td>Tx token ID
Token ID synchronization between driver and firmware for Tx management frame.</td></tr>
<tr><td>7:0</td><td>Expiry time
0 = firmware setting
Non-0 = Tx packet expiry time (TU unit)</td></tr>
</table></td></tr>
</table>

4.2.1 Per-Packet Settings

3 parameters are used to set per-packet basis:

- Transmission rate
- ACK (acknowledgement) policy
- Retry limit

5 Control Path

This section specifies the format for command packets and the command exchange protocol between the host driver and firmware.

The command packet format and command exchange protocol between the host driver and firmware are independent of the type of hardware interface. However, the mechanism for transporting the command packets across the hardware interface is hardware dependent.

5.1 Packet Format

The command packet (request or response) consists of a fixed-size header and variable-size body.

The packet header contains the command code, the size of the command packet, the sequence number, and the result. The result field is only meaningful in the response packet. The header is 8 bytes in size. The most significant bit in the command code for a request is always set to 0. The most-significant bit in the command code for a response is always set to 1.

The command body contains data for the command. The structure of the body depends on the type of command.

Table 4 shows the common structure of the command packet.

Table 4: Common Command Packet Structure

Field Name	Type	Description
CmdCode	UINT16	Command code The corresponding code for the response packet is (CmdCode | 0x8000).
Size	UINT16	Size of the packet, including the header
SeqNum	UINT8	Sequence number, set by the host
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code, set by the firmware This field is used only in the response packet. Set to 0 in request packet.
Body	UINT8[]	Command body, containing data specific to the type of command Simple commands may omit this field.

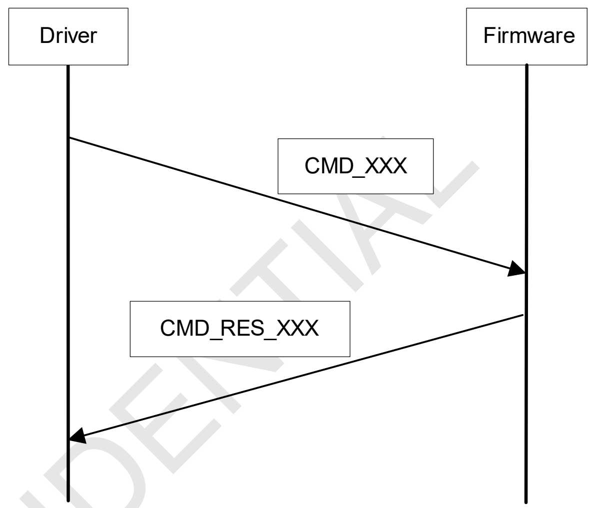
For a detailed description of all the commands, see Section 6, Host Commands, on page 18. For a detailed listing of Commands and Result Codes, see Appendix 11, Command and Event List, on page 398.

5.2 Protocol

The command exchange protocol follows the request-response model. The host driver issues a command request, and the firmware sends back a command response. The command exchange protocol is serialized. The host driver must wait until it has received a command response for the current command request before it can send the next command request.

The mechanism for transporting the command request and command response packets across the hardware interface is hardware dependent.

Figure 3: Protocol Request Response Mode



6 Host Commands

This section describes the following:

- Interaction between the host driver and firmware after card insertion
- Commands supported

Firmware that conforms to this interface specification may elect to implement only a subset of the commands described here. Firmware returns the response status code CMD_STATUS_UNSUPPORTED for any command request that it does not implement.

6.1 Initialization

Table 5 list the supported commands for reset and initialization.

Table 5: Reset and Initialization Commands

Command	Description	Page
Section 6.1.1, CMD_802_11_SNMP_MIB	Gets/sets the SNMP MIB	page 19
Section 6.1.2, CMD_802_11_MAC_ADDR	Gets/sets the Wi-Fi MAC address	page 21
Section 6.1.3, CMD_MAC_MULTICAST_ADR	Gets/sets MAC multicast filter table	page 22
Section 6.1.4.1, CMD_FUNC_INIT	Initializes data structures and hardware blocks and activates threads	page 23
Section 6.1.4.2, CMD_FUNC_SHUTDOWN	Releases resources, deactivates hardware blocks and threads, and re-initializes SDIO driver	page 24
Section 6.1.5, CMD_802_11_MGMT_IE_LIST	Gets/sets multiple user-specific IEs	page 24
Section 6.1.6, CMD_SDIO_SINGLE_PORT_RX_AGGR	Enables/disables SDIO single port Rx aggregation feature	page 25
Section 6.1.7, CMD_IPV6_MSG_OFFLOAD	Prepares and enables IPv6 address offload	page 26
Section 6.1.8, CMD_STACMD_SYS_CONFIGURE	Gets AP configuration parameters	page 27
Section 6.1.9, CMD_EMBEDDED_DHCPD_CONFIG	Gets/sets DHCP server configuration	page 28
Section 6.1.10, CMD_ADD_NEW_STATION	Adds STA information to firmware	page 29

6.1.1 CMD_802_11_SNMP_MIB

This command gets or sets the Simple Network Management Protocol (SNMP) Management Information Base (MIB) values.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SNMP_MIB = 0x0016
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read) 0x0001 = ACT_SET (write)
OID	UINT16	Object identifier as defined in Table 6
Size	UINT16	Size of OID value
Value	UINT8	Buffer to keep OID value

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SNMP_MIB | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
OID	UINT16	Object identifier
Size	UINT16	Size of OID value
Value	UINT8	Buffer to keep OID value

Table 6: List of OIDs

Name	Value	Action	Size	Description
OID_DTIM_PERIOD	0x03	R	UINT8	DTIM value
OID_RTS_THRESHOLD	0x05	RW	UINT16	Default = 2347 (decimal)
OID_SHORT_RETRY_LIMIT	0x06	RW	UINT8	Default = 0x09
OID_LONG_RETRY_LIMIT	0x07	RW	UINT8	Default = 0x04
OID_FRAG_THRESHOLD	0x08	RW	UINT16	Default = 2346 (decimal)
OID_DOT11_MULTI_DOMAIN_CAPAB_ ENABLED	0x09	RW	UINT8	0 = disable (default) 1 = enable
OID_ENABLE_11H	0x0A	RW	UINT8	Bits[7:3]: reserved Bit[2]: slaveRadarDetectionEnable Bit[1]: masterRadarDetectionEnable Bit[0]: dot11hEnable Default = 0x00 (all sub-fields disabled)
OID_DOT11_WWS_MODE	0x11	RW	UINT8	0 = disable (default) 1 = enable
OID_LINK_DETECT_PERIOD	0x24	RW	UINT32	Maximum detection period
OID_AUTO_NULL_PKT_PERIOD	0x25	RW	UINT32	Send null packet period
OID_HOST_SAFEMODE	0x26	RW	UINT8	Host safe mode
OID_WMM_TURBO_MODE	0x27	RW	UINT8	WMM turbo mode 00 = disable 01 = WMM_TURBO_MODE1 10 = WMM_TURBO_MODE2
OID_DEAUTH_FROMAP_DISABLE	0x28	RW	UINT8	Deauthenticate from AP disable
OID_MULTI_PATH_RSSI_ENABLE	0x29	RW	UINT8	Multipath RSSI
OID_ECSA_KEEP_CONNECTION	0x2A	RW	UINT8	ECSA 0 = disable 1 = enable

6.1.2 CMD_802_11_MAC_ADDR

The host driver uses this command to set or get the MAC address stored in the RAM memory of the Wi-Fi device.

This command stores the MAC address in the driver and firmware. It does not write to the EEPROM.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MAC_ADDR = 0x004D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
MACAddr	UINT8[6]	MAC address of Wi-Fi device

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MAC_ADDR | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
MACAddr	UINT8[6]	MAC address of Wi-Fi device

6.1.3 CMD_MAC_MULTICAST_ADR

This command is used to program multicast MAC address into the hardware filter table in the Wi-Fi SoC. The packets sent from these multicast MAC addresses are accepted.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_MULTICAST_ADR = 0x0010
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
NumOfAddr	UINT16	Number of multicast addresses This field is not used when Action is ACT_GET.
MACList	UINT8[32*6]	List of number of multicast addresses Maximum list size of 32. This field is not used when Action is ACT_GET.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_MULTICAST_ADR | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in the request message
NumOfAddr	UINT16	Number of multicast addresses This field is not used when Action is ACT_GET.
MACList	UINT8[32*6]	List of number of multicast addresses Maximum list size of 32. This field is not used when Action is ACT_GET.

6.1.4 Multifunction SDIO Commands

In order to utilize chipsets with both Wi-Fi and Bluetooth functionality, use these commands for driver initialization and unloading.

6.1.4.1 CMD_FUNC_INIT

This command is used during driver loading process. Upon receiving this command, firmware will initialize the data structures that are relevant to the function (that is, Wi-Fi state machine variables), initialize the relevant hardware blocks (that is, Wi-Fi MAC, baseband, and radio), and activate the relevant threads (that is, MLME thread, idle thread, and so on.).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FUNC_INIT = 0x00A9
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FUNC_INIT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = command is successful 0x0001 = command fails

6.1.4.2 CMD_FUNC_SHUTDOWN

This command is used during driver unloading process. Upon receiving this command, firmware will release the relevant resources, deactivate all relevant hardware blocks, deactivate all relevant threads, and re-initialize the interface driver for the function.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FUNC_SHUTDOWN = 0x00AA
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FUNC_SHUTDOWN | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = command is successful 0x0001 = command fails

6.1.5 CMD_802_11_MGMT_IE_LIST

This API is used to gets and sets multiple user-specific IEs.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MGMT_IE_LIST = 0x00F2
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = ACT_GET 0x0002 = ACT_SET
MgmtIEList	MrvlIETypes_MgmtIEList_t	MGMT_IE_LIST_TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MGMT_IE_LIST | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
MgmtIEList	MrvlIETypes_MgmtIEList_t	MGMT_IE_LIST_TLV

6.1.6 CMD_SDIO_SINGLE_PORT_RX_AGGR

This command is used to enable or disable Secure Digital Input/Output (SDIO) single port Rx Aggregation feature. Enabling this feature will aggregate more Rx packets on single Rx port. This feature can improve SDIO read efficiency, hence improve Wi-Fi Rx throughput over SDIO interface.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SDIO_SINGLE_PORT_RX_AGGR = 0x0223
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x0000 = ACT_GET 0x0001 = ACT_SET **NOTE:** Only support to set when no interface is active.
enable	UINT8	SDIO single port Rx aggregation 0x00 = disable 0x01 = enable
blockSize	UINT16	Reserved for CMD request

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SDIO_SINGLE_PORT_RX_AGGR | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT8	Type of action set in request message

Field Name	Type	Description
enable	UINT8	SDIO single port Rx aggregation 0x00 = disable 0x01 = enable
blockSize	UINT16	Block size of Rx data ring buffer

6.1.7 CMD_IPV6_MSG_OFFLOAD

This command is used to prepare IPv6 address for firmware and may enable IPv6 message offload.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_IPV6_MSG_OFFLOAD = 0x0238
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = get IPv6 message offload configuration 0x0001 = set IPv6 message offload configuration
Enable	UINT16	Enable IPv6 message offload 0x0000 = disable 0x0001 = enable
IPv6AddrParam	MrvlIETypes_IPv6AddrParamSet_t	IPv6 address

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_IPV6_MSG_OFFLOAD | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Enable	UINT16	Enable IPv6 message offload 0x0000 = disable 0x0001 = enable
IPv6AddrParam	MrvlIETypes_IPv6AddrParamSet_t	IPv6 internet address TLV

6.1.8 CMD_STACMD_SYS_CONFIGURE

This command gets the AP configuration parameters. Firmware parses the set of Tag Length Values (TLVs) and applies the getting of the configuration accordingly. The host driver is responsible for sending the complete set of parameters to configure the desired operational mode of the Station (STA) configuration data that it receives.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_STACMD_SYS_CONFIGURE = 0x023F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x0000 = ACT_GET
ConfigData	TLV	Configuration parameters of variable number of MrvlIETypes_ChanBandList_t TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_STACMD_SYS_CONFIGURE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT8	Type of action set in request message
ConfigData	TLV	Configuration parameters of variable number of MrvlIETypes_ChanBandList_t TLV

6.1.9 CMD_EMBEDDED_DHCPD_CONFIG

This command gets/sets the Dynamic Host Configuration Protocol (DHCP) server configuration. This command is sent by the host after the uAP BSS started to configure the DHCP server parameters, so firmware can handle the DHCP packets after the host is powered off.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_EMBEDDED_DHCPD_CONFIG = 0x0248
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Enable	UINT8	Enable embedded DHCP server in firmware 0x00 = disable 0x01 = enable
DHCP_Server_Configure	TLV	DHCP parameter type MrvlIETypes_EMBEDDED_DHCPD_CFG_t TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_EMBEDDED_DHCPD_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT8	Type of action set in request message
DHCP_Server_Configure	TLV	DHCP parameter type MrvlIETypes_EMBEDDED_DHCPD_CFG_t TLV

6.1.10 CMD_ADD_NEW_STATION

This command adds STA information to firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_ADD_NEW_STATION = 0x025F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Aid	UINT16	AID for peer
PeerMac	UINT8*6	Peer MAC address
ListenInterval	INT32	Listen interval
Capability	UINT16	STA capability
TLV(s)	TLV	1 or more of the following TLVs: • MrvlIETypes_RatesParamSet_t • MrvlIETypes_HT_Capability_t • MrvlIETypes_VHT_Capability_IE_t • MrvlIETypes_VHT_OpModeNTF_t • MrvlIETypes_QoSCapability_t • MrvlIETypes_UAP_STA_FLAGS_t • MRVL_IEEE_EXTENSION_TLV - used to indicate HE MAC Capability, HE PHY Capability, and so on. See Table 64.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_ADD_NEW_STATION | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT8	Type of action set in request message
Aid	UINT16	AID for peer
PeerMac	UINT8	Peer MAC address
ListenInterval	INT32	Listen interval

Field Name	Type	Description
Capability	UINT16	STA capability
TLV(s)	TLV	1 or more of the following TLVs: • MrvlIETypes_RatesParamSet_t • MrvlIETypes_HT_Capability_t • MrvlIETypes_VHT_Capability_IE_t • MrvlIETypes_VHT_OpModeNTF_t • MrvlIETypes_QoSCapability_t • MrvlIETypes_UAP_STA_FLAGS_t

6.2 MAC/Baseband/RF Control

Table 7 lists the supported commands for MAC, baseband, and RF control.

Table 7: MAC/Baseband/RF Control Commands

6.2.1 CMD_MAC_CONTROL

This command is used to control the hardware MAC. This command is only used to set the fields described in the action field.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_CONTROL = 0x0028
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Action</td><td>UINT16</td><td>Action
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15</td><td>Reserved (set to 0)</td></tr>
<tr><td>14</td><td>Reserved for direct forwarding of frames to the host</td></tr>
<tr><td>13</td><td>802.11g protection mode
Used to enable/disable 802.11g protection in IBSS.
0 = disable (default)
1 = enable</td></tr>
<tr><td>12:11</td><td>Reserved (set to 0)</td></tr>
<tr><td>10</td><td>Enforce protection
Used to drop received frames that are unencrypted.
0 = disable
1 = enable</td></tr>
<tr><td>9</td><td>802.11g rate protection
0 = use CTS-to-SELF
1 = use RTS/CTS (default)</td></tr>
<tr><td>8:5</td><td>Reserved (set to 0)</td></tr>
<tr><td>4</td><td>Snap header
Specifies the format Rx data packet should be returned.
0 = 802.3/802.2 SNAP (default)
1 = Ethernet II</td></tr>
<tr><td>3</td><td>WEP
0 = WEP off
1 = WEP on</td></tr>
<tr><td>2:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>Reserved</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_CONTROL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num

Field Name	Type	Description
Result	UINT16	Result code
Action	UINT16	Action <table><tr><th>Bits</th><th>Description</th></tr><tr><td>15</td><td>Reserved (set to 0)</td></tr><tr><td>14</td><td>Reserved (set to 0)</td></tr><tr><td>13</td><td>802.11g protection mode Used to enable/disable 802.11g protection in IBSS. 0 = disable (default) 1 = enable</td></tr><tr><td>12:11</td><td>Reserved (set to 0)</td></tr><tr><td>10</td><td>Enforce protection Used to drop received frames that are unencrypted. 0 = disable 1 = enable</td></tr><tr><td>9</td><td>Reserved (set to 0)</td></tr><tr><td>8:5</td><td>0 = promiscous off 1 = promiscous on</td></tr><tr><td>4</td><td>Snap header Specifies the format Rx data packet should be returned. 0 = 802.3/802.2 SNAP (default) 1 = Ethernet II</td></tr><tr><td>3</td><td>WEP 0 = WEP off 1 = WEP on</td></tr><tr><td>2:0</td><td>Reserved (set to 0)</td></tr></table>

6.2.2 CMD_802_11_RF_ANTENNA

THis command gets or sets the Tx and Rx antenna mode for 2.4 GHz/5 GHz antenna configuration. Firmware will perform error check on the setting.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RF_ANTENNA = 0x0020
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Tx Action 0x0002 = ACT_SET_TX 0x0003 = ACT_SET_BOTH 0x0008 = ACT_GET_TX 0x000C = ACT_GET_BOTH

Field Name	Type	Description
AntennaMode	UINT16	Tx Antenna mode Bits[15:8]: Tx antenna for 5 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz Bits[7:0]: Tx antenna for 2.4 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz
Action	UINT16	Rx Action 0x0001 = ACT_SET_RX 0x0003 = ACT_SET_BOTH 0x0004 = ACT_GET_RX 0x000C = ACT_GET_BOTH
AntennaMode	UINT16	Rx Antenna mode Bits[15:8]: Rx antenna for 5 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz Bits[7:0]: Rx antenna for 2.4 GHz 0x0001 = set Rx path to A on 2.4 GHz 0x0002 = set Rx path to B on 2.4 GHz 0x0003 = set Rx path to A+B on 2.4 GHz 0x0100 = set Rx path to A on 5 GHz 0x0200 = set Rx path to B on 5 GHz 0x0300 = set Rx path to A+B on 5 GHz

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RF_ANTENNA | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
AntennaMode	UINT16	Tx Antenna mode Bits[15:8]: Tx antenna for 5 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz Bits[7:0]: Tx antenna for 2.4 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz
Action	UINT16	Type of action set in request message
AntennaMode	UINT16	Rx Antenna mode Bits[15:8]: Rx antenna for 5 GHz 0x0001 = set Tx path to A on 2.4 GHz 0x0002 = set Tx path to B on 2.4 GHz 0x0003 = set Tx path to A+B on 2.4 GHz 0x0100 = set Tx path to A on 5 GHz 0x0200 = set Tx path to B on 5 GHz 0x0300 = set Tx path to A+B on 5 GHz Bits[7:0]: Rx antenna for 2.4 GHz 0x0001 = set Rx path to A on 2.4 GHz 0x0002 = set Rx path to B on 2.4 GHz 0x0003 = set Rx path to A+B on 2.4 GHz 0x0100 = set Rx path to A on 5 GHz 0x0200 = set Rx path to B on 5 GHz 0x0300 = set Rx path to A+B on 5 GHz

6.2.3 CMD_802_11_RF_CHANNEL

This command gets the current channel value.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RF_CHANNEL = 0x001D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_802_11_RF_CHANNEL | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Type of action set in request message</td></tr>
<tr><td>CurrentChannel</td><td>UINT16</td><td>Current channel number</td></tr>
<tr><td>RFType</td><td>UINT16</td><td>RF type
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5:4</td><td>Secondary channel offset
00 = no secondary
01 = secondary channel is above primary channel
10 = reserved
11 = secondary channel is below primary channel</td></tr>
<tr><td>3:2</td><td>Channel width
00 = 20 MHz channel width
01 = 40 MHz channel width
1x = reserved</td></tr>
<tr><td>1:0</td><td>Band
00 = 2.4 GHz band
01 = 5 GHz band
10 = 4 GHz band
11 = reserved</td></tr>
</table>
</td></tr>
<tr><td>Reserved</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
<tr><td>ChannelList</td><td>UINT8[32]</td><td>Not used (set to 0)</td></tr>
</table>

6.2.4 CMD_CHANNEL_TRPC_CONFIG

This command gets/sets the Transmit Rate-based Power Control (TRPC) channel configuration.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CHANNEL_TRPC_CONFIG = 0x00FB
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Reserved	UINT16	Not used (set to 0)
ChannelTRPCConfig	MrvlIETypes_ChanTRPCConfig_t	TRPC channel configuration

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CHANNEL_TRPC_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
Reserved	UINT16	Not used (set to 0)
ChannelTRPCConfig	MrvlIETypes_ChanTRPCConfig_t	TRPC channel configuration

6.2.5 CMD_802_11_REMAIN_ON_CHANNEL

This command is used to keep channel on specific channel until abort or expired.

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_802_11_REMAIN_ON_CHANNEL = 0x010D</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Action
0x0000 = ACT_GET
0x0001 = ACT_SET (start operation)
0x0004 = ACT_CLEAR (cancel operation)</td></tr>
<tr><td>Status</td><td>UINT8</td><td>Not used (set to 0)</td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>Not used (set to 0)</td></tr>
<tr><td>BandConfigType</td><td>UINT8</td><td>Band configuration type
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5:4</td><td>Secondary channel offset
00 = no secondary
01 = secondary channel is above primary channel
10 = reserved
11 = secondary channel is below primary channel</td></tr>
<tr><td>3:2</td><td>Channel width
00 = 20 MHz channel width
01 = 40 MHz channel width
1x = reserved</td></tr>
<tr><td>1:0</td><td>Band
00 = 2.4 GHz band
01 = 5 GHz band
10 = 4 GHz band
11 = reserved</td></tr>
</table></td></tr>
<tr><td>ChanNumber</td><td>UINT8</td><td>Channel number for specified band</td></tr>
<tr><td>Remain_Period</td><td>UINT32</td><td>Remain period (ms)
0 = no remain period</td></tr>
</table>

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_802_11_REMAIN_ON_CHANNEL | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Type of action set in request message</td></tr>
<tr><td>Status</td><td>UINT8</td><td>For ACT_GET:
0x00 = completed
0x01 = in progress
For ACT_SET:
0x00 = success
0x01 = reject for previous request in progress
Others = other reasons
For ACT_CLEAR:
0x00 = success
0x01 = not started yet</td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>Not used (set to 0)</td></tr>
<tr><td>BandConfigType</td><td>UINT8</td><td>Band configuration type
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5:4</td><td>Secondary channel offset
00 = no secondary
01 = secondary channel is above primary channel
10 = reserved
11 = secondary channel is below primary channel</td></tr>
<tr><td>3:2</td><td>Channel width
00 = 20 MHz channel width
01 = 40 MHz channel width
1x = reserved</td></tr>
<tr><td>1:0</td><td>Band
00 = 2.4 GHz band
01 = 5 GHz band
10 = 4 GHz band
11 = reserved</td></tr>
</table></td></tr>
<tr><td>ChanNumber</td><td>UINT8</td><td>Channel number for specified band</td></tr>
<tr><td>Remain_Period</td><td>UINT32</td><td>Remain period (ms)
0 = no remain period</td></tr>
</table>

6.2.6 CMD_CW_TONE_CTRL

This command is used to control Contention Window (CW) level on Tx antenna.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CW_TONE_CTRL = 0x0239
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = disable 0x0001 = enable
CWToneLvl	UINT16	Tx CW level

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CW_TONE_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action in request message
CWToneLvl	UINT16	Tx CW level

6.2.7 CMD_CHAN_REGION_CFG

This command configures the channel attributes and/or region information.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CHAN_REGION_CFG = 0x0242
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_CHANATTRCFG_t • MrvlIETypes_REGIONINFO_t

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CHAN_REGION_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action in request message
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_CHANATTRCFG_t • MrvlIETypes_REGIONINFO_t

6.2.8 CMD_DYN_BW

This command gets/sets the dynamic bandwidth.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_DYN_BW = 0x0252
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Dyn_bw	UINT16	Dynamic bandwidth Bits[7:4]: Reserved Bit[3]: MAC dynamic bandwidth operation mode 0 = static bandwidth operation mode 1 = dynamic bandwidth operation mode Bit[2]: TxInfo force send RTS 0 = disable 1 = enable Bit[1]: TxInfo dynamic bandwidth 0 = disable 1 = enable Bit[0]: TxInfo indicated bandwidth 0 = disable 1 = enable

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_DYN_BW | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure

Field Name	Type	Description
Action	UINT16	Type of action in request message
Dyn_bw	UINT16	Dynamic bandwidth Bits[7:4]: Reserved Bit[3]: MAC dynamic bandwidth operation mode 0 = static bandwidth operation mode 1 = dynamic bandwidth operation mode Bit[2]: TxInfo force send RTS 0 = disable 1 = enable Bit[1]: TxInfo dynamic bandwidth 0 = disable 1 = enable Bit[0]: TxInfo indicated bandwidth 0 = disable 1 = enable

6.3 Register and Memory Access

Table 8 lists the supported commands for register and memory access.

Table 8: Register and Memory Access Commands

Command	Description	Page
Section 6.3.1, CMD_BBP_REG_ACCESS	Peeks and pokes baseband processor hardware register	page 44
Section 6.3.2, CMD_RF_REG_ACCESS	Peeks and pokes RF hardware register	page 45
Section 6.3.3, CMD_MAC_REG_ACCESS	Peeks and pokes MAC hardware register	page 46
Section 6.3.4, CMD_EEPROM_ACCESS	Retrieves the EEPROM data	page 47
Section 6.3.5, CMD_PMIC_REG_ACCESS	Pokes NXP Power Management Device registers	page 48
Section 6.3.6, CMD_CAU_REG_ACCESS	Pokes NXP CAU registers	page 49

6.3.1 CMD_BBP_REG_ACCESS

This command peeks or pokes a Baseband Processor (BBP) register.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_BBP_REG_ACCESS = 0x001A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Offset	UINT16	Byte offset
Value	UINT8	Register value Only required when action is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_BBP_REG_ACCESS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (s sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in the request message
Offset	UINT16	Byte offset

Field Name	Type	Description
Value	UINT8	Register value Only required when action is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

6.3.2 CMD_RF_REG_ACCESS

This command peeks or pokes a PHY register.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_RF_REG_ACCESS = 0x001B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Offset	UINT16	Byte offset
Value	UINT8	Register value Only required when action is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_RF_REG_ACCESS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (s sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in the request message
Offset	UINT16	Byte offset
Value	UINT8	Register value Only required when action is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

6.3.3 CMD_MAC_REG_ACCESS

This command peeks or pokes a MAC register.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_REG_ACCESS = 0x001B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Offset	UINT16	Byte offset, aligned on 32-bit boundary
Value	UINT32	Register value Not used if action is ACT_GET.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC_REG_ACCESS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (s sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in the request message
Offset	UINT16	Byte offset, aligned on 32-bit boundary Present only if action is ACT_GET.
Value	UINT32	Register value Present only if action is ACT_GET.

6.3.4 CMD_EEPROM_ACCESS

This command retrieves the EEPROM data.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_EEPROM_ACCESS = 0x0059
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	ACT_GET
Offset	UINT16	Multiples of 4 Example: 0, 4, 8, 12, 16, …
ByteCount	UINT16	Multiples of 4 Example: 4, 8, 12, 16, and 20. Maximum of 20 bytes.
Value	UINT8[ByteCount]	User must provide a buffer of at least ByteCount length

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_EEPROM_ACCESS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	ACT_GET
Offset	UINT16	Multiples of 4 Example: 0, 4, 8, 12, 16, …
ByteCount	UINT16	Multiples of 4 Example: 4, 8, 12, 16, and 20. Maximum of 20 bytes.
Value	UINT8[ByteCount]	If result is ok, contains EEPROM value ordered from low to high offset

6.3.5 CMD_PMIC_REG_ACCESS

This command pokes NXP Power Management Device registers.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_PMIC_REG_ACCESS = 0x00AD
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x000 = ACT_GET 0x0001 = ACT_SET
Offset	UINT16	Byte offset
Value	UINT8	Register value Only used when Action field is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_PMIC_REG_ACCESS | 0x8000
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
Value	UINT8	Register value
Reserved	UINT8[3]	Reserved (set to 0)

6.3.6 CMD_CAU_REG_ACCESS

This command pokes NXP Common Analog Unit (CAU) registers.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CAU_REG_ACCESS = 0x00AD
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Offset	UINT16	Byte offset
Value	UINT8	Register value Only used when Action field is ACT_SET.
Reserved	UINT8[3]	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CAU_REG_ACCESS | 0x8000
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Value	UINT8	Register value Only used when Action field is ACT_GET.
Reserved	UINT8[3]	Reserved (set to 0)

6.4 RF Calibration Data

Table 9 lists the RF calibration data commands.

Table 9: RF Calibration Commands

Command	Description	Page
Section 6.4.1, CMD_802_11_CAL_DATA_EXT	Gets/sets the RF calibration data	page 50
Section 6.4.2, CMD_CFG_DATA	Downloads new optimized register settings or calibration data	page 51

6.4.1 CMD_802_11_CAL_DATA_EXT

This command gets or sets the RF calibration data. A typical application of this command is for use in initializing custom stations without an EEPROM to store RF calibration data.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CAL_DATA_EXT = 0x006D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Revision	UINT16	Revision to identify the contents of the following data
CalDataLen	UINT16	Length (bytes) of the following calibration data
CalData	UINT8[CalDataLen]	Calibration data contents

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CAL_DATA_EXT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
Revision	UINT16	Revision to identify the contents of the following data
CalDataLen	UINT16	Length (bytes) of the following calibration data
CalData	UINT8[CalDataLen]	Calibration data contents

Note: For CalData, see the RD-88W MFG Manufacturing Test Suite documentation and/or contact a NXP FAE.
On boards without an EEPROM (or when EEPROM is ignored), the default MAC address is set as 00:50:43:02:FE:01.

6.4.2 CMD_CFG_DATA

This command is to facilitate the downloading of new optimized register settings or calibration data. It is also used to configure SDIO/General Purpose Input/Output (GPIO)/RFCTRL pin pad setting in sleep mode and power down mode to optimize the current leakage. Note that firmware only supports ACT_SET, not ACT_GET.

For the register file format, see the Application Program for Optimal Register Downloading document.

For uAP usage of this command, see Section 7.8.8, CMD_CFG_DATA, on page 272.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_DATA = 0x008F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (not supported) 0x0001 = ACT_SET
Type	UINT16	0x0001 = optimized registers downloading 0x0002 = calibration data downloading 0x0003 = power management device configuration downloading 0x0005 = pad optimized registers downloading
FileLen	UINT16	Register file length
RegisterFile	UINT8[FileLen]	Register file contents

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_DATA | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Action 0x0000 = ACT_SET 0x0001 = ACT_GET (not supported)

Field Name	Type	Description
Type	UINT16	0x0001 = optimized registers downloading 0x0002 = calibration data downloading 0x0003 = power management device configuration downloading 0x0005 = pad optimized registers downloading
FileLen	UINT16	Register file length
RegisterFile	UINT8[FileLen]	Register file contents

Note: For the Optimized Registers CalData and Power Management Device configuration, see RD-88W MFG Manufacturing Test Suite documentation and/or contact a NXP FAE.
On boards without an EEPROM (or when EEPROM is ignored), the default MAC address is set as 00:50:43:02:FE:01.

6.5 Status Information

Table 10 lists the supports commands for status information.

Table 10: Status Information Commands

6.5.1 CMD_GET_HW_SPEC

This command queries hardware and other configuration data from the Wi-Fi SoC, including the MAC address, multicast address list size, and hardware version number.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_HW_SPEC = 0x0003
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
HwIfVersion	UINT16	Set to 0
Version	UINT16	Set to 0
Reserved	UINT16	Reserved (set to 0)
NumOfMCastAddr	UINT16	Set to 0
PermanentAddr	UINT8[6]	Set to 0
RegionCode	UINT16	Set to 0
NumberOfAntenna	UINT16	Set to 0
FWReleaseNumber	UINT32	Set to 0
Reserved	UINT32	Reserved (set to 0)
Reserved	UINT32	Reserved (set to 0)
Reserved	UINT32	Reserved (set to 0)
FwCapInfo	UINT32	Set to 0
Dot11nDevCap	UINT32	Set to 0
DevMCSSupported	UINT8	Set to 0
MaxMPEndPortNum	UINT16	Set to 0
MgmtIECount	UINT16	Set to 0

Field Name	Type	Description
Reserved	UINT32	Reserved (set to 0)
Reserved	UINT32	Reserved (set to 0)
Dot11acDevCap	UINT32	Set to 0
Dot11acMcsSupport	UINT32	Set to 0

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_GET_HW_SPEC | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number (as sent by the host)</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code (set to 0 for success)</td></tr>
<tr><td>HwIfVersion</td><td>UINT16</td><td>Hardware interface version number</td></tr>
<tr><td>Version</td><td>UINT16</td><td>Hardware version number</td></tr>
<tr><td>Reserved</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
<tr><td>NumOfMCastAddr</td><td>UINT16</td><td>Maximum number of multicast addresses firmware can handle</td></tr>
<tr><td>PermanentAddr</td><td>UINT8[6]</td><td>MAC address in EEPROM (0 is no EEPROM)</td></tr>
<tr><td>RegionCode</td><td>UINT16</td><td>Region code</td></tr>
<tr><td>NumberOfAntenna</td><td>UINT16</td><td>Number of antenna used</td></tr>
<tr><td>FWReleaseNumber</td><td>UINT32</td><td>Firmware release number
Example: 2.3.4.p1 = 0x01020304.</td></tr>
<tr><td>Reserved</td><td>UINT32</td><td>Reserved (set to 0)</td></tr>
<tr><td>Reserved</td><td>UINT32</td><td>Reserved (set to 0)</td></tr>
<tr><td>Reserved</td><td>UINT32</td><td>Reserved (set to 0)</td></tr>
<tr><td>FwCapInfo</td><td>UINT32</td><td>Firmware capability information:
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>30</td><td>Support 0DFS
0 = do not support
1 = support</td></tr>
<tr><td>29</td><td>Disable NAN
0 = do not support
1 = support</td></tr>
<tr><td>28</td><td>Support 802.11ax
0 = do not support
1 = support</td></tr>
<tr><td>27</td><td>Support random MAC during scan
0 = do not support
1 = support</td></tr>
<tr><td colspan="2">continues...</td></tr>
</table>
</td></tr>
</table>

Field Name	Type	Description
FwCapInfo (continued)	UINT32	continued... **26** Support MAC80211 0 = do not support 1 = support **25** Support ADHOC 0 = do not support 1 = support **24** Support 802.11p 0 = do not support 1 = support **23** Support roaming offload 0 = do not support 1 = support **22** Support embedded authenticator 0 = do not support 1 = support **21** Support embedded supplicant 0 = do not support 1 = support **20** Support MIB 0 = do not support 1 = support **19** Support cancelable scan 0 = do not support 1 = support **18** Support extend channel switch announcement 0 = do not support 1 = support **17** Support SDIO single port Tx aggregation 0 = do not support 1 = support **16** Support SDIO single port Rx aggregation 0 = do not support 1 = support continues...

Nested table (Description of FwCapInfo):

Bits	Description
continued...	
26	Support MAC80211 0 = do not support 1 = support
25	Support ADHOC 0 = do not support 1 = support
24	Support 802.11p 0 = do not support 1 = support
23	Support roaming offload 0 = do not support 1 = support
22	Support embedded authenticator 0 = do not support 1 = support
21	Support embedded supplicant 0 = do not support 1 = support
20	Support MIB 0 = do not support 1 = support
19	Support cancelable scan 0 = do not support 1 = support
18	Support extend channel switch announcement 0 = do not support 1 = support
17	Support SDIO single port Tx aggregation 0 = do not support 1 = support
16	Support SDIO single port Rx aggregation 0 = do not support 1 = support
continues...	

Field Name	Type	Description
FwCapInfo (continued)	UINT32	<table><tr><th>Bits</th><th>Description</th></tr><tr><td colspan="2">continued...</td></tr><tr><td>15</td><td>Support dynamic rapid channel switch 0 = do not support 1 = support</td></tr><tr><td>14</td><td>Support TDLS 0 = do not support 1 = support</td></tr><tr><td>13</td><td>Support 802.11ac in 5 GHz band 0 = do not support 1 = support</td></tr><tr><td>12</td><td>Support 802.11ac in 2.4 GHz band 0 = do not support 1 = support</td></tr><tr><td>11</td><td>Supports 802.11n 0 = do not support 1 = support</td></tr><tr><td>10</td><td>Supports 802.11a 0 = do not support 1 = support</td></tr><tr><td>9</td><td>Supports 802.11g 0 = do not support 1 = support</td></tr><tr><td>8</td><td>Supports 802.11b 0 = do not support 1 = support</td></tr><tr><td>7:6</td><td>Rx antenna capability for 802.11a/g/b 00 = antenna 0 only 01 = antenna 1 only 1x = diversity</td></tr><tr><td>5:4</td><td>Tx antenna capability for 802.11a/g/b 00 = antenna 0 only 01 = antenna 1 only 1x = diversity</td></tr><tr><td>3</td><td>EEPROM not exist 0 = exist 1 = does not exist</td></tr><tr><td>2</td><td>Reserved (set to 0)</td></tr><tr><td>1</td><td>PS</td></tr><tr><td>0</td><td>WPA</td></tr></table>

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Dot11nDevCap</td><td>UINT32</td><td>802.11n device capabilities
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>31</td><td>Maximum AMSDU length
0 = 3839 octets
1 = 7935 octets</td></tr>
<tr><td>30</td><td>Beamforming
0 = do not support
1 = support</td></tr>
<tr><td>29</td><td>GF preamble
0 = do not support
1 = support</td></tr>
<tr><td>28</td><td>AMPDU
0 = do not support
1 = support</td></tr>
<tr><td>27</td><td>MIMO power save
0 = do not support
1 = support</td></tr>
<tr><td>26</td><td>Rx STBC
0 = do not support
1 = support</td></tr>
<tr><td>25</td><td>Tx STBC
0 = do not support
1 = support</td></tr>
<tr><td>24</td><td>40 MHz short GI
0 = do not support
1 = support</td></tr>
<tr><td>23</td><td>20 MHz short GI
0 = do not support
1 = support</td></tr>
<tr><td>22</td><td>Receive LDPC coded packets
0 = do not support
1 = support</td></tr>
<tr><td>21:20</td><td>Number of delayed block ACK streams</td></tr>
<tr><td>19:18</td><td>Number of immediate block ACK streams</td></tr>
<tr><td>17</td><td>40 MHz channel width
0 = do not support
1 = support</td></tr>
<tr><td>16</td><td>20 MHz channel width
0 = do not support
1 = support</td></tr>
<tr><td>15</td><td>10 MHz channel width
0 = do not support
1 = support</td></tr>
<tr><td>14:8</td><td>Reserved (set to 0)</td></tr>
<tr><td>7:4</td><td>Rx antenna bitmap
Bit[7]: presence of Rx antenna D
Bit[6]: presence of Rx antenna C
Bit[5]: presence of Rx antenna B
Bit[4]: presence of Rx antenna A</td></tr>
<tr><td colspan="2">continues...</td></tr>
</table>
</td></tr>
</table>

Field Name	Type	Description
Dot11nDevCap (continued)	UINT32	Bit / Description: continued... 3:0 — Tx antenna bitmap Bit[3]: presence of Tx antenna D Bit[2]: presence of Tx antenna C Bit[1]: presence of Tx antenna B Bit[0]: presence of Tx antenna A
DevMCSSupported	UINT8	MIMO abstraction of MCSs supported by device Bits[7:4]: Tx MCSs supported by device Bits[3:0]: Rx MCSs supported by device Examples: 0x11 = MCSs for 1x1 MIMO 0x22 = MCSs for 2x2 MIMO
MaxMPEndPortNum	UINT16	This member is only applicable for SDIO interface with Multiport enabled. This will return end port number (1 to 32) that is valid with the current buffer setting in firmware. This field is reserved for other interfaces like G-SPI/USB.
MgmtIECount	UINT16	Number of management IE buffers
Reserved	UINT32	Or UINT8 HwRadioState
Reserved	UINT32	Or UINT8 WoWLANSupport
Dot11acDevCap	UINT32	802.11ac device capabilities (same definition as VHT capability information Bits[29:0])

Dot11acDevCap bits:

Bits	Description
29	Tx antenna pattern consistency 0 = do not support 1 = support
28	Rx antenna pattern consistency 0 = do not support 1 = support
27:26	Reserved
25:23	Maximum AMPDU length
22	+HTC-VHT 0 = do not support 1 = support
21	VHT TXOP PS 0 = do not support 1 = support
20	MU Rx beamformee 0 = do not support 1 = support
19	MU Tx beamformer 0 = do not support 1 = support
18:16	Number of sounding dimensions
continues...	

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Dot11acDevCap</td><td>UINT32</td><td>
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td colspan="2">continued...</td></tr>
<tr><td>15:13</td><td>Compressed steering number of beamformee antenna</td></tr>
<tr><td>12:11</td><td>Grouping set</td></tr>
<tr><td>10</td><td>SU beamformee
0 = do not support
1 = support</td></tr>
<tr><td>9</td><td>SU beamformer
0 = do not support
1 = support</td></tr>
<tr><td>8</td><td>Rx STBC
0 = do not support
1 = support</td></tr>
<tr><td>7</td><td>Tx STBC
0 = do not support
1 = support</td></tr>
<tr><td>6</td><td>160 MHz short GI
0 = do not support
1 = support</td></tr>
<tr><td>5</td><td>80 MHz short GI
0 = do not support
1 = support</td></tr>
<tr><td>4</td><td>LDPC coding
0 = do not support
1 = support</td></tr>
<tr><td>3:2</td><td>Channel width
00 = channel width 20/40/80 MHz support
01 = channel width 20/40/80/160 MHz support
10 = channel width 20/40/80/160 MHz and 80+80 MHz support</td></tr>
<tr><td>1:0</td><td>Maximum MPDU length</td></tr>
</table>
</td></tr>
<tr><td>Dot11acMcsSupport</td><td>UINT32</td><td>MCS support
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>31:16</td><td>Tx MCS support
Same definition as VHT supported MCS Set Bits[45:30]</td></tr>
<tr><td>15:0</td><td>Rx MCS support
Same definition as VHT supported MCS Set Bits[15:0]</td></tr>
</table>
</td></tr>
<tr><td>Dot11ax5gCap</td><td>TLV</td><td>Variable length TLV MRVL_IEEE_EXTENSION_TLV_ID is used to indicate 5 GHz HE MAC capabilities, HE PHY capabilities and so on. See Table 64.</td></tr>
<tr><td>Dot11ax2gCap</td><td>TLV</td><td>Variable length TLV MRVL_IEEE_EXTENSION_TLV_ID is used to indicate 2.4 GHz HE MAC capabilities, HE PHY capabilities and so on. See Table 64.</td></tr>
<tr><td>TLVs</td><td>TLV</td><td>One or more of the following TLVs
• MrvlIETypes_Api_Ver_Info_t
• MrvlIETypes_Max_Supported_STA_Count_t
• MrvlIETypes_FW_Cap_Info_t</td></tr>
</table>

6.5.2 CMD_802_11_GET_LOG

This command gets the log.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_GET_LOG = 0x000B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_GET_LOG | 0x8000
Size	UINT16	Number of bytes in response
SeqNum	UINT8	Same sequence number as in Get Log Request
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
dot11MulticastTransmittedFrameCount	UINT32	Increments when the multicast bit is set in the destination MAC address of a successfully transmitted MSDU
dot11FailedCount	UINT32	Increments when an MSDU is not transmitted successfully
dot11RetryCount	UINT32	Increments when an MSDU is successfully transmitted after 1 or more retransmissions
dot11MultipleRetryCount	UINT32	Increments when an MSDU is successfully transmitted after more than 1 retransmission
dot11FrameDuplicateCount	UINT32	Increments when a frame is received that the Sequence Control field is indicating a duplicate count
dot11RTSSuccessCount	UINT32	Increments when a CTS is received in response to an RTS
dot11RTSFailureCount	UINT32	Increments when a CTS is not received in response to an RTS
dot11ACKFailureCount	UINT32	Increments when an ACK is not received when expected
dot11ReceivedFragmentCount	UINT32	Increments for each successfully received MPDU of type Data or Management
dot11MulticastReceivedFrameCount	UINT32	Increments when a MSDU is received with the multicast bit set in the destination MAC address
dot11FCSErrorCount	UINT32	Increments when an FCS error is detected in a received MPDU
dot11TransmittedFrameCount	UINT32	Increments for each successfully transmitted MSDU
Reserved	UINT32	Reserved (set to 0)
dot11WeplcvErrCnt	UINT32[4]	Increment when WEP decryption error for key index 0–3

6.5.3 CMD_GET_TSF

This command retrieves the current Time Synchronization Function (TSF) value from the MAC.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_TSF = 0x0080
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
TSF	UINT64	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_TSF | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
TSF	UINT64	Current MAC TSF value

6.5.4 CMD_TX_PKT_STATS

This command reports the current Tx packet status.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_PKT_STATS = 0x008D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_PKT_STATS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
TxRatePktStats	TxPktStatEntry[14]	Transmit packet statistics for a given rate index: TxRatePktStats[13] = not used TxRatePktStats[12] = 54 Mbps TxRatePktStats[11] = 48 Mbps TxRatePktStats[10] = 36 Mbps TxRatePktStats[9] = 24 Mbps TxRatePktStats[8] = 18 Mbps TxRatePktStats[7] = 12 Mbps TxRatePktStats[6] = 9 Mbps TxRatePktStats[5] = 6 Mbps TxRatePktStats[4] = not used TxRatePktStats[3] = 11 Mbps TxRatePktStats[2] = 5.5 Mbps TxRatePktStats[1] = 2 Mbps TxRatePktStats[0] = 1 Mbps

Table 11: TxPktStatEntry Substructure

Field Name	Type	Description
PktInitCnt	UINT32	Number of packets whose initial rate was set to this rateId
PktSuccessCnt	UINT32	Number of packets which completed successfully using this rateId as a final rate
TxAttempts	UINT32	Number of attempts for packets using this rateId as a initial rate
RetryFailure	UINT32	Number of failures due to transmit retry exhaustion for packets using this rateId as initial rate
ExpiryFailure	UINT32	Number of failures due to MSDU lifetime expiry for packets using this rateId as initial rate

6.5.5 CMD_802_11_AVG_RATE_INFO

This command is used to get Tx or Rx last successful rate and average rate.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AVG_RATE_INFO = 0x00A3
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Ntx	UINT16	Parameter used for exponential averaging for Tx rate (needed for Set action)
Nrx	UINT16	Parameter used for exponential averaging for Rx rate (needed for Set action)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AVG_RATE_INFO | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Ntx	UINT16	Parameter used for exponential averaging for Tx rate
Nrx	UINT16	Parameter used for exponential averaging for Rx rate
TxRateLast	UINT32	Last successful Tx rate in units of 1 Kbps (only for Get action)
RxRateLast	UINT32	Last successful Rx rate in units of 1 Kbps (only for Get action)
TxAvgLast	UINT32	Average Tx rate in units of 1 Kbps (only for Get action)
RxAvgLast	UINT32	Average Rx rate in units of 1 Kbps (only for Get action)
TxDataFrames	UINT32	Number of transmitted data frames
RxDataFrames	UINT32	Number of received data frames

Note: Number of Transmitted Data Frame and Received Data Frame is counted right after the association success. It will be set to 0 when the connection to the BSSID is lost or upon receipt of OID_802_11_BSSID_RATE_RSSI_FILTER request.

6.5.6 CMD_802_11_RSSI_INFO

This command is used to get the receive signal strength. Information about instantaneous Received Signal Strength Indicator (RSSI), exponentially averaging RSSI, and Noise Floor (NF) is provided with this command.

If N is the number used for exponential averaging, the result of the new average is:

Vavg_new = (Vavg_prev*(N - 1) + Vnew)/N

where:

Vnew = new RSSI value

Vavg_prev = previous average value

Vavg_new = current average value

If N is 0, the calculation will not be performed. Firmware will keep the existing N value. The default N is 8.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RSSI_INFO = 0x00A4
Size	UINT16	Number of bytes in the command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Ndata	UINT16	Parameter used for exponential averaging for data (needed for Set action)
Nbcn	UINT16	Parameter used for exponential averaging for beacon (needed for Set action)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RSSI_INFO | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Ndata	UINT16	Parameter used for exponential averaging for data
Nbcn	UINT16	Parameter used for exponential averaging for beacon
DataRSSIlast	SINT16	Last data RSSI (dBm) (only for Get action)
DataNFlast	SINT16	Last data NF (dBm) (only for Get action)
DataRSSIAvg	SINT16	Averaging data RSSI (dBm) (only for Get action)
DataNFAvg	SINT16	Averaging data NF (dBm) (only for Get action)

Field Name	Type	Description
BcnRSSIlast	SINT16	Last beacon RSSI (dBm) (only for Get action)
BcnNFlast	SINT16	Last beacon NF (dBm) (only for Get action)
BcnRSSIAvg	SINT16	Averaging beacon RSSI (dBm) (only for Get action)
BcnNFAvg	SINT16	Averaging beacon NF (dBm) (only for Get action)
TSFbcn	UINT64	Last RSSI beacon TSF (only for Get action)

6.5.7 CMD_802_11_RSSI_INFO_EXT

This command is used to get the received signal strength for each path. Information about instantaneous RSSI, exponentially averaging RSSI, and Noise Floor is provided with this command.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RSSI_INFO_EXT = 0x0237
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Ndata	UINT16	Parameter used for exponential averaging for data (needed for Set action)
Nbcn	UINT16	Parameter used for exponential averaging for beacon (needed for Set action)
TSFbcn	UINT64	Last RSSI beacon TSF (only for Get action)
RSSI_NF	MrvlIETypes_RSSI_NF_t	Received 802.11 RSSI and Noise Floor for each path (only for Get action)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RSSI_INFO_EXT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = successful 0x0001 = not supported
Action	UINT16	Type of action set in request message
Ndata	UINT16	Parameter used for exponential averaging for data
Nbcn	UINT16	Parameter used for exponential averaging for beacon
TSFbcn	UINT64	Last RSSI beacon TSF (only for Get action)
RSSI_NF	MrvlIETypes_RSSI_NF_t	Received 802.11 RSSI and Noise Floor for each path (only for Get action)

Note: There may be multiple RSSI_NF TLVs in the command response with Get action, depending on the path number. If no TLV specified in request, then all will be returned. Otherwise, only the TLV specified in the request are returned.

6.6 LED Control

The CMD_802_11_LED_CONTROL command is used to control the operation of GPIO pins and LEDs in firmware.

6.6.1 CMD_802_11_LED_CONTROL

This command is used to control the LED functionality.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_LED_CONTROL = 0x004E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
NumLed	UINT16	Reserved (set to 0)
NOTE: Above fixed fields are followed by a variable number of TLV fields.		

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_LED_CONTROL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
NumLed	UINT16	Number of LEDs implemented by firmware
NOTE: Above fixed fields are followed by a variable number of TLV fields.		

If Action is ACT_SET, then the information in TLV fields in the request is applied to firmware. If Action is ACT_GET, then the TLV fields in the current firmware settings are returned in the response.

NumLed field is set in the response by firmware to indicate the number of LEDs that are supported. This should be set to 0 in the request, and ignored by firmware. All settings for LEDs beyond the LED indexed by this number are ignored by firmware and is set to 0 by the driver.

The following TLVs can be included with this command depending on the chip/firmware version:

- MrvlIETypes_LedGpio_t
- MrvlIETypes_LedBehavior_t

Note: See LED Configuration Application Note.

6.7 Scan

Table 12 lists the supports Scan commands.

Table 12: Scan Commands

Command	Description	Page
Section 6.7.1, CMD_802_11_SCAN	Starts the scan process	page 68
Section 6.7.2, CMD_802_11_BG_SCAN_CONFIG	Gets/sets background scan configuration	page 70
Section 6.7.3, CMD_802_11_BG_SCAN_QUERY	Gets background scan results	page 73
Section 6.7.5, CMD_802_11_SCAN_EXT	Sends scan results to host	page 76

6.7.1 CMD_802_11_SCAN

This command starts the scan process to discover existing wireless networks within proximity to the station.

For 802.11n, the MrvlIETypes_ChanListParamSet_t TLV in scan request is added with the 2-bit SecChanOffset subfield, Similarly, the MrvlIETypes_ChanBandList_t TLV in scan response is added with the 2-bit SecChanOffset subfield.

For 802.11n, the variable length IEs in the Request/Probe Response/Beacon Payload subfield in the BSSDescSet field of scan request/response will include the new elements listed below:

- High Throughput (HT) capabilities
- HT information
- Extended capabilities
- 20/40 MHz BSS coexistence
- Overlapping BSS scan parameters

For some, the following 802.11n information can be extracted:

- BSS Basic Membership Selector Set
- BSS Basic MCS Set
- HT Operational MCS Set

See the 802.11n D3.02 Specification Subclause 10.3.2.2.2.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SCAN = 0x0006
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
BssType	UINT8	BSS type 0x01 = BSS_INFRASTRUCTURE 0x02 = BSS_INDEPENDENT 0x03 = BSS_ANY
BssId	UINT8[6]	MAC address Filter on BSSID. 0s out to disable filter.
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID set parameter (optional)

Field Name	Type	Description
ChanListParamSet	MrvlIETypes_ChanListParamSet_t	Channel to scan list set parameter
OpRateSet	MrvlIETypes_RatesParamSet_t	Supported data rates set parameter (optional)
NumProbes	MrvlIETypes_NumProbes_t	Number of probes to send (optional)
WildCardSSIDParamSet	MrvlIETypes_WildCardSSIDParamSet_t	WildCard SSID parameters (optional)
WPSProbeRequest	MrvlIETypes_WPSEnrolleeProbeReqIE_t	WPS enrollee probe request contents (used for WPS enrollee)
VendorParamsSet	MrvlIETypes_VendorParamSet_t	Vendor specific type/length extended IEEE IE (The TLV ID is 0x00DD and 2 byte length) This command also has TLVs for 802.11n specific IEs like HTCap.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SCAN | 0x8000
Size	UINT16	Number of bytes in command body Indicates the entire command response size, including the TSFTable.
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
BufSize	UINT16	Scan response buffer size Stores only the scan response buffer size (BssDescSet field), and does not include the size of the TSFTable.
NumOfSet	UINT8	Number of APs in the buffer
BssDescSet	UINT8[]	Variable size buffer that keeps information related to the scanned APs
TSFTable	MrvlIETypes_TsfTimestamp_t	TSF timestamp table There are the same number of TSF values as there are NumOfSet BssDescSets in the command response. TSF Timestamp table has the firmware TSF value for each scan response that is received. The TSFTable is a TLV with a value of 0x0113.
ChannelBandTable	MrvlIETypes_ChanBandList_t	Channel/band table There are the same number of channel/band sets as there are NumOfSet BssDescSets in the command response.

Note: If the firmware buffer has been depleted, firmware will drop the lowest signal strength result currently stored to add a network with a better signal quality.

BssDescSet Format

Field Name	Type	Description
IELength	UINT16	Total IE length
BSSID	UINT8[6]	BSSID
Rssi	UINT8	RSSI value as received from peer
Probe Response/Beacon Payload	UINT8[]	Fixed and variable length IE received on probe response or beacon frames

Probe Response/Beacon Payload Format

Field Name	Type	Description
PktTimeStamp	UINT8[8]	Timestamp
BcnInterval	UINT16	Beacon interval
CapInfo	UINT16	Capabilities information
IEParameters	UINT8[]	Information element parameters

Note: Additional TLVs may optionally be returned at the end of the scan result buffer.

6.7.2 CMD_802_11_BG_SCAN_CONFIG

This command gets/sets the current background scan configuration.

For 802.11n, the MrvlIETypes_ChanListParamSet_t TLV in background scan request is added with the 2-bit SecChanOffset subfield.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BG_SCAN_CONFIG = 0x006B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (all except PPS/UAPSD configuration) 0x0001 = ACT_SET (all except PPS/UAPSD configuration) 0x0100 = ACT_GET (PPS/UAPSD configuration) 0x0101 = ACT_SET (PPS/UAPSD configuration) 0xFF01 = ACT_SET (all configuration)
Enable	UINT8	Enable background scanning 0x00 = disable 0x01 = enable
BssType	UINT8	BSS type 0x01 = BSS_INFRASTRUCTURE 0x02 = BSS_INDEPENDENT 0x03 = BSS_ANY

Field Name	Type	Description
ChannelsPerScan	UINT8	Number of channels scanned during each scan
Reserved	UINT8	Reserved (set to 0)
Reserved	UINT16	Reserved (set to 0)
ScanInterval	UINT32	Interval between consecutive scans
Reserved	UINT32	Reserved (set to 0)
ReportConditions	UINT32	Conditions to trigger report to host Bit[31]: wait for all channel scan to complete to report scan result Bit[2]: SSID match AND RSSI threshold exceeded Bit[1]: SSID match AND SNR threshold exceeded Bit[0]: SSID match
Reserved	UINT16	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BG_SCAN_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Action 0x0000 = ACT_GET (all except PPS/UAPSD configuration) 0x0001 = ACT_SET (all except PPS/UAPSD configuration) 0x0100 = ACT_GET (PPS/UAPSD configuration) 0x0101 = ACT_SET (PPS/UAPSD configuration) 0xFF01 = ACT_SET (all configuration) ACT_GET is used to query the current background scan configuration. ACT_SET is used to change the current background scan configuration.
Enable	UINT8	Enable background scanning 0x00 = disable 0x01 = enable If the background scan is disabled, it aborts the current scan in progress.
BssType	UINT8	BSS type 0x01 = BSS_INFRASTRUCTURE 0x02 = BSS_INDEPENDENT 0x03 = BSS_ANY BssType determines whether only infrastructure BSS, IBSS or all BSS descriptions are stored as a part of the background scan results.
ChannelsPerScan	UINT8	Number of channels scanned during each scan instance ChannelsPerScan determines the duration the station stays away from its operational channel, so this must be set judiciously. Specifically, the total time spent in 1 scan instance should be less than the listen interval of the station.
Reserved	UINT8	Reserved (set to 0)
Reserved	UINT16	Reserved (set to 0)
ScanInterval	UINT32	Interval between 2 consecutive scan instances (in ms)
Reserved	UINT32	Reserved (set to 0)
ReportConditions	UINT32	Conditions to trigger report to host Bit[31]: Wait for all channel scan to complete to report scan result Bit[2]: SSID match AND RSSI threshold exceeded Bit[1]: SSID match AND SNR threshold exceeded Bit[0]: SSID match Determines when firmware generates a background scan report event to the host. Any condition set in the bitmap triggers a report to the host.
Reserved	UINT16	Reserved (set to 0)

Above fields are followed by variable number of the TLV fields.

Note: If the firmware buffer has been depleted, firmware will drop the lowest signal strength result currently stored to add a network with a better signal quality.

To allow this command to be extensible, there may be 0 or more TLV fields included in the request command at the end. In order to ensure backward compatibility, any TLV field not recognized must be ignored and skipped.

The following TLV fields are currently defined:

- MrvlIETypes_ChanListParamSet_t (required)
- MrvlIETypes_SSIDParamSet_t (optional—Absent TLV matches any SSID)
- MrvlIETypes_NumProbes_t (optional)
- MrvlIETypes_BeaconLowRssiThreshold_t (optional)
- MrvlIETypes_BeaconLowSnrThreshold_t (optional)
- MrvlIETypes_WildCardSSIDParamSet_t (optional)

6.7.3 CMD_802_11_BG_SCAN_QUERY

This command gets the current background scan results. It can be issued at any time by the driver (even if firmware is currently scanning channels as part of an ongoing background scan). Firmware ensures that it generated a coherent response by suitably synchronizing access to the scan results buffer.

For 802.11n, the MrvlIETypes_ChanListParamSet_t TLV in scan request is added with the 2-bit SecChanOffset subfield. Similarly, the MrvlIETypes_ChanBandList_t TLV in scan response is added with the 2-bit SecChanOffset subfield.

For 802.11n, the variable length IEs in the Probe Response/Beacon Payload subfield in the BSSDescSet field of scan response will include the new elements listed below:

- HT capabilities
- HT information
- Extended capabilities
- 20/40 MHz BSS coexistence
- Overlapping BSS scan parameters

From some, the following 802.11n information can be extracted:

- BSS Basic Membership Selector Set
- BSS Basic MCS Set
- HT Operational MCS Set

See the 802.11n D3.02 Specification Subclause 10.3.2.2.2.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BG_SCAN_QUERY = 0x006C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Flush	UINT8	Flush 0x00 = do not flush results 0x01 = flush results Flush determines whether firmware should flush the current background scan results after responding to this command or not.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BG_SCAN_QUERY | 0x8000
Size	UINT16	Number of bytes in command body Indicates the entire command response size, including TSFTable.
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
ReportCondition	UINT32	Report condition(s) matched since last query
BuffSize	UINT16	Scan response buffer size Stores only the scan response buffer size (BssDescSet field) and does not include the size of the TSFTable.
NumOfSet	UINT8	Number of access points in the buffer
BssDescSet	UINT8[]	Variable size buffer that keeps the information related to the scanned access points
TSFTable	MrvlIETypes_TsfTimestamp_t	TSF timestamp table There are the same number of TSF values as there are NumOfSet BssDescSets in the command response. TSF timestamp table has the firmware TSF value for each scan response that is received. The TSFTable is a TLV with a value of 0x0113.
ChannelBandTable	MrvlIETypes_ChanBandList_t	Channel/band table There are the same number of channel/band sets as there are NumOfSet BssDescSets in the command response.

The response contains all the background scan results currently stored in firmware, plus the conditions that were met since the last report.

This command can be issued at any time by the driver (even if firmware is currently scanning some channel as part of the ongoing background scanning). Firmware will ensure that it generated a coherent response by suitably synchronizing access to the scan results buffer.

BssDescSet Format

Field Name	Type	Description
IELength	UINT16	Total IE length
BSSID	UINT8[6]	BSSID
Rssi	UINT8	RSSI value as received from peer
Probe Response/Beacon Payload	UINT8[]	Fixed and variable length IE received on probe response or beacon frames

Probe Response/Beacon Payload Format

Field Name	Type	Description
PktTimeStamp	UINT8[8]	Timestamp
BcnInterval	UINT16	Beacon interval

Field Name	Type	Description
CapInfo	UINT16	Capabilities information
IEParameters	UINT8[]	Information element parameters

6.7.4 Background Scan Report Event

This event is generated from firmware to the host whenever any of the report conditions are matched. The driver should issue the background scan query command when it receives this event.

> ***Note:*** See Background Scan Application Note.

6.7.5 CMD_802_11_SCAN_EXT

This command will send the scan results to host by SCAN_REPORT_BY_EVENT events which provides more scan results.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SCAN_EXT = 0x0107
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Reserved	UINT32	Reserved (set to 0)
BssIdList	MrvlIETypes_BSSIDList_t	BSSID parameter (optional)
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID set parameter (optional)
ChanListParamSet	MrvlIETypes_ChanListParamSet_t	Channel to scan list set parameter
OpRateSet	MrvlIETypes_RatesParamSet_t	Supported data rates set parameter (optional)
NumProbes	MrvlIETypes_NumProbes_t	Number of probes to send (optional)
WildCardSSIDParamSet	MrvlIETypes_WildCardSSIDParamSet_t	WildCard SSID parameters (optional)
BssMode	MrvlIEBSSScanType_t	BSS mode parameter (optional)
Tlv	MRVL_SCAN_CHAN_GAP_TLV_ID	Gap time in every scan channel set
Tlv	MRVL_WIFI_PROBERSP_ONLY_TLV_ID	Used in WPS feature

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SCAN_EXT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Reserved	UINT32	Reserved (set to 0)
ChannelStats	MrvlIEChannel_Stats_t	Channel statistics

6.8 Network Start/Stop/Join

Table 13 lists the supports network start, stop, and join commands.

Table 13: Network Start/Stop/Join Commands

Command	Description	Page
Section 6.8.1, CMD_802_11_ASSOCIATE	Initiates an association with the AP	page 77
Section 6.8.2, CMD_802_11_AD_HOC_START	Starts an Ad-Hoc network	page 82
Section 6.8.3, CMD_802_11_AD_HOC_JOIN	Joins an Ad-Hoc network	page 84
Section 6.8.4, CMD_802_11_AD_HOC_STOP	Stops Ad-Hoc network	page 85
Section 6.8.5, CMD_802_11_IBSS_COALESCING_STATUS	Combines Ad-Hoc networks	page 86
Section 6.8.6, CMD_802_11_DEAUTHENTICATE	Starts de-authentication process with the AP	page 87
Section 6.8.7, CMD_SET_BSS_MODE	Sets BSS mode	page 88
Section 6.8.8, CMD_802_11_NET_MONITOR	Sets net monitor (promiscuous) configuration	page 89
Section 6.8.9, CMD_802_11_RX_MGMT_INDICATION	Starts/stops management frame forwards to host datapath	page 90
Section 6.8.10, CMD_FAST_LINK_SYNC_QUERY	Gets fast link synchronization operation information/status	page 93

6.8.1 CMD_802_11_ASSOCIATE

This command passes all needed information from driver to firmware to build the association request packet.

For 802.11n, the MrvlIETypes_ChanListParamSet_t TLV in scan request is added with the 2-bit SecChanOffset subfield.

For 802.11n, the variable number of IEs in the IE buffer subfield of the request/response will include these new elements:

- HT capabilities
- HT information
- 20/40 MHz BSS coexistence
- Overlapping BSS scan parameters
- Extended capabilities

See the 802.11n D3.02 Specification Subclause 7.2.3.5.

To support 802.11ac, this command appends the Very High Throughput (VHT) capabilities TLV when 802.11ac is enabled.

To support 802.11ax, this command appends the IEEE_EXTENSION TLV with High Efficiency(HE) capabilities when 802.11ax is enabled.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_ASSOCIATE = 0x0012
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num

Field Name	Type	Description
Result	UINT16	Not used (set to 0)
PeerStaAddr	UINT8[6]	Peer MAC address
CapInfo	UINT16	Capability information
ListenInterval	UINT16	Listen interval
BcnPeriod	UINT16	Beacon period
DtimPeriod	UINT8	DTIM period
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID set parameter
PhyParamDSSet	MrvlIETypes_PhyParamDSSet_t	Specifies DS parameters
CfParamSet	MrvlIETypes_CfParamSet_t	Specifies CF parameters
OpRateSet	MrvlIETypes_RatesParamSet_t	Supported data rates set parameter
AuthenticationType	MrvlIETypes_AuthType_t	Specify the authentication type to be used in Auth messaging with the AP before the association attempt
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN type/length extended IEEE IE (See Section 10, TLV Usage, on page 300 only when RSN is required)
VendorParamsSet	MrvlIETypes_VendorParamSet_t	Vendor specific type/length extended IEEE IE (The TLV ID is 0x00DD and 2 byte length) This is optional for host to use.
PowerCapability	MrvlIETypes_LocalPowerCapabiity_t	Set minimum/maximum power capability (optional)
SupportedChannelsSet	MrvlIETypes_SupportedChannels_t	802.11h supported channel element Only when 802.11h is required.
PassThroughIEs	MrvlIETypes_Passthrough_t	Add IEEE IEs to the association request directly (optional)
ChanListParamSet	MrvlIETypes_ChanListParamSet_t	Specify both the band and channel for association. Required for multi-band (802.11a/g/b) Optional for 802.11g/b only
WAPI_IE	MrvlIETypes_WAPI_IE_t	WAPI IE must add when enabling WAPI
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN IE must add when embedded supplicant enabled
WPAParamSet	MrvlIETypes_VendorParamSet_t	WPA in VendorIE must add when embedded supplicant enabled
Local_Power_Constraint	MrvlIETypes_LocalPowerCapabiity_t	802.11h local power constraint information
VHT_Capability_IE	MrvlIETypes_VHT_Capability_IE_t	VHT capability IE must be added to enable 802.11ac
HE Capabilities IE	MrvlIETypes_IEEE_Extension_t	variable length TLV MRVL_IEEE_EXTENSION_TLV_ID is used to indicate HE MAC capabilities, HE PHY capabilities, and so on. See Table 64. This IE must be added to enable 802.11ax

RESPONSE (Successful Association)

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_ASSOCIATE | 0x8000
Size	UINT16	Number of bytes in command body

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number (as sent by the host)</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code</td></tr>
<tr><td>AssocRsp</td><td>IEEEtypes_AssocRsp_t (Section 7.2.3.5: IEEE 802.11—1999 Spec)</td><td>Entire IEEE (re)association response
Format is
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Capability</td><td>UINT16</td><td>If association response is received from AP:
IEEEtypes_CapInfo_t
(Section 7.3.1.4: IEEE 802.11—1999 Spec)
If association response is not received from AP:
0xFFFC = timeout waiting for AP response
0xFFFD = authentication refused
0xFFFE = authentication unhandled message
0xFFFF = internal error</td></tr>
<tr><td>StatusCode</td><td>UINT16</td><td>If association response is received from AP:
IEEEtypes_StatusCode_t
(Section 7.3.1.9: IEEE 802.11—1999 Spec)
If association response is not received from AP:
If Capability = 0xFFFC:
Reason for timeout is indicated as follows:
0x01 = association response timeout
0x02 = authentication response timeout
0x03 = network join timeout (waiting for initial beacon reception)
If Capability = 0xFFFD:
Authentication frame was received with a failure indicated. IEEE status code from the failed authentication frame sent from the AP is returned.
If Capability = 0xFFFE:
Authentication frame was received but was not handled by firmware. IEEE status code is returned, reflecting the error. Expected values are:
0x13 = unsupported authentication algorithm indicated
0x14 = invalid sequence number
If Capability = 0xFFFF:
0x01 = indicates an error processing the authentication/association in firmware.</td></tr>
<tr><td>AId</td><td>UINT16</td><td>If association response is received from AP:
IEEEtypes_AId_t
(Section 7.3.1.8: IEEE 802.11—1999 Spec)
If association response is not received from AP:
0xFFFF</td></tr>
<tr><td>IEBuffer</td><td>UINT8[]</td><td>If association response is received from AP:
The rest of the (re)association response
If association response is not received from AP:
Not present</td></tr>
</table></td></tr>
</table>

Field Name	Type	Description
VHT_ Capability_IE	MrvlIETypes_VHT_ Capability_IE_t	VHT capability IE must be added to enable 802.11ac
HE_ Capability_IE	MrvlIETypes_IEEE_ Extension_t	HE capability IE must be added to enable 802.11ax

RESPONSE (Unsuccessful Association)

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_ASSOCIATE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Capability	UINT16	When AID field in the response is 0xFF, this field will contain an error code 0xFFFF (-1) = internal error 0xFFFE (-2) = authentication frame received but unhandled 0xFFFD (-3) = authentication frame received but Authentication Refused indicated 0xFFFC (-4) = timeout
StatusCode	UINT16	Non-0 status code follow the IEEE specifications Status codes are based on Capability value: Capability (-1): 0x0001 = internal processing error Capability (-2): IEEE status code is returned reflecting the error 0x0013 = unsupported Auth Alg Indicated 0x0014 = invalid Sequence Number Capability (-3): IEEE status code from the failed authentication frame sent from the AP is returned Capability (-4): Timeout failure point indicated here as follows: 0x0001 = association response timeout 0x0002 = authentication response timeout 0x0003 = network join timeout (initial beacon reception)
AssociationID	UINT16	IEEEtypes_AId_t
IE Buffer	UINT8[]	Variable number of IEs returned in the association response following the AID
VHT_ Capability_IE	MrvlIETypes_VHT_ Capability_IE_t	VHT capability IE must be added to enable 802.11ac

Whether an association response was received or not can be determined by comparing the AId and/or Capability fields, as noted above. When an association response is received, the command response contains the exact same fields, IEs, and values as the response received from the AP. Firmware does not modify the data before passing it on to the driver. See the IEEE specification for more detailed information regarding the potential response values.

If firmware is currently in an associated state, the reception of this command will be considered as a reassociation request to the specified access point. (Even if it is the first association, the AP will accept a reassociation request in place of an association request.) Also, firmware will continue its original association if the new reassociation request fails to complete.

Make-before-break functionality is added in firmware. If firmware is associated to an AP, a new association request will be interpreted as a roam attempt. Firmware will send a powersave PM=1 notification to its associated AP and attempt an association to the AP requested in the association command. If the association fails, firmware will return to the original AP's channel and send a PM=0 notification.

A failure is still returned in the command response, but firmware remains associated to the original AP.

If a reassociation attempt is not desired, the host must first send a CMD_802_11_DEAUTHENTICATE command to firmware to break its current association before issuing this command.

6.8.2 CMD_802_11_AD_HOC_START

This command is used to start an Ad-Hoc network.

In the Ad-Hoc mode, the value of Announced Traffic Indication Message (ATIM) window controls whether power save operation is permitted in the Infrastructure BSS (IBSS) or not. If the value of ATIM window is 0, then the participating stations are not permitted to enter Power Save mode. If the value of ATIM window is non-0, then the use of Power Save mode is enabled in the IBSS. The value of ATIM window is chosen by the station that starts the IBSS, and the same value is adopted by all stations that join the IBSS.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_START = 0x002B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
SSID	UINT8[32]	SSID
BssType	UINT8	BSS type to use: BSS_INDEPENDENT
BcnPeriod	UINT16	Specify beacon period in TU
Reserved	UINT8	Reserved (set to 0)
IbssParamSet	IEEEtypes_IbssParamSet_t	ATIM window length in TU
Reserved	UINT8[4]	Reserved (set to 0)
DsParamSet	IEEEtypes_DsParamSet_t	Specifies the channel for Ad-Hoc network
Reserved	UINT8[4]	Reserved (set to 0)
Reserved	UINT16	Reserved (set to 0)
CapInfo	IEEEtypes_CapInfo_t	Capability information
Data Rate	UINT8[14]	Supported data rates Data rate are in multiples of 500 Kbps. Basic data rates have MSB set to 1, for example: 0x96 = 11 Mbps 0x8B = 5.5 Mbps 0x84 = 2 Mbps 0x82 = 1 Mbps
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN type/length IEEE extended IE (optional)

Field Name	Type	Description
VendorParamSet	MrvlETypes_VendorParamSet_t	Vendor specific type/length IEEE extended IE (The TLV ID is 0x00DD and 2 byte length) This is optional for host to use.
PassThroughIEs	MrvlIETypes_Passthrough_t	Add IEEE IEs to the Ad-Hoc beacon directly (optional)
ChanListParamSet	MrvlIETypes_ChanListParamSet_t	Channel parameters (required if 802.11a is used)
IbssDfs	MrvlETypes_Ibss_Dfs_t	Station originated IBSS DFS element with the DFS owner and recovery time from the existing IBSS (802.11h only)
QuietPeriod	MrvlETypes_Quiet_t	Quiet period for the new IBSS (802.11h only)
Local_Power_Constraint	MrvlIETypes_LocalPowerCapabiity_t	802.11h local power constraint information

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_START | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Local_Power_Constraint	MrvlIETypes_LocalPowerCapabiity_t	802.11h local power constraint information

6.8.3 CMD_802_11_AD_HOC_JOIN

This command is used to join an Ad-Hoc network.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_JOIN = 0x002C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
BssDesc	IEEEtypes_BssDesc_t	BSS descriptor
Reserved	UINT16	Reserved (set to 0)
Reserved	UINT16	Reserved (set to 0)
IbssDfs	MrvlIETypes_Ibss_Dfs_t	Station originated IBSS DFS element with the DFS owner and recovery time from the existing IBSS (802.11h only)
QuietPeriod	MrvlIETypes_Quiet_t	Quiet period for the new IBSS (802.11h only)
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN type/length IEEE extended IE (optional)
VendorParamSet	MrvlIETypes_VendorParamSet_t	Vendor specific type/length IEEE extended IE (The TLV ID is 0x00DD and 2 byte length) This is optional for host to use.
PassThroughIEs	MrvlIETypes_Passthrough_t	Add IEEE IEs to the Ad-Hoc beacon directly (optional)
ChanListParamSet	MrvlIETypes_ChanListParamSet_t	Channel parameters (required if 802.11a is used)
Local_Power_Constraint	MrvlIETypes_LocalPowerCapabiity_t	802.11h local power constraint information

REPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_JOIN | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Local_Power_Constraint	MrvlIETypes_LocalPowerCapabiity_t	802.11h local power constraint information

Table 14: IEEEtypes_BssDesc_t

Field Name	Data Type	Description
BssId	UINT8[6]	MAC address
SsId	UINT8[32]	SSID
BssType	IEEEtypes_Bss_e	BSS type 0x1 = BSS_INFRASTRUCTURE 0x2 = BSS_INDEPENDENT 0x3 = BSS_ANY (recommended for use when joining Ad-Hoc networks)
BcnPeriod	UINT16	Beacon period
DtimPeriod	UINT8	Specify DTIM period (TBTTs)
Timestamp	UINT8[8]	Timestamp
StartTs	UINT8[8]	Starting timestamp
DsParamSet	IEEEtypes_DsParamSet_t	IEEE DS parameter set element
Reserved	UINT8[4]	Reserved (set to 0)
IbssParamSet	IEEEtypes_IbssParamSet_t	IEEE IBSS parameter set
Reserved	UINT8[4]	Reserved (set to 0)
CapInfo	UINT16	Firmware capability information
DataRates	UINT8[14]	Data rates

6.8.4 CMD_802_11_AD_HOC_STOP

This command is used to stop an Ad-Hoc network. Firmware exits Ad-Hoc mode and changes to idle mode.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_STOP = 0x0040
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AD_HOC_STOP | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code

6.8.5 CMD_802_11_IBSS_COALESCING_STATUS

This command is used to configure IBSS Coalescing settings. Firmware sends an IBSS_Coalesced MAC event to the host and the host responds to the event by sending this command.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_IBSS_COALESCING_STATUS = 0x0083
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Enable	UINT16	IBSS coalescing 0x0000 = IBSS coalescing disabled 0x0001 = IBSS coalescing enabled (default)
BSSID	UINT8[6]	Not used (set to 0)
BeaconInterval	UINT16	Not used (set to 0)
ATIMWindow	UINT16	Not used (set to 0)
UseGRateProtection	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_IBSS_COALESCING_STATUS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
Enable	UINT16	IBSS coalescing 0x0000 = IBSS coalescing disabled 0x0001 = IBSS coalescing enabled (default)
BSSID	UINT8[6]	Current BSSID (only valid in Ad-Hoc mode)
BeaconInterval	UINT16	Current beacon interval in TU (1 TU = 1024 µs)

Field Name	Type	Description
ATIMWindow	UINT16	Current ATIM window length in TU
UseGRateProtection	UINT16	Current 802.11g rate protection <table><tr><th>Bits</th><th>Description</th></tr><tr><td>15:2</td><td>Reserved (set to 0)</td></tr><tr><td>1</td><td>0 = 802.11g rate protection off 1 = 802.11g rate protection on</td></tr><tr><td>0</td><td>Reserved (set to 0)</td></tr></table>

IBSS Coalescing allows 2 Ad-Hoc networks with different BSSIDs but the same SSID to join together and become a single Ad-Hoc network. The newer network (with a lower TSF value) will join with the older network.

The host must send this command to firmware in order to turn on the IBSS Coalescing feature.

When IBSS Coalescing process is finished, firmware generates an IBSS_Coalesced MAC event to the host if the BSSID is changed (merged to other IBSS network).

After the host receives the IBSS_Coalesced event, it should issue this command to obtain the BSSID, Beacon Interval, and ATIM window of the merged IBSS network.

6.8.6 CMD_802_11_DEAUTHENTICATE

This command starts the de-authentication process with the AP.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_DEAUTHENTICATE = 0x0024
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
PeerStaAddr	UINT8[6]	Peer MAC address
ReasonCode	UINT16	Reason code defined in IEEE 802.11 Specification Section 7.3.1.7 to indicate de-authentication reason

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_DEAUTHENTICATE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code

Field Name	Type	Description
PeerStaAddr	UINT8[6]	Peer MAC address
DeauthResult	UINT8	Deauthentication result 0x00 = SUCCESS 0x01 = INVALID_PARAMETERS 0x02 = TOO_MANY_RQSTS 0x03 = TIMEOUT

6.8.7 CMD_SET_BSS_MODE

This command is used to set BSS mode.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SET_BSS_MODE = 0x00F7
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Mode	UINT8	Mode 0x00 = INFRA or WIFI_DIRECT_CLIENT 0x01 = Ad-Hoc 0x02 = uAP or WIFI_DIRECT_GO

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SET_BSS_MODE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Mode	UINT8	Mode 0x00 = INFRA or WIFI_DIRECT_CLIENT 0x01 = Ad-Hoc 0x02 = uAP or WIFI_DIRECT_GO

6.8.8 CMD_802_11_NET_MONITOR

This command sets the net monitor (promiscuous) configuration. It supports all features pack (for example, FP57, FP62) except feature pack 65 due to memory usage.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_NET_MONITOR = 0x0102
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
En_NetMonitor	UINT16	Set net monitor activity from host to device 0x0000 = disable 0x0001 = enable
Filter_Flag	UINT16	Set net monitor filter flag Bit[2]: enable/disable control frame receive Bit[1]: enable/disable management frame receive Bit[0]: enable/disable data frame receive
MonitorChannel	MrvlIETypes_ChanBandList_t	Channel to monitor, only support 1 channel

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_NET_MONITOR | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
En_NetMonitor	UINT16	Set net monitor activity from host to device 0x0000 = disable 0x0001 = enable
Filter_Flag	UINT16	Set net monitor filter flag Bit[2]: enable/disable data frame receive Bit[1]: enable/disable control frame receive Bit[0]: enable/disable management frame receive
MonitorChannel	MrvlIETypes_ChanBandList_t	Channel to monitor, only support 1 channel

6.8.9 CMD_802_11_RX_MGMT_INDICATION

This command is used to start/stop the management frame forwards to host through datapath.

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_802_11_RX_MGMT_INDICATION = 0x010C</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Action
0x0000 = ACT_GET (retrieve mask settings)
0x0001 = ACT_SET (set select mask bits)</td></tr>
<tr><td>MgmtSubTypeMask</td><td>UINT32</td><td>Management Subtype Mask
If Bit X is set in mask, it means that IEEE Management Frame SubType X is to be filtered and passed through to host.
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>31:14</td><td>Reserved</td></tr>
<tr><td>13</td><td>Action frame</td></tr>
<tr><td>12:9</td><td>Reserved</td></tr>
<tr><td>8</td><td>Beacon</td></tr>
<tr><td>7:6</td><td>Reserved</td></tr>
<tr><td>5</td><td>Probe response</td></tr>
<tr><td>4</td><td>Probe request</td></tr>
<tr><td>3</td><td>Reassociation response</td></tr>
<tr><td>2</td><td>Reassociation request</td></tr>
<tr><td>1</td><td>Association response</td></tr>
<tr><td>0</td><td>Association request</td></tr>
</table>
Support multiple bits set.
0 = stop forward frame
1 = start forward frame</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_RX_MGMT_INDICATION | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
MgmtSubTypeMask	UINT32	Management Subtype Mask If Bit X is set in mask, it means that IEEE Management Frame SubType X is to be filtered and passed through to host. (see bit table below) Support multiple bits set.

Bit	Description
31:14	Reserved
13	Action frame
12:9	Reserved
8	Beacon
7:6	Reserved
5	Probe response
4	Probe request
3	Reassociation response
2	Reassociation request
1	Association response
0	Association request

6.8.9.1 Rx Management Frame Format

Table 15 shows the Tx management frame format from host through datapath.

Table 15: Tx Management Frame Format

Field Name	Data Type	Description
TxBSSType	UINT8	Type 0x00 = INFRA or ADHOC 0x01 = uAP 0x02 = WIFI_DIRECT_CLIENT or WIFI_DIRECT_GO
TxBSSNum	UINT8	BSS number
TxPacketLength	UINT16	Number of bytes in transmit data packet
TxPacketOffset	UINT16	Offset to beginning of transmit data packet
TxPacketType	UINT16	0x5E = PKT_TYPE_MGT
TxControl	UINT32	Set to 0
...	UINT8[]	Offset
TxMgmtFrame	mgmtFrame_t	Must be 4-byte alignment (see Table 17, mgmtFrame_t Format, on page 92)

- Host is expected to fill the following fields:
  - Type in Frame Control
  - Subtype in Frame Control
  - Addr 1
  - Addr 2
  - Addr 3
- TxPacketOffset shall align on word (4 bytes) boundary to aid firmware to transmit the frame successfully
- Duration and SeqCtl is filled by firmware appropriately

6.8.9.2 Rx Management Frame Format

Table 16 shows the RxPD format.

Table 16: RxPD Format

Field Name	Data Type	Description
RxBSSType	UINT8	Type 0x00 = INFRA or ADHOC 0x01 = uAP 0x02 = WIFI_DIRECT_CLIENT or WIFI_DIRECT_GO
RxBSSNum	UINT8	BSS number
RxPacketLength	UINT16	Number of bytes in transmit data packet
RxPacketOffset	UINT16	Offset to beginning of transmit data packet
RxPacketType	UINT16	0x005E = PKT_TYPE_MGT
...	UINT8[]	Offset
RcvdMgmtFrame	mgmtFrame_t	See Table 17, mgmtFrame_t Format, on page 92

Table 17: mgmtFrame_t Format

Field Name	Data Type	Description
FrmBodyLen	UINT16	Number of bytes
FrmCtl	UINT16	Frame control
Duration	UINT16	--
Address_1	UINT8[6]	--
Address_2	UINT8[6]	--
Address_3	UINT8[6]	--
SeqCtl	UINT16	Sequence control
Address_4	UINT8[6]	Reserved (set to 0)
FrameBody	UINT8[]	Management frame contents

6.8.10 CMD_FAST_LINK_SYNC_QUERY

This command gets the fast link synchronization operation information/status.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FAST_LINK_SYNC_QUERY = 0x0244
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Type	UINT16	Type 0x0001 = INTERFACE_STATUS 0x0002 = STA_INFO 0x0004 = PORT_INFO 0x0008 = STA_LIST_INFO 0x0010 = UAP_INFO

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FAST_LINK_SYNC_QUERY | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Type	UINT16	Type 0x0001 = INTERFACE_STATUS 0x0002 = STA_INFO 0x0004 = PORT_INFO 0x0008 = STA_LIST_INFO 0x0010 = UAP_INFO
Tlv_buffer	TLV	TLV buffer See Table 18, TLV Buffer Type, on page 94.

Table 18: TLV Buffer Type

<table>
<tr><th>Type</th><th>Description</th></tr>
<tr><td>INTERFACE_STATUS</td><td>Interface basic information:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Interface_basic_info</td><td>MrvlIETypes_FAST_LINK_INTERFACE_BASIC_INFO_t</td><td>Interface basic information</td></tr>
</table></td></tr>
<tr><td>STA_INFO</td><td>STA information:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Ssid</td><td>MrvlIETypes_SSIDParamSet_t</td><td>Ext_AP SSID information</td></tr>
<tr><td>Bssid</td><td>MrvlIETypes_BSSIDList_t</td><td>Ext_AP BSSID information</td></tr>
<tr><td>ChanBand</td><td>MrvlIETypes_ChanBandList_t</td><td>Ext_AP channel information</td></tr>
</table></td></tr>
<tr><td>PORT_INFO</td><td>Port information:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Port_info</td><td>MrvlIETypes_FAST_LINK_DATA_PORT_INFO_t</td><td>Port information</td></tr>
</table></td></tr>
<tr><td>STA_LIST_INFO</td><td>STA list information (1 or more):
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Sta_in_uap_info</td><td>MrvlIETypes_FAST_LINK_EXT_STATION_INFO_t</td><td>Information of STA connected to uAP</td></tr>
</table></td></tr>
<tr><td>UAP_INFO</td><td>Micro AP (uAP) information:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Ht_capbility</td><td>IEEEtypes_HTCap_t</td><td>HT capability of connected STA</td></tr>
<tr><td>Wmm_params</td><td>IEEEtypes_WMM_ParamElement_t</td><td>WMM parameters</td></tr>
<tr><td>Vht_capability</td><td>IEEEtypes_VHTCap_t</td><td>VHT capability of connected STA</td></tr>
<tr><td>ChanBand</td><td>MrvlIETypes_ChanBandListParamSet_t</td><td>Ext_AP channel information</td></tr>
<tr><td>APMacAddress</td><td>MrvlIETypes_ApMacAddr_t</td><td>Local MAC address</td></tr>
</table></td></tr>
</table>

6.9 Security

Table 19 lists the supports security commands.

Table 19: Security Commands

Command	Description	Page
Section 6.9.1, CMD_802_11_KEY_MATERIAL	Gets/sets key material used to do Tx encryption or Rx decryption	page 95
Section 6.9.2, CMD_802_11_CRYPTO	Provides AES cipher engine API between firmware and driver	page 97
Section 6.9.3, CMD_SUPPLICANT_PMK	Configures embedded supplicant information	page 99
Section 6.9.4, CMD_SUPPLICANT_PROFILE	Configures supplicant profile	page 101
Section 6.9.5, CMD_CRYPTO	Sets crypto format	page 102

6.9.1 CMD_802_11_KEY_MATERIAL

This command is a generic command used to set/get the key material used to do Tx encryption or Rx decryption of unicast, multicast, and broadcast data packets for:

- Wi-Fi Protected Access (WPA)
- Wi-Fi Protected Access 2 (WPA2)
- Wired Equivalent Privacy (WEP)
- Wi-Fi Authentication and Privacy Infrastructure (WAPI)
- Advanced Encryption Standard (AES) Cipher-based Message Authentication Code (CMAC)

Since this command makes use of variable length IE, it is possible to include multiple KeyParamSetCfg in a single command. It is assumed the driver validates the API command before passing it down to firmware. If there are conflicts in 2 or more KeyParamSetCfg, the latter of the parameter set in the chain takes precedence.

For example, if the command contains a KeyParamSetCfg of type Temporal Key Integrity Protocol (TKIP) (unicast) and a KeyParamSetCfg of type AES (unicast), firmware encrypts/decrypts data packets using the AES key.

This command is used when authenticator is implemented on host.

- Unless the key is configured, no packets will be received from or transmitted to the specified MAC address. The host will use this command to set the unicast and multicast keys.
- See Section 6, Host Commands, on page 18 for details on the definitions of existing TLVs in the command.
- MAC address will be set to all 0xFF for group key.
- GET action will not have any TLVs added by the host. Firmware will return all configured keys.
- To SET multiple keys, the TLV list should be a list of pairs of {KeyParamSet, MacAddress}.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_KEY_MATERIAL = 0x005E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

Field Name	Type	Description
Action	UINT16	Action 0x0000 = ACT_GET: returns all the available keys 0x0001 = ACT_SET
KeyParamSetCfg	KeyParamSet_v1_t or KeyParamSet_v2_t	Supports 2 different TLV configuration See Table 20, KeyParamSet_v1_t Format, on page 96 and Table 21, KeyParamSet_v2_t Format, on page 96.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_KEY_MATERIAL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET 0x0004 = ACT_REMOVE
KeyParamSetCfg	KeyParamSet_v1_t or KeyParamSet_v2_t	Supports 2 different TLV configuration See Table 20 and Table 21.

Table 20: KeyParamSet_v1_t Format

Field Name	Type	Description
KeyParamSet	MrvlIETypes_KeyParamSet_t	Key parameter set(s)
StationMacAddress	MrvlIETypes_StaMacAddr_t	Station MAC address

Table 21: KeyParamSet_v2_t Format

Field Name	Type	Description
KeyParamSet	MrvlIETypes_KeyParamSet_v2_t	Key parameter set(s)

6.9.2 CMD_802_11_CRYPTO

This command provides the AES cipher engine API between firmware and driver. This command also includes the RC4 functionality.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CRYPTO = 0x0078
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
EncDec	UINT16	Encryption 0x0000 = decrypt 0x0001 = encrypt
Algorithm	UINT16	Algorithm 0x0001 = RC4 0x0002 = AES 0x0003 = AES_KEY_WRAP Others = reserved
KeyIVLength	UINT16	Length of KeyIV (bytes)
KeyIV	UINT8[32]	Key IV
KeyLength	UINT16	Length of key (bytes)
Key	UINT8[32]	Key
Data	MrvlIETypes_Crypto_Data_t	Data 0 = cipher text data if EncDec 1 = plain text data if EncDec

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CRYPTO | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result Code
EncDec	UINT16	Encryption 0x0000 = decrypt 0x0001 = encrypt
Algorithm	UINT16	Algorithm 0x0001 = RC4 0x0002 = AES 0x0003 = AES_KEY_WRAP Others = reserved
KeyIVLength	UINT16	Length of KeyIV (bytes)

Field Name	Type	Description
KeyIV	UINT8[32]	Key IV
KeyLength	UINT16	Length of key (bytes)
Key	UINT8[32]	Key
Data	MrvlIETypes_Crypto_Data_t	Data 0 = cipher text data if EncDec 1 = plain text data if EncDec

For AES algorithms:

- KeyIVLength and KeyIV is ignored.
- KeyLength needs to be 16, 24, or 32. Any other values return an error.
- MrvlIETypes_Crypto_Data_t payload length needs to be 16. Any other value returns an error.
- For AES_KEY_WRAP algorithms:
- KeyIVLength should be 8. Any other value returns an error.
  - ENCRYPT (AES Key Wrap) = default KeyIV value can be set to 0xA6A6A6A6A6A6A6A6 per RFC3394 section 2.2.3.1.
  - DECRYPT (AES Key Unwrap) = CMD_802_11_CRYPTO request, the first 8 byte of Ciphertext should be put in the KeyIV field and the rest of the Ciphertext should be put in the Data field. After decryption, it is the host side's responsibility to check the correctness of KeyIV for data integrity in CMD_802_11_CRYPTO response.
- KeyLength needs to be 16, 24, or 32. Any other values return an error.
- MrvlIETypes_Crypto_Data_t payload length needs to be multiple of 8 (bytes). Any other value returns an error.

For RC4 algorithm:

- 1 <= (KeyIVLength + KeyLength) <= 256
- If RC4 is used to decrypt key data from first group key EAPOL-Key frame, both KeyIVLength and KeyLength are 16.
- The payload length in data field is 256 + length of key material to extract.
- After decryption, data payload field should be plaintext key material pre-pended with 256 byte decrypted filler. It is the host side's responsibility to discard this 256-byte decrypted filler and keep the plain text key material.

6.9.3 CMD_SUPPLICANT_PMK

This command is used to configure embedded supplicant.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SUPPLICANT_PMK = 0x00C4
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET 0x0004 = ACT_CLEAR
CacheResult	UINT16	Not used (set to 0)
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID set parameter
BSSIDList	MrvlIETypes_BSSIDList_t	BSSID list will have a single BSSID
PMK	MrvlIETypes_PMK_t	PMK information
Passphrase	MrvlIETypes_Passphrase_t	Passphrase information

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SUPPLICANT_PMK | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
CacheResult	UINT16	Cache operation result (0 for success)
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID set parameter
BSSIDList	MrvlIETypes_BSSIDList_t	BSSID list will have a single BSSID
PMK	MrvlIETypes_PMK_t	PMK information
Passphrase	MrvlIETypes_Passphrase_t	Passphrase information

Accepted TLV Combinations for ACT_SET

The command will allow the following combinations of TLVs. Any set command will enable the embedded supplicant.

- Passphrase TLV—When only this TLV is present, the passphrase used for the Pairwise Master Key (PMK) applies to all SSIDs. For this configuration, firmware will generate a new PMK when the SSID of the target AP changes. The generation of a PMK is a time consuming process and could add considerable delay to an association. If this configuration is used, the command timeout for the association will have to be increased in the driver.
- Passphrase and SSID TLV—When these TLVs are present, firmware will immediately generate a PMK from the SSID and passphrase. The PMK generated will be used for associations matching the SSID.
- PMK and BSSID list TLV—When these TLVs are present, the PMK will be stored and will be sued when associating to the given SSID.
- PMK and SSID TLV—When these TLVs are present, the Pre-Shared Key (PSK) will be stored and will be used when associating to the given SSID.
- No TLVs present—This command will enable the supplicant for use with an enterprise supplicant where the PMK is not yet known. Once enabled, the embedded supplicant stays enabled until explicitly disabled. Enabling the embedded supplicant before the PMK is available is required so the embedded supplicant can parse and generate the WPA or Robust Secure Network (RSN) IE required in the association message and 4-way handshake.

Accepted TLV Combinations for ACT_GET

- BSSID list TLV present—Firmware will access the PMK cache and if the BSSID is found, returns a PMK TLV, otherwise, no TLVs will be added to the result.
- SSID LTV present—Firmware will access the PSK cache and if the SSID is found, return a PMK TLV with the PSK filled in, otherwise no TLVs will be added to the result.
- Passphrase TLV present—Firmware will return the value of the global PSK passphrase.

Accepted TLV Combinations for ACT_CLEAR

- No TLV present—Firmware will clear all cached PMK and PSK values. After the clear, the embedded supplicant will be disabled.
- BSSID list TLV present—Firmware will access the PMK cache and if the BSSID is found, clear the entry. This will leave the embedded supplicant enabled.
- SSID TLV present—Firmware will access the PSK cache and if the SSID is found, clear the entry. This will leave the embedded supplicant enabled.
- Passphrase TLV present—The ASCII passphrase stored in firmware will be cleared.

6.9.4 CMD_SUPPLICANT_PROFILE

This command is used to configure supplicant profile.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SUPPLICANT_PROFILE = 0x00C5
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET 0x0005 = ACT_GET_CURRENT **NOTE:** For ACT_SET, both WPA Protocol TLV and Cipher TLV must be present. For ACT_GET and ACT_GET_CURRENT, the driver sends the command down with no TLVs present and firmware will add a WPA Protocol TLV and a Cipher TLV.
Reserved	UINT16	Not used (set to 0)
CipherInfo	MrvlIETypes_Cipher_t	Cipher suites information
ProtocolInfo	MrvlIETypes_EncryptionProtocol_t	Protocol used information

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SUPPLICANT_PROFILE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
Reserved	UINT16	Not used (set to 0)
CipherInfo	MrvlIETypes_Cipher_t	Cipher suites information
ProtocolInfo	MrvlIETypes_EncryptionProtocol_t	Protocol used information

Association Command Usage:

- If the embedded supplicant is in a quiet period due to counter measures, all association attempts during the quiet time will return an association response with a Join Failure result.
- When the embedded supplicant is used, if no RSN or WPA IE is provided by the enterprise supplicant, the driver must add both the WPA and RSN IE, if available, from the scan result to the association command.
- The 4 way handshake handled by the embedded supplicant is guarded by a 10 seconds guard timer. If the handshake fails to complete, at the expiration of the timer, the embedded supplicant will send a Deauth message to the AP as well as a Deauth Indication to the driver. Both messages will have the reason indicating 4-way Handshake Failed.
- Embedded supplicant cache will support 1 PMK minimum. For the initial implementation, the PMK cache supports 10 entries.
- After the group key has been negotiated, firmware generates a controlled port open event to the driver to indicate that data can be sent on the connection.

6.9.5 CMD_CRYPTO

This command sets the crypto format. This command follows the host driver command with the sub command contained in its payload. The sub command starts with the sub command code, followed by sub command payload with variable TLV data.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CRYPTO = 0x025E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = request crypto processing
Sub Cmd Code	UINT8	Sub command code 0x01 = CRYPTO_SUBCMD_PRF_HMAC_SHA1 0x02 = CRYPTO_SUBCMD_HMAC_SHA1 0x03 = CRYPTO_SUBCMD_HMAC_SHA256 0x04 = CRYPTO_SUBCMD_SHA256 0x05 = CRYPTO_SUBCMD_RIJNDAEL 0x06 = CRYPTO_SUBCMD_RC4 0x07 = CRYPTO_SUBCMD_MD5 0x08 = CRYPTO_SUBCMD_MRVL_5 0x09 = CRYPTO_SUBCMD_SHA256_KDF
Sub Cmd	Variable	Sub command data

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CRYPTO | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number

Field Name	Type	Description
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set as in request message
Sub Cmd Code	UINT8	Sub command code 0x01 = CRYPTO_SUBCMD_PRF_HMAC_SHA1 0x02 = CRYPTO_SUBCMD_HMAC_SHA1 0x03 = CRYPTO_SUBCMD_HMAC_SHA256 0x04 = CRYPTO_SUBCMD_SHA256 0x05 = CRYPTO_SUBCMD_RIJNDAEL 0x06 = CRYPTO_SUBCMD_RC4 0x07 = CRYPTO_SUBCMD_MD5 0x08 = CRYPTO_SUBCMD_MRVL_5 0x09 = CRYPTO_SUBCMD_SHA256_KDF
Output Len	UINT16	Output length of crypto operation
Output	Variable	Specified length of output data required in corresponding sub crypto command

6.9.5.1 CRYPTO_SUBCMD_PRF_HMAC_SHA1

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_PRF_HMAC_SHA1 = 0x01
Output Len	UINT16	Placeholder for command response output length
Ctypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Prefix_TLV	Variable	TLV_TYPE_CRYPTO_PREFIX
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_PRF_HMAC_SHA1 = 0x01
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Hashed data of Output Len (octets)

6.9.5.2 CRYPTO_SUBCMD_HMAC_SHA1

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_HMAC_SHA1 = 0x02
Output Len	UINT16	Placeholder for command response output length
Data_Blcks_nr	UINT16	Number of data blocks
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK This TLV is repeated Data_Blcks_nr times.

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_HMAC_SHA1 = 0x02
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Hashed data of Output Len (octets)

6.9.5.3 CRYPTO_SUBCMD_HMAC_SHA256

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_HMAC_SHA256 = 0x03
Output Len	UINT16	Placeholder for command response output length
Data_Blcks_nr	UINT16	Number of data blocks
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK This TLV is repeated Data_Blcks_nr times.

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_HMAC_SHA256 = 0x03
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Hashed data of Output Len (octets)

6.9.5.4 CRYPTO_SUBCMD_SHA256

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_SHA256 = 0x04
Output Len	UINT16	Placeholder for command response output length
Data_Blcks_nr	UINT16	Number of data blocks
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK This TLV is repeated Data_Blcks_nr times.

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_SHA256 = 0x04
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Hashed data of Output Len (octets)

6.9.5.5 CRYPTO_SUBCMD_RIJNDAEL

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_RIJNDAEL = 0x05
Output Len	UINT16	Placeholder for command response output length
Sub Command Action	UINT8	Sub command action 0x00 = decrypt (not used) 0x01 = set key for decrypt 0x02 = encrypt (not used) 0x03 = set key for encrypt
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_RIJNDAEL = 0x05
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Encrypted/decrypted data of Output Len (octets)

6.9.5.6 CRYPTO_SUBCMD_RC4

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_RC4 = 0x06
Output Len	UINT16	Placeholder for command response output length
SkipBytes	UINT16	Iteration times for RC4 key mixing, normally it is 256
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_KeyIV_TLV	Variable	TLV_TYPE_CRYPTO_KEY_IV
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_RC4 = 0x06
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Encrypted/decrypted data of Output Len (octets)

6.9.5.7 CRYPTO_SUBCMD_MD5

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_MD5 = 0x07
Output Len	UINT16	Placeholder for command response output length
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_MD5 = 0x07
Output Len	UINT16	Required output length equals output length in command
Output	Variable	128-bit hashed data

6.9.5.8 CRYPTO_SUBCMD_MRVL_5

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_MRVL_5 = 0x08
Output Len	UINT16	Placeholder for command response output length
Iterations	UINT32	Number of times of HMAC_SHA1 processing
Count	UINT32	Integer value of hashed along with data
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_MRVL_5 = 0x08
Output Len	UINT16	Required output length equals output length in command
Output	Variable	160-bit hashed data

6.9.5.9 CRYPTO_SUBCMD_SHA256_KDF

REQUEST

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_SHA256_KDF = 0x09
Output Len	UINT16	Placeholder for command response output length
Crypto_Key_TLV	Variable	TLV_TYPE_CRYPTO_KEY
Crypto_Prefix_TLV	Variable	TLV_TYPE_CRYPTO_PREFIX
Crypto_Data_BLK_TLV	Variable	TLV_TYPE_CRYPTO_DATA_BLK

RESPONSE

Field Name	Type	Description
Sub Cmd Code	UINT8	CRYPTO_SUBCMD_SHA256_KDF = 0x09
Output Len	UINT16	Required output length equals output length in command
Output	Variable	Hashed data of Output Len (octets)

6.10 Rate Adaptation

Table 19 lists the rate adaptation commands.

Table 22: Rate Adaptation Commands

Command	Description	Page
Section 6.10.1, CMD_TX_RATE_QUERY	Reports the current Tx rate of the first packet associated with rate adaptation	page 108
Section 6.10.2, CMD_TX_RATE_CFG	Gets/sets 802.11n transmission data rate	page 111
Section 6.10.3, CMD_PACKET_AGGR_CTRL	Gets/sets software Tx/Rx aggregation configuration parameters in firmware	page 112

6.10.1 CMD_TX_RATE_QUERY

This command reports the current Tx rate of the first packet associated with rate adaptation.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_RATE_QUERY = 0x007F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
TxRate	UINT8	Not used (set to 0)
TxRate_Info	UINT8	Not used (set to 0)
HeRateInfo	UINT8	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_RATE_QUERY | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
TxRate	UINT8	Return Tx data rate value

Field Name	Type	Description
TxRate_Info	UINT8	Return Tx packet rate information
HeRateInfo	UINT8	Bits[0]: DCM Bits[1:3]: tone mode 000: 26 tone 001: 52 tone 002: 106 tone 003: 242 tone 004: 484 tone 005: 996 tone Bits[7:4]: Reserved

Table 23: Tx packet HT/VHT/HE information

Bits	Description
7	[Bit7][Bit4] 11ax GI 00 = 1xHELTF + GI0.8us 01 = 2xHELTF + GI0.8us 10 = 2xHELTF + GI1.6us 11 = 4xHELTF + GI0.8us if DCM = 0 and STBC = 0 11 = 4xHELTF + GI3.2us, otherwise
6	LDPC support
5	STBC support
4	HT/VHT guard interval 0 = long guard interval 1 = short guard interval
3:2	HT/VHT bandwidth 00 = BW20 01 = BW40 10 = BW80 11 = BW160
1:0	RxRate format 00 = legacy rates 01 = HT rates 10 = VHT rate 11 = HE rate

Table 24: Tx data rate value

Value	Data Rate (TxRateInfo Bits[1:0] = 00)	Data Rate (TxRateInfo Bits[1:0] = 01)
0x0	1 Mbps	MCS0
0x1	2 Mbps	MCS1
0x2	5.5 Mbps	MCS2
0x3	11 Mbps	MCS3
0x4	Reserved	MCS4
0x5	6 Mbps	MCS5
0x6	9 Mbps	MCS6
0x7	12 Mbps	MCS7

Table 24: Tx data rate value (Continued)

Value	Data Rate (TxRateInfo Bits[1:0] = 00)	Data Rate (TxRateInfo Bits[1:0] = 01)
0x8	18 Mbps	MCS8
0x9	24 Mbps	MCS9
0xA	36 Mbps	MCS10
0xB	48 Mbps	MCS11
0xC	54 Mbps	MCS12
0xD and up	Reserved	MCS13 ... MCSn

Table 25: VHT and HE Rate Value

TxRate[7:4] (TxRateInfo Bits[1:0] = 10/11)	Number of Spatial Stream	TxRate[3:0] (TxRateInfo Bits[1:0] = 10)	VHT MCS Index	TxRate[3:0] (TxRateInfo Bits[1:0] = 11)	HE MCS index
0x0	Nss=1	0x0	MCS0	0x0	MCS0
0x1	Nss=2	0x1	MCS1	0x1	MCS1
0x2	Reserved	0x2	MCS2	0x2	MCS2
0x3	Reserved	0x3	MCS3	0x3	MCS3
0x4	Reserved	0x4	MCS4	0x4	MCS4
0x5	Reserved	0x5	MCS5	0x5	MCS5
0x6	Reserved	0x6	MCS6	0x6	MCS6
0x7	Reserved	0x7	MCS7	0x7	MCS7
0x8	Reserved	0x8	MCS8	0x8	MCS8
0x9	Reserved	0x9	MCS9	0x9	MCS9
0xA	Reserved	0xA	Reserved	0xA	MCS10
0xB	Reserved	0xB	Reserved	0xB	MCS11
0xC and up	Reserved	0xC and up	Reserved	0xC and up	Reserved

6.10.2 CMD_TX_RATE_CFG

This command gets or sets the 802.11n transmission data rate.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_RATE_CFG = 0x00D6
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Reserved	UINT16	Reserved (set to 0)
RateScope	MrvlIE_RateScope_t	Bitmap of HR/DSSS rates Bitmap of OFDM rates Bitmap of HT-MCSs allowed for initial rate Bitmap of VHT-MCSs allowed for initial rate Bitmap of HE-MCSs
RateDropPattern	MrvlRateDropPattern_t	Rate drop table
TxRateCfg	MRVL_TX_RATE_ CFG_TLV_ID	Tx packet parameter configuration - Preamble type, STBC and so on

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TX_RATE_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Reserved	UINT16	Reserved (set to 0)
RateScope	MrvlIE_RateScope_t	Bitmap of HR/DSSS rates Bitmap of OFDM rates Bitmap of HT-MCSs allowed for initial rate Bitmap of VHT-MCSs allowed for initial rate Bitmap of HE-MCSs
RateDropPattern	MrvlRateDropPattern_t	Rate drop table
TxRateCfg	MRVL_TX_RATE_ CFG_TLV_ID	Tx packet parameter configuration - Preamble type, STBC, GI duration and so on

6.10.3 CMD_PACKET_AGGR_CTRL

This command gets/sets the software Tx/Rx aggregation configuration parameters in firmware. This command is specifically being used for USB specific builds only.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_PACKET_AGGR_CTRL = 0x0251
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SWTx/RxAggregationEnable	UINT16	Software Tx/Rx aggregation enable Bits[15:2]: Reserved Bit[1]: Software Rx aggregation enable Bit[0]: Software Tx aggregation enable
SWTxAggregationMaxSize	UINT16	Maximum Tx aggregation size
SWTxAggregationMaxPktCnt	UINT16	Maximum Tx aggregation packet count
SWTxAggregationAlign	UINT16	Software Tx aggregation alignment

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_PACKET_AGGR_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
SWTx/RxAggregationEnable	UINT16	Software Tx/Rx aggregation enable Bits[15:2]: Reserved Bit[1]: Software Rx aggregation enable Bit[0]: Software Tx aggregation enable
SWTxAggregationMaxSize	UINT16	Maximum Tx aggregation size
SWTxAggregationMaxPktCnt	UINT16	Maximum Tx aggregation packet count
SWTxAggregationAlign	UINT16	Software Tx aggregation alignment

6.11 Transmit Power Control

The NXP enhanced Transmit Power Control (TPC) algorithm minimizes power consumption while transmitting at the highest rate.

The power level can be configured by the TPC algorithm dynamically adapts between these power levels.

The power is dynamically adjusted only at the highest transmit rate allowed by the network and the CMD_TX_RATE_CFG command (typically 11 Mbps for 802.11b and 54 Mbps for 802.11a/g networks). When the transmit rate is less than the highest rate, the highest transmit power is always used.

The key metrics used in the algorithm are the Packet Error Rate (PER) and Signal to Noise Ratio (SNR) indicators. Firmware maintains an exponentially averaged value of SNR based on the SNR of all received frames.

The following thresholds are defined:

- PER_BAD_Th—PER threshold, value used to increase rate and reduce power
- PER_GOOD_Th—PER threshold, value used to increase rate and reduce power

If at any point, the PER is worse than PER_BAD_Th, the rate is reduced and power is increased. The PER will be re-evaluated.

Also, after every second or after a certain large number of packets are transmitted (whichever happens earlier), if the PER is better than PER_GOOD_Th, the rate is increased and the power is reduced.

Table 26 lists the supported power configuration command.

Table 26: Power Configuration Command

Command	Description	Page
Section 6.11.1, CMD_REGION_POWER_CFG	Configures region information and/or power table	page 113

6.11.1 CMD_REGION_POWER_CFG

This command configures the region information and/or power table.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_REGION_POWER_CFG = 0x0249
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_REGIONINFO_t • MrvlIETypes_POWERTABLE_t

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_REGION_POWER_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_REGIONINFO_t • MrvlIETypes_POWERTABLE_t

6.12 Event Subscription

Table 27 lists the supported event subscription command.

Table 27: Event Subscription Commands

Command	Description	Page
Section 6.12.1, CMD_802_11_SUBSCRIBE_EVENT	Subscribes to events and sets thresholds	page 115

6.12.1 CMD_802_11_SUBSCRIBE_EVENT

This command allows the host driver to subscribe to a variety of different events and dynamically set thresholds specific to each event.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SUBSCRIBE_EVENT = 0x0075
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET 0x0002 = ACT_BITWISE_SET (set only those events for which bit is set in the Events bit field) 0x0003 = ACT_BITWISE_CLR (clear those events for which bit is set in the Events bit field, it also does not modify the settings for other events)
Events	UINT16	Bitmap of subscribed events

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SUBSCRIBE_EVENT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Events	UINT16	Current bitmap of subscribed events

ACT_SET is used to change the set of subscribed events. ACT_GET is used to query the current set of subscribed events.

The Events field in the request is the bitmap of subscribed events. Events field in the response is the current bitmap of subscribed events. The default value of the Events bitmap is 0x0008, which means only BeaconLoss event subscribed.

The bits in the bitmap correspond to the events according to Table 28. Bit[0] is the Least Significant Bit (LSb).

Table 28: Subscribed Events Bitmap Bits

Bit	Event
15:12	Reserved (set to 0)
11	Pre_Beacon_Lost This event is generated once the number of consecutive beacons missed since the last traffic reaches the configured value. After this event is triggered, the event will be disabled and the host will need to re-enable it if needed. The configured number should be smaller than LINK_LOST counts. If it is larger than LINK_LOST, this event will not be triggered.
10	Link_Quality This event is generated based on 3 parameters: • LQ_SNR • LQ_TX_RATE • LQ_TX_LATENCY If there is no Tx traffic in a certain interval, then received beacon's SNR is monitored. This event is generated if the value of the received beacon's SNR falls below the threshold for consecutive 'n' number of times (frequency parameter). OR In the case where there is Tx traffic, then the event generation logic is slightly different. It also takes into consideration the rate fall back before generating the event. In this case, this event is only generated if the received beacon's SNR falls below the threshold for consecutive 'n' number of times (frequency parameter), as well as the Tx rate falls to the value less than or equal to the value set in the threshold for consecutive 'n' number of times (frequency parameter for the rate). OR This event is also generated when the Tx latency for the packet falls below the threshold for consecutive 'n' number of times (frequency parameter).
9	Data SNR_HIGH This event is generated when the average received SNR in data packets goes above a threshold. The threshold is specified by a TLV in the request.
8	Data RSSI_HIGH This event is generated when the average received RSSI in data packets goes above a threshold. The threshold is specified by a TLV in the request.
7	Data SNR_LOW This event is generated when the average received SNR in data packets goes below a threshold. The threshold is specified by a TLV in the request.
6	Data RSSI_LOW This event is generated when the average received RSSI in data packets goes below a threshold. The threshold is specified by a TLV in the request.
5	Beacon SNR_HIGH This event is generated when the average received SNR in beacons goes above a threshold. The threshold is specified by a TLV in the request.
4	Beacon RSSI_HIGH This event is generated when the average received RSSI in beacons goes above a threshold. The threshold is specified by a TLV in the request.
3	BEACON_LOSS/LINK_LOSS This event is generated when the number of consecutive beacons missed exceeds a threshold. The threshold is specified by a TLV in the request.

Table 28: Subscribed Events Bitmap Bits (Continued)

Bit	Event
2	MAX_FAIL This event is generated when the number of consecutive transmit failures exceeds a threshold. The threshold is specified by a TLV in the request.
1	Beacon SNR_LOW This event is generated when the average received SNR in beacons goes below a threshold. The threshold is specified by a TLV in the request.
0	Beacon RSSI_LOW This event is generated when the average received RSSI in beacons goes below a threshold. The threshold is specified by a TLV in the request.

To allow this command to be extensible, there can be variable length TLV fields after the aforementioned fixed fields in the command request. The allowed TLVs are:

- MrvlIETypes_BeaconLowRssiThreshold_t (see Section 10.4.4 on page 321)
- MrvlIETypes_BeaconLowSnrThreshold_t (see Section 10.4.5 on page 322)
- MrvlIETypes_FailureCount_t (see Section 10.4.6 on page 322)
- MrvlIETypes_BeaconsMissed_t (see Section 10.4.7 on page 323)
- MrvlIETypes_BeaconHighRssiThreshold_t (see Section 10.4.17 on page 328)
- MrvlIETypes_BeaconHighSnrThreshold_t (see Section 10.4.18 on page 328)
- MrvlIETypes_DataLowRssiThreshold_t (see Section 10.4.27 on page 332)
- MrvlIETypes_DataLowSnrThreshold_t (see Section 10.4.28 on page 332)
- MrvlIETypes_DataHighRssiThreshold_t (see Section 10.4.29 on page 333)
- MrvlIETypes_DataHighSnrThreshold_t (see Section 10.4.30 on page 333)
- MrvlIETypes_LinkQualityThreshold_t (see Section 10.4.26 on page 331)
- MrvlIETypes_PreBeaconMissed_t (see Section 10.4.36 on page 337)

6.13 Power Management

This section describes the various power management modes supported by NXP devices.

Table 29 lists the supported power management commands.

Table 29: Power Management Commands

Command	Description	Page
Section 6.13.3, Power Management Commands		
Section 6.13.3.1, CMD_802_11_SLEEP_PARAMS	Gets/sets sleep parameters	page 120
Section 6.13.3.2, CMD_802_11_INACTIVITY_TIMEOUT_EXT	Gets/sets inactivity timeout value	page 122
Section 6.13.3.3, CMD_802_11_FW_WAKE_METHOD	Firmware wake method	page 124
Section 6.13.3.4, CMD_802_11_AUTO_TX	Auto-transmission	page 125
Section 6.13.3.5, CMD_SDIO_GPIO_INT_CONFIG	Gets/sets SDIO/GPIO interrupt	page 126
Section 6.13.3.6, CMD_802_11_PS_MODE_ENH	Enables/disables power save	page 128
Section 6.13.3.7, CMD_802_11_HS_ENH	Configures wakeup semantics for host driver	page 131
Section 6.13.3.8, CMD_HS_WAKEUP_REASON	Gets host wakeup reason from firmware	page 133
Section 6.13.8, APSD		
Section 6.13.8.2, CMD_802_11_SLEEP_PERIOD	Specifies the sleep period	page 137

A station (client) can be in 1 of the 2 Power Management (PM) modes:

- Active mode—stays in Awake state
- Power Save (PS) mode—stays in either Awake or Doze state

The AP always needs to be aware of the current power management mode that the station is in. For this reason, the station notifies the AP every time it switches modes. It sends a Null data frame with the PM bit set to 0 when entering Active mode or 1 when entering PS mode. It retries the Null frame until it is successfully acknowledged by the AP. If the station needs to switch to Active mode and there is some data to transmit, the station uses the PM bit in the data frame to indicate the mode switch to the AP and avoids generating an extra Null frame.

The AP buffers all unicast traffic for a station in PS mode and indicates the presence of buffered traffic through Traffic Indication Map (TIM) Information Element (IE) in the beacon frame. Also, if any associated station is in PS mode, the AP delivers multicast traffic immediately following a Delivery Traffic Indication Message (DTIM) beacon only.

By default, the NXP station always stays in Active mode. If the host enables PS mode, it activates an algorithm which switches between Active and PS mode as required. One of the parameters provided by the host driver is the number of DTIM periods the station is allowed to sleep before waking up to hear a beacon.

The algorithm operates as follows:

- When the station is in Active mode to start with, and if there is no transmit or receive activity for 1 beacon interval, the station decides to switch to PS mode and sends a Null packet to the AP to signal that activity.
- When the station is in PS mode, the station computes the total sleep time so that it wakes up a *wakeup offset* before the TBTT corresponding to the desired DTIM beacon. The total sleep time is based on the number of DTIM periods set by the host driver.
- The wakeup offset depends on the total time required to power up the device and the accuracy of the low power oscillator which is used during power down mode.

- When the station wakes up, it waits to receive a beacon. If the incoming beacon indicates unicast or multicast traffic or the host generates transmit packets, the station transitions to Active mode and sends a Null packet to the AP to signal that activity. If none of these conditions occur, the station again computes the sleep time and goes back to sleep.
- This process repeats as long as PS mode stays enabled. If the host disables PS mode, then the algorithm is deactivated and the station always stays in the Active mode.

6.13.1 Entering Power Save Mode

The host driver sends the command CMD_802_11_PS_MODE_ENH with the Action field set to PS_CMD_ENTER. After sending the command, the host driver must not send any data or commands to firmware.

When firmware enters PS mode, a number of hardware blocks are turned off (hardware MAC, baseband processor, and RF chip). A wakeup timer starts that wakes up the Wi-Fi device at the next expected beacon transmit time from the AP.

When in PS mode, the hardware bus interface to the host is turned off, as well. It is important that the host driver does not try to send commands or data to firmware when it is sleeping.

When the listening interval equals DTIM or multiples of DTIM, the device achieves the lowest current consumption. The listening interval does not need to equal multiples of DTIM. Listening interval is implemented as follows:

- Listening interval equals DTIM or multiples of DTIM:
  - Wi-Fi device wakes up on every DTIM.
- Listening interval is less than 1 DTIM interval (for example, DTIM = 10, LI = 3):
  - DTIM count is numbered as 0,9,8,7,6,5,4,3,2,1,0,9,8,7,6,5,4,3,2,1,0.
  - Wi-Fi device wakes up at DTIM count = 0.
  - Wi-Fi device sleep listening interval (LI = 3) before next wake up.
- Listening interval is larger than 1 DTIM interval
  - Wi-Fi device wakes up at DTIM count = 0.
  - Listening interval is not used.
- Wi-Fi device waking up on multiple DTIM interval
  - Listening interval is not used. Wi-Fi device only wakes up on the specified DTIM interval.

6.13.2 Exiting Power Save Mode

To exit PS mode, the host driver sends the command CMD_802_11_PS_MODE_ENH with subcommand PS_CMD_EXIT to firmware. After exiting PS mode, the Wi-Fi device returns to normal operation.

Note: See the following documents for further information:

- Power Save FAQ Application Note
- Ad-Hoc Power Save Mode Application Note
- Power Management Architecture Specification v1.14 or later

6.13.3 Power Management Commands

This section describes the APIs between the driver and firmware which enable the different power management modes.

6.13.3.1 CMD_802_11_SLEEP_PARAMS

This command is used to set or get the various sleep parameters used for the correct power save operation.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SLEEP_PARAMS = 0x0066
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Error	UINT16	Sleep clock error (ppm) Error is the sleep clock error in parts per million (ppm). The valid range is 0 to 65535 ppm. Default value is 5000.
Offset	UINT16	Wakeup offset (µs) Offset is the wakeup from the TBTT in microseconds (µs). This denotes the total time from when the stable clock is available to when the wireless system is ready to receive packets (RF fully settles down). The valid range is 0 to 65535 µs. Default value is 500 µs.
StableTime	UINT16	Clock stabilization time (µs) StableTime is the time taken by the oscillator to stabilize (µs). This determines the amount of time that the PMU state machine inserts between enabling the oscillator and enabling the PLL. This should be set to the maximum time elapsed between the clock enable being asserted and a stable oscillator output being available. The valid range is 0 to 65535 µs. Default value is 4000 µs.
CalControl	UINT8	Control periodic calibration CalControl determines whether the periodic calibration is turned on or not. The valid range is 0 to 2. If this is set to 1, the periodic calibration is enabled. If this is set to 2, the periodic calibration is disabled. Disable periodic calibration when the high accuracy external sleep clock is being used. The default value is 1, which means that the periodic calibration is enabled by default.
ExternalSleepClk	UINT8	Control the use of external sleep clock ExternalSleepClk determines whether the external sleep clock is used or not. The valid range of values is 0 to 2. If this is set to 1, the internal sleep clock is used. If this is set to 2, the external sleep clock is used. The default value is 1, which means that the internal sleep clock is used by default. The host driver must be careful to set this to 2 only when an external sleep clock is connected.
FrequencyUnit	UINT16	To retrieve the sleep clock frequency settings, the user can specify the units of the results 0x0000 = 1 Hz All other values are reserved.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SLEEP_PARAMS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
Error	UINT16	Sleep clock error (ppm) Error is the sleep clock error in parts per million (ppm). The valid range is 0 to 65535 ppm. Default value is 5000.
Offset	UINT16	Wakeup offset (µs) Offset is the wakeup from the TBTT in microseconds (µs). This denotes the total time from when the stable clock is available to when the wireless system is ready to receive packets (RF fully settles down). The valid range is 0 to 65535 µs. Default value is 500 µs.
StableTime	UINT16	Clock stabilization time (µs) StableTime is the time taken by the oscillator to stabilize (µs). This determines the amount of time that the PMU state machine inserts between enabling the oscillator and enabling the PLL. This should be set to the maximum time elapsed between the clock enable being asserted and a stable oscillator output being available. The valid range is 0 to 65535 µs. Default value is 4000 µs.
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ExternalSleepClk	UINT8	Control the use of external sleep clock ExternalSleepClk determines whether the external sleep clock is used or not. The valid range of values is 0 to 2. If this is set to 1, the internal sleep clock is used. If this is set to 2, the external sleep clock is used. The default value is 1, which means that the internal sleep clock is used by default. The host driver must be careful to set this to 2 only when an external sleep clock is connected.
SleepclkFrequency	UINT16	Value of the sleep clock frequency (units of FrequencyUnit)

Setting any of the above parameters to 0 keeps the value of that parameter in firmware unchanged.

This command can be invoked by the driver at any time and the change in any value(s) takes effect immediately.

6.13.3.2 CMD_802_11_INACTIVITY_TIMEOUT_EXT

Inactivity Timeout Enhancement is enabled as soon as firmware accepts this host command with valid parameters. Without receiving this command, firmware should behave the same as the existing implementation.

The default infrastructure power save algorithm allows the device to go to sleep only after at least 1 beacon interval of inactivity. This command is an enhancement to eliminate this limitation of the basic power save algorithm.

In Infrastructure Power Save mode, the station periodically wakes up to listen to beacons. If the TIM information element in the beacon indicates no broadcast or unicast traffic, and there is no uplink traffic to be transmitted, then the station goes back to sleep right away. However, if there is either uplink traffic to transmit or downlink traffic to receive, the station stays awake to transmit and/or receive. The Inactivity Timeout is used to determine how long the station stays awake after it transmits or receives any packet. The station starts or restarts an already running timer with this timeout value at the end of each transmit or receive. When the timer expires, the station goes back to sleep.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_INACTIVITY_TIMEOUT_EXT = 0x008A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = general error
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
TmoUnit	UINT16	In microseconds (μs) 0x0000 means 1000 μs (1 ms)
UnicastTmo	UINT16	Inactivity timeout for unicast data (UIT) to be set by user
MulticastTmo	UINT16	Inactivity timeout for multicast data (MIT) to be set by user
PsEntryTmo	UINT16	Inactivity timeout after Null PM=1 completion to AP to detect further Tx from the AP before sending a sleep event to the host
Reserved	UINT16	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_INACTIVITY_TIMEOUT_EXT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = general error
Action	UINT16	Type of action set in request message
TmoUnit	UINT16	0x0000 means 1000 µs (1 ms)
UnicastTmo	UINT16	Inactivity timeout for unicast data (UIT) to be set by user
MulticastTmo	UINT16	Inactivity timeout for multicast data (MIT) to be set by user
PsEntryTmo	UINT16	Inactivity timeout after NullPM=1 completion to AP to detect further Tx from the AP before sending a sleep event to the host
Reserved	UINT16	Reserved (set to 0)

6.13.3.3 CMD_802_11_FW_WAKE_METHOD

This command is used by the host to indicate to firmware how it wishes to wakeup when firmware is in Power Save mode or Deep Sleep mode. This command is issued by the host prior to enabling firmware into any Power Save mode. This command is enabled once. Firmware remembers this setting until it resets.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_FW_WAKE_METHOD = 0x0074
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Method	UINT16	Method to wakeup firmware Indicates the method by which the host intends to wakeup firmware. This can be 1 of the following: 0x0001 = firmware wakeup through the command interrupt 0x0002 = firmware wakeup through the GPIO pin Default method for firmware wakeup is through command interrupt for SDIO interface.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_FW_WAKE_METHOD | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
Method	UINT16	Same as in Request

6.13.3.4 CMD_802_11_AUTO_TX

This command is a power saving feature that reduces host current consumption by minimizing the host-awake time. When enabled, firmware will autonomously transmit packets periodically. It applies to host sleep mode but not to deep sleep mode of Wi-Fi operation. It also works in the following modes: Infrastructure and Ad-Hoc modes, Normal, IEEE-PS, WMM_UAPSD, No-SEC, WEP, WPA, and WPA2. The Wi-Fi station will not send an Auto-Tx frame unless it is associated with the BSS.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AUTO_TX = 0x0082
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET_Tx 0x0001 = ACT_SET_Tx
MrvlIETypes_AutoTxMacFrame_t	UINT16	As defined in Section 10.4.19, MrvlIETypes_AutoTxMacFrame_t, on page 329

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_AUTO_TX | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = general error
Action	UINT16	Type of action set in request message
MrvlIETypes_AutoTxMacFrame_t	UINT16	As defined in Section 10.4.19, MrvlIETypes_AutoTxMacFrame_t, on page 329

The host driver will register the Auto-Tx frame by issuing a “Set Auto-Tx Command” with a non-0 MrvlIETypes_AutoTxMacFrame_t interval. Only the last Auto-Tx frame is saved.

When in power-save mode, the auto-transmission occurs immediately after the device wakes up to receive the beacon. Therefore, the timing of each Auto-TX packet will not be exactly the same as the configured interval.

If the host driver wants to de-register, it needs to send a “Set Auto-Tx Command” without a MrvlIETypes_AutoTxMacFrame_t interval. The Wi-Fi station will immediately stop transmission of the Auto-Tx packets after de-registration. It will only stop auto-transmission only when it is disassociated or de-authenticated. There is no additional notification event.

If auto-transmission packets are not acknowledged, there will be a MAC layer re-transmission until the max retries count expires. It will then be discarded silently.

Note: This command can only be executed when connected to an AP. Association implicitly begins AutoTx if it is already configured during a previous association. Any subsequent association does not require the re-configuration of AutoTx.

6.13.3.5 CMD_SDIO_GPIO_INT_CONFIG

This command is applicable to the SDIO host interface on boards using the NXP Monahans family of processors.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SDIO_GPIO_INT_CONFIG = 0x0088
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
GPIOPin	UINT16	Wi-Fi SoC GPIO pin to interrupt host 0xFFFF = disable GPIO interrupt mode (default) Others = GPIO pin assigned to generate pulse to host
GPIOInterruptEdge	UINT16	GPIO interrupt pulse type (to host) 0x0000 = low-to-high-to-low 0x0001 = high-to-low-to-high Ignored if GPIO pin = 0xFFFF
GPIOPulseWidth	UINT16	Pulse width for GPIO interrupt (μs) Minimum = 2 μs Maximum = 64000 μs (64 ms) Ignored if GPIO pin = 0xFFFF

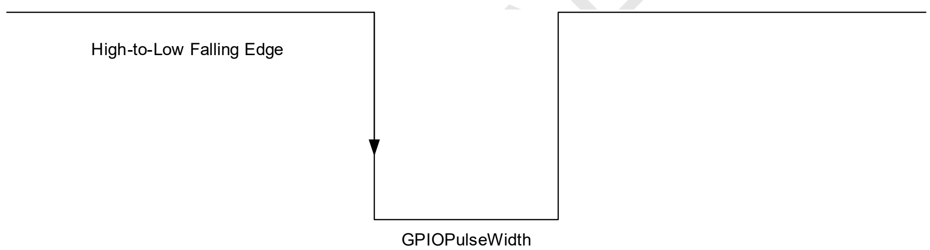
RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SDIO_GPIO_INT_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
GPIOPin	UINT16	Wi-Fi SoC GPIO Pin to Interrupt Host 0xFFFF = disable GPIO interrupt mode (default) Others = GPIO pin assigned to generate pulse to host

Field Name	Type	Description
GPIOInterruptEdge	UINT16	GPIO interrupt pulse type (to host) 0x0000 = low-to-high-to-low 0x0001 = high-to-low-to-high Ignored if GPIO pin = 0xFFFF.
GPIOPulseWidth	UINT16	Pulse width for SDIO GPIO interrupt (µs) Minimum = 2 µs Maximum = 64000 µs (64 ms) Ignored if GPIO pin = 0xFFFF.

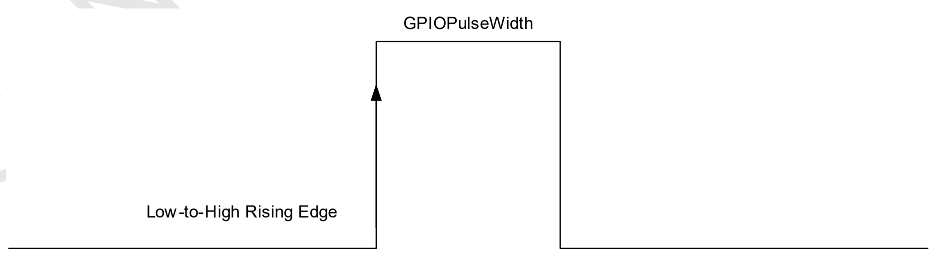
For GPIO high-to-low-to-high trigger, the GPIO pin usually remains in high state. When firmware generates a SDIO interrupt to host, a high-to-low-to-high transition pulse is generated as shown in Figure 4.

Figure 4: GPIO High-to-Low-to-High (Falling) Edge Trigger



For GPIO low-to-high-to-low trigger, the GPIO pin usually remains in low state. When firmware generates a SDIO interrupt to host, a low-to-high-to-low transition pulse will be generated as shown in Figure 5.

Figure 5: GPIO Low-to-High-to-low (Rising) Edge Trigger



Note: Since this configuration affects the SDIO interrupt generation behavior in the client card, it should be the first command issued from the host/driver for the SDIO host interface platform after firmware has downloaded. Right after issuing this command, the host/driver should be able to handle the interrupt based on the method defined starting from the command's response.

6.13.3.6 CMD_802_11_PS_MODE_ENH

This API is used to enable or disable Power Save (PS) in IEEE PS, UAPSD, PPS, and Auto deep sleep mode. It implements PS_MODE_ENH command in the Power Management Specification v2.5.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_PS_MODE_ENH = 0x00E4
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = gets PS information from firmware (the return format used will be similar to auto PS action mode, 0x00FF, which includes fixed field to identity what all PS (auto deep sleep, STA, uAP PS) scheme are enabled in firmware and their respective TLVs containing PS parameters) 0x0005 = PS sleep confirm 0x00FE = disable Auto PS mode (this action would allow disabling auto deep sleep, STA, uAP, APSTA PS in firmware.) 0x00FF = auto PS save (this action would allow firmware to run uAP, STA, APSTA, auto deep sleep PS based on firmware connection state.)

Field Name	Type	Description
Data[]	UINT8	Action specific data

Action	Data	Type
Get PS information from firmware	PS Bitmap Get power save parameters as per bitmap: Bits[10:15]: reserved Bit[9]: uAP periodic DTIM PS Bit[8]: uAP inactivity based PS Bits[5:7]: reserved Bit[4]: STA power save Bits[3:1]: reserved Bit[0]: auto deep sleep mode A bit set (1) in the PS bitmap indicates the corresponding power save mode is enabled in firmware. The following TLVs follows PS bitmap: • TLV1: (MrvlIETypes_Auto_DS_Param_t 0x0171): Sets/gets auto deep sleep parameters • TLV2: (MrvlIETypes_Enh_Sta_PS_t 0x172): Sets/gets station power save parameters • TLV3: (MrvlIETypes_ApSleepParamsInact_t 0x016B): Sets/gets uAP inactivity based power save parameters • TLV4: (MrvlIETypes_ApSleepParams_t 0x016A): Sets/gets uAP periodic DTIM PS parameters	UINT16
Enable enhanced power save	NullPktInterval	UINT16
	NumDtims	UINT16
	BeaconMisstimeour	UINT16
	LocalListenInterval	UINT16
	AdhocAwakePeriod	UINT16
	Mode 0x01 = auto mode (firmware automatically chooses PS_POLL or NULL mode) 0x02 = PS_POLL mode 0x03 = NULL packet mode	UINT16
	DelayToPS (ms) Specifies the delay before firmware initiates the power sleep handshake to host driver in infrastructure mode: • Delay starts right after association response is taken by host driver in open or WEP mode if WMM is disabled. • Delay starts upon EVENT_WMM_STATUS_CHANGE is taken by host driver after association in open or WEP mode if WMM is disabled. • Delay starts upon GTK is set after association in WPA/WPA2 mode.	UINT16
Auto deep sleep in firmware	DSInactTO Inactivity timeout (ms) before firmware generates PS_SLEEP event to driver in auto deep sleep mode	UINT16
PS sleep mode	RespCtrl 0x0000 = response from firmware to driver is not needed for this command 0x0001 = response is needed	UINT16
continued...		

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td></td><td></td><td>
<table>
<tr><th>Action</th><th>Data</th><th>Type</th></tr>
<tr><td colspan="3">continued...</td></tr>
<tr><td>Enable Auto PS (uAP, Auto DS, STA, APSTA PS)</td><td>PS Bitmap
Get power save parameters as per bitmap:
Bits[10:15]: reserved
Bit[9]: uAP periodic DTIM PS
Bit[8]: uAP inactivity based PS
Bits[5:7]: reserved
Bit[4]: STA power save
Bits[3:1]: reserved
Bit[0]: auto deep sleep mode
Based on PS bitmap, the following TLVs are passed to configure various power save parameters:
• TLV1: (MrvlIETypes_Auto_DS_Param_t 0x0171): Sets/gets auto deep sleep parameters
• TLV2: (MrvlIETypes_Enh_Sta_PS_t 0x172): Sets/gets station power save parameters
• TLV3: (MrvlIETypes_ApSleepParamsInact_t 0x016B): Sets/gets uAP inactivity based power save parameters
• TLV4: (MrvlIETypes_ApSleepParams_t 0x016A): Sets/gets uAP periodic DTIM PS parameters.</td><td>UINT16</td></tr>
<tr><td>Disable PS mode</td><td>PS Bitmap
Disable power save parameters as per bitmap:
Bits[10:15]: reserved
Bit[9]: disable uAP periodic DTIM PS
Bit[8]: disable uAP inactivity based PS
Bits[5:7]: reserved
Bit[4]: disable STA power save
Bits[3:1]: reserved
Bit[0]: disable auto deep sleep mode</td><td>UINT16</td></tr>
</table>
</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_PS_MODE_ENH | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set as in request message
Data[]	UINT8	Same as request

6.13.3.7 CMD_802_11_HS_ENH

This API is used to configure the wakeup semantics before it goes to sleep and to activate host sleep after reception of HS_ACT_REQ event. It implements HS_CFG_ENH and HS_SLEEP_CONFIRM commands in power management specifications v2.5.

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_802_11_HS_ENH = 0x00E5</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Action
0x0001 = configure enhanced host sleep mode
0x0002 = activate enhanced host sleep mode in full power mode</td></tr>
<tr><td>Data[]</td><td>UINT8</td><td>Action specific data
<table>
<tr><th>Action</th><th>Data</th><th>Type</th></tr>
<tr><td rowspan="4">Configure enhanced host sleep mode</td><td>Criteria
Bit[6]: wake up host when management frame is received
Bits[5:4]: reserved
Bit[3]: multicast data
Client will wake the host when multicast data from connected BSS is received.
Bit[2]: MAC event
Client will wake host when a MAC event is received (for example, Link Loss).
Bit[1]: unicast data
Client will wake host when unicast data from connected BSS is received.
Bit[0]: broadcast data
Client will wake host when broadcast data from connected BSS is received.</td><td>UINT32</td></tr>
<tr><td>GPIO
GPIO number to wake up host or interface (0xFF) to wake up host</td><td>UINT8</td></tr>
<tr><td>Gap
Delay (ms) before sending data/event to host. 0xFF is the special value which firmware will wait for the host to acknowledge before sending data/event to host.</td><td>UINT8</td></tr>
<tr><td>Others (optional HostSleepFilter IE)</td><td>--</td></tr>
<tr><td>Activate enhanced host sleep mode</td><td>RespCtrl
0x0000 = response from firmware to driver is not needed for this command
0x0001 = response is needed</td><td>UINT16</td></tr>
</table>
</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_HS_ENH | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Data[]	UINT8	Same as request

For UAPSD and PPS, the existing sleep APIs shall be used to program sleep period.

Note: The following features could cause the chip to wake up the host:

- Background scan function, when operational, could wake up the host when 1 of the report conditions, set in the ReportConditions parameter of the background scan configuration, matches.
- Memory Efficient Filtering (MEF) is an additional feature that could cause the chip to wake up the host.

6.13.3.8 CMD_HS_WAKEUP_REASON

This command gets the host wakeup reason from firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_HS_WAKEUP_REASON = 0x0116
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Wakeup Reason	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_HS_WAKEUP_REASON | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Wakeup Reason	UINT16	Wakeup reason code 0x0000 = unknown 0x0001 = broadcast data matched 0x0002 = multicast data matched 0x0003 = unicast data matched 0x0004 = maskable event matched 0000x5 = non-maskable event matched 0x0006 = non-maskable condition matched EAPOL rekey 0x0007 = magic pattern matched 0x0008 = control frames matched 0x0009 = management frame matched 0x000A = GTK rekey failure 0x000B = management frame filter extension matched Others = reserved

6.13.4 Power Save Events

Firmware sends the following events to the host driver when in PS mode:

- Awake event
- Sleep event
- Host Sleep Activate Request event

6.13.4.1 Awake Event

This Power Save Awake event is generated to notify the host driver that firmware is woken up in power save mode. The exact semantics of this event depend on the context in which this is generated.

6.13.4.2 Sleep Event

This Power Save Sleep event is generated in power save mode to notify the host driver that firmware wishes to put the device to sleep.

6.13.4.3 Host Sleep Activate Request Event

This Host Sleep Activate Request event is generated in full power mode to notify the host driver that firmware is ready to activate host sleep mode.

6.13.5 Driver/Firmware Interaction

Firmware will not send Awake event to driver unless firmware is woken up by firmware. The driver needs to wake up firmware before sending any data or command.

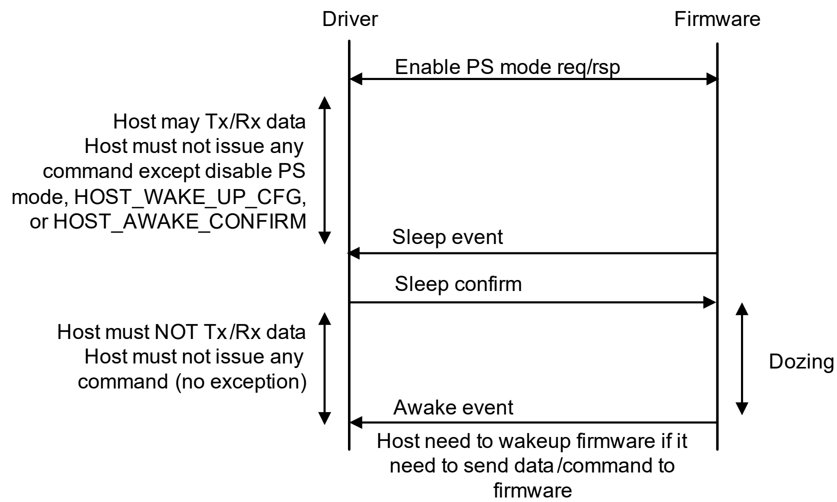
Every time firmware decides to put the system to sleep, it sends an event notification to the driver to indicate an action to put the system to sleep. Firmware then waits for a confirmation message from the host driver before it actually puts the system to sleep.

6.13.6 Assumptions

The host must not issue any command (except the disable PS mode command) after it enables PS mode. The host may, however, continue to process data transmit and receive.

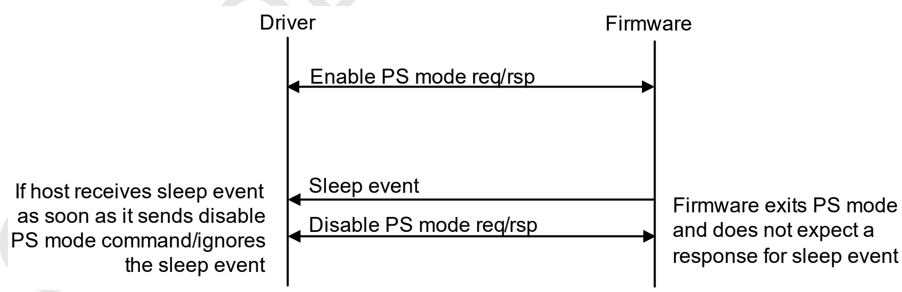
After sending the sleep confirm message, the host can wake up firmware through wakeup method before sending any data or command (see Figure 6).

Figure 6: PS Mode—Assumption 1



The sleep event (from firmware) and disable PS command (from the host) may be generated simultaneously. In this case, firmware honors the disable PS command and exits PS mode. The host ignores the sleep event in this case (see Figure 7).

Figure 7: PS Mode—Assumption 2



6.13.7 AutoDeep Sleep Mode

When the host enables auto deep sleep mode, firmware will enter deep sleep when idle timer expired. The host and firmware will go through PS_SLEEP/SLEEP CONFIRM before the device enters sleep mode.

6.13.7.1 Auto Deep Sleep Mode Assumptions

The host only needs to issue CMD_802_11_PS_MODE_ENH command (Action = 0x0003) to enable auto deep sleep mode in firmware once. Firmware will automatically decide when issue PS_SLEEP event to the host. After the host gets CMD_802_11_PS_MODE_ENH command (Action = 0x0005) response, the host will need to wake up the device before sending data or command to firmware.

6.13.8 APSD

Automatic Power Save Delivery (APSD) refers to the enhanced mechanisms introduced in the 802.11e specification to allow power save operation in the presence of periodic traffic.

6.13.8.1 APSD Overview

These additional mechanisms can only be used if APSD is supported by both the client (stations) and the AP. There are 2 modes of operations for APSD:

- Scheduled
- Unscheduled

Scheduled APSD requires negotiation of a delivery schedule between the AP and the client. The schedule specification includes service start time and service interval. A scheduled Service Period (SP) starts at the service start time and every service interval later thereafter. The client must be ready to receive traffic from the AP at the start of each SP.

Unscheduled APSD (also referred to as UPSD) does not require setting up an explicit schedule. However, the client may choose to enable only certain Access Categories (AC) to be serviced using this mechanism. Specifically, the client may choose to trigger-enable certain ACs and delivery-enable certain other ACs. If an AC is trigger-enabled, then any uplink QoS Data/Null frame belonging to that AC triggers an unscheduled SP. If an AC is delivery-enabled, then it is serviced by the AP during an unscheduled SP (in the order of priority).

The End of Service Period (EOSP) bit in the MAC header of the transmitted packets indicates the end of a service period (for both scheduled and unscheduled SP). When the SP ends, the client may power down until the start of the next SP.

The number of frames the AP is allowed to send in a single SP is controlled by the client. It chooses between 2, 4, 6, or all of the frames buffered at the AP to be delivered during a single SP. This information is conveyed to the AP during the association.

The next section describes the design for the UPSD mode of operation. The scheduled version of APSD is not being implemented currently.

6.13.8.2 CMD_802_11_SLEEP_PERIOD

The default infrastructure power save algorithm allows the device to go to sleep only after at least 1 beacon interval of inactivity. Also, once it puts the device to sleep, the device is only woken up before the designated TBTT (depending on listen interval and DTIM period).

This section describes some enhancements to the basic power save algorithm to allow power savings in the presence of frequent periodic traffic.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SLEEP_PERIOD = 0x0068
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SleepPeriod	UINT16	Sleep period (ms)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_SLEEP_PERIOD | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
SleepPeriod	UINT16	Sleep period (ms)

The SleepPeriod field specifies the interval between consecutive wakeup instances of the device (ms). Typically, this interval is much less than the beacon interval. The valid range for the interval is [10, 60]. A value of 0 indicates that the sleep period is disabled. In other words, the device sleeps and wakes up for DTIM beacons as per the legacy power save procedure. The default value is 0.

6.13.8.3 APSD Initialization

For APSD initialization, WMM must be enabled in a BSS before the UPSD is enabled. The AP advertises its support for the UPSD by setting the UPSD bit in the QoS Info field of the WMM IE in the beacons and probe responses transmitted by it.

If the AP does not support the UPSD, the client cannot use the mechanisms provided by the UPSD. If the AP supports the UPSD, then the client may choose to utilize the mechanisms provided by the UPSD.

The client includes the WMM IE in the (re)association request frame to indicate to the AP that it wishes to use the WMM. Within the WMM IE, it uses the QoS Info field to request the use of the UPSD.

The format of QoS Info field is as listed in Table 30.

Table 30: QoS Information

Bit	Description
7	Reserved
6:5	Maximum SP length
4	Reserved
3	AC_BEU_APSD flag
2	AC_BKU_APSD flag
1	AC_VIU_APSD flag
0	AC_VOU_APSD flag

The WMM IE may be included in the association command sent by the driver to firmware.

The driver sets this field by:

- Setting the reserved fields to 0
- Setting the Max SP Length field to 0 (recommended)
- To enable UPSD, set all 4 UPSD flags
- UPSD is not enabled when no flags are set

The current design does not support scenarios where 1, 2, or 3 UPSD flags are set, because the implementation is significantly more complex. These configurations do not enable any compelling use cases that are not covered by what is supported.

Therefore, when UPSD is enabled, all 4 UPSD flags must be set.

In this case, all the downlink traffic from the AP is delivery-enabled, and all the uplink traffic from the client is trigger-enabled. Any uplink transmission triggers the delivery of downlink packets (in the order of priority).

The TIM bit as well as the More Data bit indicates the presence of more buffered traffic at the AP belonging to any AC.

The behavior of firmware in this case depends on the value of the sleep period that has been set (using the sleep period API) by the driver. The 2 cases are described in the following sections.

Zero Sleep Period

If the sleep period is set to 0, there is no periodic traffic (such as a voice call) in progress. This operates very similar to the legacy PS behavior.

Each time the device wakes up to receive a beacon, it inspects the TIM bit in the beacon to determine whether there is any traffic buffered at the AP or not. If the TIM bit is set, it generates a QoS Null frame, which triggers the delivery of all the frames buffered at the AP. It also monitors the EOSP bit in the received frames and goes back to sleep after either the More Data or the EOSP bit goes to 0.

If the TIM indicates buffered multicast/broadcast frames, it waits to receive all multicast/broadcast frames as well.

During this entire exchange the device stays in PS state, which is indicated by the PM bit being set in all frames transmitted by the device (including the QoS Null).

Non-Zero Sleep Period

If the sleep period is set to a non-0 value, there is some sort of periodic traffic (such as a voice call) in progress. In this case, the operation is as follows.

Firmware wakes up with the period specified by the host. In addition, it may wakeup at the DTIM beacons (if required by the host) in order to receive any multicast traffic. It may also periodically wakeup to receive beacons in order to synchronize the value of local TSF counter.

Each time it wakes up, it sends an Awake event to the host driver.

Then, it waits to receive a packet from the host. If the host does not have any packets ready at that time, it is recommended that the host generate a Null frame to minimize the time firmware stays up and consequently maximize the power savings. Section 4.2, Transmit Packet Descriptor, on page 13 describes how the driver generates a Null frame.

When receiving, firmware transmits a packet (or Null Frame) as it's received. Firmware continues as long as it keeps receiving data with the EOSP bit cleared. When it receives the EOSP indication, firmware prepares to go back to sleep.

Before going back to sleep, firmware computes time until next wakeup instant, and programs sleep timer accordingly.

6.13.8.4 APSD Last Packet Indication

Firmware implementation depends on the driver supporting the last packet indication in the transmit packets.

When the driver has no more buffered packets to transfer to firmware, it sets the last packet bit in the transmit info header of the packet. Firmware always waits to receive the last packet indication (except when time-out occurs) before it initiates the sequence for going back to sleep.

The driver must guarantee that it does not transfer any more packets to firmware (after it sends this indication) until it receives another awake event. This allows firmware to determine the last trigger sent and determine the last EOSP indication that received from the AP. See Section 4.2, Transmit Packet Descriptor, on page 13 for the API setting of the last packet indicator.

6.13.8.5 APSD Driver Requirements

The optimal operation of the UPSD depends on co-operation between driver and firmware. Steps the driver takes to ensure maximum power savings is obtained include:

- Respond to sleep request with sleep confirm in a timely manner:
  - If the driver is in the middle of a packet download when it receives a sleep request, it waits until that transaction is complete, and then issues a sleep confirm (without initiating another transaction).
  - If the driver is waiting for a command response when it receives a sleep request, it waits until it receives the response, and then issues a sleep confirm (without initiating another transaction).
  - If the driver is in the middle of receiving a packet when it receives a sleep request, it waits until that transaction is complete, and then issues a sleep confirm (without initiating another transaction).
- When receiving an Awake event, it promptly transfers all the queued packets to firmware.
- Sets the last packet indicator in the last packet it transfers to firmware.
- After setting the last packet indicator, it should not transfer packets to firmware until the next awake event (even if received before the sleep request from firmware).
- If there are no queued packets when the awake event is received, sends a Null frame with the last packet indicator set.

  Note: See APSD Application Note.

6.14 Bluetooth Coexistence

The driver/application configures the Wi-Fi MAC block thru firmware for Bluetooth Coexistence Arbitration (BCA) by using the existing command. See Section 6.3.3, CMD_MAC_REG_ACCESS, on page 46 (Hardware MAC Register Access) to peek and poke Wi-Fi MAC hardware registers directly. While using this method, the driver/application level must have enough knowledge to program the low level Wi-Fi MAC registers for BCA functionality. In addition, CMD_802_11_BCA_CONFIG_TIMESHARE is used to configure the BCA timeshare interval and percentage of time in this timeshare interval.

Table 31: Bluetooth Coexistence Commands

Command	Description	Page
Section 6.14.1, CMD_802_11_BCA_CONFIG_TIMESHARE	Configures the BCA timeshare interval and percentage of time in this timeshare interval	page 141
Section 6.14.2, CMD_ROBUST_COEX	Sets the infrastructure power save mode and period	page 143

6.14.1 CMD_802_11_BCA_CONFIG_TIMESHARE

This command configures the BCA timeshare interval and percentage of time in the timeshare interval.

In this command, the BTTime is further defined:

For Bluetooth and Wi-Fi coexistence, Wi-Fi traffic always wins arbitration when low priority Bluetooth and low priority Wi-Fi traffic are both subjected during an arbitration window. In addition, Wi-Fi traffic always wins arbitration when high priority Bluetooth and high priority Wi-Fi traffic is subjected to arbitration. The high priority Wi-Fi traffic is limited to frame types of very short duration and so should cause minimal interference to Bluetooth traffic. However, low priority Wi-Fi traffic includes data frames of a long duration and high throughput. This may cause low priority Bluetooth traffic to stall as Wi-Fi is always preferred by the arbiter by default.

The host driver controls access to allow Bluetooth traffic. Using the command, the host driver can specify the total time for the timeshare interval and percentage of time (in the timeshare interval) to be given to Bluetooth traffic in cases where equal priority Bluetooth and Wi-Fi traffic is seen by the arbiter.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BCA_CONFIG_TIMESHARE = 0x0069
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET

Field Name	Type	Description
TrafficType	UINT16	Type of traffic for which to configure this timeshare interval <table><tr><th>Type</th><th>Description</th></tr><tr><td>Mode 0 = 00</td><td>Toggle the bit(s) for Lo/Lo (Tx) arbitration</td></tr><tr><td>Mode 1 = 01</td><td>Toggle the bit(s) for Hi/Hi (Tx) arbitration</td></tr><tr><td>Mode 2 = 10</td><td>Toggle the bit(s) for Me/Me (Tx) arbitration (only applies to 88W88688)</td></tr><tr><td>Mode 3 = 11</td><td>Toggle the bit(s) for Mh/Mh (Tx) arbitration (only applies to 88W88688)</td></tr><tr><td>Mode 0xFFFF</td><td>Reset fairshare: Disable all modes above that are running, and restore arbitration table register values to before the user enabled any of the above fairshare modes. (In this mode, TimeshareInterval and BTTime are ignored.)</td></tr></table>
TimeshareInterval	UINT32	Total timeshare interval (ms) Valid range from 20 ms to 60000 ms in multiples of 10 ms. If value specified is not in multiples of 10 then a floor value (multiple of 10) is used. ACT_SET = configures timeshare interval ACT_GET = value ignored
BTTime	UINT32	Bluetooth time (ms) Time interval within TimeshareInterval where the PTA arbiter selects Bluetooth traffic over Wi-Fi traffic, in the case where both Bluetooth and Wi-Fi traffic have equal priority. This value should be in multiples of 10 with a valid range from 0 to TimeshareInterval. If the specified value is not multiples of 10 then a floor value (multiple of 10) is used. ACT_SET = configures Bluetooth time interval ACT_GET = value ignored When BTTime = 0, timeshare mode is off with priority given to Wi-Fi. When BTTime = TimeshareInterval, timeshare mode is off with priority given to Bluetooth.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_BCA_CONFIG_TIMESHARE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 on success)
Action	UINT16	Type of action set in request message
TrafficType	UINT16	Type of traffic for which to configure this timeshare interval <table><tr><th>Type</th><th>Description</th></tr><tr><td>Mode 0 = 00</td><td>Toggle the bit(s) for Lo/Lo (Tx) arbitration</td></tr><tr><td>Mode 1 = 01</td><td>Toggle the bit(s) for Hi/Hi (Tx) arbitration</td></tr><tr><td>Mode 2 = 10</td><td>Toggle the bit(s) for Me/Me (Tx) arbitration (only applies to 88W88688)</td></tr><tr><td>Mode 3 = 11</td><td>Toggle the bit(s) for Mh/Mh (Tx) arbitration (only applies to 88W88688)</td></tr><tr><td>Mode 0xFFFF</td><td>Reset fairshare: Disable all modes above that are running, and restore arbitration table register values to before the user enabled any of the above fairshare modes. (In this mode, TimeshareInterval and BTTime are ignored.)</td></tr></table>

Field Name	Type	Description
TimeshareInterval	UINT32	Total timeshare interval (ms) ACT_SET = configures timeshare interval ACT_GET = returns timeshare value currently used by firmware
BTTime	UINT32	Bluetooth time (ms) ACT_SET = configures Bluetooth time interval ACT_GET = returns Bluetooth time value currently used by firmware When BTTime = 0, timeshare mode is off with priority given to Wi-Fi. When BTTime = TimeshareInterval, timeshare mode is off with priority given to Bluetooth.

Note: Different values of TimeshareInterval and BTTime can be set for each TrafficMode. The settings for each TrafficMode are independent of each other.
ACT_GET on mode 0xFFFF is invalid.

6.14.2 CMD_ROBUST_COEX

For host command, this command sets the infrastructure power save mode and the period when the station is connected and Bluetooth is active. The default mode is enabled, with default RobustCoexMode set to 20 ms.

For uAP, this command is used to configure the Robust Bluetooth Coexistence settings. These are optional settings that can be done by the host. Firmware automatically adapts to choose the correct settings for optimal coexistence in case of SCO/ACL.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_ROBUST_COEX = 0x00E0
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Reserved	UINT16	Not used (set to 0)
RobustCoexMode	Various	Included TLVs

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_ROBUST_COEX | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num

Field Name	Type	Description
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in the request message
Reserved	UINT16	Not used (set to 0)
RobustCoexMode	Various	Included TLVs

For host commands, RobustCoexMode TLVs includes the following:

- MrvlIETypes_RobustCoex_t
- MrvlIETypes_RobustCoex_Period_t

For uAP, TLV buffer can contain the following TLVs:

- MrvlIETypes_ApBTCoexCommonConfig_t
- MrvlIETypes_ApBTCoexScoConfig_t
- MrvlIETypes_ApBTCoexAclConfig_t
- MrvlIETypes_ApBTCoexStats_t

6.15 Wi-Fi Multimedia

Table 32 lists the supported Wi-Fi Multimedia (WMM) commands.

Table 32: WMM Commands

Command	Description	Page
Section 6.15.1, CMD_WMM_GET_STATUS	Retrieves current WMM state	page 145
Section 6.15.2, CMD_WMM_ADDTS_REQ	Sends an ADDTS frame to the associated AP	page 146
Section 6.15.3, CMD_WMM_DELTS_REQ	Sends a DELTS to the associated AP	page 147
Section 6.15.4, CMD_WMM_QUEUE_CONFIG	Gets/sets/defaults the configuration of a WMM queue	page 148
Section 6.15.5, CMD_WMM_QUEUE_STATS	Manages WMM AC queues statistics collection	page 149
Section 6.15.6, CMD_WMM_TS_STATS_CMD_802_11	Retrieves current state of given Traffic Stream (TSID) from firmware	page 150
Section 6.15.7, CMD_WMM_PARAM_CONFIG	Configures WMM parameters from host	page 151

WMM support is enabled in firmware based on the presence of the WMM Parameter IE in the association response from the AP.

6.15.1 CMD_WMM_GET_STATUS

This command is issued when the driver/host wants to retrieve the current WMM state in firmware. The driver layer is required to issue this command in response to a WMM Status Change Event (Event code 23, see Section 8, MAC Events, on page 278).

This command does not take any input arguments, but must provide sufficient buffer space for the expected response. The command response is TLV based and returns a Queue Status TLV for each queue as well as a TLV for the WMM Parameter IE if it has been received by firmware while associated to a BSS.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_GET_STATUS = 0x0071
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
ResponseBuffer	UINT8[]	Not used (set to 0) Provide sufficient buffer space for the WMM IE Parameter TLV and the 4 Queue Status TLVs as shown in the response.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_GET_STATUS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
QueueStatus	MrvlIETypes_WmmQStatus_t[4]	Queue status TLV for each AC queue See Section 10.4.11, MrvlIETypes_WmmQStatus_t, on page 324 for details.
WMM Parameter IE	MrvlIETypes_VendorParamSet_t	Vendor specific IEEE IE with extended 16-bit type and length fields This vendor specific IE contains the WMM Parameter IE as described by the WMM Specification.

The result from this command when issued in response to a WMM Status Change event is 4 Queue Status TLVs (1 for each AC), followed by the WMM Parameter IE without any intermediate pad bytes or reserved fields.

The WMM Parameter IE is a NXP extended version of the standard IEEE IE. The fields and layout of the TLV match the IEEE IE exactly except for the 16-bit element ID (0xDD; 221) and the 16-bit element length (currently 24 bytes by the standard).

6.15.2 CMD_WMM_ADDTS_REQ

This command sends an ADDTS frame to the associated AP. Firmware transmits the Traffic Specification (TSPEC) element followed by any additional IEs present in the command to the associated AP for the purpose of call admission control.

Firmware waits for an ADDTS response and enables ACM designated queues in the event of a successful ADDTS response. ADDTS response failures and timeouts are indicated to the calling layer and the given queue state is left unchanged.

A successful TSPEC negotiation is indicated by both a successful TspecResult and IeeeStatusCode in the command response. In addition, firmware sends a WMM Status Change event when the operation results in a change of queue status (see Table 70, Event Support, on page 278).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_ADDTS_REQ = 0x006E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
TspecResult	UINT8	Not used (set to 0)
Timeout	UINT32	Timeout for ADDTS response to be received (ms)
DialogToken	UINT8	Used to track the ADDTS request as described in the WMM specification
IeeeStatusCode	UINT8	Not used (set to 0)
TspecData	UINT8[63]	Standard TSPEC element with header and length as described in the WMM specification
ExtraIeData	UINT8[256]	Extra buffer space for additional IEEE formatted IEs that follow the TSPEC element in the ADDTS request **NOTE:** This field may be omitted entirely. The length of the TspecData and ExtraIeData are derived from the passed Size field.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_ADDTS_REQ | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
TspecResult	UINT8	Result of the TSPEC operation 0x00 = ADDTS request success 0x01 = execution failure 0x02 = timeout waiting for response 0x03 = data invalid
Timeout	UINT32	Same as request

Field Name	Type	Description
DialogToken	UINT8	Same as request or the dialog token in the ADDTS response if successfully received (should also match input)
IeeeStatusCode	UINT8	IEEE status code in the ADDTS response frame Only valid if the TspecResult field indicates a successful ADDTS request.
TspecData	UINT8[63]	Standard TSPEC element with header and length as described in the WMM Specification returned in the TSPEC response. May vary from the request as described by the specification.
ExtraIeData	UINT8[256]	Extra buffer space for any returned IEs present in the ADDTS response from the AP The length of the TspecData and ExtraIeData are derived from the passed Size field.

6.15.3 CMD_WMM_DELTS_REQ

This command sends a DELTS to the associated AP for the indicated TSPEC. Since DELTS messages to the AP do not receive a response, firmware also disables any previously added flows.

In addition, firmware sends a WMM Status Change event when the operation results in a change of queue status (see Table 70, Event Support, on page 278).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_DELTS_REQ = 0x006F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
TspecResult	UINT8	Not used (set to 0)
DialogToken	UINT8	Used to track ADDTS request as described in the WMM specification
IeeeReasonCode	UINT8	Reason code sent in the DELTS to the associated AP
TspecData	UINT8[63]	Standard TSPEC element with header and length as described in the WMM specification

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_DELTS_REQ | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
TspecResult	UINT8	Result of the TSPEC operation 0x00 = DELTS success 0x01 = execution failure 0x02 = reserved 0x03 = data invalid

Field Name	Type	Description
DialogToken	UINT8	Same as request
IeeeReasonCode	UINT8	Same as request
TspecData	UINT8[63]	Standard TSPEC element with header and length as described in the WMM specification

6.15.4 CMD_WMM_QUEUE_CONFIG

This sets, gets, or defaults the configuration of a WMM queue. Currently only 1 parameter is configurable through this command (MSDU Lifetime Expiry, as defined by the 802.11e MIB dot11EDCATableMSDULifetime attribute).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_QUEUE_CONFIG = 0x0070
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x00 = ACT_GET 0x01 = ACT_SET 0x02 = ACT_DEFAULT
AccessCategory	UINT8	Access category 0x00 = AC_BK 0x01 = AC_BE 0x02 = AC_VI 0x03 = AC_VO
MsduLifetimeExpiry	UINT16	ACT_SET = lifetime expiry of a MSDU in the AC queues for this access category (TU) (0 is ignored.) ACT_GET, ACT_DEFAULT = not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_QUEUE_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT8	Type of action set in request message
AccessCategory	UINT8	Same as request
MsduLifetimeExpiry	UINT16	Current MSDU lifetime expiry setting for this access category

6.15.5 CMD_WMM_QUEUE_STATS

This command manages the queue statistics collection for the WMM AC queues:

- Turn on or off statistics collection for a given AC
- Retrieve and clear the gathered statistics for a given AC

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_QUEUE_STATS = 0x0081
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x00 = START_STATS 0x01 = STOP_STATS 0x02 = GET_CLR_STATS
AccessCategory	UINT8	Access category 0x00 = AC_BK 0x01 = AC_BE 0x02 = AC_VI 0x03 = AC_VO
PktCount	UINT16	Not used (set to 0)
PktLoss	UINT16	Not used (set to 0)
AvgQueueDelay	UINT32	Not used (set to 0)
AvgTxDelay	UINT32	Not used (set to 0)
UsedTime	UINT16	Medium time used in units of 32 µs
PolicedTime	UINT16	Amount of time downgraded in units of 32 µs due to medium time violation
DelayHistogram	UINT16[7]	Not used (set to 0)
Reserved1	UINT16	Reserved (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_QUEUE_STATS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT8	Type of action set in request message
AccessCategory	UINT8	Same as request
PktCount	UINT16	Number of packets processed on the queue Retries not included.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>PktLoss</td><td>UINT16</td><td>Number of packets not successfully transmitted</td></tr>
<tr><td>AvgQueueDelay</td><td>UINT32</td><td>Average time the successfully transmitted packets (PktCount - PktLoss) spent in the queue (µs)</td></tr>
<tr><td>AvgTxDelay</td><td>UINT32</td><td>Not used (set to 0)</td></tr>
<tr><td>UsedTime</td><td>UINT16</td><td>Medium time used in units of 32 µs</td></tr>
<tr><td>PolicedTime</td><td>UINT16</td><td>Amount of time downgraded in units of 32 µs due to medium time violation</td></tr>
<tr><td>DelayHistogram</td><td>UINT16[7]</td><td>Histogram of successfully transmitted packets that fall into specific queue delay ranges:
<table>
<tr><th>Index</th><th>Description</th></tr>
<tr><td>[6]</td><td>50 ms <= delay</td></tr>
<tr><td>[5]</td><td>40 ms <= delay < 50 ms</td></tr>
<tr><td>[4]</td><td>30 ms <= delay <40 ms</td></tr>
<tr><td>[3]</td><td>20 ms <= delay < 30 ms</td></tr>
<tr><td>[2]</td><td>10 ms <= delay < 10 ms</td></tr>
<tr><td>[1]</td><td>5 ms <= delay < 10 ms</td></tr>
<tr><td>[0]</td><td>0 ms <= delay < 5 ms</td></tr>
</table></td></tr>
<tr><td>Reserved1</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
</table>

6.15.6 CMD_WMM_TS_STATS_CMD_802_11

This command retrieves the current state of a given traffic stream (TSID) from firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_TS_STATS_CMD_802_11 = 0x005D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET
TID	UINT8	Traffic stream identifier from 0 to 7

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_TS_STATS_CMD_802_11 | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result Code
Action	UINT16	Type of action set in request message

Field Name	Type	Description
TID	UINT8	Traffic stream identifier from 0 to 7
Valid	UINT8	TSID validity 0x00 = TSID has not been provisioned 0x01 = TSID has been successfully provisioned
AccessCategory	UINT8	AC TSID that was assigned to via UserPriority in the ADDTS Request
UserPriority	UINT8	UserPriority that was specified in the ADDTS Request
PowerSaveMode	UINT8	Power save mode 0x00 = legacy powersave (0) 0x01 = UAPSD set in the ADDTS Request
FlowDirection	UINT8	Flow direction 0x00 = upstream 0x01 = downlink 0x02 = reserved 0x03 = bi-directional
MediumTime	UINT16	MediumTime granted by the AP for the TSID in units of 32 µs

6.15.7 CMD_WMM_PARAM_CONFIG

This command is used to configure WMM parameters from host. It should be issued when the device (all interfaces) is in idle state. By default, STA/GC follows the WMM parameters of ext-AP/ext-GO after connected, uAP/GO use the default parameters in firmware. When the command issued, the parameters specified in the command will be applied to all the interfaces since there is only 1 WMM queue used by all interfaces for each AC.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_PARAM_CONFIG = 0x023A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
AcParam	Parameter records for BE, BK, VI, and VO	As defined in IEEE specification except reusing ACM bit (see note)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WMM_PARAM_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num

Field Name	Type	Description
Result	UINT16	Result code 0x0000 = successful 0x0001 = not supported
Action	UINT16	Type of action set in request message
AcParam	Parameter records for BE, BK, VI, and VO	As defined in IEEE specification except reusing ACM bit (see note)
AciAifsn	UINT8	Bit[7]: reserved Bits[6:5]: Aci Bit[4]: UCM (rename ACM bit to UCM) Bits[3:0]: Aifsn
EcwMinMax	UINT8	Bits[7:4]: EcwMax Bits[3:0]: EcwMin
TxopLimit	UINT16	Maximum Txop duration

Note: 4 sets of parameters are included in the *AcParam* field as the definition in EDCA Parameter Set element. Reuse the Bit[4] of *AciAifsn* field to control whether firmware should use the parameters of this AC as below:

- If UCM is set to 0, the default setting/behavior in firmware will be used.
- If UCM is set to 1, the corresponding setting in this command for this AC will be used.

6.16 802.11d

Table 33 lists the supported 802.11d command.

Table 33: 802.11d Command

Command	Description	Page
Section 6.16.1, CMD_802_11_D_DOMAIN	Configures 802.11d domain	page 153

6.16.1 CMD_802_11_D_DOMAIN

This command enables 802.11d controlled by the CMD_802_11_SNMP_MIB command with OID_DOT11_MULTI_DOMAIN_CAPAB_ENABLED. Writing a 1 enables 802.11d and writing a 0 disables 802.11d. In addition, when 802.11d is enabled, the country information needs to be set and that is done using this command.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_D_DOMAIN = 0x005B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read) 0x0001 = ACT_SET (write)
Domain	MrvlIETypes_DomainParam_t	802.11 domain parameter TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_D_DOMAIN | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Domain	MrvlIETypes_DomainParam_t	802.11 domain parameter TLV

Firmware does not perform any validity checks on the domain information. That is the responsibility of the driver/application. Firmware always assumes that the channels specified in the subband are adjacent channels (5 MHz apart). If the AP specifies a different channel spacing in the country IE, the driver must convert the channel representation in the country IE to 5 MHz spacing.

If the Domain TLV is not specified along with the SET command, then the existing country information will be “cleared” by firmware.

6.17 802.11h

The APIs in this section support the 802.11h protocol Dynamic Frequency Selection (DFS) protocol extensions. The 802.11h protocol is used for radar detection for 802.11a channels in Europe. 802.11h support in firmware is controlled by the SNMP command with OID 10. Writing a value of 0x00 disables 802.11h and writing 0x01 enables 802.11h.

Table 34 lists the supported 802.11h commands.

Table 34: 802.11h Commands

Command	Description	Page
Section 6.17.1, CMD_802_11_MEASUREMENT_REQUEST	Sends measurement request	page 154
Section 6.17.2, CMD_802_11_MEASUREMENT_REPORT	Gets measurement response report frame	page 156
Section 6.17.3, CMD_802_11_CHAN_SW_ANN	Broadcasts a channel switch announcement	page 157
Section 6.17.4, CMD_DFS_REPEATER_MODE	Enables/disables DFS repeater mode	page 158

6.17.1 CMD_802_11_MEASUREMENT_REQUEST

Using this command, the driver can send a measurement request to firmware. The driver must indicate the peer MAC address of the station that needs to make the measurement. If the MAC address is that of the current station, firmware makes the measurement and reply with a measurement response. If it is intended for another station, firmware sends the measurement request to the intended station and sends an event to the driver when it gets a response.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MEASUREMENT_REQUEST = 0x0062
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Peer Station Address	UINT8[6]	MAC address of the peer entity to which the request is sent
Dialog Token	UINT8	Same as the IEEE measurement request action frame dialog token
Measurement Request Set	Variable	Measurement request set See Table 35, Measurement Request Set, on page 155.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MEASUREMENT_REQUEST | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code CMD_STATUS_ERROR—measurement request could not be completed
Peer Station Address	UINT8[6]	MAC address of the peer entity to which the request is sent
Dialog Token	UINT8	Same as the IEEE measurement request action frame dialog token
Measurement Request Set	Variable	Measurement request set See Table 35, Measurement Request Set, on page 155.

Measurement Request Set

This measurement set will include measurement mode, measurement type, and measurement request. The measurement request has a start time and a duration time. In an IBSS, the driver may also couple a measurement request with a quiet element to quiet the other stations in the network. But it may do so only if it is the creator of the Ad-Hoc network.

Table 35: Measurement Request Set

Field Name	Type	Description
Request Mode	UINT8	Request mode
Type	UINT8	Measurement type
Measurement Request	Variable	Measurement request • basic report • CCA report • RPI report

There are 4 cases when the measurement request and measurement responses are exchanged.

- The driver may send a measurement request to firmware for a measurement to be done by firmware. In this case, firmware responds with a measurement response after the measurement is done.
- The driver may send a measurement request to firmware for a measurement to be done by the peer station. In this case, firmware sends the same measurement request to the peer station. When the peer station responds with a measurement report, firmware sends the same to the driver.
- On detecting radar, if the peer station accepts autonomous reports and if firmware is not the DFS owner of an IBSS, firmware sends an autonomous measurement report to the peer station. This report is not acknowledged by the driver. There is no measurement request in this transaction.
- The driver may choose to send a measurement report to a peer station. It does so by sending a measurement report command to firmware with the peer station address equal to the MAC address of the peer station. There is no measurement request in this transaction.

6.17.2 CMD_802_11_MEASUREMENT_REPORT

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MEASUREMENT_REPORT = 0x0063
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Peer Station Address	UINT8[6]	MAC Address of the peer entity from which the report came from
Dialog Token	UINT8	Same as the IEEE Measurement Report Action Frame Dialog Token
Measurement Report Set	Variable	Measurement report set See Table 35, Measurement Request Set, on page 155.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_MEASUREMENT_REPORT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Peer Station Address	UINT8[6]	Same as request
Dialog Token	UINT8	Same as request
Measurement Report Set	Variable	Measurement report set See Table 35, Measurement Request Set, on page 155.

The sequence of messages between driver and firmware is:

1. Firmware sends a CBP_EV_MEASUREMENT_RESPONSE_RDY event to the driver.
2. Driver should reply with this command with the Peer Station Address set to 0.
3. Firmware responds by filling this command with the measurement response report frame. Firmware also fills the peer station address with the MAC address of the station that generates the report.

6.17.3 CMD_802_11_CHAN_SW_ANN

If the driver chooses to broadcast a channel switch announcement, it can send this command to firmware. This is to be used for testing purposes only—in an IBSS mode, firmware manages any necessary channel switch announcements while in infrastructure mode, only the AP can send a channel switch announcement.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CHAN_SW_ANN = 0x0061
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Ch_Sw_Mode	UINT8	Set to 1 if transmissions are to cease pending the channel switch
ChNum	UINT8	New channel
ChSwCnt	UINT8	Number of TBTTs until the channel switch (as defined by IEEE)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_CHAN_SW_ANN | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Ch_Sw_Mode	UINT8	Same as request
ChNum	UINT8	Same as request
ChSwCnt	UINT8	Same as request

6.17.4 CMD_DFS_REPEATER_MODE

This command is used to enable/disable DFS repeater mode. DFS repeater mode is disabled by default.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_DFS_REPEATER_MODE = 0x012B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
mode	UINT16	Repeater mode 0x0000 = disable 0x0001 = enable

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_DFS_REPEATER_MODE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
mode	UINT16	Sane as set in request message

6.18 802.11n

The APIs in this section support the 802.11n protocol.

Table 34 lists the supported 802.11n commands.

Table 36: 802.11h Commands

6.18.1 CMD_11N_CFG

This command gets or sets the 802.11n configuration.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_CFG = 0x00CD
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>HTTxCapability</td><td>UINT16</td><td>HT Tx capability
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:7</td><td>Reserved (set to 0)</td></tr>
<tr><td>6</td><td>Short GI capability for 40 MHz transmit
0 = disable
1 = enable</td></tr>
<tr><td>5</td><td>Short GI capability for 20 MHz transmit
0 = disable
1 = enable</td></tr>
<tr><td>4</td><td>Greenfield capability for transmit
0 = disable
1 = enable</td></tr>
<tr><td>3:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1</td><td>Support channel width</td></tr>
<tr><td>0</td><td>LDPC enable/disable</td></tr>
</table></td></tr>
<tr><td>HTTxInfo</td><td>UINT16</td><td>HT Tx information
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:4</td><td>Reserved (set to 0)</td></tr>
<tr><td>3</td><td>RIFS mode for transmit
0 = disable RIFS
1 = enable RIFS</td></tr>
<tr><td>2:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>MiscConfig</td><td>UINT16</td><td>Miscellaneous Configuration
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1:0</td><td>Band selected
00 = both 2.4 GHz and 5 GHz (default) (ACT_GET returns 2.5 GHz setting only)
01 = 2.4 GHz
10 = 5 GHz
11 = reserved</td></tr>
</table></td></tr>
</table>

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_11N_CFG | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code (set to 0 for success)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Type of action set in request message</td></tr>
<tr><td>HTTxCapability</td><td>UINT16</td><td>HT Tx capability
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:7</td><td>Reserved (set to 0)</td></tr>
<tr><td>6</td><td>Short GI capability for 40 MHz transmit
0 = disable
1 = enable</td></tr>
<tr><td>5</td><td>Short GI capability for 20 MHz transmit
0 = disable
1 = enable</td></tr>
<tr><td>4</td><td>Greenfield capability for transmit
0 = disable
1 = enable</td></tr>
<tr><td>3:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1</td><td>Support channel width</td></tr>
<tr><td>0</td><td>LDPC enable/disable</td></tr>
</table></td></tr>
<tr><td>HTTxInfo</td><td>UINT16</td><td>HT Tx information
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:4</td><td>Reserved (set to 0)</td></tr>
<tr><td>3</td><td>RIFS mode for transmit
0 = disable RIFS
1 = enable RIFS</td></tr>
<tr><td>2:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>MiscConfig</td><td>UINT16</td><td>Miscellaneous Configuration
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1:0</td><td>Band selected
00 = both 2.4 GHz and 5 GHz (default) (ACT_GET returns 2.5 GHz setting only)
01 = 2.4 GHz
10 = 5 GHz
11 = reserved</td></tr>
</table></td></tr>
</table>

6.18.2 CMD_11N_ADDBA_REQ

This command is executed on the originator node and adds a Block Ack (acknowledgement) agreement with a recipient node. The driver provides only RA (if IBSS) and TID. For querying an existing Block Ack agreement (ACT_GET case), the driver sets RA and TID, and sets 0 for other subfields of BlockAckParamSet, BlockAckTmo, and SSN.

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_11N_ADDBA_REQ = 0x00CE</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>AddReqResult</td><td>UINT8</td><td>Not Used (set to 0)</td></tr>
<tr><td>PeerMACAddr</td><td>UINT8[6]</td><td>Peer MAC address
If 0 is infrastructure, firmware substitutes with AP’s MAC address.</td></tr>
<tr><td>DialogToken</td><td>UINT8</td><td>To distinguish between successive ADDBA requests due to retries
Should always be a non-0 value.</td></tr>
<tr><td>BlockAckParamSet</td><td>UINT16</td><td>Block Ack parameter set
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:6</td><td>Buffer size</td></tr>
<tr><td>5:2</td><td>TID</td></tr>
<tr><td>1</td><td>Block Ack policy
0 = delay Block Ack policy
1 = immediate Block Ack policy</td></tr>
<tr><td>0</td><td>AMSDU supported
0 = no AMSDU supported under this Block Ack agreement
1 = AMSDU supported under this Block Ack agreement</td></tr>
</table></td></tr>
<tr><td>BlockAckTmo</td><td>UINT16</td><td>Block Ack timeout value</td></tr>
<tr><td>SSN</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_ADDBA_REQ | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
AddReqResult	UINT8	Result of ADDBA request operation 0x00 = success 0x01 = execution failure 0x02 = timeout waiting for response 0x03 = data invalid

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>PeerMACAddr</td><td>UINT8[6]</td><td>Peer MAC address</td></tr>
<tr><td>DialogToken</td><td>UINT8</td><td>To distinguish between successive ADDBA requests due to retries</td></tr>
<tr><td>StatusCode</td><td>UINT16</td><td>See Section 7.3.1.9 in the 802.11 Specification</td></tr>
<tr><td>BlockAckParamSet</td><td>UINT16</td><td>Block Ack parameter set
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:6</td><td>Buffer size</td></tr>
<tr><td>5:2</td><td>TID</td></tr>
<tr><td>1</td><td>Block Ack policy
0 = delay Block Ack policy
1 = immediate Block Ack policy</td></tr>
<tr><td>0</td><td>AMSDU supported
0 = no AMSDU supported under this Block Ack agreement
1 = AMSDU supported under this Block Ack agreement</td></tr>
</table></td></tr>
<tr><td>BlockAckTmo</td><td>UINT16</td><td>Block Ack timeout value</td></tr>
<tr><td>SSN</td><td>UINT16</td><td>Starting sequence number
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:4</td><td>Starting sequence number</td></tr>
<tr><td>3:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
</table>

Note: The response of CMD_11N_ADDBA_REQ format is the same as that of request since the frame is sent from peer side as response.

6.18.3 CMD_11N_ADDBA_RSP

This command is executed on the recipient node, instructs firmware to send ADDBA response action frame to the originator.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_ADDBA_RSP = 0x00CF
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
AddReqResult	UINT8	Result of ADDBA request operation 0x00 = success 0x01 = execution failure 0x02 = timeout waiting for response 0x03 = data invalid
PeerMACAddr	UINT8[6]	Peer MAC address
DialogToken	UINT8	To distinguish between successive ADDBA requests due to retries
StatusCode	UINT16	See Section 7.3.1.9 in the 802.11 Specification

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>BlockAckParamSet</td><td>UINT16</td><td>Block Ack parameter set
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:6</td><td>Buffer size</td></tr>
<tr><td>5:2</td><td>TID</td></tr>
<tr><td>1</td><td>Block Ack policy
0 = delay Block Ack policy
1 = immediate Block Ack policy</td></tr>
<tr><td>0</td><td>AMSDU supported
0 = no AMSDU supported under this Block Ack agreement
1 = AMSDU supported under this Block Ack agreement</td></tr>
</table></td></tr>
<tr><td>BlockAckTmo</td><td>UINT16</td><td>Block Ack timeout value</td></tr>
<tr><td>SSN</td><td>UINT16</td><td>Starting sequence number</td></tr>
</table>

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_11N_ADDBA_RSP | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>AddReqResult</td><td>UINT8</td><td>Not Used (set to 0)</td></tr>
<tr><td>PeerMACAddr</td><td>UINT8[6]</td><td>Peer MAC address
If 0 is infrastructure, firmware substitutes with AP’s MAC address.</td></tr>
<tr><td>DialogToken</td><td>UINT8</td><td>To distinguish between successive ADDBA requests due to retries
The value should be same as the command received.</td></tr>
<tr><td>StatusCode</td><td>UINT16</td><td>See Section 7.3.1.9 in the 802.11 Specification</td></tr>
<tr><td>BlockAckParamSet</td><td>UINT16</td><td>Block Ack parameter set
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:6</td><td>Buffer size</td></tr>
<tr><td>5:2</td><td>TID</td></tr>
<tr><td>1</td><td>Block Ack policy
0 = delay Block Ack policy
1 = immediate Block Ack policy</td></tr>
<tr><td>0</td><td>AMSDU supported
0 = no AMSDU supported under this Block Ack agreement
1 = AMSDU supported under this Block Ack agreement</td></tr>
</table></td></tr>
<tr><td>BlockAckTmo</td><td>UINT16</td><td>Block Ack timeout value</td></tr>
<tr><td>SSN</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
</table>

6.18.4 CMD_11N_DELBA

This command instructs firmware to delete a Block Ack agreement.

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_11N_DELBA = 0x00D0</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code (set to 0 for success)</td></tr>
<tr><td>DelResult</td><td>UINT8</td><td>Not used (set to 0)</td></tr>
<tr><td>PeerMACAddr</td><td>UINT8[6]</td><td>Peer MAC address</td></tr>
<tr><td>DelBaParamSet</td><td>UINT16</td><td>DELBA parameter set
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:12</td><td>TID</td></tr>
<tr><td>11</td><td>Initiator
0 = recipient
1 = originator</td></tr>
<tr><td>10:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>ReasonCode</td><td>UINT16</td><td>Reason code sent for DELBA</td></tr>
</table>

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_DELBA | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
DelResult	UINT8	Result of DELBA request operation 0x00 = success 0x01 = execution failure 0x02 = timeout waiting for response 0x03 = data invalid
PeerMACAddr	UINT8[6]	Peer MAC address

Field Name	Type	Description
DelBaParamSet	UINT16	DELBA parameter set **Bits** / **Description** 15:12 — TID 11 — Initiator 0 = recipient 1 = originator 10:0 — Reserved (set to 0)
ReasonCode	UINT16	Reason code sent for DELBA

6.18.5 CMD_RECONFIGURE_TX_BUFF

This command gets/sets the Tx buffer size.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_RECONFIGURE_TX_BUFF = 0x00D9
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
BuffSize	UINT16	Buffer size If Action = Set and BuffSize = 0xFFFF, it indicates to firmware to reclaim all Tx buffers from HID, and reassign them according to the updated buffer limits.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_RECONFIGURE_TX_BUFF | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
BuffSize	UINT16	Buffer size If Action = GET, then current firmware Tx buffer size. Otherwise, value is same as Request.

6.18.6 CMD_AMSDU_AGGR_CTRL

Firmware is able to generate event along with maximum aggregation size to indicate if AMSDU is recommender or not. This feature is disabled in firmware by default. This command enables/disables this feature.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_AMSDU_AGGR_CTRL = 0x00DF
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Enable	UINT16	0x0000 = disable 0x0001 = enable
MaxAggrSize	UINT16	Not used

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_AMSDU_AGGR_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Enable	UINT16	0x0000 = disable 0x0001 = enable
MaxAggrSize	UINT16	Maximum AMSDU aggregation size 0x0000 = AMSDU aggregation is not recommended

6.18.7 CMD_11N_2040COEX

This command is used for sending 20/40 Mz coexistence management action to AP after Overlapping Basic Service Set (OBSS) scan is done.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_2040COEX = 0x00E9
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
CoexElement	MrvlIETypes_2040Coex_t	20/40 MHz coexistence element
ChannelReport	MrvlIETypes_2040BSSIntolerantChannelReport_t	Intolerant channel report element

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_11N_2040COEX | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
CoexElement	MrvlIETypes_2040Coex_t	20/40 MHz coexistence element
ChannelReport	MrvlIETypes_2040BSSIntolerantChannelReport_t	Intolerant channel report element

6.18.8 CMD_RX_PKT_COALESC_CFG

This command gets or sets the Rx packet coalescing. This feature enables packet aggregation in firmware rather than interrupting host for each Rx packet. Rx packets from MAC would be sent to host when either packet count reaches certain threshold or timer which is started at first packet reception expires (whichever is earlier).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_RX_PKT_COALESC_CFG = 0x012C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
packet_threshold	UINT32	Packet threshold
delay	UINT16	Timeout value

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_RX_PKT_COALESC_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
packet_threshold	UINT32	Packet threshold
delay	UINT16	Timeout value

6.18.9 CMD_802_11_EAPOL_AMSDU_INDICATION

This command is used by the host to send GTK frame to firmware. This command needs to be handled by the embedded supplicant on the host side. It cannot be used with host supplicant.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_EAPOL_AMSDU_INDICATION = 0x012E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = ACT_SET
tlv_buffer	MrvlIETypes_Eapol_Pkt_t	Embedded WPA EAPOL packet

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_EAPOL_AMSDU_INDICATION | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type action set in request message
tlv_buffer	MrvlIETypes_Eapol_Pkt_t	Embedded WPA EAPOL packet

6.19 802.11ac

The APIs in this section support the 802.11ac protocol.

Table 37 lists the supported 802.11ac command.

Table 37: 802.11ac Commands

Command	Description	Page
Section 6.19.1, CMD_11AC_CFG	Gets/sets 802.11ac VHT Tx/Rx capability and MCS set	page 171

6.19.1 CMD_11AC_CFG

This command gets/sets the 802.11ac VHT Tx/Rx capability and supported MCS set.

In STA mode, it can be used to control VHT Tx option (BW80, LDPC, short GI for BW80, and STBC).

In NXP Mobile Hotspot (MMH) mode, the command is used to configure the 802.11ac parameters MMH should advertise and use, that is, Tx and Rx Options. Currently not supported in firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_11AC_CFG = 0x0112
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
BandConfig	UINT8	Band configuration Bits[7:2]: reserved (set to 0) Bit[1]: 5 GHz select Bit[0]: 2.4 GHz select **NOTE:** For ACT_GET, only 1 bit must be set, otherwise the command will return error.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>MiscConfig</td><td>UINT8</td><td>Miscellaneous configuration
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:3</td><td>Reserved (set to 0)</td></tr>
<tr><td>2</td><td>BW80 Tx if Bit[0] = 1
0 = follow BW from 11N_CFG
1 = follow BW from VHTCap
NOTE: STBC and LDPC are always taken from VHT Capability Information field.
GI is dependent of BW. For 80/160 MHz rates, GI configuration in VHTCap is used. For 20/40 MHz rates, GI configuration in 11N_CFG is used.</td></tr>
<tr><td>1</td><td>VHT configuration for Rx
0 = disable
1 = enable</td></tr>
<tr><td>0</td><td>VHT configuration for Tx
0 = disable
1 = enable</td></tr>
<tr><td colspan="2">NOTE: For action ACT_GET, only 1 bit must be set, otherwise the command will return an error.</td></tr>
</table></td></tr>
<tr><td>VHTCapabilityInfo</td><td>UINT32</td><td>VHT capability information
Same definition as VHT Capability Information field in 802.11ac Specifications.
For VHT Tx capability in STA mode:
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7</td><td>STBC
0 = disable
1 = enable</td></tr>
<tr><td>6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5</td><td>Short GI for BW80 Tx
0 = disable
1 = enable</td></tr>
<tr><td>4</td><td>LDPC
0 = disable
1 = enable</td></tr>
<tr><td>3:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>VHTSupportMCSSet</td><td>UINT64</td><td>VHT supported MCS set
Same definition as VHT Supported MCS Set field in 802.11ac specifications.
NOTE: This field is ignored in STA mode.</td></tr>
</table>

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_11AC_CFG | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code (set to 0 for success)</td></tr>
<tr><td>Action</td><td>UINT16</td><td>Type of action set in request message</td></tr>
<tr><td>BandConfig</td><td>UINT8</td><td>Band configuration
Bits[7:2]: reserved (set to 0)
Bit[1]: 5 GHz select
Bit[0]: 2.4 GHz select
NOTE: For action ACT_GET, configuration parameters for the band requested are returned.</td></tr>
<tr><td>MiscConfig</td><td>UINT8</td><td>Miscellaneous configuration
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>7:3</td><td>Reserved (set to 0)</td></tr>
<tr><td>2</td><td>BW80 Tx if Bit[0] = 1
0 = follow BW from 11N_CFG
1 = follow BW from VHTCap
NOTE: STBC and LDPC are always taken from VHT Capability Information field.
GI is dependent on BW. For 80/160 MHz rates, GI configuration in VHTCap is used. For 20/40 MHz rates, GI configuration in 11N_CFG is used.</td></tr>
<tr><td>1</td><td>VHT configuration for Rx
0 = disable
1 = enable</td></tr>
<tr><td>0</td><td>VHT configuration for Tx
0 = disable
1 = enable</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>VHTCapabilityInfo</td><td>UINT32</td><td>VHT capability information
Same definition as VHT Capability Information field in 802.11ac specifications.
For VHT Tx capability in STA mode:
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>7</td><td>STBC
0 = disable
1 = enable</td></tr>
<tr><td>6</td><td>Reserved (set to 0)</td></tr>
<tr><td>5</td><td>Short GI for BW80 Tx
0 = disable
1 = enable</td></tr>
<tr><td>4</td><td>LDPC
0 = disable
1 = enable</td></tr>
<tr><td>3:0</td><td>Reserved (set to 0)</td></tr>
</table></td></tr>
<tr><td>VHTSupportMCSSet</td><td>UINT64</td><td>VHT supported MCS set
Same definition as VHT Supported MCS Set field in 802.11ac specifications.
NOTE: This field is ignored in STA mode.</td></tr>
</table>

6.20 Memory Efficient Filtering

Memory Efficient Filtering (MEF) is a generic Wi-Fi packet filtering function.

The driver should use the CMD_MEF_CFG command to send 0 or more MEF entries to firmware.

Note: For more information on how to form the MEF expressions, see the Memory Efficient Filtering Application Note.

6.20.1 CMD_MEF_CFG

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MEF_CFG = 0x009A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Criteria	UINT32	Criteria for frame types that firmware receives in host sleep mode
NumEntries	UINT16	Number of MEF entries
MEFEntry	MEFEntry	0 or more entries of Table 38, MEFEntry

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MEF_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Criteria	UINT32	Criteria for frame types that firmware receives in host sleep mode
NumEntries	UINT16	Number of MEF entries
MEFEntry	MEFEntry	0 or more entries of Table 38, MEFEntry

Table 38: MEFEntry

Field Name	Type	Description
Mode	UINT8	Mode of receiving packet
Action	UINT8	Action on matching filter
ExprSize	UINT16	Number of bytes in MEF expression
Expr[ExprSize]	UINT8	MEF expression

6.21 Small Debug Print

This feature provides a mechanism to monitor firmware debug messages (vitals and trace memory) over the air or through the host. There are 2 commands to support this feature.

Table 39: Debug Print Commands

Command	Description	Page
Section 6.21.1, CMD_DBGS_CFG	Configures small debug print facility	page 176
Section 6.21.2, CMD_GET_MEM	Retrieves firmware vitals or trace memory	page 178

6.21.1 CMD_DBGS_CFG

This command configures the Small Debug Print facility. It configures:

- Output destination(s)
- Selection of debug prints as enabled based on IDs

REQUEST

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_DBGS_CFG = 0x008B</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Not used (set to 0)</td></tr>
<tr><td>Destination</td><td>UINT8</td><td>Debug output destination
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:3</td><td>Reserved (set to 0)</td></tr>
<tr><td>2</td><td>1 = enable vitals to air</td></tr>
<tr><td>1</td><td>1 = debug packet to air</td></tr>
<tr><td>0</td><td>1 = debug packet to debug viewer on host</td></tr>
</table></td></tr>
<tr><td>Reserved</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
<tr><td>En_NumEntries</td><td>UINT8</td><td>Number of enable entries
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7</td><td>0 = let subsequent enable entries specify IDs to enable
1 = enable all debug prints, ignore subsequent enable entries</td></tr>
<tr><td>6:0</td><td>Number of enable entries
Valid range: 0 through 2.</td></tr>
</table></td></tr>
</table>

Enable Entries (0 or more of the following)

Enable Entry	Type	Description
EnableMode_MaskOrID	UINT16	Bits[15:13]: EnableMode 0x0002 = enable range method of IDs specified by Mask and Base
		Bits[12:0]: MaskOrID Mask when EnableMode uses range method.
BaseOrID	UINT16	Bits[15:13]: Reserved (set to 0)
		Bits[12:0]: BaseOrID Base when EnableMode uses range method.

An Enable Entry is used to selectively specify IDs to enable their corresponding debug prints.

Note:

- In determining the enabling of each Debug Print calling point, ID matching is first performed in the ascending order starting from global enable subfield.
- A match quits the ascending traversal.
- A match is followed by the print action per enable/disable setting.

RESPONSE

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>CmdCode</td><td>UINT16</td><td>CMD_DBGS_CFG | 0x8000</td></tr>
<tr><td>Size</td><td>UINT16</td><td>Number of bytes in command body</td></tr>
<tr><td>SeqNum</td><td>UINT8</td><td>Command sequence number (same as that in Request)</td></tr>
<tr><td>BSS</td><td>UINT8</td><td>Bits[7:4]: BSS_Type
Bits[3:0]: BSS_Num</td></tr>
<tr><td>Result</td><td>UINT16</td><td>Result code (set to 0 for success)</td></tr>
<tr><td>Destination</td><td>UINT8</td><td>Debug output destination
Same as that in Request.</td></tr>
<tr><td>Reserved</td><td>UINT16</td><td>Reserved (set to 0)</td></tr>
<tr><td>En_NumEntries</td><td>UINT8</td><td>Number of enable entries
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7</td><td>0 = let subsequent enable entries specify IDs to enable
1 = enable all debug prints, ignore subsequent enable entries</td></tr>
<tr><td>6:0</td><td>Number of enable entries
Valid range: 0 through 2.</td></tr>
</table>
</td></tr>
</table>

6.21.2 CMD_GET_MEM

This command is used by the host to retrieve the firmware vitals or trace memory. This command is designed for getting a memory segment, even if the memory segment is larger than the command response buffer. It uses the model of the requested memory broken up into fragments with each command response carrying a fragment to the host.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_MEM = 0x008C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
StartAddr	UINT32	Specifies the starting address of firmware memory segment to get It must not be 0 and must be 4-byte aligned. Special handling cases specify these values for StartAddr field 0x00000001 = get vitals in firmware 0x00000002 = get trace memory in firmware
Len	UINT16	Number of bytes to get For the special handling cases, set the length to 0.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_MEM | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (same as that in Request)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
NextAddr	UINT32	0x00000000 if last fragment, else address of next fragment
RemainingLen	UINT16	Remaining length of segment

Note: To get the next fragment, the host should send a subsequent request with:

- StartAddr = previous Response's NextAddr
- Len = previous Response's RemainingLen

The last response (last fragment) from firmware will contain NextAddr=0.

6.22 Wi-Fi Direct

Wi-Fi Direct is the framework for device to device Wi-Fi connectivity. A Wi-Fi Direct device supports the following mechanisms:

- Discovery (Device Discovery and Service Discovery)
- Pairing (including Group Formation and Point-to-Point (P2P) Invitation)
- Connectivity
- Power Management
- Group Management
- Coexistence
- Legacy

Table 40 lists the commands that support this feature.

Table 40: Wi-Fi Commands

Command	Description	Page
Section 6.22.1, CMD_WIFI_DIRECT_PARAMS_CONFIG	Sets/modifies/deletes Wi-Fi Direct protocol related parameters	page 179
Section 6.22.2, CMD_WIFI_DIRECT_MODE_CONFIG	Sets device mode externally	page 186
Section 6.22.3, CMD_802_11_ACTION_FRAME	Sends different types of Wi-Fi Direct Action Frames	page 187
Section 6.22.4, CMD_WIFI_DIRECT_SERVICE_ DISCOVERY_PACKET	Sends GAS packet sent by driver to Wi-Fi	page 190

6.22.1 CMD_WIFI_DIRECT_PARAMS_CONFIG

This command is used to set/modify/delete Wi-Fi Direct protocol related parameters like minDiscoverableInterval, maxDiscoverableInterval, Wi-Fi Direct device state transition, and so on, as well as to trigger some specific operations.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_PARAMS_CONFIG = 0x00EA
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read data) 0x0001 = ACT_SET (write data)
ConfigParams	TLV	ConfigParams consists of variable number of TLVs

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_PARAMS_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Type of action set in request message
ConfigParams	TLV	ConfigParams consists of variable number of TLVs

The following TLVs can be included in the ConfigParams:

- WIFI_DIRECT_DISC_PERIOD_TLV_ID
- WIFI_DIRECT_SCAN_ENABLE_TLV_ID
- WIFI_DIRECT_SCAN_PEER_DEVICE_ID_TLV_ID
- WIFI_DIRECT_SCAN_REQ_DEVICE_TYPE_TLV_ID
- WIFI_DIRECT_DEVICE_STATE_TLV_ID
- WIFI_DIRECT_INTENT_VALUE_TLV_ID
- WIFI_DIRECT_CAPABILITY_TLV_ID
- WIFI_DIRECT_NOA_TLV_ID
- WIFI_DIRECT_OPP_PS_TLV_ID
- WIFI_DIRECT_INVITATION_LIST_TLV_ID
- WIFI_DIRECT_LISTEN_CHANNEL_TLV_ID
- WIFI_DIRECT_OPERATING_CHANNEL_TLV_ID
- WIFI_DIRECT_PERSISTENT_GROUP_TLV_ID
- WIFI_DIRECT_PERSISTENT_GROUP_INVOKE_TLV_ID
- WIFI_DIRECT_PRESENSE_REQ_PARAMS_TLV_ID
- WIFI_DIRECT_EXTENDED_LISTEN_TIME_TLV_ID
- WIFI_DIRECT_PROVISIONING_PARAMS_TLV_ID
- WIFI_DIRECT_WPS_PARAMS_TLV_ID

A Wi-Fi Direct Host utility can be developed to send these TLVs individually (as separate commands) or collectively (through a config file). The individual command usage is indicated wherever possible.

6.22.1.1 WIFI_DIRECT_DISC_PERIOD_TLV_ID

This TLV is used to set/get the minimum and maximum discovery periods.

Field Name	Type	Description
Type	UINT16	Type ID = 0x017C
Length	UINT16	Length of payload (4 bytes)
MinDiscInterval	UINT16	Minimum discovery period
MaxDiscInterval	UINT16	Maximum discovery period

6.22.1.2 WIFI_DIRECT_SCAN_ENABLE_TLV_ID

This TLV is used to enable Wi-Fi Direct scan.

Field Name	Type	Description
Type	UINT16	Type ID = 0x017D
Length	UINT16	Length of payload
Enable	UINT8	Scan mode 0 = disable 1 = enable

6.22.1.3 WIFI_DIRECT_SCAN_PEER_DEVICE_ID_TLV_ID

This TLV is used to set a particular peer's device ID in directed Wi-Fi Direct scan.

Field Name	Type	Description
Type	UINT16	Type ID = 0x017E
Length	UINT16	Length of payload (6 bytes)
PeerDeviceId	UINT8[6]	Peer's device ID in P2P IE

6.22.1.4 WIFI_DIRECT_SCAN_REQ_DEVICE_TYPE_TLV_ID

This TLV is used to specify the request device type attribute in the WPS IE during directed Wi-Fi Direct scan.

Field Name	Type	Description
Type	UINT16	Type ID = 0x017F
Length	UINT16	Length of payload (8 bytes)
PeerDeviceId	UINT8[8]	Requested device type attribute in WPS IE

6.22.1.5 WIFI_DIRECT_DEVICE_STATE_TLV_ID

This TLV is used to get the current device state.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0180
Length	UINT16	Length of payload (2 bytes)
CurrDeviceState	UINT16	Device's current state

6.22.1.6 WIFI_DIRECT_INTENT_VALUE_TLV_ID

This TLV is used to get/set the intent value.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0181
Length	UINT16	Length of payload (1 byte)
IntentValue	UINT8	Intent value

6.22.1.7 WIFI_DIRECT_CAPABILITY_TLV_ID

This TLV is used to get/set the persistent capability setting.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0182
Length	UINT16	Length of payload (2 bytes)
deviceCapability	UINT8	Device capability bitmap
groupCapability	UINT8	Group capability bitmap
NOTE: The uAP command SYS_CFG_PKT_FWD_CTL can be used to control intra-BSS distribution.		

6.22.1.8 WIFI_DIRECT_NOA_TLV_ID

This TLV is used to get/set the Group Owner's NOA settings.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0183
Length	UINT16	Length of payload (4 bytes)
Enable/Disable	UINT8	0 = disable 1 = enable
Index	UINT16	Index of NoA schedule
NoACount	UINT8	NoA schedule count
NoADuration	UINT32	Duration of NoA (in TUs)
NoAInterval	UINT32	Interval of NoA (in TUs)

6.22.1.9 WIFI_DIRECT_OPP_PS_TLV_ID

This TLV is used to get/set the Group Owner's Opportunistic Power Save settings.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0184
Length	UINT16	Length of payload (2 bytes)
Enable/Disable	UINT8	0 = disable 1 = enable
CT Window	UINT8	CT window value

6.22.1.10 WIFI_DIRECT_INVITATION_LIST_TLV_ID

This TLV is used to get/set peer whose invitation request should be accepted when received.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0185
Length	UINT16	Length of payload (6 bytes)
InvPeer	UINT8[6]	Peer device ID

6.22.1.11 WIFI_DIRECT_LISTEN_CHANNEL_TLV_ID

This TLV is used to get/set the listen channel value.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0186
Length	UINT16	Length of payload (5 bytes)
Country String	UINT8[3]	Country string
Operating Class	UINT8	Operating class
ListenChannel	UINT8	Listen channel

6.22.1.12 WIFI_DIRECT_OPERATING_CHANNEL_TLV_ID

This TLV is used to get/set the operating channel value.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0187
Length	UINT16	Length of payload (2 bytes)
Country String	UINT8[3]	Country string
Operating Class	UINT8	Operating class
OpChannel	UINT8	Operating channel

6.22.1.13 WIFI_DIRECT_PERSISTENT_GROUP_TLV_ID

This TLV is used to get/set the persistent group record.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0188
Length	UINT16	Length of payload
RecordINdex	UINT8	Index of persistent group in database
DeviceRole	UINT8	Device role 0 = client 1 = group owner (GO)
GroupBssid	UINT8[6]	Group BSS ID
GroupDev	UINT8[6]	Device ID in GroupId of persistent group
GroupIdSsidLen	UINT8	GroupId SSID length
GroupSsid	UINT8[]	Group SSID length 0 to 32 characters.
PSKLen	UINT8	PSK length
PSK	UINT8[]	Hex input (32 octets) characters ASCII input (minimum 8, maximum 63) characters
NumClients	UINT8	Number of clients, when DeviceRole is GO
PeerMacAddr1	UINT8[6]	Peer MAC address, if present, when DeviceRole is GO
PeerMacAddr2	UINT8[6]	Peer MAC address, if present, when DeviceRole is GO
...	...	...
PeerMacAddrN	UINT8[6]	Peer MAC address, if present, when DeviceRole is GO

6.22.1.14 WIFI_DIRECT_PERSISTENT_GROUP_INVOKE_TLV_ID

This TLV is used to get/set the persistent group index to be invoked while sending invitation request. Based on the Length field value, the intent of the command—to get/set persistent group record or to invoke an existing persistent group record—is interpreted.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0188
Length	UINT16	Length of payload
groupIndex	UINT8	Persistent group record index

6.22.1.15 WIFI_DIRECT_PRESENSE_REQ_PARAMS_TLV_ID

This TLV is used to get/set the presence request parameters that the device wants to invoke while sending presence request.

Field Name	Type	Description
Type	UINT16	Type ID = 0x018D
Length	UINT16	Length of payload (9 bytes)
Type	UINT8	Presence request type 0x01 = preferred values 0x02 = acceptable values
Duration	UINT8[4]	Preferred (or minimal acceptable) presence period duration
Interval	UINT8[4]	Referred (or maximum acceptable) interval between presence periods

6.22.1.16 WIFI_DIRECT_EXTENDED_LISTEN_TIME_TLV_ID

This TLV is used to get/set the extended listen timing parameters.

Field Name	Type	Description
Type	UINT16	Type ID = 0x018E
Length	UINT16	Length of payload (4 bytes)
AvailabilityPeriod	UINT16	Extended listen time availability period
AvailabilityInterval	UINT16	Extended listen time interval

6.22.1.17 WIFI_DIRECT_PROVISIONING_PARAMS_TLV_ID

This TLV is used to get/set the WPS provisioning protocol related parameters used by the device.

Field Name	Type	Description
Type	UINT16	Type ID = 0x018F
Length	UINT16	Length of payload (6 bytes)
Action	UINT16	Action 0x0000 = request parameters 0x0001 = response parameters
ConfigMethod	UINT16	Configuration method used during provision discovery 0x0008 = display pin 0x0080 = push button 0x0100 = keypad 0xFFFF = none
DevicePasswordId	UINT16	Device password ID used during GO negotiation 0x0001 = user specified 0x0004 = push button 0x0005 = registrar specified 0xFFFF = none

The request parameters are specified by the host application. The response parameters are decided by firmware based on the provision discovery request/GO negotiation request sent by the peer device.

While sending Provision Discovery Request, the device will request that the peer uses the Configuration Method configured by this API. The Device Password ID is automatically decided by firmware based on the requested Configuration Method.

While sending GO negotiation Request or Response, the device will use Device Password ID configured by this API.

6.22.1.18 WIFI_DIRECT_WPS_PARAMS_TLV_ID

This TLV is used to get/set the information about the PIN/PBC availability. This API mainly provides a way for the host to indicate to firmware if the PIN/PBC information is available or entered by the user. Firmware uses this information to correctly respond to the peer device during GO negotiation.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0190
Length	UINT16	Length of payload (2 bytes)
Action	UINT16	Action 0x0000 = PIN available 0x0004 = button pushed 0xFFFF = none

6.22.2 CMD_WIFI_DIRECT_MODE_CONFIG

This command is used to externally set the Device Mode and/or to initiate device discovery related actions.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_MODE_CONFIG = 0x00EB
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read data) 0x0001 = ACT_SET (write data)
Mode	UINT16	Mode 0x0000 = disable WIFI_DIRECT, clear existing parameters 0x0001 = device in WIFI_DIRECT configured state 0x0002 = GO mode 0x0003 = WIFI_DIRECT client mode 0x0004 = device find phase start 0x0005 = stop find phase (if any) and place device in listen state 0x0006 = disable extended listen state 0x0009 = enable extended listen state 0x000C = place device in disabled state, configuration is preserved 0x000D = place device in listen state 0x000E = disable listen state 0x001D = place device in fast listen mode (Windows only) Others = reserved

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_MODE_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result 0x0000 = success 0x0001 = failure

Field Name	Type	Description
Action	UINT16	Type of action set in request message
Mode	UINT16	Mode 0x0000 = disable WIFI_DIRECT, clear existing parameters 0x0001 = device in WIFI_DIRECT configured state 0x0002 = GO mode 0x0003 = WIFI_DIRECT client mode 0x0004 = device find phase start 0x0005 = stop find phase (if any) and place device in listen state 0x0006 = disable extended listen state 0x0009 = enable extended listen state 0x000C = place device in disabled state, configuration is preserved 0x000D = place device in listen state 0x000E = disable listen state 0x001D = place device in fast listen mode (Windows only) Others = reserved

6.22.3 CMD_802_11_ACTION_FRAME

This command is used to send different types of Wi-Fi Direct Action Frames to the peer Wi-Fi Direct Device.

The P2P IE subelements and/or WPS IE parameters needed for the P2P specific Action Frame are passed as part of this command. If the subelements have already been passed earlier using a different command (for example, SYS_CONFIG or CMD_802_11_MGMT_IE_LIST command) and they need no update, then the IE can be skipped. The IE sublelements sent by this command will be updated in firmware. The command handler of this command will form the Action Frame and send it out.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_ACTION_FRAME = 0x00F4
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0) Peer's channel (set to non-0)
PeerMacAddr	UINT8[6]	Peer MAC address
Category	UINT8	Category 0x04 = Public Action Frame 0x7F = vendor specific
Action	UINT8	For Public Action Frame: 0x09 = vendor specific usage Others = reserved
OUI	UINT8[3]	50 6F 9A = WFA specific OUI
OUIType	UINT8	Identify type or version of Action Frame 0x09 = WFA P2P

Field Name	Type	Description
OUISubtype	UINT8	Public Action Frame subtype—WFA P2P: 0x00 = GO negotiation request 0x01 = GO negotiation response 0x02 = GO negotiation confirmation 0x03 = P2P invitation request 0x04 = P2P invitation response 0x05 = device discoverability request 0x00 = device discoverability response 0x07 = provision discovery request 0x08 = provision discovery response Vendor specific subtypes—P2P Action Frame: 0x00 = notice of absence 0x01 = P2P presence request 0x02 = P2P presence response 0x03 = GO discoverability request
DialogToken	UINT8	Dialog Token 0 = ignore
IEList	TLV	Public Action Frame subtype—WFA P2P: • Peer Operating or Listen Channel TLV • Passthrough Buffer TLV • Wi-Fi Direct TLVs Vendor specific subtypes—P2P Action Frame: • Peer Operating or Listen Channel TLV • Passthrough Buffer TLV • Wi-Fi Direct TLVs

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_ACTION_FRAME | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result 0x0000 = success 0x0001 = failure
PeerMacAddr	UINT8[6]	Peer MAC address
Category	UINT8	Category 0x04 = Public Action Frame 0x7F = vendor specific
Action	UINT8	Type of action set in request message
OUI	UINT8[3]	OUI
OUIType	UINT8	Identify type or version of Action Frame 0x09 = WFA P2P

Field Name	Type	Description
OUISubtype	UINT8	Public Action Frame subtype—WFA P2P: 0x00 = GO negotiation request 0x01 = GO negotiation response 0x02 = GO negotiation confirmation 0x03 = P2P invitation request 0x04 = P2P invitation response 0x05 = device discoverability request 0x00 = device discoverability response 0x07 = provision discovery request 0x08 = provision discovery response Vendor specific subtypes—P2P Action Frame: 0x00 = notice of absence 0x01 = P2P presence request 0x02 = P2P presence response 0x03 = GO discoverability request
DialogToken	UINT8	Dialog Token 0 = ignore
PeerMacAddr	UINT8[6]	Peer MAC address
IEList	TLV	TLVs if present in the command

6.22.4 CMD_WIFI_DIRECT_SERVICE_DISCOVERY_PACKET

This command is used to send Generic Advertisement Service (GAS) packet sent by the driver to Wi-Fi. It is assumed that the host will construct the entire GAS packet and send it to firmware. The host takes care of the SD packet fragmentation.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_SERVICE_DISCOVERY_PACKET = 0x00EC
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0) Peer's channel (set to non-0)
PeerMacAddr	UINT8[6]	MAC address of the peer to whom SD packet is to be sent
SDPacket	UINT8[]	Service Discovery packet to be sent to Wi-Fi

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_WIFI_DIRECT_SERVICE_DISCOVERY_PACKET | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result 0x0000 = success 0x0001 = failure

If the host interface is SDIO, the general format of an event data packet for SDIO interface is as follows:

Event Data Packet
SDIO packet header (4 bytes)
Event ID (4 bytes)
Event data (variable length)

6.23 Data Flow Control

Table 41 lists the data flow control command.

Table 41: Data flow Control Command

Command	Description	Page
Section 6.23.1, CMD_CFG_TX_DATA_PAUSE	Controls Tx data packets buffering by firmware	page 191
Section 6.23.2, CMD_COALESCE_CFG	Sets rules to filter and buffer certain type of packet	page 192

6.23.1 CMD_CFG_TX_DATA_PAUSE

This command is used to control Tx data packet buffering by firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_TX_DATA_PAUSE = 0x0103
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
PauseTxEn	UINT8	If Action = 1: 0x00 = disable pause Tx events from firmware to host 0x01 = enable pause Tx events from firmware to host
PauseTxCnt	UINT8	Maximum number of Tx buffers allowed for all PS clients

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_TX_DATA_PAUSE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
PauseTxEn	UINT8	If Action = 1: 0x00 = disable pause Tx events from firmware to host 0x01 = enable pause Tx events from firmware to host
PauseTxCnt	UINT8	Maximum number of Tx buffers allowed for all PS clients

6.23.2 CMD_COALESCE_CFG

This command sets the rules to filter and buffer certain type of packet, which will reduce the number of received interrupt to host. It can improve the processing overhead and the power consumption.

Received packets that match the receive filters are cached or coalesced in firmware. The adapter does not generate a receive interrupt for coalesced packets right away. Instead, the adapter interrupts the host when another event occurs.

Firmware will generate a receive interrupt when 1 of the following events occurs:

- The expiration of a timer whose expiration time is set to a maximum coalescing delay value of the matching receive filter
- The available space within the coalescing buffer reaches a specified low-water mark
- A packet is received that does not match a coalescing filter
- Another interrupt event has occurred

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_COALESCE_CFG = 0x010A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
NumRules	UINT16	Number of rules 0x0000 = clear all previous rules Others = number of rules added into firmware (maximum number of rules is 8)
Rules	MrvlCoalesceRule_t	Rule to filter and set condition to report coalesce packet

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_COALESCE_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in the request message
NumRules	UINT16	Number of rules 0x0000 = clear all previous rules Others = number of rules added into firmware (maximum number of rules is 8)
Rules	MrvlCoalesceRule_t	Rule to filter and set condition to report coalesce packet

6.24 Multi-Channel Operation

Table 42 lists the multi-channel operation commands.

Table 42: Multi-channel Operation Commands

Command	Description	Page
Section 6.24.1, CMD_MULTI_CHAN_TIMESLICE_STATISTICS	Gets timeslice and channel switching statistics	page 193
Section 6.24.2, CMD_MULTI_CHAN_CONFIG	Configures multi-channel operation	page 193
Section 6.24.3, CMD_MULTI_CHAN_POLICY	Gets/sets multi-channel policy	page 196
Section 6.24.4, CMD_MULTI_CHAN_TIMESLICE_CONFIG	Configures timeslice of each channel for DRCS	page 197
Section 6.24.5, CMD_DMCS_CONFIG	Gets/sets status of DMCS agent	page 198

6.24.1 CMD_MULTI_CHAN_TIMESLICE_STATISTICS

This command is used to get the timeslice and channel switching statistics.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_TIMESLICE_STATISTICS = 0x011D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_TIMESLICE_STATISTICS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	0x0000 = no error 0x0001 = command failed 0x0002 = command is not supported
TimeSlice_Cycle_Statistcs	TimeSlice_Cycle_Statistics_t	Structure contains switching information and timeslice information of all the multi-channels

Table 43: TimeSlice_Cycle_Statistics_t

Field Name	Type	Description
RadioTemperatureCelsiusTenths	UINT32	Operating temperature of the device
MinSwitchingTimeUs	UINT32	Minimum channel switching time
MaxSwitchingTimeUs	UINT32	Maximum channel switching time
AvgSwitchingTimeUs	UINT32	Average channel switching time
NoOfSwitches	UINT32	Total number of channel switches occurred
NoOfTimeSlices	UINT16	Number of timeslices present in cycle - 2 Maximum simultaneous channels of operation supported (presently fixed to 2).
Reserved	UINT16	Not used (set to 0)
TimeSlice_Statistics[0]	TimeSlice_Statistics_t	Contains timeslice information of the first active channel
...	...	...
TimeSlice_Statistics[N-1]	TimeSlice_Statistics_t	Contains timeslice information of the N^{th} active channel

If no active channel is present then TimeSlice_Statistics_t structure will not be present in the command response.

Table 44: TimeSlice_Statistics_t

Field Name	Type	Description
MinTimeSliceLenUs	UINT32	Minimum timeslice length (µs)
MaxTimeSliceLenUs	UINT32	Maximum timeslice length (µs)
AvgTimeSliceLenUs	UINT32	Average timeslice length (µs)
TimeSliceCount	UINT32	Number of timeslices occurred in this channel
TimeSliceDurLastSec	UINT32	Total Time (µs) spent on this timeslice in last second
AvgTxRate	UINT32	Average transmission rate (in 500 Kbps)

6.24.2 CMD_MULTI_CHAN_CONFIG

This command is used to configure multi-channel operation.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_CONFIG = 0x011E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read) 0x0001 = ACT_SET (write)
ChannelTime	UINT32	Channel time (µs) for interface
BufferWeight	UINT8	Buffer weight for interface

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	0x0000 = no error 0x0001 = command failed 0x0002 = command is not supported
Action	UINT16	Type of action set in request message
ChannelTime	UINT32	Channel time (µs) for interface
BufferWeight	UINT8	Buffer weight for interface
Multi_Chan_Info	MrvlIETypes_Multi_Chan_Info	Multi-channel information TLV

6.24.3 CMD_MULTI_CHAN_POLICY

This command is used to get/set the multi-channel policy.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_POLICY = 0x0121
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read) 0x0001 = ACT_SET (write)
MultiChanPolicy	UINT16	Multi-channel policy 0x0000 = 2 or more interfaces should operate on common channel 0x0001 = 2 or more interfaces can operate on different channels

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_POLICY | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	0x0000 = no error 0x0001 = command failed 0x0002 = command is not supported
Action	UINT16	Type of action set in request message
MultiChanPolicy	UINT16	Multi-channel policy 0x0000 = 2 or more interfaces should operate on common channel 0x0001 = 2 or more interfaces can operate on different channels

6.24.4 CMD_MULTI_CHAN_TIMESLICE_CONFIG

This command configures the timeslice of each channel for Dynamic Rapid Channel Switching (DRCS).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_TIMESLICE_CONFIG = 0x024A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Channel Time-Slicing	UINT8[]	DRCS time slicing parameter for each channel of MrvlIETypes_MULTI_CHAN_TIME_SLICE_t TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MULTI_CHAN_TIMESLICE_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
Channel Time-Slicing	UINT8[]	DRCS time slicing parameter for each channel of MrvlIETypes_MULTI_CHAN_TIME_SLICE_t TLV

6.24.5 CMD_DMCS_CONFIG

This command gets/sets the status of Dual MAC Connection Service (DMCS) agent.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_DMCS_CONFIG = 0x0260
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SubCmd	UINT16	Sub command 0 = disable/enable dynamic mapping 1 = get status of DMCS
Data[]	Structure	Data Every sub command has its own structure. Some sub command uses TLV format while others do not. See Table 45, Sub Command Structure, on page 199 for details of the structures.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_DMCS_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
SubCmd	UINT16	Sub command 0 = disable/enable dynamic mapping 1 = get status of DMCS
Data[]	Structure	Data Every sub command has its own structure. Some sub command uses TLV format while others do not. See Table 45, Sub Command Structure, on page 199 for details of the structures.

Table 45: Sub Command Structure

<table>
<tr><th>SubCmd</th><th>Description</th></tr>
<tr><td>0</td><td>Disable/Enable dynamic mapping
It has the following structure:
REQUEST:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>MappingPolicy</td><td>UINT8</td><td>0 = direct mapping
1 = dynamic mapping</td></tr>
</table>
RESPONSE:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>MappingPolicy</td><td>UINT8</td><td>0 = direct mapping
1 = dynamic mapping</td></tr>
</table>
</td></tr>
<tr><td>1</td><td>Get status of DMCS
It has the following structure:
REQUEST:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Data[]</td><td>Variable</td><td>Not used</td></tr>
</table>
RESPONSE:
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Data[]</td><td>Variable</td><td>MrvlIETypes_DMCS_STATUS_t</td></tr>
</table>
</td></tr>
</table>

6.25 TDLS

Table 46 lists the Tunneled Direct Link Setup (TDLS) commands.

Table 46: TDLS Commands

Command	Description	Page
Section 6.25.1, CMD_802_11_TDLS	Supports embedded supplicant TDLS features	page 200
Section 6.25.2, CMD_TDLS_OPERATIONS	Setup or teardown TDLS link with each peer	page 206

6.25.1 CMD_802_11_TDLS

This command is used to support embedded supplicant TDLS features, including TDLS-Channel Switch and TDLS-UAPSD that is not supported by supplicant based TDLS command 0x0122.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_TDLS = 0x0100
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Specifies the action of this command See Table 47, Action Code, on page 201.
Data[]	UINT8	Action specific data See Table 47, Action Code, on page 201.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11_TDLS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Specifies the action of this command See Table 47, Action Code, on page 201.
Data[]	UINT8	Action specific data See Table 47, Action Code, on page 201.

Table 47: Action Code

Action		Description	
0x0000	SET_TDLS_CONFIG	Enable or disable TDLS on the device	
		Argument	Data[1:0] 0 = disable 1 = enable
		Return	0 = success
0x0001	SET_TDLS_INFO	Setting the capabilities of TDLS Station	
		Argument	CapInfo (UINT16)
			tlv_buffer[] (UINT8) May include the following: • MRVL_RATES_TLV_ID • MRVL_QOSCAPABILITY_TLV_ID • MRVL_HT_CAPABILITY_TLV_ID • MRVL_HT_INFORMATION_TLV_ID • MRVL_2040_BSS_COEX_TLV_ID • MRVL_EXTENDED_CAP_TLV_ID • MRVL_RSN_TLV_ID • MRVL_SUPPORTEDCHANNELS_TLV_ID • MRVL_COUNTRY_TLV_ID
		Return	0 = success
0x0002	TDLS_DISCOVERY_REQ	Request TDLS discovery	
		Argument	Data[5:0]: MAC address of peer
		Return	When 0 = success is returned, Data field contains the following: Data[8]: RSSI (UINT8) Data[7:2]: peer MAC address Peer address from which the discovery response is received. Data[1:0]: BufSize (UINT16) Contains the length of remaining fields. Bufsize of 0 indicates no response received. Following by the buffer containing frame (TDLS Discovery Response) receive from the other end starting with the CapInfo field. When 1 = error is returned, the Data field contains the following: Data[5:0]: peer MAC address

Table 47: Action Code (Continued)

Action		Description	
0x0003	TDLS_SETUP_REQ	Request TDLS setup request Before invoking this command, driver should make sure no packets destined for the MAC address given in this command are given to firmware after issuing this command and before TDLS Link Established event is received for this address.	
		Argument	Data[13:10]: time (s) indicating the lifetime of security key, if security is being used Data[9:6]: time (ms) to wait for setup response from peer Data[5:0]: MAC address of peer with which TDLS link is to be setup
		Return	When 0 = success is returned, Data field contains the following: Data[5:0]: peer MAC address When 1 = error is returned, Data field contains the following: Data[7:6]: reason 1 – INTERNAL_ERROR 2 – MAX_TDLS_LINKS_EST 3 – TDLS_SETUP_IN_PROGRESS 4 – TDLS_LINK_EXISTS 5 – TDLS_PROHIBITED_BY_BSS Data[5:0]: peer MAC address
0x0004	TDLS_TEARDOWN_REQ	Request teardown of TDLS link	
		Argument	Data[7:6]: reason Data[5:0]: MAC address of peer with which TDLS link is to be torn down
		Return	When 0 = success is returned, Data field contains the following: Data[5:0]: peer MAC address
0x0005	TDLS_POWERMODE	Request setting of power mode for TDLS link	
		Argument	Data[6] 0 = active 1 = power save Data[5:0]: MAC address of peer with which TDLS link power mode is to be changed
		Return	When 0 = success is returned, Data field contains the following: Data[5:0]: peer MAC address

Table 47: Action Code (Continued)

Action		Description	
0x0006	TDLS_INITIATE_ CHANNEL_SWITCH	Request for initiating channel switch on a direct link	
		Argument	Data[14]: periodicity 0 = not periodic 1 = periodic Data[13]: target regulatory class Data[12:11]: switch timeout Data[10:9]: switch time Data[8]: band 0 = 2.4 GHz band 1 = 5 GHz band Data[7]: secondary channel offset 0x0 = no secondary channel 0x1 = secondary channel above 0x3 = secondary channel below Data[6]: primary channel Data[5:0]: MAC address of peer
		Return	When 0 = success is returned, Data field contains the following: Data[5:0]: peer MAC address When 1 = error is returned, Data field contains the following: Data[7:6]: reason 1 – Internal error 6 – TDLS link non-existent 10 – Channel switch prohibited by AP Data[5:0]: peer MAC address
0x0007	TDLS_STOP_CHANNEL_ SWITCH	Request for stopping periodic channel switch on direct link	
		Argument	Data[5:0]: peer MAC address
		Return	When 0 = success is returned, Data field contains the following: Data[5:0]: peer MAC address When 1 = error is returned, Data field contains the following: Data[7:6]: reason 6 – TDLS link non-existent 11 – No channel switch active for peer Data[5:0]: peer MAC address **NOTE:** On issuing this command, if STA is on base channel, no more attempts will be made to move to off-channel. If STA is on off-channel, firmware will complete the channel switch state matching and once back to base channel, no more attempts will be made to move to off-channel.
0x0008	TDLS_CONFIG_CS_ PARAMS	Request for configuring CS related parameters	
		Argument	Data[2]: threshold for link intending to perform channel switch (in number of packets per unit time) Data[1]: threshold for links other than the link intending to perform CS (in number of packets per unit time) Data[0]: unit time (calculated in multiples of 10 ms) Example: If host provides value 2, unit time will be considered as 20 ms.
		Return	--

Table 47: Action Code (Continued)

Action		Description	
0x000A	TDLS_GET_ACTIVE_ LINKS	Request for obtaining data about active TDLS links, if any	
		Argument	Data[5:0]: Optional peer MAC address
		Return	When 0 = success, Data field contains the following: Data[(31+k+1):29]: key Data[28]: key length (k) Data[27:24]: key lifetime Data[23]: security used 0 = none 1 = AES/CCMP If B2 of Data[9] is set, the above fields are present. Data[22]: Tx packet HT information Variable structure for each record out of n (if n > 0). Data[21]: Tx data rate value Data[20:19]: data NF average Data[18:17]: data RSSI average Data[16:15]: data NF last Data[14:13]: data RSSI last Data[12]: active channel number Data[11]: successive Tx failure count Data[10]: buffered traffic status B0: AC_BK B1: AC_BE B2: AC_VI B3: AC_VO For B0 to B3, value of 1 indicates traffic is buffered for the particular AC. B4 to B7: reserved Data[9]: flags B0: self initiated link 0 = false 1 = true B1: security enabled 0 = false 1 = true B2: self power save mode 0 = active 1 = sleep B3: peer power save mode 0 = active 1 = sleep B4: channel switch supported 0 = no 1 = yes B5: current channel 0 = base 1 = off B6 to B7: reserved Data[8:3]: peer MAC address Data[2]: number of active TDLS links (n) Fixed structure for each record out of n (if n > 0). Data[1:0]: total length When 1 = error, TDLS is not supported

Table 47: Action Code (Continued)

Action		Description	
0xFFF9	TDLS_STOP_RX	Request to stop Rx path, so that hardware does not ACK any incoming frame This is required for making the STA unreachable for TDLS teardown test case.	
		Argument	Data[0] 0 = start Rx path again 1 = stop Rx path
		Return	0 = success (no data in response field)
0xFFFA	TDLS_IGNORE_KEY_ LIFETIME_EXPIRY	Request to ignore the expiry of security key lifetime and do not send out a teardown frame	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)
0xFFFB	TDLS_HIGHER_LOWER_ MAC	Request to send a setup request to a peer as soon as a setup is received from the peer for a new TDLS link	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)
0xFFFC	TDLS_ALLOW_SETUP_ WITH_PROHIBITED	Request to allow sending of setup request even if AP advertises TDLS prohibited bit	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)
0xFFFD	TDLS_ALLOW_SETUP_ ON_EXISTING_LINK	Request to allow sending of setup request on an existing direct link	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)
0xFFFE	TDLS_FAIL_SETUP_ CONFIRM	Request sending of status code 37 in setup confirm frame even if setup response received was ok	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)

Table 47: Action Code (Continued)

Action		Description	
0xFFFF	TDLS_WRONG_BSSID	Request sending of wrong BSSID in link identifier for setup request and discovery request	
		Argument	Data[0] 0 = disable 1 = enable Others = reserved
		Return	0 = success (no data in response field)

6.25.2 CMD_TDLS_OPERATIONS

This command is used to setup or teardown TDLS link with each peer. This command is used for supplicant based TDLS.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TDLS_OPERATIONS = 0x0122
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = TDLS_DELETE (PeerMacAddress only) 0x0001 = TDLS_CREATE (PeerMacAddress only) 0x0002 = TDLS_CONFIG Others = reserved
Reason	UINT16	Reserved (set to 0)
PeerMacAddress	UINT8[6]	TDLS peer MAC address
CapInfo	UINT16	Capability information
QoS_Capability	MrvlIEType_QosCapability_t	QoS capability exchanged in TDLS handshake
OpRateSet	MrvlIETypes_RateParamSet_t	Rates used in TDLS link
HTCap	MrvlIETypes_HT_Capability_t	HT capability information type/length extended IEEE IE
HTInfo	MvrlIETypes_HT_Information_t	HT information type/length extended IEEE IE
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN type/length extended IEEE IE
VHT Cap	MrvlIETypes_VHTCap_t	VHT capability information type/length extended IEEE IE
VHT Info	MrvlIETypes_VHTOprat_t	VHT information type/length extended IEEE IE
TDLS Idle Timeout	MrvlIETypes_TDLS_Idle_Timeout_t	TDLS idle timeout value

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TDLS_OPERATIONS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code
Action	UINT16	Type of action set in request message
Reason	UINT16	Operation result reason 0x0000 = success 0x0001 = internal error 0x0002 = exceed maximum TDLS support 0x0003 = TDLS link existed 0x0004 = TDLS link not existed Others = reserved
PeerMacAdress	UINT8[6]	TDLS peer MAC address
CapInfo	UINT16	Capability information
QoS_Capability	MrvlIEType_QosCapability_t	QoS capability exchanged in TDLS handshake
Extended Capability	MrvlIEType_ExtCap_t	Extended capability to announce power save or channel switch
OpRateSet	MrvlIETypes_RateParamSet_t	Rates used in TDLS link
HTCap	MrvlIETypes_HT_Capability_t	HT capability information type/length extended IEEE IE
HTInfo	MrvlIEType_HT_Information_t	HT information type/length extended IEEE IE
RsnParamSet	MrvlIETypes_RsnParamSet_t	RSN type/length extended IEEE IE
VHT Cap	MrvlIETypes_VHTCap_t	VHT capability information type/length extended IEEE IE
VHT Info	MrvlIETypes_VHTOprat_t	VHT information type/length extended IEEE IE
TDLS Idle Timeout	MrvlIETypes_TDLS_Idle_Timeout_t	TDLS idle timeout value

Note: When received deauthentication/deassociation frame from AP, the related TDLS connections will be kept since teardown frame needs to be sent through direct link.

6.26 Neighbor Awareness Network

Table 48 lists the Neighbor Awareness Network (NAN) commands.

Table 48: NAN Commands

Command	Description	Page
Section 6.26.1, CMD_NAN_PARAM_CONFIG	Gets/sets NAN configuration parameters used by NAN MAC layer	page 208
Section 6.26.2, CMD_NAN_MODE_CONFIG	Gets/sets NAN mode	page 209
Section 6.26.3, CMD_NAN_TRANSMIT_SDFRAME	Sends packet formed by NAN discovery engine to publish or subscribe NAN services	page 210
Section 6.26.4, CMD_NAN_UPDATE_SERVICE_HASH_DB	Adds/removes service hashes	page 211
Section 6.26.5, CMD_NAN_STATE_CONFIG	Sets runtime parameters of NAN state	page 212
Section 6.26.6, CMD_NAN_FLUSH_SD_QUEUE	Flushes all SD frames queued in SD queue to stop advertising of SD frames	page 213

6.26.1 CMD_NAN_PARAM_CONFIG

This command is used to get and set NAN configuration parameters which are used by NAN MAC layer. NAN need to be configured using this command before it is started. The body of the command consists of 1 or more TLVs.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_PARAM_CONFIG = 0x0228
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read data) 0x0001 = ACT_SET (write data)
ConfigParams	TLVs	ConfigParams consists of variable number of TLVs

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_PARAM_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure

Field Name	Type	Description
Action	UINT16	Type of action set in request message
ConfigParams	TLVs	ConfigParams consists of variable number of TLVs NAN TLV 0x1D2 to 0X1D9.

6.26.2 CMD_NAN_MODE_CONFIG

This command is used to get and set NAN mode. Upon receiving this command, firmware will reset, start or stop NAN device.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_MODE_CONFIG = 0x0229
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read data) 0x0001 = ACT_SET (write data)
Mode	UINT16	Mode 0x0000 = reset NAN 0x0001 = stop NAN 0x0002 = start NAN

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_MODE_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Mode	UINT16	NAN mode as passed in request

6.26.3 CMD_NAN_TRANSMIT_SDFRAME

This command is used to send the packet formed by NAN discovery engine to publish or subscribe NAN services. It is assumed that the NAN discovery engine at host side will construct the entire GAS packet and send it to firmware. The Service Discovery (SD) packet attribute in the command will start from category field and it is as per format of SD frame defined in NAN Specification Section 5.3.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_TRANSMIT_SDFRAME = 0x022A
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
PeerMacAddr	UINT8[6]	MAC address of peer to whom SD packet is to be sent
TimeToLive	UINT32	Not used (set to 0)
TxType	UINT8	Type of Tx frame 0x01 = NAN action frame 0x02 = further availability Tx frame
SDPAcket	UINT8[]	Service discovery packet to be sent to Wi-Fi

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_TRANSMIT_SDFRAME | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure

6.26.4 CMD_NAN_UPDATE_SERVICE_HASH_DB

This command is used by the driver to add and remove service hashes.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_UPDATE_SERVICE_HASH_DB = 0x022B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = add service hash 0x0002 = remove service hash
Service Index Buffers	UINT8[]	Buffers of N*6 bytes N is number of service hashes.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_UPDATE_SERVICE_HASH_DB | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Service Index Buffers	UINT8[]	Buffers of N*6 bytes N is number of service hashes.

6.26.5 CMD_NAN_STATE_CONFIG

This command is used to set/get runtime parameters of NAN state, such as master preference, role, hop count, and so on.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_STATE_CONFIG = 0x022C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = get (read data) 0x0001 = get runtime 0x0002 = set (write data)
HoldRole	UINT8	Hold role
CurRole	UINT8	Role of NAN device anchor, master, non master sync/non-sync
HoldMasterPref	UINT8	Hold master preference flag
MasterPref	UINT8	Master preference held
HoldRfactor	UINT8	Hold RF factor flag
HoldHopCount	UINT8	Hold hop count flag
HopCount	UINT8	Current hop count flag held
Disable2G	UINT8	Disable beacon transmission for 2.4 GHz operation

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_STATE_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
HoldRole	UINT8	Hold role
CurRole	UINT8	Role of NAN device anchor, master, non master sync/non-sync
holdMstPref	UINT8	Hold master preference flag
curMstPref	UINT8	Hold current master preference flag
holdRfactor	UINT8	Hold RF factor flag
curRfactor	UINT8	Current RF factor flag

Field Name	Type	Description
HoldHc	UINT8	Hold hop count flag
curHc	UINT8	Current hop count flag
Disable2G	UINT8	Disable beacon transmission for 2.4 GHz operation

6.26.6 CMD_NAN_FLUSH_SD_QUEUE

This command is used to flush all SD frames queued in the SD queue to stop advertising of SD frames.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_FLUSH_SD_QUEUE = 0x022D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_NAN_FLUSH_SD_QUEUE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure

6.27 802.11k

Table 50 lists the 802.11k command.

Table 49: 802.11k Command

Command	Description	Page
Section 6.27.1, CMD_GET_NEIGHBORLIST	Gets Neighbor list	page 214

6.27.1 CMD_GET_NEIGHBORLIST

This command is used by the driver to get Neighbor list. Upon receiving this command, firmware sends Neighbor report request to the associated AP and sends the received Neighbor report response as an event.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_NEIGHBORLIST = 0x0231
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET
DialogToken	UINT8	Token for this command

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_NEIGHBORLIST | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
DialogToken	UINT8	Token for this command

6.28 Smart Mode Configuration

Table 50 lists the Smart Mode command.

Table 50: Smart Mode Command

Command	Description	Page
Section 6.28.1, CMD_SMART_MODE_CFG	Configures smart configuration mode	page 215

6.28.1 CMD_SMART_MODE_CFG

This command is used to configure the smart configuration mode. Upon receiving this command, firmware will perform actions like get or set configuration parameters or start/stop smart configuration mode.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SMART_MODE_CFG = 0x012D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (return current configuration) 0x0001 = ACT_SET (set parameters) 0x0002 = ACT_START (start Smart configuration mode) 0x0004 = ACT_STOP (stop Smart configuration mode)
ConfigParams	TLVs	ConfigParams consists of variable number of TLVs

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SMART_MODE_CFG = 0x812D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
ConfigParams	TLVs	ConfigParams consists of variable number of TLVs • MrvlIETypes_SSIDParamSet_t • MrvlIETypes_Beacon_Period_t • MrvlIETypes_ChanListParamSet_t • MrvlIETypes_Passthrough_t • MrvlIETypes_SMCAddrRange_t • MrvlIETypes_SMCFrameFilter_t

6.29 ED-MAC Control

Table 50 lists the ED-MAC command.

Table 51: ED-MAC Command

Command	Description	Page
Section 6.29.1, CMD_ED_CTRL	Supports ED_MAC control feature	page 216

6.29.1 CMD_ED_CTRL

This command helps to set/get EDMAC unit control parameters.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_ED_CTRL = 0x0130
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
ed_ctrl_2g	UINT16	0x0000 = disable EU adaptivity for 2.4GHz band 0x0001 = enable EU adaptivity for 2.4GHz band
ed_offset_2g	SINT8	0x00 = Default Energy Detect threshold offset value Range: -128 (0x80) to 127 (0x7F)
Reserved	UINT8	Not used (set to 0)
ed_ctrl_5g	UINT16	0x0000 = disable EU adaptivity for 5GHz band 0x0001 = enable EU adaptivity for 5GHz band
ed_offset_5g	SINT8	0x00 = Default Energy Detect threshold offset value Range: -128 (0x80) to 127 (0x7F)
Reserved	UINT8	Not used (set to 0)
ed_ctrl_txg_lock	UINT32	This field is meant for FW internal usage. DO NOT change this line

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_ED_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set in request message
ed_ctrl_2g	UINT16	0x0000 = disable EU adaptivity for 2.4 GHz band 0x0001 = enable EU adaptivity for 2.4 GHz band

Field Name	Type	Description
ed_offset_2g	SINT8	0x00 = Default Energy Detect threshold offset value Range: -128 (0x80) to 127 (0x7F)
Reserved	UINT8	Not used (set to 0)
ed_ctrl_5g	UINT16	0x0000 = disable EU adaptivity for 5 GHz band 0x0001 = enable EU adaptivity for 5 GHz band
ed_offset_5g	SINT8	0x00 = Default Energy Detect threshold offset value Range: -128 (0x80) to 127 (0x7F)
Reserved	UINT8	Not used (set to 0)
ed_ctrl_txg_lock	UINT32	This field is meant for FW internal usage. DO NOT change this line

6.30 Temperature Sensor

Table 50 lists the temperature sensor command.

Table 52: Temperature Sensor Command

Command	Description	Page
Section 6.30.1, CMD_GET_SENSOR_TEMPERATURE	Gets temperature from temperature sensor	page 218

6.30.1 CMD_GET_SENSOR_TEMPERATURE

This command is used to get the temperature from the temperature sensor.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_SENSOR_TEMPERATURE = 0x0227
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Temperature	UINT32	Radio temperature Celsius tenths

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_GET_SENSOR_TEMPERATURE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Temperature	UINT32	Radio temperature Celsius tenths

6.31 Bridge Mode Control

Table 53 lists the bridge mode command.

Table 53: Bridge Mode Command

Command	Description	Page
Section 6.31.1, CMD_BRIDGE_MODE_CTRL	Controls bridge mode for bridge device	page 219

6.31.1 CMD_BRIDGE_MODE_CTRL

This command is used to control bridge mode for a bridge device. The first SSIDParamSet and passphrase are for the AP. The second SSIDParamSet and passphrase are for the bridge in-AP.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_BRIDGE_MODE_CTRL = 0x022E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = get firmware bridge configuration 0x0001 = set firmware bridge configuration
Enable	UINT8	Enable firmware bridge 0x00 = disable 0x01 = enable
BridgeParamSet	MrvlIETypes_BridgeParamSet_t	Bridge parameters

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_BRIDGE_MODE_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set in request message
Enable	UINT8	Enable firmware bridge 0x00 = disable 0x01 = enable
BridgeParamSet	MrvlIETypes_BridgeParamSet_t	Bridge parameters

6.32 802.1AS Timesync

Table 54 lists the 802.11AS timesync command.

Table 54: Timesync Command

Command	Description	Page
Section 6.32.1, CMD_TIMESYNC_CFG	Synchronizes host (system) and firmware (device) clocks	page 220

6.32.1 CMD_TIMESYNC_CFG

This command synchronizes the host (system) and firmware (device) clocks. On receiving this command, firmware establishes a relationship between the baseband unit clock (on 28 nm SoCs) or derived clock (on 40 nm SoCs, using TSF and baseband unit) and host (system) clocks.

Firmware uses this established relationship to derive other clocks when 1 type of clock is known (for example, derive system clock from hardware clock). In the response of this command, firmware passes the relationship to the host, in case the host wants to perform further adjustments.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TIMESYNC_CFG = 0x0246
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = get (for TimeSync) 0x0001 = set (for HW_Timestamping)
HostClockValue	UINT64	System clock (ns) since epoch
HwClockValue	UINT64	Not used

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TIMESYNC_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message

Field Name	Type	Description
HostClockValue	UINT64	System clock (ns) since epoch Returned as is from request.
HwClockValue	UINT64	Hardware clock (ns) For 28 nm SoCs, converted from baseband counter. For 40 nm SoCs, derived by firmware (considering TSF and baseband unit counter if available).

6.33 Wi-Fi Location

Table 55 lists the Wi-Fi location commands.

Table 55: Wi-Fi Location Commands

Command	Description	Page
Section 6.33.1, CMD_FTM_CONFIG_SESSION_ PARAMS	Configures Wi-Fi location session configuration parameters for Initiator and Responder	page 222
Section 6.33.2, CMD_FTM_SESSION_CTRL	Triggers/stops FTM procedure by requesting firmware to initiate/stop FTM session	page 223
Section 6.33.3, CMD_FTM_FEATURE_CTRL	Configures Wi-Fi location feature responsibilities between host application and firmware	page 224
Section 6.33.4, CMD_FTM_CONFIG_RESPONDER	Gets RTT Responder information	page 225

6.33.1 CMD_FTM_CONFIG_SESSION_PARAMS

This command configures the Wi-Fi location session configuration parameters for Initiator and Responder. This command accepts 1 or more TLVs for the configuration of session parameters.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_CONFIG_SESSION_PARAMS = 0x024D
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = get configuration data 0x0001 = set configuration data
ConfigParams	TLVs	Configuration parameters of variable number of TLVs of the following: • MrvlIETypes_FTM_SESSION_CFG_INITIATOR_t • MrvlIETypes_FTM_SESSION_CFG_RESPONDER_t • MrvlIETypes_FTM_SESSION_CFG_LOCATION_CIVIC_t • MrvlIETypes_FTM_SESSION_CFG_LCI_t

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_CONFIG_SESSION_PARAMS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
ConfigParams	TLVs	Configuration parameters of variable number of TLVs of the following: • MrvlIETypes_FTM_SESSION_CFG_INITIATOR_t • MrvlIETypes_FTM_SESSION_CFG_RESPONDER_t • MrvlIETypes_FTM_SESSION_CFG_LOCATION_CIVIC_t • MrvlIETypes_FTM_SESSION_CFG_LCI_t

6.33.2 CMD_FTM_SESSION_CTRL

This command triggers/stops the Fine Time Measurement (FTM) procedure by requesting firmware to initiate/stop FTM session.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_SESSION_CTRL = 0x024E
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = start FTM session 0x0002 = abort ongoing FTM session
MAC address	UINT8	MAC address of peer with whom FTM session is required
Channel	UINT8	Channel on which FTM must be started
Civic_request	UINT8	Include civic request
Lci_request	UINT8	Include location civic request

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_SESSION_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
MAC address	UINT8	MAC address of peer with whom FTM session is required
Channel	UINT8	Channel on which FTM must be started
Civic_request	UINT8	Include civic request
Lci_request	UINT8	Include location civic request

6.33.3 CMD_FTM_FEATURE_CTRL

This command configures the Wi-Fi Location feature responsibilities between the host application and firmware.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_FEATURE_CTRL = 0x024F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = start FTM session 0x0002 = abort ongoing FTM session
Offload_to_FW	UINT8	Bits[7:2]: Reserved Bit[1]: If set, firmware prepares and processes Neighbor report Bit[0]: If set, firmware prepares and processes range request and report frames (in this case, firmware carries FTM session internally and sends the range report back to AP)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_FEATURE_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
Offload_to_FW	UINT8	Bits[7:2]: Reserved Bit[1]: If set, firmware prepares and processes Neighbor report Bit[0]: If set, firmware prepares and processes range request and report frames (in this case, firmware carries FTM session internally and sends the range report back to AP)

6.33.4 CMD_FTM_CONFIG_RESPONDER

This command gets the Round Trip Time (RTT) Responder information.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_CONFIG_RESPONDER = 0x0255
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT8	Action 0x00 = GET_RESPONDER_INFO 0x01 = SET_RESPONDER_ENABLE 0x02 = SET_RESPONDER_DISABLE 0x03 = SET_REPSONDER_LCI 0x04 = SET_REPSONDER_LCR
tlv_buffer	UINT8[]	Not used (set to 0).

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_FTM_CONFIG_RESPONDER | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
tlv_buffer	UINT8[]	TLV buffer Firmware returns MrvlIETypes_RTT_RESPONDER_INFO_ID_t.

6.34 Channel State Information

Table 56 lists the Channel State Information (CSI) command.

Table 56: CSI Command

Command	Description	Page
Section 6.34.1, CMD_CSI	Informs firmware to start/stop CSI collection	page 226

6.34.1 CMD_CSI

This command informs firmware to start/stop CSI collection.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CSI = 0x025B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = start 0x0002 = stop
CSI filter count	UINT16	Number of CSI filters
CSI filter	Structure	CSI filters Maximum 16 filters. See Table 57, CSI Filter Structure, on page 227 for the structure for CSI filter.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CSI | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set as in request message
CSI filter count	UINT16	Number of CSI filters
CSI filter	Structure	CSI filters Maximum 16 filters. See Table 57, CSI Filter Structure, on page 227 for the structure for CSI filter.

Table 57: CSI Filter Structure

Field Name	Type	Description
Address	UINT8[6]	MAC address
Packet Type	UINT8	Packet type If set to 0xFF, firmware will not filter packets using packet type.
Packet Subtype	UINT8	Packet Subtype If set to 0xFF, firmware will not filter packets using packet subtype.
Flags	UINT8	Bits[7:3]: Reserved Bit[2]: Send CSI error event when timeout Bit[1]: Wait for trigger Bit[0]: Check sounding flag

6.35 802.11P

Table 58 lists the 802.11p commands.

Table 58: 802.11p Commands

Command	Description	Page
Section 6.35.1, CMD_802_11P_CONFIG	Gets/sets 802.11p specific parameter configuration	page 228
Section 6.35.2, CMD_802_11P_STATS	Gets/sets 802.11p status	page 229

6.35.1 CMD_802_11P_CONFIG

This command gets/sets the 802.11p specific parameter configuration.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11P_CONFIG = 0x024B
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_11P_MODE_CONFIG_t • MrvlIETypes_11P_TDM_CONFIG_t

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11P_CONFIG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
ConfigParams	TLV	Configuration parameters of variable number of the following TLVs: • MrvlIETypes_11P_MODE_CONFIG_t • MrvlIETypes_11P_TDM_CONFIG_t

6.35.2 CMD_802_11P_STATS

This command gets/sets the 802.11p status. This command is used when the application wants to get or reset the various radio of Decentralized Congestion Control (DCC) counters firmware maintains.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11P_STATS = 0x024C
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
TlvBuffer	TLV(s)	Configuration parameters of variable number of the MrvlIETypes_11P_RADIO_COUNTERS_t TLV

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11P_STATS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
TlvBuffer	TLV(s)	Configuration parameters of variable number of the MrvlIETypes_11P_RADIO_COUNTERS_t TLV

6.36 Independent Reset

Table 59 lists the independent reset command.

Table 59: Independent Reset Command

Command	Description	Page
Section 6.36.1, CMD_INDEPENDENT_RESET	Configures Wi-Fi OOB independent reset mode and GPIO pin	page 230

6.36.1 CMD_INDEPENDENT_RESET

This command configures the Wi-Fi Out-of-Band (OOB) independent reset mode and GPIO pin.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_INDEPENDENT_RESET = 0x0243
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = return current configuration 0x0001 = set parameters
OOBMode	UINT8	OOB independent reset mode 0x00 = disable 0x01 = enable
GPIOPin	UINT8	GPIO pin 0xFF = default GPIO pin pre-defined in firmware Others = new GPIO pin assignment

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_INDEPENDENT_RESET | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message

Field Name	Type	Description
OOBMode	UINT8	OOB independent reset mode 0x00 = disable 0x01 = enable
GPIOPin	UINT8	GPIO pin 0xFF = default GPIO pin pre-defined in firmware Others = new GPIO pin assignment

6.37 mac80211

Table 60 lists the mac80211 command.

Table 60: mac80211 Command

Command	Description	Page
Section 6.37.1, CMD_MAC80211_STA_ADD	Stores peer information during connection through mac80211 feature	page 232

6.37.1 CMD_MAC80211_STA_ADD

This command stores the peer information during connection through mac802.11 feature.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC80211_STA_ADD = 0x0253
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = ACT_SET
add	UINT32	STA states 00 = remove STA 01 = add STA 10 = associated STA
Peer_mac	UINT8[6]	MAC address pf peer STA
Aid	UINT16	AID value assigned after connection is established between 2 STAs
Bitmap_supp_rate	UINT32	Common bitmap of supported rates
mcs_mask	UINT8[4]	Common MCS Rx mask
ht_supported	UINT8	HT enable flag
ht_cap	UINT16	Common HT capabilities
bw	UINT8	Bandwidth
vht_supported	UINT8	VHT enable flag
vht_cap	UINT16	Common VHT capabilities
vht_rx_mcs	UINT16	Common VHT Rx MCS mask
max_mpdu_le	UINT16	Common maximum MPDU length

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_MAC80211_STA_ADD | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
add	UINT32	STA states 00 = remove STA 01 = add STA 10 = associated STA
Peer_mac	UINT8[6]	MAC address pf peer STA
Aid	UINT16	AID value assigned after connection is established between 2 STAs
Bitmap_supp_rate	UINT32	Common bitmap of supported rates
mcs_mask	UINT8[4]	Common MCS Rx MCS mask
ht_supported	UINT8	HT enable flag
ht_cap	UINT16	Common HT capabilities
bw	UINT8	Bandwidth
vht_supported	UINT8	VHT enable flag
vht_cap	UINT16	Common VHT capabilities
vht_rx_mcs	UINT16	Common VHT Rx mask
max_mpdu_le	UINT16	Common maximum MPDU length

6.38 SoC Spectral Analysis Unit

Table 61 lists the SoC Spectral Analysis Unit (SSU) feature command.

Table 61: SSU Command

Command	Description	Page
Section 6.38.1, CMD_SSU	Passes buffer address, length, and other baseband parameters to device for SSU feature	page 234

6.38.1 CMD_SSU

This command passes the buffer address, length, and other baseband parameters to the device for the SSU feature.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_SSU = 0x0259
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0001 = ACT_SET
nskip	UINT32	Number of Fast Fourier Transform (FFT) sample to skip
nsel	UINT32	Number of FFT sample to dump
adcdownsample	UINT32	Down sample Analog-to-Digital Converter (ADC) input for buffering
mask_adc_pkt	UINT32	Mask out ADC data from spectral packet
out_16bits	UINT32	Enable 16-bit FFT output data precision in spectral packet
spec_pwr_enable	UINT32	Enable power spectrum (dB) for spectral packet
rate_deduction	UINT32	Enable spectral packet rate reduction (dB) output format
n_pkt_avg	UINT32	Number of spectral packets over which spectral data is to be averaged
fft_len	UINT32	Calculated FFT length in DW
adc_len	UINT32	Calculated ADC length in DW
rec_len	UINT32	Calculated record length in DW
buffer_base_addr[2]	UINT32*2	Mapped address of DMA buffer
buffer_pool_size	UINT32	Total size of allocated buffer for SSU DMA
number_of_buffers	UINT32	Calculated buffer numbers
buffer_size	UINT32	Calculated buffer size in bytes for each descriptor

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_SSU | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Type of action set as in request message
nskip	UINT32	Number of FFT sample to skip
nsel	UINT32	Number of FFT sample to dump
adcdownsample	UINT32	Down sample ADC input for buffering
mask_adc_pkt	UINT32	Mask out ADC data from spectral packet
out_16bits	UINT32	Enable 16-bit FFT output data precision in spectral packet
spec_pwr_enable	UINT32	Enable power spectrum (dB) for spectral packet
rate_deduction	UINT32	Enable spectral packet rate reduction (dB) output format
n_pkt_avg	UINT32	Number of spectral packets over which spectral data is to be averaged
fft_len	UINT32	Calculated FFT length in DW
adc_len	UINT32	Calculated ADC length in DW
rec_len	UINT32	Calculated record length in DW
buffer_base_addr[2]	UINT32*2	Mapped address of DMA buffer
buffer_pool_size	UINT32	Total size of allocated buffer for SSU DMA
number_of_buffers	UINT32	Calculated buffer numbers
buffer_size	UINT32	Calculated buffer size in bytes for each descriptor

6.39 Virtual Dynamic Library Loading

Table 61 lists the Virtual Dynamic Library Loading (VDLL) command.

Table 62: SSU Command

Command	Description	Page
Section 6.39.1, CMD_VDLL	Downloads VDLL block as requested by firmware through event	page 236

6.39.1 CMD_VDLL

This command downloads a VDLL block as requested by firmware through EVT_VDLL_IND event.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_VDLL = 0x0240
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
Result	UINT16	Not used (set to 0)
Payload	UINT8*n	VDLL block

RESPONSE

None

6.40 802.11ax

Table 63 lists the 802.11ax command.

Table 63: 802.11ax Command

Command	Description	Page
Section 6.40.1, CMD_802_11AX_CFG	Gets/sets 802.11ax HE configuration	page 238
Section 6.40.2, CMD_TWT	Gets/sets TWT.setup and teardown parameters	page 240
Section 6.40.3, CMD_802_11AX_CMD	Gets/sets 802.11ax HE configuration using sub-commands	page 243

6.40.1 CMD_802_11AX_CFG

This command gets/sets the 802.11ax HE configuration. This command can be used to set the 802.11ax specific configuration, such as HE MAC capability, HE Phy capability, and so on.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11AX_CFG =0x0266
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Select2G:5G	UINT8	Bits[7:2]: Reserved Bit[1]: 5 GHz Bit[1]: 2.4 GHz
ConfigParams	TLV	Consists of variable length TLV MRVL_IEEE_EXTENSION_TLV_ID is used to indicate HE MAC Capability, HE PHY Capability, and so on.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11AX_CFG | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
Select2G:5G	UINT8	Bits[7:2]: Reserved Bit[1]: 5 GHz Bit[1]: 2.4 GHz
ConfigParams	TLV	Consists of variable length TLV MRVL_IEEE_EXTENSION_TLV_ID is used to indicate HE MAC Capability, HE PHY Capability, and so on. See Table 64.

Table 64: Payload for TLV

Field Name	Type	Description
ElementId	UINT16	ELEM_ID_EXTENSION = 255
Len	UINT16	Length of payload
ElementIdExt	UINT8	ELEM_ID_EXT_HE_CAPABILITIES = 35
HeMacCap	UINT8*6	HE MAC capabilities as per the 802.11ax-2021 standard
HePhyCap	UINT8*11	HE PHY capabilities as per the 802.11ax-2021 standard
HeMcsNss	UINT16*2	HE MCS map for RxMcs and TxMcs
Var	UINT8*28	Variable length fields, such as PPE threshold

6.40.2 CMD_TWT

This command gets/sets TWT setup and teardown parameters.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_TWT = 0x0270
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SubId	UINT16	SubId for either setup or teardown parameters 0x114 - DOT11AX_CMDID_TWT_SETUP 0x115 - DOT11AX_CMDID_TWT_TEARDOWN
SubId contents	UINT8*7	Buffer to hold SubId contents When SubId = 0x114 - DOT11AX_CMDID_TWT_SETUP field contains TWT setup parameters as described in Table 65 When SubId = 0x115 - DOT11AX_CMDID_TWT_TEARDOWN field includes TWT teardown parameters as mentioned in Table 66

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_TWT | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SubId	UINT16	SubID contains either setup or teardown parameters 0x114 - DOT11AX_CMDID_TWT_SETUP 0x115 - DOT11AX_CMDID_TWT_TEARDOWN
Value	UINT8*7	Buffer to hold SubId contents When SubId = 0x114 - DOT11AX_CMDID_TWT_SETUP field contains TWT setup parameters as described in Table 65 When SubId = 0x115 - DOT11AX_CMDID_TWT_TEARDOWN field includes TWT teardown parameters as mentioned in Table 66

Table 65: 802.11ax TWT setup parameters

Field Name	Type	Description
Implicit	UINT8	Indicates if TWT session is implicit or explicit 0: Explicit 1: Implicit
Announced	UINT8	Indicates if TWT session is announced or unannounced 1: Announced TWT 0: Unannounced
TriggerEnable	UINT8	Indicates if the trigger frames to schedule STA's transmissions are needed or not 0: Non-Trigger enabled 1: Trigger enabled TWT
TWT info disabled	UINT8	Indicates if TWT information frames are acceptable or not during the TWT session 0: TWT information enabled 1: TWT information disabled
Negotiation type	UINT8	Indicates if TWT session is Individual or Broadcast 0: Future Individual TWT SP start time 1: Next Wake TBTT time 3: B-TWT Support
TWT Wakeup duration	UINT8	Indicates the time after which the TWT requesting STA can transition to doze state Range: [0-sizeof(UINT8)]
Flow identifier	UINT8	As per specification, up to 8 implicit TWT sessions can exist in parallel based on different flow identifiers Range: [0-7]
Hard contraint	UINT8	Identifies if firmware can internally change any of the TWT parameters in case AP sends TWT_REJECT to TWT Setup. 0: FW can tweak the TWT setup parameters if it is rejected by AP. 1: Firmware should not tweak any parameters.
TWTExponent	UINT8	Range [0:63]
TWTMantissa	UINT16	TWTExponent and TWTMantissa determine the time interval between successive TWT SP for individual TWT according to the formula Mantissa * (2^Exponent) Range: [0-sizeof(UINT16)]
TWT request	UINT8	0: REQUEST_TWT 1: SUGGEST_TWT Range: [0-7]
TWT setupState	UINT8	Not used. Set to 0
Reserved	UINT8	Not used. Set to 0
Reserved	UINT8	Not used. Set to 0

Table 66: 802.11ax TWT teardown parameters

Field Name	Type	Description
Flow identifier	UINT8	TWT Flow Identifier. Range: [0-7]
Negotiation type	UINT8	0: Future Individual TWT SP start time 1: Next Wake TBTT time
Tear down All TWT	UINT8	1: To teardown all TWT, 0 otherwise
TWT teardown state	UINT8	Not used. Set to 0.
Reserved	UINT8	Not used. Set to 0.

6.40.3 CMD_802_11AX_CMD

This command gets/sets various 802.11ax related parameters through subIDs.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11AX_CMD = 0x026d
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SubId	UINT16	SubId as mentioned in Table 67.
Value	UINT8	Buffer for SubId contents as per Table 67

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_802_11AX_CMD | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0x0000 = success 0x0001 = failure
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
SubId	UINT16	SubId as mentioned in Table 67.
Value	UINT8	Buffer for SubId contents as per Table 67

Table 67: List of Sub IDs in CMD_802_11AX_CMD

Name	Value	Size	SubId contents
DOT11AX_CMDID_SR	0x102	Variable	Passes Spatial Reuse (SR) related parameters using the following TLVs: MRVL_DOT11AX_OBSS_PD_OFFSET_TLV_ID MRVL_DOT11AX_ENABLE_SR_TLV_ID
DOT11AX_CMDID_BEAM_CHANGE	0x103	UINT8	0x00 - disable beam change 0x01 - enable beam change
DOT11AX_CMDID_HTC	0x104	UINT8	0x00 - disable transmission of HTC 0x01 - enable transmission of HTC
DOT11AX_CMDID_TXOMI	0x105	UINT16	OMI value is used for OMI transmission in MAC header HTC+ field Bits[15:12]: Reserved Bits[11] : DL MU Data Disable Bits[10] : DL MU MIMO Resound Bits[9] : Er SU Disable Bits[8:6] : Tx NSTS Bits[5] : UL MU Disable Bits[4:3] : Channel width Bits[2:0] : Rx NSS
DOT11AX_CMDID_OBSSNBRU	0x106	UINT32	Set/get OBSS narrow-band RU tolerance time value in seconds Valid value range [1-3600] in seconds. Any value more than 3600 will disable this feature
DOT11AX_CMDID_TXOP_RTS_THRESHOLD	0x108	UINT16	This subcmd applies to uapX only. uAP uses it to manage the RTS/CTS used by associated non-AP HE STA Bit[16:10]: Reserved Bit[9:0]: TXOP RTS Threshold ; if value > 1023, disable TXOP duration RTS
DOT11AX_CMDID_RUPOWER	0x117	Variable	Sets Tx Power Limit in dBm for each RU (RU=26,52,106,242,484,996) Range from -64dBm to +63dBm in steps of 1dBm Channel number, channel band and RU TxPower information for each channel is passed through MrvlIETypes_ChanRuPwrCfg _t TLV

7 Micro Access Point

This section describes the high level software design and firmware API for the Micro Access Point (uAP) feature in embedded Wi-Fi products.

uAP Definition:

- The same Wi-Fi firmware can support both uAP function and client function. Either function uAP or client function or both functions can be activated simultaneously.
- The WLAM firmware will support a command interface for the uAP function in addition to the command interface for client function. The host will use this command interface to configure the uAP parameters (for example: SSID, encryption mode and key, MAC address filtering table, channel number, and so on).
- The host must perform 802.11d bridging if it is required.
- The number of clients permitted varies for different chip/feature-pack. See the product release notes.
- The uAP will support client MAC address filtering, that is, association control list.
- The uAP will have limited buffering available for client stations that are in IEEE power save.
- The uAP will have limited buffering available for station to station packet forwarding.
- uAP will support 802.11n/a/g/b modes. Some chip/feature-pack do not have 802.11a mode support. See the product release notes.
- Multi-BSS support.
- Automatic Channel Selection (ACS) feature allows uAP to automatically identify least congested channel for BSS operation.
- The uAP will only support basic access point functionalities as follows:
  - ERP protection mechanism
  - WEP encryption
  - Open system authentication
  - WEP shared key authentication
  - WPA and WPA2 security
  - Embedded WPA/WPA2-PSK authenticator
  - External authenticator support (PSK and 802.1x)
  - IEEE power save
  - Hidden SSID feature
  - Aging of inactive client stations and the aging timer is user configurable
  - WMM and WMM-PS
  - 802.11n

The uAP is restricted in the following:

- 802.11n/a/g/b modes operation only
- With a host system
- SDIO or USB host interface (depends on the interface that the chip supports)

7.1 Software Architecture

Figure 8 shows the high level uAP software block diagram, and Figure 9 shows a typical uAP software architecture in the Linux host system.

Figure 8: uAP Software Block Diagram

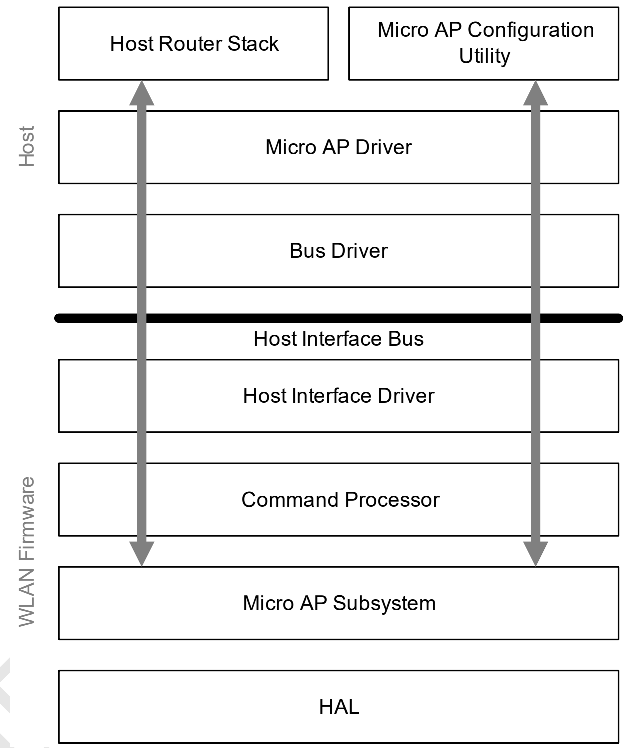
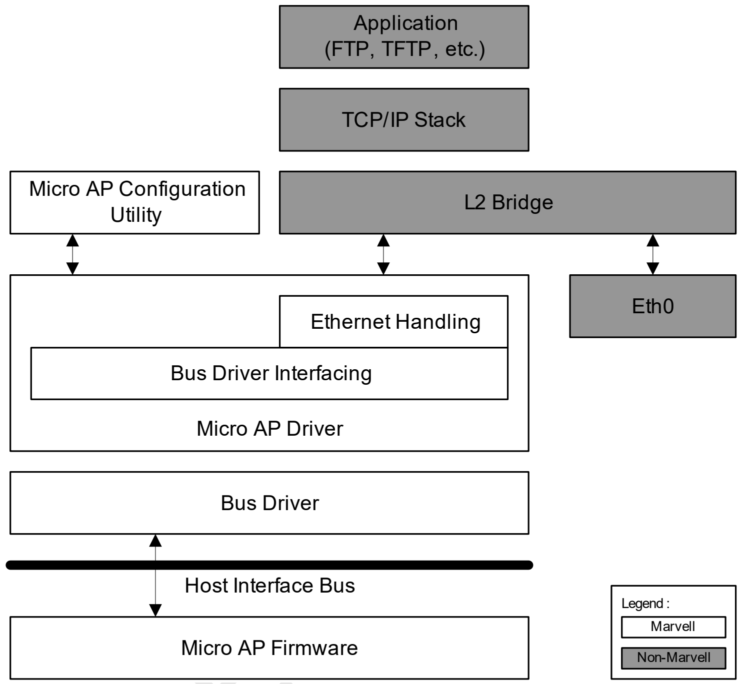


Figure 9: Typical uAP Software Architecture in Linux Host System



7.2 Data Path

The uAP firmware uses the same data path protocol as the Wi-Fi client firmware. The data protocol and the format of the data packets are described in Section 4, Data Path, on page 8.

The structure of the TxP and RxPD in uAP firmware is the same as in Wi-Fi client firmware. See Section 4.1, Receive Packet Descriptor and Section 4.2, Transmit Packet Descriptor for details.

7.3 Command Path

The uAP firmware uses the same command path protocol as Wi-Fi client firmware. See Section 6, Host Commands, on page 18.

7.4 Micro AP Event Indication

During operation, the uAP can send asynchronous event indications, such as when a client station changes association status, to the host driver.

The mechanism for sending the event to the host driver is the same as in the Wi-Fi client firmware.

Each event indication also carries with it event data, and the structure of the event data depends on the host interface type and the event type.

The general format of an event data packet for SDIO interface is as follows:

Event Data Packet
SDIO packet header (4 bytes)
Event ID (2 bytes)
BSS Number (1 byte)
BSS Type (1 byte)
Event data (variable length)

7.5 Micro AP Configuration Data

The host application is responsible for providing the uAP firmware with all the AP configuration data necessary for normal operation, such as SSID, security settings, client MAC filter table, and so on. Typically, users should be able to view the current AP configuration, change it, and save it in Flash memory.

Upon system startup, the uAP has the default configuration:

Parameter	Default Value
SSID	"NXP Micro AP"
Beacon period	100 TU (1 TU = 1024 μs)
Channel	6 (ACS disabled)
Secondary channel	Secondary channel: none
Basic rate set	1, 2, 5.5, 11 Mbps
Operational rate set	6, 9, 12, 18, 24, 36, 48, 54 Mbps
Transmit power level	13 dBm
Broadcast SSID	Enabled
Preamble type	Short
RTS threshold	2347
Fragmentation threshold	2346
DTIM	1
Antenna mode	Antenna A (Tx and Rx)
Radio state	Off
Transmit data rate	Auto (unicast and multicast)
Security settings	No security and open authentication
WEP keys	Set to 0
Custom IEs	Set to 0
Packet forwarding mode	Firmware bridges packets from a client station to another station in the same BSS
Client station MAC address filter table	Disabled, filled with 0s
Aging timer	3 minutes
PS aging timer	40s
MAC address	Use MAC address programmed in EEPROM If that fails, default to 00:50:43:02:FE:01 (see below)
Maximum station count	Varies between chip/feature-pack (see product release notes)
Tx retry limit (long and short)	7
Scan channels	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
Key management mode	PSK
WPA pass phrase	Not configured (empty)
RSN replay protection	Disabled
Pairwise and groupwise handshake retries	3

<table>
<tr><th>Parameter</th><th>Default Value</th></tr>
<tr><td>Pairwise and groupwise handshake timeout</td><td>100 ms</td></tr>
<tr><td>Group cipher re-key time</td><td>86400 seconds (1 day)
This will be used only when using WPA/WPA2.</td></tr>
<tr><td>Power mode</td><td>Disabled</td></tr>
<tr><td>Power save parameters (varies from chip to chip)</td><td>Protected = enabled
Sleep duration = 17 ms (minimum and maximum)
Inactivity timeout = 200 ms
Awake period = 2 ms (minimum and maximum)</td></tr>
<tr><td>802.11d settings</td><td>802.11d state disabled and country code not set</td></tr>
<tr><td>HT capability information</td><td>802.11n mode = enabled
HT capability information = 0x111C
AMPDU parameters = 0x3</td></tr>
<tr><td>20/40 BSS Coexistence</td><td>Enabled</td></tr>
<tr><td>ADDBA parameters</td><td>Timeout = 65535
Txwinsize = 32
Rxwinsize = 16</td></tr>
<tr><td>AMPDU/AMSDU priority table</td><td>1 1 2 2 0 0 3 3 4 4 5 5 255 255 255 255</td></tr>
<tr><td>ADDBA reject table</td><td>0 0 0 0 0 0 0 0</td></tr>
<tr><td>WMM parameters</td><td>QoS information = 0x80
<table>
<tr><th></th><th>AIFSN</th><th>CW MAX</th><th>CW MIN</th><th>TXOP</th></tr>
<tr><td>BE</td><td>3</td><td>10</td><td>4</td><td>0</td></tr>
<tr><td>BK</td><td>7</td><td>10</td><td>4</td><td>0</td></tr>
<tr><td>VI</td><td>2</td><td>4</td><td>3</td><td>94</td></tr>
<tr><td>VO</td><td>2</td><td>3</td><td>2</td><td>47</td></tr>
</table></td></tr>
<tr><td>802.11h settings</td><td>802.11h disabled</td></tr>
</table>

The MAC address of the uAP is determined as follows:

1. If the EEPROM exists and the MAC address field in the EEPROM is not blank (that is, filled with 0xFF), firmware will use it for the AP MAC address.
2. Else, firmware will use the default MAC address that is hard coded in firmware.
3. After system startup, the host driver can query the AP MAC address by issuing the command APCMD_SYS_CONFIGURE.GET. The AP MAC address is returned in MrvlIETypes_ApMacAddr_t.
4. The host driver can set a new MAC address by issuing the command APCMD_SYS_CONFIGURE.SET (MrvlIETypes_ApMacAddr_t).

7.6 Micro AP Use Cases

This section describes the use cases for the uAP.

7.6.1 Startup Micro AP

This use case is about how to boot up the uAP and bring it into operation.

A typical startup sequence to create a functional uAP consists of the following steps (see Figure 10):

1. The host driver downloads the uAP firmware into the Wi-Fi module.
2. Firmware initializes and enters the idle state.

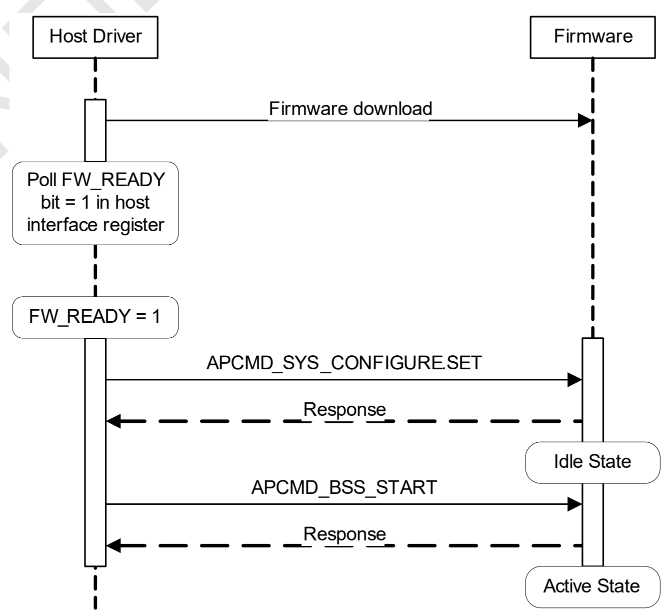
   In the idle state:

   a) Hardware is reset and initialized.
   b) Radio is turned off, that is, not transmitting and not receiving.
   c) No BSS is configured.

   After firmware initializes, it sets the FW_READY bit in a host interface register, and the host driver should poll the FW_READY bit and wait for it to be set before continuing with the uAP start up process. The first Wi-Fi host command (except SHUTDOWN command) turns radio on.
3. The host driver sends the command APCMD_SYS_CONFIGURE.SET to set the AP parameters, such as SSID, basic rates, radio channel, security, and so on. The driver can override the AP MAC address with a new one here.
4. The host driver activates the BSS by sending the command APCMD_BSS_START.
5. On receiving the command to start the BSS, firmware performs the following:

   a) Start beaconing.
   b) Enter the active state.

Figure 10: Sequence uAP Startup



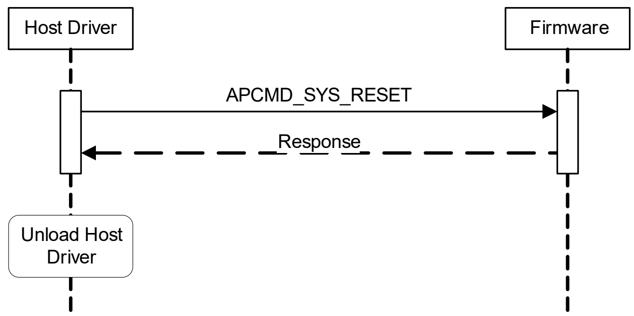
7.6.2 Shutdown Micro AP

This use case happens when the host system terminates the uAP application for any reason, like switching the Wi-Fi module's role from AP function to client function.

The shutdown sequence (Figure 11) is as follows:

1. The host driver sends the command APCMD_SYS_RESET.
2. On receiving the command, firmware performs the following:
   a) De-authenticate all client stations that are associated with the AP.
   b) Reset the Wi-Fi hardware to initial conditions.
   c) Reset all internal variables to default values.
   d) Enter the idle state.
   e) Send the response message to the host driver.
3. The host system unloads the host driver.

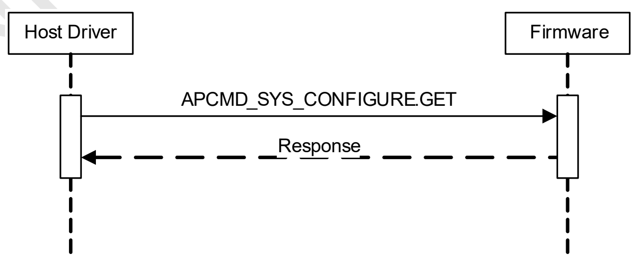
Figure 11: Sequence uAP Shutdown



7.6.3 Query Micro AP Configuration

The host application can query the current AP configuration by sending the command APCMD_SYS_CONFIGURE.GET.

Figure 12: Sequence Query Current AP Configuration



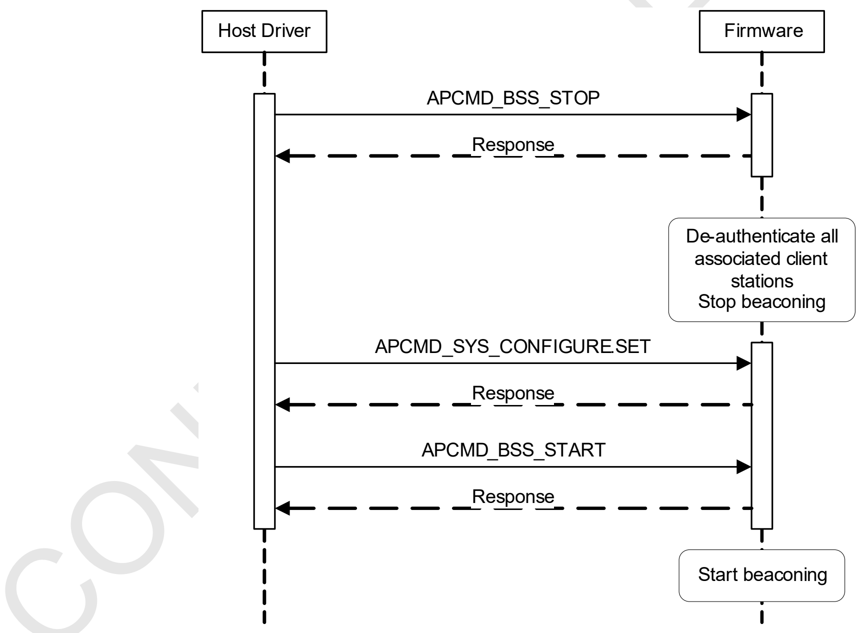
7.6.4 Change Micro AP Configuration

This use case happens when the user wishes to change the parameters of the current BSS, such as changing channel, or security settings. In general, this use case does not require resetting the uAP.

The sequence is as follows:

1. The host driver sends the command APCMD_BSS_STOP.
2. On receiving the command, firmware de-authenticates all client stations that are associated and stops beaconing.
3. The host driver sends new AP parameters via the command APCMD_SYS_CONFIGURE.SET.
4. On receiving the command, firmware applies the new settings specified.
5. The host driver sends the command APCMD_BSS_START.
6. On receiving the command, firmware starts beaconing again. The beacons will reflect the AP configuration changes where applicable.

Figure 13: Sequence Change AP Configuration



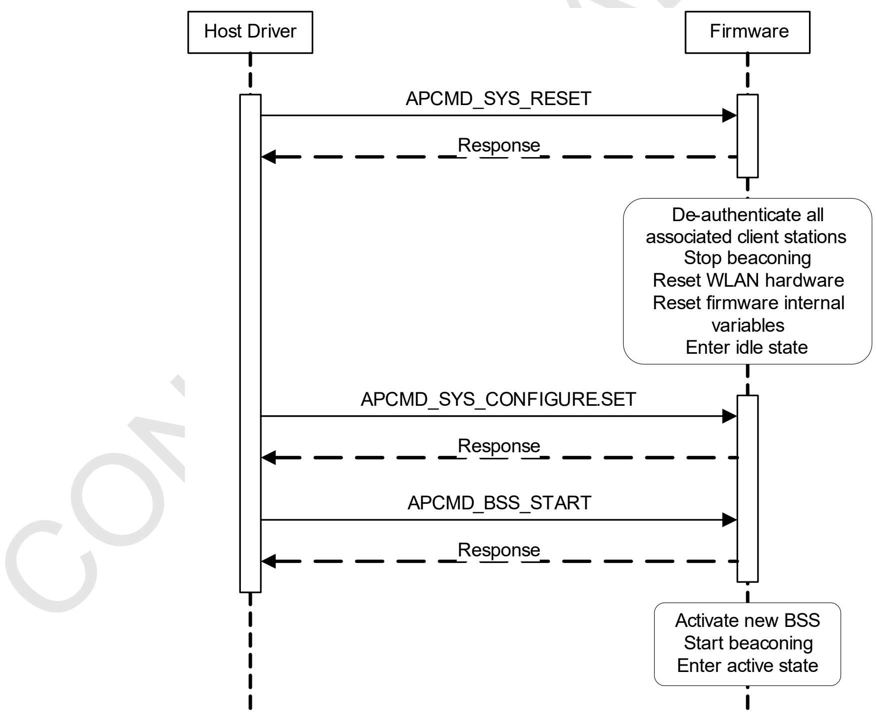
7.6.5 Create New BSS

This use case happens when the user wishes to create a new BSS. In this case, the current BSS is terminated and the new BSS is created.

The sequence is as follows:

1. The host driver sends the command APCMD_SYS_RESET.
2. On receiving the command, firmware resets the system as described in Section 7.6.2, Shutdown Micro AP, on page 251.
3. The host driver sends all the parameters for the new BSS via command APCMD_SYS_CONFIGURE.SET.
4. On receiving the command, firmware will initializes the hardware and firmware internal variables to create the new BSS.
5. The host driver then sends the command APCMD_BSS_START to activate the new BSS.
6. On receiving the command, firmware starts beaconing.

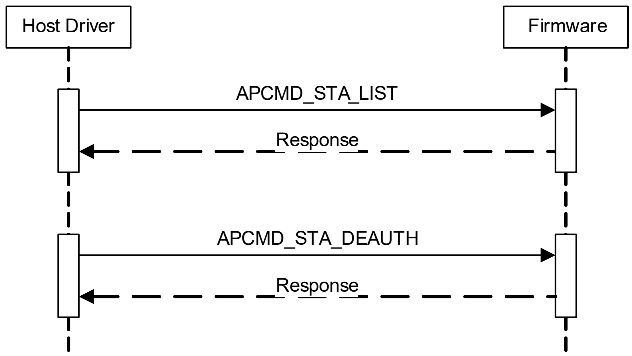
Figure 14: Sequence Create New BSS



7.6.6 Manage Client Stations

This use case is about managing the client stations, which typically involves getting the list of client stations that are currently associated with the AP or de-authenticating a client station for any reason.

Figure 15: Sequence Query List of Client Stations and De-authenticate Client Station



7.6.7 External Authenticator

Micro AP supports internal and external authenticator. If MRVL_WPA_PASSPHRASE_TLV_ID is configured, the internal authenticator is to be used. Otherwise, uses the external authenticator.

The sequence to bring up uAP in external authenticator is as follows:

1. Set required configurations.
2. Issue APCMD_BSS_START command.
3. Download GTK through CMD_802_11_KEY_MATERIAL command (download once only).
4. Download individual PTK for each associated stations.

The external authenticator also issues the APCMD_BSS_STOP command when the authenticator application is terminated.

7.7 Micro AP Events

This section describes the events from uAP firmware to host driver that may occur during operation. Table 68 lists the supported events.

Table 68: Event Support

Event	EventID	Description
PS_AWAKE	0x000A	Generated when hardware device wakes up
PS_ASLEEP	0x000B	Generated when hardware device goes to sleep
MICRO_AP_EV_ID_STA_DEAUTH	0x002C	Generated when a client station leaves the BSS
MICRO_AP_EV_ID_STA_ASSOC	0x002D	Generated when a client station joins in the BSS
MICRO_AP_EV_ID_BSS_START	0x002E	Generated when BSS is started
DOT11N_ADDBA	0x0033	Sent by firmware to driver upon receipt of ADDBA request action frame (see Section 8.3.2.1, DOT11N_ADDBA, on page 293)
DOT11N_DELBA	0x0034	Sent by firmware to driver upon receipt of DELBA request action frame (see Section 8.3.2.2, DOT11N_DELBA, on page 293)
BA_STREAM_TIMEOUT	0x0037	Used for Block Ack inactivity timeout (see Section 8.3.2.3, BA_STREAM_TIMEOUT, on page 294)
MICRO_AP_EV_ID_BSS_IDLE	0x0043	Generated when BSS is idle
MICRO_AP_EV_ID_BSS_ACTIVE	0x0044	Generated when BSS is active
HS_ACT_REQ	0x0047	Generated in Full Power mode to notify host that firmware is ready to activate enhanced HS (host sleep) mode (see Section 8.3.1.6, HS_ACT_REQ, on page 283)
MICRO_AP_EV_ID_MIC_COUNTERMEASURES	0x004C	Generated when firmware starts or stops MIC countermeasure
MICRO_AP_EV_ID_RSN_CONNECT	0x0051	Generated when STA successfully completed 4-way handshake
RADAR_DETECTED	0x0053	Generated if master mode radar detection mode is enabled in firmware and if firmware detects radar on the current operating channel or firmware receives a measurement report/autonomous measurement report which indicate that radar is detected on the current operating channel
CHANNEL_REPORT_RDY	0x0054	Generated when channel report measurement is completed
TX_DATA_PAUSE	0x0055	Generated when Tx Pause state for a client has changed (see Section 8.3.1.10, TX_DATA_PAUSE, on page 286)
EV_RXBA_SYNC	0x0059	Triggered by firmware to synchronize Rx BA windows bitmap (see Section 8.3.2.6, EV_RXBA_SYNC, on page 295)
MICRO_AP_EV_ID_BSS_STOP	0x0079	Triggered to indicate to driver that int-STA disconnects from ext-AP and uAP will stop automatically depending on setting.

7.7.1 MICRO_AP_EV_ID_STA_DEAUTH

This event is triggered when a client station leaves the BSS, that is, gets de-authenticated, for a number of reasons. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	MICRO_AP_EV_ID_STA_DEAUTH = 0x002C
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ReasonCode	UINT16	Reason for leaving BSS: 0x00 = unspecified 0x01 = client station leaving network 0x02 = client station aged out 0x03 = client station de-authenticated by user's request 0x04 = client station authentication failed 0x05 = client station association failed 0x06 = client station MAC address is blocked by filter table 0x07 = reached maximum station count limit that uAP supports 0x08 = 4-way handshake timeout 0x09 = group key handshake timeout Others = reserved
StaMacAddress	UINT8[6]	MAC address of client station that gets de-authenticated

7.7.2 MICRO_AP_EV_ID_STA_ASSOC

This event is triggered when a client station joins in the BSS. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	MICRO_AP_EV_ID_STA_ASSOC = 0x002D
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT8[2]	Reserved
StaMacAddress	UINT8[6]	MAC address of client station that has associated
WapiInfoSet	MrvlIETypes_WapiInfoSet_t	Used when WAPI is enabled
MgmtFrameSets	MrvlIETypes_MgmtFrameSet_t	1 or more management frames
VHTCapabilityIE	MrvlIETypes_VHT_Capability_IE_t	Used when VHT feature is enabled

Upon successful association of a VHT ext-STA, firmware will add MrvlIETypes_VHT_Capability_IE_t to this event, which is generated for that ext-STA. Driver may use the presence of this TLV to configure certain VHT-specific parameters for the ext-STA.

This TLV will not be added to this event if ext-STA does not support VHT feature set.

> ***Note:*** Same ID and usage as FORWARD_MGMT_FRAME.

7.7.3 MICRO_AP_EV_ID_BSS_START

This event is triggered when the BSS is started by using the BSS start command. The structure of the event data is:

Field Name	Type	Description
EventId	UINT16	MICRO_AP_EV_ID_BSS_START = 0x002E
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT8[2]	Reserved
apMacAddress	UINT8[6]	MAC address of the AP
TLV List	UINT8[]	There could be 1 or more TLV fields that are present in the event payload

The following TLVs are included:

- MrvlIETypes_HT_Capability_t if 802.11n is enabled
- MrvlIETypes_VHT_Capability_IE_t if 802.11ac is enabled or supported
  (If MMH is configured to support VHT features, firmware will add this TLV to this event. Driver may use the presence of this TLV to configure certain VHT-specific parameters for that MMH interface. This TLV will not be added to this event if VHT feature set is not supported or enabled in MMH.)
- MrvlIETypes_PKT_Fwd_Ctrl_t
- MrvlIETypes_VendorSpecific_t if WMM is enabled

7.7.4 MICRO_AP_EV_ID_BSS_IDLE

This event is triggered when the BSS is idle. This event does not have any payload. The structure of the event data is:

Field Name	Type	Description
EventId	UINT16	MICRO_AP_EV_ID_BSS_IDLE = 0x0043
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT8[2]	Reserved

7.7.5 MICRO_AP_EV_ID_BSS_ACTIVE

This event is triggered when the BSS is active. This event does not have any payload. The structure of the event data is:

Field Name	Type	Description
EventId	UINT16	MICRO_AP_EV_ID_BSS_ACTIVE = 0x0044
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT8[2]	Reserved

The definition of why and when BSS idle and BSS active will be triggered is as follows:

1. Default carrier state will be OFF (this will mean that no data should be sent for Tx to firmware).
2. BSS_ACTIVE event turn the carrier ON (when carrier is ON, data can be sent to firmware for Tx).
3. BSS_ACTIVE event will be triggered under the following conditions:
   a) First STA associates (that is, AssocCount transitions from 0 to 1).
   b) RADIO_ON TLV in SYS_CONFIG command received and AssocCount > 0.
   c) BSS_PAUSE event previously triggered by firmware to notify host that it cannot temporarily accept any data frames from host and firmware is now ready to accept the data packets.
4. BSS_IDLE event will turn the carrier OFF.
5. BSS_IDLE event will be triggered by firmware under the following conditions:
   a) Last STA disconnects (that is, AssocCount transitions from 1 to 0) and RADIO Status in ON.
   b) BSS_STOP command received and AssocCount > 0.
   c) SYS_RESET command received and AssocCount > 0.
   d) RADIO_OFF TLV in SYS_CONFIG command received and AssocCount > 0.

7.7.6 MICRO_AP_EV_ID_MIC_COUNTERMEASURES

This event is triggered when firmware starts or stops MIC countermeasures. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	MICRO_AP_EV_ID_MIC_COUNTERMEASURES = 0x004C
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Status	UINT16	Status 0 = countermeasures inactive (default) 1 = countermeasures active

The host should use this event as follows:

- When countermeasures are active, firmware dis-associates all stations connected to it; hence host should delete any station specific data and should not send any data packets for Tx.
- When countermeasures are indicated as inactive and authenticator is implemented on host, a new GTK must be programmed on firmware using the key material API.

Firmware will trigger “inactive” event to host if it had triggered event with “active” state earlier. Firmware will trigger the event irrespective of whether the “authenticator” feature is on device or host.

7.7.7 MICRO_AP_EV_ID_RSN_CONNECT

This event has is triggered when a STA successfully completes a 4-way handshake if it is associating in WPA2 mode or upon successful group key exchange if it is using WPA security mode. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	MICRO_AP_EV_ID_RSN_CONNECT = 0x0051
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT8[2]	Reserved
StaMacAddress	UINT8[6]	MAC address of the client station that has completed handshake
TLV List	UINT8[]	There could be 1 or more TLV fields that are present in the event payload

Currently, either WPA or RSN IE is added as TLV. Below is the format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0030 (RSN) or 0x00DD (WPA)
Length	UINT16	Length of payload
IE Buffer	UINT8[]	WPA or RSN IE’s content

7.7.8 RADAR_DETECTED

This event has an EventID = 0x0053 and is generated if the master mode radar detection mode is enabled in firmware and if firmware detects radar on the current operating channel or firmware receives a measurement report/autonomous measurement report which indicates that radar is detected on the current operating channel. Master mode radar detection mode will be enabled in firmware if current operating channel is a DFS channel. This event does not have any payload.

7.7.9 CHANNEL_REPORT_RDY

This event is generated when channel report measurement is completed. Channel report API is made asynchronous. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	CHANNEL_REPORT_RDY = 0x0054
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
CommandResult	UINT32	Result of channel report request
startTsf	UINT64	TSF at which measurement is started
Duration	UINT32	Duration of measurement
tlvBuffer	UINT8[]	1 or more TLVs

The following TLVs are included:

- MrvlIETypes_Chanrpt_Chan_Load_t
- MrvlIETypes_Chanrpt_Noise_Hist_t
- MrvlIETypes_Chanrpt_11H_Basic_t

7.7.10 MICRO_AP_EV_ID_BSS_STOP

When the int-STA disconnects from the ext-AP, the uAP will be stopped automatically if “Control” is set to 0x0001 or 0x0002. This event is then triggered to indicate to the driver. The structure of the event data is:

Field Name	Type	Description
EventID	UINT16	MICRO_AP_EV_ID_BSS_STOP = 0x007B
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ReasonCode	UINT16	Reason code of BSS STOP event 0x0001 = micro AP stops due to int-STA disconnect, will not restart automatically 0x0002 = micro AP stops and restarts automatically due to int-STA disconnect

7.8 Micro AP Firmware Commands

This section describes the set of firmware commands for uAP.

Due to the limited amount of memory in the Wi-Fi module, the host driver should not expect firmware to validate the data that it receives. The host driver shall be responsible to ensure that the data that it sends to firmware in the commands is correct.

Firmware will perform basic error checking on the parameters in a command, for example, checking that the parameter is within a valid range. Complicated error checking, such as ensuring that the value of a certain parameter is consistent with other configuration data, is the responsibility of the host system.

Table 69 lists the supported commands.

Table 69: Supported uAP Commands

Command	Description	Page
Section 7.8.1, APCMD_SYS_INFO	Returns system information	page 263
Section 7.8.2, APCMD_SYS_RESET	Resets the uAP to its initial state	page 264
Section 7.8.3, APCMD_SYS_CONFIGURE	Gets/sets AP configuration parameters	page 265
Section 7.8.4, APCMD_BSS_START	Starts BSS	page 268
Section 7.8.5, APCMD_BSS_STOP	Stops active BSS	page 269
Section 7.8.6, APCMD_STA_LIST	Gets list of client stations currently associated with the AP	page 270
Section 7.8.7, APCMD_STA_DEAUTH	De-authenticates a client station	page 271
Section 7.8.8, CMD_CFG_DATA	Facilitates downloading of calibration data	page 272
Section 7.8.9, CMD_CHAN_REPORT_ REQUEST	Performs channel load, channel noise histogram, basic report measurement on a given channel	page 273
Section 7.8.10, CMD_PMF_PARAMS	Gets/sets PMR support on uAP with embedded authenticator	page 274
Section 7.8.11, CMD_APCMD_ACS_SCAN	Supports ACS feature on DFS channels for uAP	page 275
Section 7.8.12, CMD_UAP_OPERATION_ CTRL	Gets/sets uAP operation control parameters in non-DRCS case	page 276
CMD_802_11_MAC_ADDR	Gets/sets the Wi-Fi MAC address (see Section 6.1.2 on page 21)	--
CMD_CHANNEL_TRPC_CONFIG	Gets/sets TRPC channel configuration (see Section 6.2.4 on page 37)	--
CMD_802_11_KEY_MATERIAL	Gets/sets key material used to do Tx encryption or Rx decryption (see Section 6.9.1 on page 95)	--
CMD_TX_RATE_CFG	Gets/sets 802.11n transmission date rate (see Section 6.10.2 on page 111)	--
CMD_802_11_D_DOMAIN	Configures 802.11d domain (see Section 6.16.1 on page 153)	--
CMD_11N_ADDBA_REQ	Sends ADDBA request to a peer (see Section 6.18.2 on page 162)	--
CMD_11N_ADDBA_RSP	Sends ADDBA response to a peer (see Section 6.18.3 on page 163)	--
CMD_11N_DELBA	Deletes a Block Ack agreement (see Section 6.18.4 on page 165)	--
CMD_RECONFIGURE_TX_BUFF	Gets/sets Tx buffer size (see Section 6.18.5 on page 166)	--

Table 69: Supported uAP Commands (Continued)

Command	Description	Page
CMD_AMSDU_AGGR_CTRL	Enables/disables AMSDU agreement (see Section 6.18.6 on page 167)	--
CMD_802_11_PS_MODE_ENH	Enables/disables AP/client power save (see Section 6.13.3.6 on page 128)	--
CMD_802_11_HS_ENH	Configures wakeup semantics of host sleep (see Section 6.13.3.7 on page 131)	--

7.8.1 APCMD_SYS_INFO

This command returns the system information, such as firmware version number and hardware information.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_INFO = 0x00AE
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_INFO | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = command is successful 1 = command fails
SysInfo	UINT8[64]	Wi-Fi hardware information in null-terminated ASCII string (device type, revision number, and other hardware information) and firmware version number.

7.8.2 APCMD_SYS_RESET

The host driver should use this command whenever there is a need to get the uAP back to its initial state, for example, to recover from a serious error, or before creating a new BSS.

This command has the following effects:

- Wi-Fi hardware MAC is reset
- All MIB variables are initialized to their respective default values
- Firmware internal variables are reset to their respective default values
- Firmware state machines are reset to their respective initial states

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_RESET = 0x00AF
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_RESET | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by the host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = command is successful 1 = command fails

7.8.3 APCMD_SYS_CONFIGURE

This command gets or sets the AP configuration parameters.

The body of the command contains a variable number of TLV constructs (tag, length, value format). Firmware parses the set of TLV and applies the setting accordingly.

The host driver is responsible for sending the complete set of parameters to configure the desired operational mode of the AP properly. Firmware does not check for the correctness, consistency, and the applicability of the set of AP configuration data that it receives. For example, if the host driver does not specify in the command the set of basic rates that it desires to use then firmware will use the basic rates setting that it has at the time, which could be anything.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_CONFIGURE = 0x00B0
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACTION_GET (read) 0x0001 = ACTION_SET (write)
ConfigData	TLV	The configuration data consists of a variable number of TLV constructs.

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_SYS_CONFIGURE | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success (only when Action succeeds for ALL TLVs) 1 = failure (if Action fails for 1 or more of the TLVs)
Action	UINT16	Type of action set in request message
ConfigData	TLV	The configuration data consists of a variable number of TLV constructs.

The behavior of the GET action has some special cases listed below:

- APCMD_SYS_CONFIGURE + ACTION_GET + no TLV append
  In this case, firmware will return all the TLVs currently been configured in firmware.
- APCMD_SYS_CONFIGURE + ACTION_GET + specific TLV (with length 0) append
  In this case, firmware will return the specific TLV only.
- APCMD_SYS_CONFIGURE + ACTION_GET + more than 1 specific TLV (with length 0) append
  In this case, firmware will return all the specific TLVs.

List of possible TLV included in ConfigData:

Name	"On-the-fly" Update
MRVL_SSID_TLV_ID	N
MRVL_RATES_TLV_ID	N
MRVL_PHYPARAMDSSET_TLV_ID (deprecated)	N
MrvlIETypes_HTInfo_t	N
MrvlIETypes_ApMacAddr_t	N
MrvlIETypes_Beacon_Period_t	N
MrvlIETypes_DTIM_Period_t	N
MrvlIETypes_ApTx_Power_t	Y
MrvlIETypes_ApBCast_SSID_Ctrl_t	N
MrvlIETypes_Preamble_Ctrl_t	N
MrvlIETypes_Antenna_Ctrl_t	Y
MrvlIETypes_PKT_Fwd_Ctrl_t	N
MrvlIETypes_RTS_Threshold_t	Y
MrvlIETypes_Frag_Threshold_t	Y
MrvlIETypes_TxData_Rate_t	Y
MrvlIETypes_STA_Ageout_t	N
MrvlIETypes_WEP_Key_t	N
MrvlIETypes_AuthType_t	N
MrvlIETypes_STA_MAC_Filter_t	N
MrvlIETypes_Encryption_Protocol_t	N
MrvlIETypes_Max_STA_Counts_t	N
MrvlIETypes_Retry_Limit_t	Y
MrvlIETypes_ChanBandList_t	N
MrvlIETypes_ChanListParamSet_t	N
MrvlIETypes_MCBC_Data_Rate_t	Y
MrvlIETypes_AKMP_t	N
MrvlIETypes_Cipher_t	N
MrvlIETypes_GTK_Rekey_t	N
MrvlIETypes_WPA_Passphrase_t	N
MrvlIETypes_RSN_Replay_t	Y
MrvlIETypes_Mgmt_Passthru_Mask_t	Y
MrvlIETypes_PTK_Timeout_t	N
MrvlIETypes_PTK_Retry_t	N
MrvlIETypes_GTK_Timeout_t	N
MrvlIETypes_GTK_Retry_t	N
MrvlIETypes_PS_STA_Ageout_t	N
MrvlIETypes_PTK_Cipher_t	N

Name	"On-the-fly" Update
MrvlIETypes_GTK_Cipher_t	N
MrvlIETypes_WAPI_t	N
MrvlIETypes_HT_Capability_t	N
MrvlIETypes_VendorSpecific_t	N
MrvlIETypes_ApBSS_Status_t	N
MrvlIETypes_Sticky_Tim_Config_t	Y
MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t	Y
MrvlIETypes_2040_BSS_Coex_Ctrl_t	N
IEEEtypes_WMM_ParamElement_t	Y

Of these, the SET operation is allowed while BSS is active only for TLVs that are marked as "Y" for "On-the-fly" update above. For the rest, the BSS needs to be in inactive state for the SET action to be performed. If a set operation is performed on any TLV not allowed for On-the-fly update, the result will always be a failure.

Note:

- MrvlIETypes_Antenna_Ctrl_t has Rx and Tx antenna setting as part of the "which antenna field' in the TLV, which is part of the value. As a result, when a GET operation is performed on this TLV, the response will include 2 entries with the same TLV in which 1 entry will have Rx antenna setting and the other will have the Tx antenna setting. Differentiating on the basis of the 'which antenna' field in the value part of the TLV.
- MrvlIETypes_Preamble_Ctrl_t is a "read-only" TLV. The SET operation for this will fail on uAP. The AP will always support short and long preamble both, but since at run-time, the preamble mode will depend on the capabilities of stations connected to it. The GET operation will show the mode that is currently being used (that is, long or short).

7.8.4 APCMD_BSS_START

This command starts the BSS. Prior to starting the BSS, the host driver must have sent the command APCMD_SYS_CONFIGURE to set the desired BSS parameters.

Note:

- In case WPA/WPA2 security mode is used and the passphrase is configured as ASCII string of length 8 to 63 characters, firmware needs to convert the passphrase to a 256-bit key value using the IEEE specified hash function. This is CPU intensive and depending on the embedded processor on which this firmware is running, it will delay the response to the APCMD_BSS_START command. The host application should take this into account if there is a timeout configured for the command sent to firmware. This can takes from 5 to 15 seconds, depending on the chip being used. To reduce this time, host processor can perform the hash and program the entire 256-bit key value as a 64 character ASCII string (that is, hexadecimal converted to ASCII).
- In case ACS (automatic channel selection) is enabled, the command response to APCMD_BSS_START would be delayed further by 2 seconds. Firmware will need to scan across configured channels to select the best channel. Firmware will scan all channels in scan channel list within 2 seconds. When ACS is enabled and BSS is configured to operate in HT2040 mode (802.11n enabled), secondary channel will also be automatically selected during the ACS process.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_BSS_START = 0x00B1
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_BSS_START | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = invalid BSS parameters

7.8.5 APCMD_BSS_STOP

This command stops the active BSS. It is typically sent by the host before making certain configuration changes in the AP. This command will cause firmware to de-authenticates all associated client stations and stop beaconing.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_BSS_STOP = 0x00B2
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_BSS_STOP | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = generic error

7.8.6 APCMD_STA_LIST

This command returns the list of client stations that are currently associated with the AP.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_STA_LIST = 0x00B3
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_STA_LIST | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = failure
StaCount	UINT16	Number of client stations associated with the AP
StaList	TLVs	Sequence of TLVs that contain client station information—MrvlIETypes_STA_Info_t

7.8.7 APCMD_STA_DEAUTH

This command de-authenticates a client station. This command has been extended to take the IEEE reason code to be sent in the de-authenticate message transmitted to the STA.

If the reason is not specified, firmware will use code 0x2 (prior authentication invalid).

REQUEST

Field Name	Type	Description
CmdCode	UINT16	APCMD_STA_DEAUTH = 0x00B5
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
StaMacAddress	UINT8[6]	MAC address of client station to de-authenticate
Reason	UINT16	IEEE reason code to be used for de-authenticate message

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	APCMD_STA_DEAUTH | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = failure
StaMacAddress	UINT8[6]	MAC address of client station to de-authenticate
Reason	UINT16	IEEE reason code to be used for de-authenticate message

7.8.8 CMD_CFG_DATA

This command is used to facilitate downloading of calibration data.

For host command usage, see Section 6.4.2, CMD_CFG_DATA, on page 51.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_DATA = 0x008F
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET (read) 0x0001 = ACT_SET (write)
Type	UINT16	0x0001 = reserved 0x0002 = calibration data Others = reserved
PayloadLen	UINT16	Payload length
Payload	UINT8[]	Payload (as per length specified in PayloadLen)

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CFG_DATA | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number (as sent by host)
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = failure
Action	UINT16	Type of action set in request message
Type	UINT16	0x0001 = reserved 0x0002 = calibration data Others = reserved
PayloadLen	UINT16	Payload length
Payload	UINT8[]	Payload (as per length specified in PayloadLen)

7.8.9 CMD_CHAN_REPORT_REQUEST

This command performs channel load, channel noise histogram, basic report measurement on a given channel. The command response will be sent immediately after performing error checking. The channel report is sent as an event (CHANNEL_REPORT_RDY) after the measurement is completed.

This command is also used to start the radar detection on 0DFS interface. BSS type BSS_TYPE_DFS is used for this purpose.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_CHAN_REPORT_REQUEST = 0x00DD
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
ChanDesc	MrvlIEChannelDesc_t	Details of channel on which measurement is to be carried out
MillisecDwellTime	UINT32	Duration for which the measurement is to be carried out
tlv_buffer	Variable	1 or more of the following TLVs: • MrvlIETypes_ZERO_DFS_t • MrvlIETypes_Chanrpt_11H_Basic_t

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_CHAN_REPORT_REQUEST | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = success 1 = failure
ChanDesc	MrvlIEChannelDesc_t	Details of channel on which measurement is to be carried out
MillisecDwellTime	UINT32	Duration for which the measurement is to be carried out
tlv_buffer	Variable	1 or more of the following TLVs: • MrvlIETypes_ZERO_DFS_t • MrvlIETypes_Chanrpt_11H_Basic_t

7.8.10 CMD_PMF_PARAMS

This command is used by the driver to get or set Protected Management Frame (PMF) support on uAP with embedded authenticator. It cannot be used with host supplicant.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_PMF_PARAMS = 0x0131
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET
PMFParams	UINT8	PMF parameters for embedded authenticator Bits[7:2]: Reserved Bit[1]: MFPR dot11RSNAUnprotectedMgmtFramesAllowed = !MFPR 0 = PMF is not required 1 = PMF is required Bit[0]: MFPC dot11RSNAProtectedMgmtFramesEnabled = MFPC 0 = no PMF capability 1 = PMF capability

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_PMF_PARAMS | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Action	UINT16	Type of action set is request message
PMFParams	UINT8	PMF parameters for embedded authenticator Bits[7:2]: Reserved Bit[1]: MFPR dot11RSNAUnprotectedMgmtFramesAllowed = !MFPR 0 = PMF is not required 1 = PMF is required Bit[0]: MFPC dot11RSNAProtectedMgmtFramesEnabled = MFPC 0 = no PMF capability 1 = PMF capability

7.8.11 CMD_APCMD_ACS_SCAN

This command supports ACS feature on DFS channels for uAP.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_APCMD_ACS_SCAN = 0x0224
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Not used (set to 0)
Band	UINT8	Band
Channel	UINT8	Channel

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_APCMD_ACS_SCAN | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code (set to 0 for success)
Band	UINT8	Band
Channel	UINT8	Channel

7.8.12 CMD_UAP_OPERATION_CTRL

This command is used to set or get the uAP operation control parameters in non-DRCS case. It controls the behavior of uAP when int-STA is disconnected from the ext-AP. See Control and ChanOpt fields for options.

REQUEST

Field Name	Type	Description
CmdCode	UINT16	CMD_UAP_OPERATION_CTRL = 0x0233
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type identifier on which command is issued Bits[3:0]: BSS_Num identifier on which command is issued
Result	UINT16	Result code 0 = successful 1 = not supported
Action	UINT16	Action 0x0000 = ACT_GET 0x0001 = ACT_SET **NOTE:** Only support to set when no interface is active.
Control	UINT16	Control The following operation is defined for uAP operation when int-STA is disconnected from ext-AP if multi channel policy was disabled in firmware. 0x0000 = no action (uAP keeps running as it is. Default.) 0x0001 = uAP stops automatically and indicates a BSS STOP event to host **NOTE:** Control with 0x0001 is optional, it is not included in current implementation. 0x0002 = uAP stops automatically and indicates a BSS STOP event to host, then restarts automatically on channel specified in ChanOpt parameter **NOTE:** This auto stop/start behavior only take effect when host already started uAP. If host side did not start uAP yet, when int-STA disconnects, firmware will not auto start the uAP.
ChanOpt	UINT16	Channel option Defines which channel should be used when uAP is restart automatically (that is, when Control was set to 0x0002). 0x0001 = uAP restarts on default (2.4 GHz/channel 6) 0x0002 = uAP restarts on band/channel configured by driver previously 0x0003 = uAP restarts on band/channel specified in TLV in this command, firmware returns band/channel information based on ChanOpt setting.
ChanBand_t	MrvlIETypes_ ChanBandList_t	Band/Channel 1 = on Set action, driver may include this ChanBandList TLV to specify which band/channel uAP restarts on when ChanOpt was set to 0x0003 2 = on Get action, driver will not include this TLV in command request, firmware returns band/channel information based on ChanOpt setting in command response

RESPONSE

Field Name	Type	Description
CmdCode	UINT16	CMD_UAP_OPERATION_CTRL | 0x8000
Size	UINT16	Number of bytes in command body
SeqNum	UINT8	Command sequence number
BSS	UINT8	Bits[7:4]: BSS_Type Bits[3:0]: BSS_Num
Result	UINT16	Result code 0 = successful 1 = not supported
Action	UINT16	Type of action set in request message
Control	UINT16	Control The following operation is defined for uAP operation when int-STA is disconnected from ext-AP if multi channel policy was disabled in firmware. 0x0000 = no action (uAP keeps running as it is. Default.) 0x0001 = uAP stops automatically and indicates a BSS STOP event to host **NOTE:** Control with 0x0001 is optional, it is not included in current implementation. 0x0002 = uAP stops automatically and indicates a BSS STOP event to host, then restarts automatically on channel specified in ChanOpt parameter **NOTE:** This auto stop/start behavior only take effect when host already started uAP. If host side did not start uAP yet, when int-STA disconnects, firmware will not auto start the uAP.
ChanOpt	UINT16	Channel option Define which channel should be used when uAP is restart automatically (that is, when Control was set to 0x0002). 0x0001 = uAP restarts on default (2.4 GHz/channel6) 0x0002 = uAP restarts on band/channel configured by driver previously 0x0003 = uAP restarts on band/channel specified in TLV in this command, firmware returns band/channel information based on ChanOpt setting.
ChanBand_t	MrvlIETypes_ ChanBandList_t	Band/Channel 1 = on set action, driver may include this ChanBandList TLV to specify which band/channel uAP restarts on when ChanOpt was set to 0x0003 2 = on get action, driver will not include this TLV in command request, firmware returns band/channel information based on ChanOpt setting in command response

8 MAC Events

The Wi-Fi SoC device firmware generates events to notify the host driver of certain MAC events that occurred during normal operation. MAC events do not carry data. Only the event type is communicated to the host driver. The mechanism for generating events is dependent upon the hardware interface.

8.1 Events Supported

Table 70 lists the supported events.

Table 70: Event Support

Events	EventID	Description
Dummy Host Wakeup Signal	0x0001	Dummy event used to generate an interrupt to wakeup the host through the host interface if configured
Beacon Loss/Link Loss	0x0003	Beacon loss/link loss
Link Sense	0x0004	In the IBSS network, if the total number of joined stations (including self) changes from 1 station to more than 1 station, Link Sense is issued
Deauthenticate	0x0008	Generated when firmware receives a 802.11 deauthenticate management frame from the AP (see Section 8.2, SDIO Event Extended Format, on page 279)
Disassociate	0x0009	Generated when firmware receives a 802.11 disassociate management frame from the AP (see Section 8.2, SDIO Event Extended Format, on page 279)
PS Awake	0x000A	Generated when firmware first takes the Wi-Fi SoC device out of Sleep mode
Enter Sleep Mode	0x000B	Generated when firmware is about to put the Wi-Fi SoC device in Sleep mode
Group MIC Error	0x000C	MIC error generated for Broadcast packets
UNICAST MIC Error	0x000D	MIC error generated for Unicast packets
Ad-Hoc Beacon Lost	0x0011	In the IBSS network, if the total number of joined stations (including self) changed from more than 1 station to only 1 station, Ad-Hoc Beacon Lost is issued.
Stop Transmit	0x0013	Generated to indicate firmware is not ready to receive data packets. Generation of this event can occur during measurements, quiet periods, or channel switches.
Start Transmit	0x0014	Generated to indicate firmware is ready to receive data packets
Measurement Report Ready	0x0016	Generated when firmware has a measurement report waiting for driver retrieval using Section 6.17.1, CMD_802_11_MEASUREMENT_REQUEST, on page 154.
WMM Status Change	0x0017	Generated when firmware monitors a change in the WMM state. This is a result of an AP change in the WMM Parameter IE or a change in the operational state of 1 of the AC queues (see Section 6.15.1, CMD_WMM_GET_STATUS, on page 145).
Background Scan Report	0x0018	Generated as a result of a completed background scan report (see Section 6.7.2, CMD_802_11_BG_SCAN_CONFIG, on page 70)
BeaconRSSI Low	0x0019	Subscribed event (see Section 6.12, Event Subscription, on page 115)
BeaconSNR Low	0x001A	Subscribed event (see Section 6.12, Event Subscription, on page 115)

Table 70: Event Support (Continued)

Events	EventID	Description
MAX Failures	0x001B	Subscribed event (see Section 6.12, Event Subscription, on page 115)
BeaconRSSI High	0x001C	Subscribed event (see Section 6.12, Event Subscription, on page 115)
BeaconSNR High	0x001D	Subscribed event (see Section 6.12, Event Subscription, on page 115)
IBSS Coalesced	0x001E	Generated when the IBSS Coalescing process is finished and the BSSID has changed (see Section 6.8.5, CMD_802_11_IBSS_COALESCING_STATUS, on page 86)
BG Scan Timeout	0x001F	Generated when firmware is unable to find a WPS registrar within the timeout duration specified by (see Section 10.4.20, MrvlIETypes_WPSEnrolleeProbeReqIE_t, on page 329)
Data RSSI Low	0x0024	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Data SNR Low	0x0025	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Data RSSI High	0x0026	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Data SNR High	0x0027	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Link Quality Indication	0x0028	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Pre Beacon Lost	0x0031	Subscribe event (see Section 6.12, Event Subscription, on page 115)

8.2 SDIO Event Extended Format

For SDIO interface only. The event packet generated by firmware has additional information.

Table 71: SDIO Event Extended Format

Field Name	Type	Description
Length	UINT16	Number of bytes in the packet including this field In an SDIO Event that does not include any extra information, the length would be set to 8 (fields Length, Type and EventID only). When there is extra information, the Length field represents the actual number of bytes included in the event packet.
Type	UINT16	3 (EVENT)
EventID	UINT16	Enumerated identifier for the event (see Section 8.1, Events Supported, on page 278)
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ReasonCode	UINT16	IEEE Reason Code () as described in the 802.11 standard
MacAddress	UINT8[6]	Peer STA Address

8.3 802.11 ac/n/a/g/b Events

Table 72: 802.11 Events

Event	EventID	Description	Page
General Events			
Section 8.3.1.1, ADHOC_STATION_CONNECT	0x0020	Generated when a station is Ad-Hoc connected	page 281

Table 72: 802.11 Events

Event	EventID	Description	Page
Section 8.3.1.2, ADHOC_STATION_ DISCONNECT	0x0021	Generated when a station is Ad-Hoc disconnected	page 282
Section 8.3.1.3, PORT_RELEASE	0x002B	Triggered when controlled port are released	page 282
Section 8.3.1.4, FORWARD_MGMT_FRAME	0x002D	Generated when a management frame is forwarded	page 282
Section 8.3.1.5, EVT_WEP_ICV_ERR	0x0046	Triggered by WEP decryption error	page 283
Section 8.3.1.6, HS_ACT_REQ	0x0047	For enhanced power management	page 283
Section 8.3.1.7, MEF_HOST_WAKEUP	0x004F	Triggered to wakeup host when MEF condition matched	page 283
Section 8.3.1.8, CHANNEL_SWITCH_ ANNOUNCEMENT	0x0050	Reports channel switch	page 284
Section 8.3.1.9, TDLS_GENERIC	0x0052	Triggered in firmware when it detects TDLS link does not exist and informs driver to perform a teardown	page 285
Section 8.3.1.10, TX_DATA_PAUSE	0x0055	Generated when a uAP or GO client in PS exceeds allowed PS buffering, or when client has buffering available	page 286
Section 8.3.1.11, SCAN_REPORT_BY_EVENT	0x0058	Used to scan results for CMD_802_11_ SCAN_EXT	page 287
Section 8.3.1.12, MULTI_CHAN_INFO	0x006A	Triggered whenever any interfaces becomes active or inactive	page 287
Section 8.3.1.13, EVENT_FW_DUMP_INFO	0x0073	Triggered when firmware dumps memory for debug usage	page 288
Section 8.3.1.14, EVT_TX_STATUS_REPORT	0x0074	Triggered by firmware to return tx_status_ report for requested host Tx management frame fro mdriver	page 288
Section 8.3.1.15, EVENT_CLASS_NAN	0x0075	Used to send all type of NAN events to the host	page 289
Section 8.3.1.16, EVT_BTCOEX_WLAN_PARA_ CHANGE	0x0076	Sent to driver in 2 cases: • GO/MMH+SCO • GC+OPP_TX	page 289
Section 8.3.1.17, EV_SMC_GENERIC	0x0077	Generated by firmware while Smart mode configuration is under progress	page 290
Section 8.3.1.18, EV_NEIGHBOR_REPORT	0x0078	Triggered when driver issues CMD_GET_ NEIGHBORLIST command	page 291
Section 8.3.1.19, EV_BSS_TRANSITION_ REQUEST	0x0079	Used for DOT11V, voice enterprise certification	page 292
Section 8.3.1.20, EVT_CHAN_REG_INFO	0x0082	Generated by firmware when channel attributes and/or region information has been changed	page 292
Section 8.3.1.21, EVT_SSU_DUMP_DMA	0x008C	Sent to driver by firmware when SSU DMA complete or error detected	page 292
802.11n Events			

Table 72: 802.11 Events

Event	EventID	Description	Page
Section 8.3.2.1, DOT11N_ADDBA	0x0033	Generated when ADDBA request action frame is received	page 293
Section 8.3.2.2, DOT11N_DELBA	0x0034	Generated when DELBA request action frame is received	page 293
Section 8.3.2.3, BA_STREAM_TIMEOUT	0x0037	Block Ack inactivity timeout	page 294
Section 8.3.2.4, AMSDU_AGGR_CTRL	0x0042	Reports aggregation size available for host	page 294
Section 8.3.2.5, DOT11N_BWCHANGE_EVENT	0x0048	Reports bandwidth change	page 294
Section 8.3.2.6, EV_RXBA_SYNC	0x0059	Triggered by firmware to synchronize Rx BA window bitmap	page 295
Section 8.3.2.7, REMAIN_ON_CHANNEL_ EXPIRED	0x005F	Informs host about CMD_802_11_ REMAIN_ON_CHANNEL command expired	page 295

8.3.1 General Events

8.3.1.1 ADHOC_STATION_CONNECT

This event is generated when a station is Ad-Hoc connected.

Field Name	Type	Description
EventID	UINT16	ADHOC_STATION_CONNECT = 0x0020
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT16	Reserved (set to 0)
StaMacAddr	UINT8[6]	MAC address of peer side

8.3.1.2 ADHOC_STATION_DISCONNECT

This event is generated when a station is Ad-Hoc disconnected.

Field Name	Type	Description
EventID	UINT16	ADHOC_STATION_CONNECT = 0x0021
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT16	Reserved (set to 0)
StaMacAddr	UINT8[6]	MAC address of peer side

Note: In Ad-Hoc mode, when the last station left, only link lost event will be sent. The Ad-Hoc disconnect event will not be sent.

8.3.1.3 PORT_RELEASE

This event has an EventID = 0x002B. This event will be sent to host for the following scenario:

- No security—Event sent after associate is completed
- WEP mode—Event sent after associate is completed
- WPA/WPA2/WAPI modes—Event sent after unicast and multicast keys are downloaded

8.3.1.4 FORWARD_MGMT_FRAME

This event is generated when a management frame is forwarded.

Field Name	Type	Description
EventID	UINT16	FORWARD_MGMT_FRAME = 0x002D
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reserved	UINT16	Reserved (set to 0)
FromMacAddr	UINT8[6]	MAC address of sender
MgmtFrameSet	MrvlIETypes_MgmtFrameSet_t	Management frame TLVs

8.3.1.5 EVT_WEP_ICV_ERR

This event is triggered by WEP decryption error.

Field Name	Type	Description
EventID	UINT16	EVT_WEP_ICV_ERR = 0x0046
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ReasonCode	UINT16	Reason to send this event Not used (set to 0).
SrcMacAddr	UINT8[6]	Source MAC address
WepKeyIdx	UINT8	WEP decryption used key
WepKeyLen	UINT8	Key length
Key	UINT8[WepKeyLen]	WEP key

Trigger Rules:

- Event is triggered when decryption error packet is received
- Event is sent only once for each MAC address plus key index
- Event history for same MAC address will be reset for all key indices if:
  - Packet is successfully decrypted
  - Used key ID is changed
  - Connection is lost
  - Station disappeared from the beacon monitor for over 5 seconds (Ad-Hoc only)

8.3.1.6 HS_ACT_REQ

This event has an EventID = 0x0047. This event is for enhanced power management based on the PM document v2.6. This event is generated in Full Power mode to notify the host that firmware is ready to activate enhanced HS (host sleep) mode. This needs to be defined for new enhanced host sleep mode.

8.3.1.7 MEF_HOST_WAKEUP

This event has an EventID = 0x004F. This event is triggered to wakeup the host when MEF condition is matched.

8.3.1.8 CHANNEL_SWITCH_ANNOUNCEMENT

This event is used to report the channel switch with new event content included when the channel switch announcement action frame or channel switch, that is, seen in the beacon frame.

Field Name	Type	Description
EventID	UINT16	CHANNEL_SWITCH_ANNOUNCEMENT = 0x0050
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
SwitchMode	UINT8	Switch mode 0x00 = does not impose any requirement on the receiving STA 0x01 = STA in a BSS to which the frame contains the element is addressed shall transmit no further frames within the BSS until the scheduled channel switch A STA in an IBSS may treat a Channel Switch Mode field to as advisory.
RegClass	UINT8	Not used for 802.11h (set to 0)
ChannelNumber	UINT8	New channel number which the STA is moving
SwitchCount	UINT8	Channel switch count number

When this event is received, the host can have the following actions:

- Send deauth command with broadcast destination address to disconnect silently. Firmware does not send deauth frame to AP in this case.
- Send deauth command with AP's MAC address to firmware. Firmware sends deauth frame to AP to disconnect.
- Stop traffic and wait for link lost.

After that host can scan and then associate to AP.

8.3.1.9 TDLS_GENERIC

This event is triggered in firmware when it detects the TDLS link does not exist and informs the driver to perform a teardown.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>EventID</td><td>UINT16</td><td>TDLS_GENERIC = 0x0052</td></tr>
<tr><td>BssNum</td><td>UINT8</td><td>BSS number</td></tr>
<tr><td>BssType</td><td>UINT8</td><td>BSS type</td></tr>
<tr><td>Data</td><td>UINT8[]</td><td>Event argument
Data[1:0]: Event type
<table>
<tr><th>Value</th><th>Description</th></tr>
<tr><td>1</td><td>Indicates failure of TDLS setup
Data[9:8]: reason
0x1 = internal error
0x7 = timeout waiting for response
0x8 = setup RSP status non-0
Data[7:2]: MAC address</td></tr>
<tr><td>2</td><td>TDLS setup request received from peer
Data[7:2]: MAC address
Action required in receiving this event:
Driver should make sure that no packets destined for the MAC address given in this event are given to firmware after receiving this event and before “TDLS Link Established” event is received for this address.</td></tr>
<tr><td>3</td><td>TDLS link torndown
Data[9:8]: reason code from peer
0x03 = TDLS peer station leaving BSS
0x19 = TDLS peer station is unreachable on direct link
0x1A = unspecified reason
0xD3 = setup received from TDLS peer
0xD4 = clearing TDLS connections because of deauthentication/ disassociation
Data[7:2]: MAC address</td></tr>
<tr><td>4</td><td>TDLS link established
Data[37:10]: HT Cap IE received from peer
Data[9:8]: length of HT Cap IE
Data[7:2]: MAC address</td></tr>
<tr><td>5</td><td>TDLS debug print
Data[24] onwards: actual format string with null termination
Data[23:22]: length of format string with null
Data[21:18]: third 32-bit argument
Data[17:14]: second 32-bit argument
Data[13:10]: first 32-bit argument
Data[9:2]: timestamp</td></tr>
<tr><td>6</td><td>TDLS raw event
Data[n:4]: raw packet data
Data[3:2]: length of raw packet data</td></tr>
<tr><td colspan="2">continue...</td></tr>
</table></td></tr>
</table>

Field Name	Type	Description
		continued...
		Value 7: TDLS channel switch result Data[10]: reason code if status is non-0 01 = timeout for channel switch response 10 = rejected by peer 11 = peer not available on off-channel Data[9]: status 0 = success 1 = fail Data[8]: current channel 0 = base channel 1 = off-channel Data[7:2]: MAC address
		Value 8: TDLS channel switch initiated Data[7:2]: MAC address
		Value 9: TDLS channel switch stopped Data[8]: reason Data[7:2]: MAC address This event is given to host by firmware when channel switch is stopped internally by firmware. It is NOT given in response to TDLS_STOP_CHANNEL_SWITCH command.

8.3.1.10 TX_DATA_PAUSE

This event is sent by firmware to the driver when firmware determines the Tx pause state for a client has changed. The pause event can have several MrvlIETypes_Tx_Data_Pause_t TLVs.

Field Name	Type	Description
EventID	UINT16	TX_DATA_PAUSE = 0x0055
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Tx_Data_Pause	MrvlIETypes_Tx_Data_Pause_t	Tx Data Pause TLVs

8.3.1.11 SCAN_REPORT_BY_EVENT

This event is used to scan results for CMD_802_11_SCAN_EXT.

Field Name	Type	Description
EventID	UINT16	SCAN_REPORT_BY_EVENT = 0x0058
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
More_Event	UINT8	0x00 = no more event 0x01 = more events
Reserved	UINT8[3]	Reserved (set to 0)
BufSize	UINT16	Size of BSS Description data
NumOfSet	UINT8	Number of BSS scan response returned
Bss_ScanRsp[0]	Bss_Scan_Rsp_t	Variable number of other TLV IDs associated with the BSS
...	...	...
BssScanRsp[NumOfSet-1]	Bss_Scan_Rsp_t	Variable number of other TLV IDs associated with the BSS

Table 73: Bss_Scan_Rsp_t

Field Name	Type	Description
BssContent	MrvlIETypes_Bss_Scan_Rsp_t	BSS content
BssInfo	MrvlIETypes_Bss_Scan_Info_t	BSS information
SrcMacAddr	MrvlIETypes_ApMacAddr_t	Sender MAC address (optional)

8.3.1.12 MULTI_CHAN_INFO

This event is triggered whenever any interfaces becomes active or inactive. Event payload is MrvlIETypes_Multi_Chan_Info TLVs.

Field Name	Type	Description
EventID	UINT16	MULTI_CHAN_INFO = 0x006A
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Multi_Chan_Info	MrvlIETypes_Multi_Chan_Info	Multiple channel information TLVs

8.3.1.13 EVENT_FW_DUMP_INFO

This event is triggered when firmware dumps memory for debug usage.

Field Name	Type	Description
EventID	UINT16	EVENT_FW_DUMP_INFO = 0x0073
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
SeqNum	UINT16	Sequence number of firmware memory dump
Reserved	UINT16	Reserved
Type	UINT16	Type of firmware memory dump in this event
Len	UINT16	Length of memory dump in this event

8.3.1.14 EVT_TX_STATUS_REPORT

This event is triggered by firmware to return tx_status_report for the requested host Tx management frame from driver.

Field Name	Type	Description
EventID	UINT16	EVT_TX_STATUS_REPORT = 0x0074
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
PacketType	UINT8	Same as requested Tx packet type
Tx_token_id	UINT8	Same as requested Tx packet tx_token_id field
status	UINT8	Status 0x00 = success 0x01 = failed 0x02 = watchdog timeout

8.3.1.15 EVENT_CLASS_NAN

This event is used to send all types of NAN events to the host.

It is a single event sent by firmware to the host for the following 3 sub events handled:

- SUBTYPE_SD_EVENT – Allow the host/application to know about the received Service Discovery Frame (SDF)
- SUBTYPE_NAN_STARTED – Open a NAN interface at the host
- SUBTYPE_SDF_TX_DONE – Make the application send the next SDF

Field Name	Type	Description
EventID	UINT16	EVENT_CLASS_NAN = 0x0075
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
subType	UINT16	Sub type 0x0000 = SUBTYPE_SD_EVENT 0x0001 = SUBTYPE_NAN_STARTED 0x0002 = SUBTYPE_SDF_TX_DONE
PeerMacAddr	UINT8[6]	MAC address of the peer from whom SD packet is received
RSSI	UINT32	Received RSSI of a SDF
SDPacket	UINT8[]	SD packet received from the peer

8.3.1.16 EVT_BTCOEX_WLAN_PARA_CHANGE

This event is sent to the driver in 2 cases:

- GO/MMH+SCO
  Send event with MRVL_BTCOEX_WLAGGER_WINSIZE_TLV_ID to driver in MMH+SCO Coexistence case to reduce Rx WinSize. This will reduce the aggregation size for incoming frames. Send event again on exit MMH+SCO Coexistence case, to restore default Rx Win.
- GC+OPP_TX
  Send event with MRVL_BTCOEX_WLSCANTIME_TLV_ID to driver in STA/GC+ACL Coexistence case to increase WL Scan time. Send event again on exit STA/GC+ACL Coexistence case, to restore default WL Scan times.

Field Name	Type	Description
EventID	UINT16	EVT_BTCOEX_WLAN_PARA_CHANGE = 0x0076
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ApBTCoexScoWinSize	MrvlIETypes_ApBTCoexScoWinSize_t	Optional
BtCoexScanTime	MrvlIETypes_BTCoexScanTime_t	Optional

8.3.1.17 EV_SMC_GENERIC

This event is a main/master event generated by firmware while Smart Mode configuration is in progress. There are multiple sub-events that can be generated with this event. SubType differentiates about these sub-events. All the sub-events will follow current channel configuration after its sub type.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>EventID</td><td>UINT16</td><td>EV_SMC_GENERIC = 0x0077</td></tr>
<tr><td>BssNum</td><td>UINT8</td><td>BSS number</td></tr>
<tr><td>BssType</td><td>UINT8</td><td>BSS type</td></tr>
<tr><td>SubType</td><td>UINT16</td><td>Sub-event type
<table>
<tr><th>Type</th><th>Description</th></tr>
<tr><td>0x0000</td><td>EV_SUBTYPE_CHANNEL_SWITCHED
This is generated every time new channel is switched. Disabled by default.
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>ChanInfo</td><td>ChanInfo_t</td><td>Current channel configuration</td></tr>
</table></td></tr>
<tr><td>0x0001</td><td>EV_SUBTYPE_FIRST_PACKET
This is generated on receiving of first relevant packet. For example, if multicast address range is defined then first multicast packet which is from this multicast address range would generate this event.
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>ChanInfo</td><td>ChanInfo_t</td><td>Current channel information</td></tr>
</table></td></tr>
<tr><td>0x0002</td><td>EV_SUBTYPE_END_OF_MAX_DWELL
This is generated when current channel was dwelling in MAX_SCAN_TIME and is about to end its dwell by switching to next channel in the list.
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>ChanInfo</td><td>ChanInfo_t</td><td>Current channel information</td></tr>
</table></td></tr>
</table></td></tr>
</table>

Table 74: ChanInfo_t

Field Name	Type	Description
BandConfigType	UINT8	Band configuration type Bits[7:6]: reserved (set to 0) Bits[5:4]: secondary channel offset 00 = no secondary channel 01 = secondary channel above 10 = reserved 11 = secondary channel below Bits[3:2]: channel width 00 = 20 MHz 01 = 10 MHz 10 = 40 MHz 11 = 80 MHz Bits[1:0]: band information 00 = 2.4 GHz 01 = 5 GHz 10 = 4 GHz 11 = reserved
ChanNumber	UINT8	Channel number for specific band
ScanType	UINT8	Scan type bitmap Bits[7:2]: reserved (set to 0) Bit[1]: disable channel filter Bit[0]: passive scan
MinScanTime	UINT16	Minimum scan time (ms)
MaxScanTime	UINT16	Maximum scan time (ms)

8.3.1.18 EV_NEIGHBOR_REPORT

This event is triggered when driver issues CMD_GET_NEIGHBORLIST command, firmware sends Neighbor report request to the associated AP. It then sends the received Neighbor report response as an event to the driver.

Field Name	Type	Description
EventID	UINT16	EV_NEIGHBOR_REPORT = 0x0079
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Length	UINT16	Length of reportElem in bytes
reportElem	NeighborRprtTlv_t	Report Neighbor TLV

8.3.1.19 EV_BSS_TRANSITION_REQUEST

This event is used for 802.11v, voice enterprise certification.

Field Name	Type	Description
EventID	UINT16	EV_BSS_TRANSITION_REQUEST = 0x007A
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Length	UINT16	Length of BssTransReqElem in bytes
BssTransReqElem	MRVL_BSS_TRANSITION_REQ_TLV_ID	BSS transition request elemet

8.3.1.20 EVT_CHAN_REG_INFO

This event is generated by firmware when the channel attributes and/or the region information in firmware has been changed. It will specify the cause of the event, along with a variable number of TLVs reporting the changed information to the host.

Field Name	Type	Description
EventID	UINT16	EVT_CHAN_REG_INFO = 0x0082
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
cause	UINT8	Value denoting cause of event
bufSize	UINT16	Size of TLV buffer
tlvBuffer	UINT8[]	TLV buffer Array containing the first byte of the TLV buffer. May contain 1 or more of the following TLVs: • MrvlIETypes_CHANATTRCFG_t • MrvlIETypes_REGIONINFO_t

8.3.1.21 EVT_SSU_DUMP_DMA

This event is sent to the driver by firmware when firmware detects a SSU DMA complete or SSU DMA error.

Field Name	Type	Description
EventID	UINT16	EVT_SSU_DUMP_DMA = 0x008C
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
ssuHeader	UINT8[256]	SSU header of 256 bytes

8.3.2 802.11n Events

8.3.2.1 DOT11N_ADDBA

This event is sent by firmware to driver upon the receipt of ADDBA request action frame.

Field Name	Type	Description
EventID	UINT16	DOT11N_ADDBA = 0x0033
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
CommandResult	UINT8	Not used (set to 0)
PeerMACAddr	UINT8[6]	Peer MAC address
DialogToken	UINT8	To distinguish between successive ADDBA requests due to retries
BlockAckParamSet	UINT16	Block Ack parameter set Bits[15:6]: buffer size Bits[5:2]: TID Bit[1]: Block Ack policy 0 = delayed Block Ack policy 1 = immediate Block Ack policy Bit[0]: AMSDU supported 0 = No AMSDU supported under this Block Ack agreement 1 = AMSDU supported under this Block Ack agreement
BlockAckTmo	UINT16	Block Ack timeout value
SSN	UINT16	Start sequence number Bits[15:4]: starting sequence number Bits[3:0]: reserved (set to 0)

8.3.2.2 DOT11N_DELBA

This event is sent by firmware to driver upon receipt of DELBA request action frame.

Field Name	Type	Description
EventID	UINT16	DOT11N_DELBA = 0x0034
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
CommandResult	UINT8	Not used (set to 0)
PeerMACAddr	UINT8[6]	Peer MAC address
DelParamSet	UINT16	Bits[15:12]: TID Bit[11]: initiator 0 = recipient 1 = originator Bits[10:0]: reserved (set to 0)
ReasonCode	UINT16	Reason code sent for DELBA request

8.3.2.3 BA_STREAM_TIMEOUT

This event is used to Block Ack inactivity timeout.

Field Name	Type	Description
EventID	UINT16	BA_STREAM_TIMEOUT = 0x0037
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
TID	UINT8	Transaction ID
PeerMACAddr	UINT8[6]	Peer MAC address
Originator	UINT8	Originator 0x00 = AP 0x01 = UUT

8.3.2.4 AMSDU_AGGR_CTRL

This event is used to report the aggregation size available for host.

Field Name	Type	Description
EventID	UINT16	AMSDU_AGGR_CTRL = 0x0042
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
AmsduLastReportAggrSize	UINT16	Report AMSDU aggregation size

8.3.2.5 DOT11N_BWCHANGE_EVENT

This event is reported when the associated AP switches the channel bandwidth among 20/40/80 MHz and 20 MHz in 802.11n mode. Firmware will handle the channel re-configuration. No action is required in the host driver.

Field Name	Type	Description
EventID	UINT16	DOT11N_BWCHANGE_EVENT = 0x0048
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
CurrentBandwidth	UINT8	Current bandwidth 0x00 = bandwidth changed to 20 MHz 0x01 = bandwidth changed to 40 MHz 0x02 = bandwidth changed to 80 MHz

8.3.2.6 EV_RXBA_SYNC

This event is triggered by firmware to synchronize the Rx Block Acknowledgement (BA) window bitmap and to fill any holes that may have been created because of firmware filtering and dropping any packets received on the Rx BA link. Such filtering can take place for numerous reasons such as received packet not matching any host wakeup conditions and other filters when host is sleeping or MEF filters or firmware receiving some packets which are unconditionally filtered regardless of host-sleep status (for example, ARP SA = DA packet).

If host is asleep and the Rx BA bitmap is not empty, firmware will forcefully wake up host and send this event only when absolutely necessary. Otherwise, send this event to synchronize only when host needs to be woken up for sending some other event or data packet or host wakes up on its own. Exceptional cases where host is forcefully woken up if required are:

- Reception of packet which is out of current window or reception of BAR frame
- Rx BA is getting deleted by host or peer

There will be multiple events triggered in case the Rx BAs (which need to “sync” with host) are in use across different interfaces. Each event will group the Rx BA on that interface.

Field Name	Type	Description
EventID	UINT16	EV_RXBA_SYNC = 0x0059
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
TLVs	TLVs	Payload will consist of 1 or more TLVs. Currently, the following TLV is defined for this event: MrvlIETypes_RxBASync_t (1 TLV for each Rx BA that is setup on this interface and which needs Rx BA synchronization with host)

8.3.2.7 REMAIN_ON_CHANNEL_EXPIRED

This event is used to inform the host about CMD_802_11_REMAIN_ON_CHANNEL command expired.

Field Name	Type	Description
EventID	UINT16	REMAIN_ON_CHANNEL_EXPIRED = 0x005F
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Reason	UINT16	Reason 0x0000 = operation completed 0x0001 = operation cancel Others = reserved

8.4 Wi-Fi Direct Events

Table 75: Wi-Fi Direct Events

Event	Description	Page
Section 8.4.1, WIFI_DIRECT_EVENT	Generated by firmware to send data received from peer to driver and application	page 296
Section 8.4.2, WIFI_DIRECT_SERVICE_DISCOVERY_EVENT	Passes received GAS service discovery packet to host	page 297

8.4.1 WIFI_DIRECT_EVENT

This event is generated by firmware to send the data (P2P IE and WPS IE) received from the peer to the driver and application.

Field Name	Type	Description
Length	UINT16	Number of bytes in the event data including this field
Wi-Fi Direct Event Type	UINT16	0x0000 = WIFI_DIRECT_NEGOTIATION_REQUEST_EVENT 0x0001 = WIFI_DIRECT_NEGOTIATION_RESPONSE_EVENT 0x0002 = WIFI_DIRECT_NEGOTIATION_RESULT_EVENT 0x0003 = WIFI_DIRECT_INVITATION_REQUEST_EVENT 0x0004 = WIFI_DIRECT_INVITATION_RESPONSE_EVENT 0x0005 = WIFI_DIRECT_DEVICE_DISCOVERABILITY_REQUEST_EVENT 0x0006 = WIFI_DIRECT_DEVICE_DISCOVERABILITY_RESPONSE_EVENT 0x0007 = WIFI_DIRECT_PROVISION_DISCOVERY_REQUEST_EVENT 0x0008 = WIFI_DIRECT_PROVISION_DISCOVERY_RESPONSE_EVENT 0x0009 = WIFI_DIRECT_NEGOTIATION_RESPONSE_TX_EVENT 0x000A = WIFI_DIRECT_NEGOTIATION_CFRM_TX_EVENT 0x000E = WIFI_DIRECT_PEER_DETECTED_EVENT 0x000F = WIFI_DIRECT_CLIENT_ASSOCIATED_EVENT 0x0010 = WIFI_DIRECT_PRESENCE_REQUEST_EVENT 0x0011 = WIFI_DIRECT_PRESENCE_RESPONSE_EVENT

Field Name	Type	Description

Wi-Fi Direct Event Subtype	UINT16	0x0000 = ignore (by default, except for the following EventType)
		WIFI_DIRECT_NEGOTIATION_RESULT_EVENT
		    0x0 = no role
		    0x1 = GO role
		    0x2 = client role
		    Others = reserved
		WIFI_DIRECT_INVITATION_REQUEST_EVENT
		WIFI_DIRECT_INVITATION_RESPONSE_EVENT
		    Device Role Flag (2 bits):
		        0x0 = no role
		        0x1 = GO role
		        0x2 = client role
		    Packet Process State (3 bits):
		        0x0 = none
		        0x1 = processing
		        0x2 = insufficient information, dropped
		        0x3 = success
		        0x4 = fail
		    Invitation Flag (1 bit)
		    Reserved (10 bits)
		WIFI_DIRECT_PROVISION_DISCOVERY_REQUEST_EVENT
		WIFI_DIRECT_PROVISION_DISCOVERY_RESPONSE_EVENT
		    0x0000 = no configuration method selected
		    0x0008 = display
		    0x0080 = push button
		    0x0100 = keypad

PeerMacAddr	UINT8[6]	Peer MAC address
Index	UINT32	Index of entry in persistent group table This field is present only when EventType is WIFI_DIRECT_INVITATION_REQUEST_EVENT or WIFI_DIRECT_INVITATION_RESPONSE_EVENT.
IEList	TLV	IE list Peer P2P IE, peer WPS IE, DS channel IE, peer Wi-Fi display IE, and so on.

WIFI_DIRECT_PEER_DETECTED_EVENT – This event is generated in the device discovery phase when a Probe Request or Probe Response with the remote device's P2P IE is received. The GO will generate this event when a Probe Request with P2P IE is received and responded with Probe Response. The P2P Client will generate this event when the Probe Response frame with P2P IE is received.

Some of these events are used to trigger application level responses.

Example: Event of type WIFI_DIRECT_NEGOTIATION_RESULT_EVENT with subtype GO ROLE will trigger the WPS Registrar application.

8.4.2 WIFI_DIRECT_SERVICE_DISCOVERY_EVENT

This event passes received GAS packet service discovery packet to the host.

Field Name	Type	Description
Length	UINT16	Number of bytes in the event data including this field
PeerMacAddr	UINT8[6]	MAC address of peer from whom SD packet is received
SDPacket	UINT8[]	Service discovery packet received from peer

8.5 VDLL Event

Table 76: VDLL Event

Event	Description	Page
Section 8.5.1, EVT_VDLL_IND	Sent to host by firmware in 2 cases: • Case 1 – Inform driver the VDLL image offset in firmware during initialization • Case 2 – Request VDLL block at any time	page 298

8.5.1 EVT_VDLL_IND

This event is sent to the host by firmware in 2 cases:

- Case 1: Inform the driver of the offset of the VDLL image in firmware image during firmware initialization

  In this case, the driver should save the VDLL image starting from the specified offset in a local memory after receiving the this event with *Type* = 1.

- Case 2: Request a VDLL block at any time

  In this case, the driver should download a VDLL block starting from the specified offset and with the specified *BlockLength* based on the saved VDLL image, using the CMD_VDLL command.

Field Name	Type	Description
EventID	UINT16	EVT_VDLL_IND = 0x0081
BssNum	UINT8	BSS number
BssType	UINT8	BSS type
Type	UINT16	Type 0x0000 = VDLL block request 0x0001 = VDLL offset
Rsved	UINT16	Reserved
Offset	UINT32	VDLL block start position If Type = 0, specify the offset of VDLL block in VDLL image. If Type = 1, specify the length of VDLL image in firmware image.
BlockLength	UINT16	VDLL block length Not used if Type = 1.

9 WPA, QoS, and RSSI Support

9.1 WPA Support

WPA is supported by the following:

- WPA related commands
- Tx descriptor for each packet contains bit fields to specify the per packet encryption method

9.1.1 WPA-PSK Setup Procedure

The procedure for setting up WPA-PSK is:

1. Set firmware to Infrastructure mode.
2. Set firmware to open authentication.
3. Enable RSN in the firmware.
4. Send Association command to firmware.
5. Supplicant sends PWK to driver and the driver forwards the PWK to the firmware.
6. Supplicant sends GWK to driver and the driver forwards the GWK to firmware.

See Section 6.9, Security, on page 95 for additional information.

9.2 QoS Support

Quality of Service (QoS) support is based on the WMM subset of the 802.11e specification. WMM features are supported by the following:

- WMM related commands (see Section 6.15, Wi-Fi Multimedia, on page 144)
- Per packet control provided by the Tx Descriptors (see Section 4, Data Path, on page 8)

9.3 RSSI and Noise Floor Support

The firmware supports the reporting of RSSI and Noise Floor values:

- RSSI can be derived by adding Noise Floor to SNR value
- Noise Floor is returned in each received packet through the NF field in the Rx Descriptor (dBm).
- Signal Quality is determined by RSSI and SNR.

See Section 6.5, Status Information, on page 53 for additional information.

10 TLV Usage

This section describes the functions of the TLV in API commands.

10.1 TLV Format

All arrays (with exception of MAC Address) in the APIs follow the NXP Information Element Parameter Set as defined in Table 77 and Table 78.

Table 77 shows the structure of a standard IEEE IE type. NXP IE types (see Table 78) that are translations of IEEE types are exact duplicates of the IEEE type (except for the 16-bit extension of the Type and Length fields).

Table 77: IEEE IE Type Format

Field Name	Type	Description
Type	UINT8	Type ID
Length	UINT8	Length of payload
Payload	UINT8[]	Data

Table 78: MrvlIEType Format

Field Name	Type	Description
Type	UINT16	Type ID
Length	UINT16	Length of payload
Payload	UINT8[]	Data

NXP IE Parameter Set is defined with a Type member of UINT16 because the lower byte value is reserved for IEEE 802.11 standard IE type definitions. The reserved value greater than 0xFF is for NXP Specific IE types.

NXP IE Parameter Set is defined with a Length member of UINT16 (no payload limit at 255 bytes of data) because there is a need for large payload.

10.2 TLV Defined Types

Some TLVs in commands are required only for special extensions to the API. Cases where the functionality of the TLV is not required, the TLV can be omitted from the command. TLVs are not required to be in a specific order in the command structure. TLVs are always contiguous in memory and do not include pad bytes unless expressly indicated in the TLV structure itself. In APIs where non-TLV arguments are specified, TLVs are appended directly after the fixed field arguments without any pad bytes.

Table 79: IEEE 802.11 Standard IE Translated to NXP IE

TLV Type (UINT16)	Description
0x0000	MrvlIETypes_SSIDParamSet_t
0x0001	MrvlIETypes_RatesParamSet_t
0x0002	MrvlIETypes_PhyParamFHSet_t (Frequency Hopping not supported)
0x0003	MrvlIETypes_PhyParamDSSet_t
0x0004	MrvlIETypes_CfParamSet_t
0x0006	MrvlIETypes_IbssParamSet_t
0x0007	MrvlIETypes_DomainParam_t
0x0020	MrvlIETypes_LocalPowerConstraint_t
0x0021	MrvlIETypes_LocalPowerCapabiity_t
0x0024	MrvlIETypes_SupportedChannels_t
0x0028	MrvlIETypes_Quiet_t
0x0029	MrvlIETypes_Ibss_Dfs_t
0x0030	MrvlIETypes_RsnParamSet_t
0x0048	MrvlIETypes_2040Coex_t
0x0049	MrvlIETypes_2040BSSIntolerantChannelReport_t
0x00DD	MrvlIETypes_VendorParamSet_t
0x00FF	MrvlIETypes_IEEE_Extension_t

Note: These IEEE 802.11 standard IEs, when extended as NXP IEs, the Type and Length fields are expanded per Section 10.1, TLV Format, on page 300.

Table 80: NXP Proprietary IE

TLV Type (UINT16)	Description
0x0100	MrvlIETypes_KeyParamSet_t
0x0101	MrvlIETypes_ChanListParamSet_t
0x0102	MrvlIETypes_NumProbes_t
0x0103	Reserved
0x0104	MrvlIETypes_BeaconLowRssiThreshold_t
0x0105	MrvlIETypes_BeaconLowSnrThreshold_t
0x0106	MrvlIETypes_FailureCount_t
0x0107	MrvlIETypes_BeaconsMissed_t
0x0108	MrvlIETypes_LedGpio_t

Table 80: NXP Proprietary IE (Continued)

TLV Type (UINT16)	Description
0x0109	MrvlIETypes_LedBehavior_t
0x010A	MrvlIETypes_Passthrough_t
0x010B to 0x010F	Reserved
0x0110	MrvlIETypes_WmmQStatus_t
0x0111	MrvlIETypes_Crypto_Data_t
0x0112	MrvlIETypes_WildCardSSIDParamSet_t
0x0113	MrvlIETypes_TsfTimestamp_t
0x0114	MrvlIETypes_PowerAdapt_Group_t
0x0115	MrvlIETypes_HostSleepFilterType1
0x0116	MrvlIETypes_BeaconHighRssiThreshold_t
0x0117	MrvlIETypes_BeaconHighSnrThreshold_t
0x0118	MrvlIETypes_AutoTxMacFrame_t
0x011B	MrvlIETypes_WPSEnrolleeProbeReqIE_t
0x011C	MrvlIETypes_WPSRegistrarBeaconIE_t
0x011D	MrvlIETypes_WPSRegistrarProbeRespIE_t
0x011E	Reserved
0x011F	MrvlIETypes_AuthType_t
0x0120	MrvlIETypes_StaMacAddr_t
0x0121 to 0x0122	Reserved
0x0123	MrvlIETypes_BSSIDList_t
0x0124	MrvlIETypes_LinkQualityThreshold_t
0x0125	Reserved
0x0126	MrvlIETypes_DataLowRssiThreshold_t
0x0127	MrvlIETypes_DataLowSnrThreshold_t
0x0128	MrvlIETypes_DataHighRssiThreshold_t
0x0129	MrvlIETypes_DataHighSnrThreshold_t
0x012A	MrvlIETypes_ChanBandList_t
0x012B to 0x013F	Reserved
0x0140	MrvlIETypes_EncryptionProtocol_t
0x0141	Reserved
0x0142	MrvlIETypes_Cipher_t
0x0143	Reserved
0x0144	MrvlIETypes_PMK_t
0x0145	MrvlIETypes_Passphrase_t
0x0146 to 0x0148	Reserved
0x0149	MrvlIETypes_PreBeaconMissed_t
0x014A to 0x0150	Reserved

Table 80: NXP Proprietary IE (Continued)

TLV Type (UINT16)	Description
0x0151	MrvlRateDropPattern_t
0x0152	Reserved
0x0153	MrvlIE_RateScope_t
0x0154	MrvlIETypes_Power_Group_t
0x0155	Reserved
0x0156	MrvlIETypes_Bss_Scan_Rsp_t
0x0157	MrvlIETypes_Bss_Scan_Info_t
0x0158 to 0x015D	Reserved
0x015E	MrvlIETypes_WAPI_IE_t
0x015F	Reserved
0x0160	MrvlIETypes_RobustCoex_t
0x0161	MrvlIETypes_RobustCoex_Period_t
0x0162 to 0x0166	Reserved
0x0167	MrvlIETypes_WapiInfoSet_t
0x0168	MrvlIETypes_MgmtFrameSet_t
0x0169	MrvlIETypes_MgmtIEList_t
0x016A	MrvlIETypes_ApSleepParams_t
0x016B	MrvlIETypes_ApSleepParamsInact_t
0x016C	MrvlIETypes_ApBTCoexCommonConfig_t
0x016D	MrvlIETypes_ApBTCoexScoConfig_t
0x016E	MrvlIETypes_ApBTCoexAclConfig_t
0x016F	MrvlIETypes_ApBTCoexStats_t
0x0170	Reserved
0x0171	MrvlIETypes_Auto_DS_Param_t
0x0172	MrvlIETypes_Enh_Sta_PS_t
0x0173 to 0x0188	Reserved
0x0189	MrvlIETypes_ChanTRPCConfig_t
0x018A to 0x0198	Reserved
0x0199	MrvlIETypes_RxBASync_t
0x019A	MrvlCoalesceRule_t
0x019B	Reserved
0x019C	MrvlIETypes_KeyParamSet_v2_t
0x019D to 0x01B6	Reserved
0x01B7	MrvlIETypes_Multi_Chan_Info
0x01B8	MrvlIETypes_Chan_Group_Info
0x01B9 to 0x01C4	Reserved
0x01C5	MRVL_SCAN_CHAN_GAP_TLV_ID

Table 80: NXP Proprietary IE (Continued)

TLV Type (UINT16)	Description
0x01C6	MrvlIEChannelStats_t
0x01C7 to 0x1C9	Reserved
0x01CA	MrvlIETypes_ApBTCoexScoWinSize_t
0x01CB	MrvlIETypes_BTCoexScanTime_t
0x01CC	MrvlIETypes_SMCAddrRange_t
0x01CD	Reserved
0x01CE	MrvlIEBSSScanType_t
0x01CF	MrvlIETypes_Eapol_Pkt_t
0x01D0	IEEEtypes_WMM_ParamElement_t
0x01D1	MrvlIETypes_SMCFrameFilter_t
0x01D2	NAN_MASTER_OPERATION_MODE_TLV_ID
0x01D3	NAN_DISC_BEACON_PERIOD_TLV_ID
0x01D4	NAN_SCAN_DWELL_TIME_TLV_ID
0x01D5	NAN_MASTER_PREFERENCE_TLV_ID
0x01D6	NAN_SDFRAME_TRS_TLV_ID
0x01D7	MrvlIETypes_NAN_FAM_ATTRIBUTE_ID_TLV_ID
0x01D8	NAN_OPERATING_CHANNEL_TLV_ID
0x01D9	NAN_WARMUP_TIME_TLV_ID
0x01DA	Reserved
0x01DB	MrvlIETypes_NAN_UNALIGNED_SCHED_t
0x01DC	Reserved
0x01DD	MrvlIETypes_NAN_PMF_AND_SECURE_DATA_PATH_t
0x01DE	NeighborRprtTlv_t
0x01DF	MRVL_BSS_TRANSITION_REQ_TLV_ID
0x01E0	MrvlIETypes_BridgeParamSet_t
0x01E1 to 0x1E4	Reserved
0x01E5	MrvlIETypes_RSSI_NF_t
0x01E6	MrvlIETypes_IPv6AddrParamSet_t
0x1E7 to 0x1E8	Reserved
0x01E9	MRVL_WIFI_PROBERSP_ONLY_TLV_ID
0x01EA to 0x1EC	Reserved
0x01ED	MrvlIETypes_CHANATTRCFG_t
0x01EE	MrvlIETypes_REGIONINFO_t
0x01EF to 0x01F1	Reserved
0x01F2	MrvlIETypes_FAST_LINK_INTERFACE_BASIC_INFO_t
0x01F3	MrvlIETypes_FAST_LINK_EXT_STATION_INFO_t
0x01F4	MrvlIETypes_FAST_LINK_DATA_PORT_INFO_t

Table 80: NXP Proprietary IE (Continued)

TLV Type (UINT16)	Description
0x01F5 to 0x0200	Reserved
0x0201	MrvlIETypes_EMBEDDED_DHCPD_CFG_t
0x0202 to 0x0205	Reserved
0x0206	MrvlIETypes_POWERTABLE_t
0x0207	MrvlIETypes_MULTI_CHAN_TIME_SLICE_t
0x0208 to 0x020D	Reserved
0x020E	MrvlIETypes_FTM_SESSION_CFG_LCI_t
0x020F	MrvlIETypes_FTM_SESSION_CFG_LOCATION_CIVIC_t
0x0210	MrvlIETypes_FTM_SESSION_CFG_RESPONDER_t
0x0211	MrvlIETypes_FTM_SESSION_CFG_INITIATOR_t
0x0212	MrvlIETypes_11P_MODE_CONFIG_t
0x0213	MrvlIETypes_11P_TDM_CONFIG_t
0x0214	MrvlIETypes_11P_RADIO_COUNTERS_t
0x0215 to 0x0227	Reserved
0x0228	MrvlIETypes_RTT_RESPONDER_INFO_ID_t
0x0229 to 0x238	Reserved
0x0239	MrvlIETypes_UAP_STA_FLAGS_t
0x023A	MrvlIETypes_DMCS_STATUS_t
0x023B	MrvlIETypes_ZERO_DFS_t

Table 81: NXP Micro AP TLVs

TLV Type (UINT16)	Description
0x002D	MrvlIETypes_HT_Capability_t
0x00DD	MrvlIETypes_VendorSpecific_t
0x012B	MrvlIETypes_ApMacAddr_t
0x012C	MrvlIETypes_Beacon_Period_t
0x012D	MrvlIETypes_DTIM_Period_t
0x012F	MrvlIETypes_ApTx_Power_t
0x0130	MrvlIETypes_ApBCast_SSID_Ctrl_t
0x0131	MrvlIETypes_Preamble_Ctrl_t
0x0132	MrvlIETypes_Antenna_Ctrl_t
0x0133	MrvlIETypes_RTS_Threshold_t
0x0135	MrvlIETypes_TxData_Rate_t
0x0136	MrvlIETypes_PKT_Fwd_Ctrl_t
0x0137	MrvlIETypes_STA_Info_t
0x0138	MrvlIETypes_STA_MAC_Filter_t
0x0139	MrvlIETypes_STA_Ageout_t

Table 81: NXP Micro AP TLVs (Continued)

TLV Type (UINT16)	Description
0x013B	MrvlIETypes_WEP_Key_t
0x013C	MrvlIETypes_WPA_Passphrase_t
0x0140	MrvlIETypes_Encryption_Protocol_t
0x0141	MrvlIETypes_AKMP_t
0x0142	MrvlIETypes_Cipher_t
0x0146	MrvlIETypes_Frag_Threshold_t
0x0147	MrvlIETypes_GTK_Rekey_t
0x0155	MrvlIETypes_Max_STA_Counts_t
0x0159	MrvlIETypes_Chanrpt_Chan_Load_t
0x015A	MrvlIETypes_Chanrpt_Noise_Hist_t
0x015B	MrvlIETypes_Chanrpt_11H_Basic_t
0x015D	MrvlIETypes_Retry_Limit_t
0x015E	MrvlIETypes_WAPI_t
0x0162	MrvlIETypes_MCBC_Data_Rate_t
0x0164	MrvlIETypes_RSN_Replay_t
0x0167	MrvlIETypes_WapiInfoSet_t
0x0170	MrvlIETypes_Mgmt_Passthru_Mask_t
0x0175	MrvlIETypes_PTK_Timeout_t
0x0176	MrvlIETypes_PTK_Retry_t
0x0177	MrvlIETypes_GTK_Timeout_t
0x0178	MrvlIETypes_GTK_Retry_t
0x017B	MrvlIETypes_PS_STA_Ageout_t
0x0191	MrvlIETypes_PTK_Cipher_t
0x0192	MrvlIETypes_GTK_Cipher_t
0x0193	MrvlIETypes_ApBSS_Status_t
0x0194	MrvlIETypes_Tx_Data_Pause_t
0x0196	MrvlIETypes_Sticky_Tim_Config_t
0x0197	MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t
0x0198	MrvlIETypes_2040_BSS_Coex_Ctrl_t
0x01C7	MrvlIETypes_Api_Ver_Info_t
0x0217	MrvlIETypes_Max_Supported_STA_Count_t
0x023E	MrvlIETypes_FW_Cap_Info_t

10.3 NXP Extended IEEE IE Formats

NXP Extended IEEE IEs maintain the same data and structure as the standard IEEE IEs, except for the 16-bit Type and Length fields. The additional upper byte in these fields is always set to 0x00 for IEEE replacement IEs.

This section displays examples of the IEEE IEs in cases where non-intuitive functionality in the firmware may be triggered from the IE parameters. See the IEEE documentation for field layout of types not explicitly defined here.

Table 82: NXP Extended IEEE IE Formats

IE Format	Description	Page
Section 10.3.1, MrvlIETypes_DomainParam_t	802.11 domain parameters TLV	page 307
Section 10.3.2, MrvlIETypes_LocalPowerConstraint_t	802.11h local power constraint information	page 308
Section 10.3.3, MrvlIETypes_LocalPowerCapabiity_t	802.11h power capability element	page 308
Section 10.3.4, MrvlIETypes_2040Coex_t	Configures 20/40 MHz BSS coexistence element	page 309
Section 10.3.5, MrvlIETypes_ 2040BSSIntolerantChannelReport_t	Configures BSS intolerant channel report	page 309
Section 10.3.6, MrvlIETypes_HTInfo_t	Retrieves IEEE HT Operation Information Element	page 310

10.3.1 MrvlIETypes_DomainParam_t

This TLV specifies the 802.11 domain parameters. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0007
Length	UINT16	Length of payload
CountryCode	UINT8[3]	Country string
FirstChannel	UINT8	First channel in a sub-band
NumChannels	UINT8	Number of channels in sub-band
MaxTxPower	UINT8	Power for each channel (dBm)
...	...	...
FirstChannel	UINT8	First channel in a sub-band
NumChannels	UINT8	Number of channels in sub-band
MaxTxPower	UINT8	Power for each channel (dBm)

Firmware does not enable channel validity checks on the 802.11d domain information provided in Set Domain command. The driver is responsible for ensuring the channels provided in the Domain Information command are valid.

Firmware always assumes the channels specified in a sub-band triplet are adjacent channels (5 MHz apart). If the AP specifies a different channel spacing in the Country IE, the driver must convert the channel representation in the Country IE to a default 5 MHz spacing, expected by firmware, before issuing the Set Domain command to firmware.

For instance, if the AP's Country IE contains a channel triplet (36, 4, 20) in the 5 GHz band, firmware interprets this as channels 36, 37, 38, and 39 with 20 dBm maximum power even though channels 37, 38, and 39 do not exist. To correct this problem, the driver must convert this to 4 triplets:

- 36, 1, 20
- 40, 1, 20
- 44, 1, 20
- 48, 1, 20

10.3.2 MrvlIETypes_LocalPowerConstraint_t

This TLV specifies the 802.11h local power constraint. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0020
Length	UINT16	Length of TLV excluding Type and Length fields
Reserved	UINT8	Not used
Power Constraint	UINT8	Local power constraint (dBm) Power constraint value received in beacon/probe response frames or a user specified/hard coded value when starting a BSS in Ad-Hoc mode.

10.3.3 MrvlIETypes_LocalPowerCapabiity_t

This TLV specifies the 802.11h power capability element. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0021
Length	UINT16	Length of TLV excluding Type and Length fields
Minimum Power	SINT8	Signed value indicating minimum power the station is capable of transmitting (dBm)
Maximum Power	SINT8	Signed value indicating maximum power of the station (dBm) If the driver specifies a value that exceeds the regulatory domain maximum minus any local channel constraint, then the firmware limits the power. If the value specified is less than the regulatory maximum minus any local channel constraint, the value specified by driver is used as is.

A zero length local power constraint (or power capability) element removes any previously set local power constraint (or power capability) in firmware.

10.3.4 MrvlIETypes_2040Coex_t

This TLV configures the 20/40 MHz BSS coexistence element. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0048
Length	UINT16	Length of payload (excluding Type and Length fields)
CoexElement	UINT8	20/40 MHz coexistence value Bits[7:5]: reserved (set to 0) Bit[4]: OBSS scanning grant (not for STA) Bit[3]: OBSS scanning exemption request (not used) Bit[2]: 20 MHz BSS width request (this bit is set when a 40 MHz intolerant AP is detected as a part of OBSS scan) Bit[1]: 40 MHz intolerant (this bit is set when STA = 40 MHz intolerant) Bit[0]: information request (not valid for STA)

10.3.5 MrvlIETypes_2040BSSIntolerantChannelReport_t

This TLV configures the 20/40 MHz BSS intolerant channel report. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0049
Length	UINT16	Length of payload (excluding Type and Length fields)
RegulatoryDomain	UINT8	Regulatory domain under which reported channels fall in
ChannelList	UINT8[]	Channel number for legacy AP Variable size, each octet contains 1 channel number for legacy AP.

10.3.6 MrvlIETypes_HTInfo_t

This TLV is used with command APCMD_SYS_CONFIGURE to retrieve IEEE HT Operation Information Element that is currently being used by the uAP. Only GET operation is allowed on this TLV; SET operation is not supported. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0169
Length	UINT16	Length of payload (excluding Type and Length fields)
PrimaryChan	UINT8	Channel number of the primary channel
HTOperationInfo	UINT8[5]	HT operation information Bits[39:36]: reserved Bits[35]: PCO phase 0 = switch to or continue 20 MHz phase 1 = switch or continue 40 MHz phase Bit[35]: PCO active 0 = PCO is not active in BSS 1 = PCO is active in BSS Bit[33]: L-SIG TxOP protection full support Bit[32]: STBC beacon 0 = primary beacon 1 = STBC beacon Bit[31]: dual CTS protection 0 = dual CTS protection is not required 1 = dual CTS protection is required Bit[30]: dual beacon 0 = no STBC beacon is transmitted 1 = STBC beacon is transmitted by the AP Bits[29:23]: reserved Bits[12]: OBSS non-HT STAs present 0 = 1 or more non-HT OBSSs present 1 = otherwise Bit[11]: reserved Bit[10]: Non-Greenfield HT STAs present 0 = all associated HT STAs are HT-Greenfield capable 1 = 1 or more HT STAs that are not HT-Greenfield capable are associated Bits[9:8]: HT protection 00 = no protection 01 = non-member protection mode 10 = 20 MHz protection mode 11 = non-HT mixed mode Bits[7:4]: Reserved Bit[3]: RIFS mode 0 = use of RIFS prohibited 1 = use of RIFS permitted Bit[2]: STA channel width 0 = 20 MHz channel width 1 = both 20/40 MHz channel width Bits[1:0]: secondary channel offset 00 = (SCN) no secondary channel present 01 = (SCA) secondary channel above primary channel 10 = reserved 11 = (SCB) secondary channel below primary channel
BasicMCSSet	UINT8[16]	MCS values supported by AP

10.3.7 MrvlIETypes_IEEE_Extension_t

This generic TLV indicates the payload for various IEEE 802.11 Element ID extensions. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x00FF
Length	UINT16	Length of payload (excluding Type and Length fields)
Params	Variable	Payload based on IEEE 802.11 Element ID extension

10.4 NXP Proprietary IE Formats

This section lists the format for various NXP proprietary IE used in firmware API commands.

Table 83: NXP Proprietary IE Formats

Table 83: NXP Proprietary IE Formats (Continued)

Table 83: NXP Proprietary IE Formats (Continued)

Table 83: NXP Proprietary IE Formats (Continued)

10.4.1 MrvlIETypes_KeyParamSet_t

This TLV sets the key parameters. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0100
Length	UINT16	Length of payload
KeyTypeID	UINT16	Type of key 0x0000 = WEP 0x0001 = TKIP 0x0002 = AES 0x0003 = WAPI 0x0004 = AES CMAC
KeyInfo	UINT16	Key control info specific to a KeyTypeID
KeyLen	UINT16	Length of key
Key	UINT8[]	Key material of size KeyLen

10.4.1.1 WEP Key Type

Bit definition of KeyInfo for the WEP key type material is:

Field Name	Type	Description
Reserved	Bits[15:3]	Reserved (set to 0)
IsKeyEnabled	Bit[2]	Key enable and valid for use
IsUnicastKey	Bit[1]	Key used as the unicast key
IsMulticastKey	Bit[0]	Key used as the multicast key

- Setting a WEP key will keep the WEP key active until a successful association that requires WPA/WPA2 or does not require security at all.
- Disconnecting from a network (de-authentication command, de-authentication indication, link loss) will not clear the WEP key. Reconnecting to the network would not require a re-key.

The key materials in WEP consist of 7 octets for 40 bits and 15 octets for 104 bits in length. It contains 1 octet WepKepIndex and 1 octet to indicate the Tx default key or not and contains 5 octets key or 13 octets key.

Field Name	Type	Description
WepKeyIndex	UINT8	WEP key index 0 to 3
isDefaultTxKey	UINT8	0x00 = not default Tx key 0x01 = default Tx key
WepKey	UINT8[]	WEP encryption/decryption key

10.4.1.2 TKIP Key Type

Bit definition of KeyInfo for the TKIP key type material is:

Field Name	Type	Description
keyIndex	Bits[15:14]	Key index number used in transmitted group frames
Reserved	Bits[13:3]	Reserved (set to 0)
isKeyEnabled	Bit[2]	Key enabled and valid for use
isUnicastKey	Bit[1]	Key used as the unicast key
isMulticastKey	Bit[0]	Key used as the multicast key

If it is required that the KeyInfo be updated for options such as enabling/disabling the key, the command can be sent down with only KeyInfo and the key length set to 0. This way only the KeyInfo is updated and not the key value.

If the connection with a peer that made use of this key is lost during the course of operation, the key is automatically disabled by the firmware. Firmware notifies the driver of connection lost via a link lost event. Upon receiving the link lost event, the driver assumes that all key(s) currently being used are disabled.

The key materials in TKIP consist of 32 octets in length. It contains a TKIP key of 16 octets and 2 MIC keys of 8 octets each.

Field Name	Type	Description
TkipKey	UINT8[16]	TKIP encryption/decryption key
TkipTxMicKey	UINT8[8]	TKIP transmission MIC key
TkipRxMicKey	UINT8[8]	TKIP receive MIC key

10.4.1.3 AES Key Type

Bit definition of KeyInfo for the AES key type material is:

Field Name	Type	Description
keyIndex	Bits[15:14]	Key index number used in transmitted group frames
Reserved	Bits[13:3]	Reserved (set to 0)
IsKeyEnabled	Bit[2]	Key enable and valid for use
IsUnicastKey	Bit[1]	Key used as the unicast key
IsMulticastKey	Bit[0]	Key used as the multicast key

If it is required that the KeyInfo be updated for options such as enabling/disabling the key, the command can be sent down with KeyInfo only and the key length set to 0. This way only the KeyInfo is updated and not the key value.

If the connection with a peer that made use of this key is lost during the course of operation, the key is automatically disabled by firmware. Firmware notifies the driver of connection lost via a link lost event. Upon receiving the link lost event, the driver assumes that all key(s) currently being used are disabled.

The key materials in AES consisted of 16 octets in length.

Field Name	Type	Description
AesKey	UINT8[16]	AES encryption/decryption key

Note: Firmware should filter out MIC errors in AES mode.

10.4.1.4 WAPI Key Type

Bit definition of KeyInfo for the WAPI key type material is:

Field Name	Type	Description
Reserved	Bits[15:3]	Reserved (set to 0)
IsKeyEnabled	Bit[2]	Key enable and valid for use
IsUnicastKey	Bit[1]	Key used as the unicast key
IsMulticastKey	Bit[0]	Key used as the multicast key

If it is required that the KeyInfo be updated for options such as enabling/disabling the key, the command can be sent down with KeyInfo only and the key length set to 0. This way only the KeyInfo is updated and not the key value.

If the connection with a peer that made use of this key is lost during the course of operation, the key is automatically disabled by firmware. Firmware notifies the driver of connection lost via a link lost event. Upon receiving the link lost event, the driver assumes that all key(s) currently being used are disabled.

The key materials in WAPI consisted of 50 octets in length.

Field Name	Type	Description
wapiKeyIdx	UINT8	WAPI key index
isDefaultKey	UINT8	0x00 = not default Tx key 0x01 = default Tx key
wapiKey	UINT8[16]	WAPI encryption/decryption key
micKey	UINT8[16]	WAPI MIC key
RxPN	UINT8[16]	Rx initial packet number

- Firmware should filter out MIC errors in WAPI mode.
- The first key should set isDefaultKey to 1.
- The key of rekey should set the isDefaultKey to 0.

10.4.1.5 AES CMAC Key Type

Bit definition of KeyInfo for the AES CMAC key type material is:

Field Name	Type	Description
Reserved	Bits[15:11]	Reserved (set to 0)
isIGTKKey	Bit[10]	Key used as IGTK key
Reserved	Bits[9:3]	Reserved
IsKeyEnabled	Bit[2]	Key enable and valid for use
IsUnicastKey	Bit[1]	Key used as the unicast key
IsMulticastKey	Bit[0]	Key used as the multicast key

The key materials in AES CMAC consist of 24 octets in length. Set the Key_Len to 24.

Field Name	Type	Description
IPN	UINT8[8]	IGTK peer side Rx packet number [0 to 5] octets Bytes[7:6]: Reserved (set to 0)
AesCmacKey	UINT8[16]	AES CMAC encryption/decryption key

10.4.2 MrvlIETypes_ChanListParamSet_t

This TLV is used in the request of the following commands:

- CMD_802_11_SCAN
- CMD_802_11_BG_SCAN_CONFIG
- CMD_802_11_ASSOCIATE
- CMD_802_11_AD_HOC_START
- CMD_802_11_AD_HOC_JOIN
- APCMD_SYS_CONFIGURE

This TLV, if included in APCMD_SYS_CONFIGURE, specifies the channel scan list for uAP. This channel list is used when ACS is enabled to select the best channel for uAP operation.

By specifying a secondary channel, the user implicitly defines the channel width to be 40 MHz. This TLV has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0101
Length	UINT16	Length of payload
Multiple instances of the following fields, 1 set for each channel to be scanned: (maximum of 14 instances)		
BandConfigType	UINT8	Band configuration type Bits[7:6]: reserved (set to 0) Bits[5:4]: secondary channel offset 00 = no secondary channel 01 = secondary channel is above primary channel 10 = reserved 11 = secondary channel is below primary channel Bits[3:2]: channel width 00 = 20 MHz channel width 01 = 10 MHz channel width 10 = 40 MHz channel width 11 = 80 MHz channel width Bits[1:0]: band information 00 = 2.4 GHz band 01 = 5 GHz band 10 = 4 GHz band 11 = reserved
ChanNumber	UINT8	Channel number for the specified band
ScanType	UINT8	Scan type bitmap Bits[7:2]: reserved (set to 0) Bit[1]: disable channel filter Bit[0]: passive scan
MinScanTime	UINT16	Minimum scan time (ms)
MaxScanTime	UINT16	Maximum scan time (ms)

10.4.3 MrvlIETypes_NumProbes_t

This TLV sets the number of times each transmitted probe request frame is replicated. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0102
Length	UINT16	Length of payload (always 2 bytes)
NumProbes	UINT16	Number of probes to be sent Indicates the number of times each transmitted probe request frame is replicated. The valid range is [1:2]. The default value is 1.

10.4.4 MrvlIETypes_BeaconLowRssiThreshold_t

This TLV specifies the value of low beacon RSSI threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0104
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Absolute value of RSSI threshold value (dBm)
Frequency	UINT8	Reporting frequency

The Value field specifies the absolute value of the RSSI threshold. The actual value of the RSSI threshold is always negative. By default, the RSSI threshold of 0 is used by the firmware. RSSI threshold can be set high (0 or 1), so that background scan does not stop because of RSSI threshold match condition. This could be a desired setting in certain situations where background scan should continue until the device receives packets with certain characteristics. The Beacon RSSI_LOW event is triggered when the average RSSI in received beacons falls below the actual value.

The Frequency field specifies the reporting frequency for this event. If it is set to 0, the event is reported only once, and then automatically unsubscribed. If it is set to 1, the event is reported every time it occurs. If it is set to N (N > 1), an event is generated only when the condition happens N consecutive times.

10.4.5 MrvlIETypes_BeaconLowSnrThreshold_t

This TLV specifies the low beacon SNR threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0105
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	SNR threshold value (dB)
Frequency	UINT8	Reporting frequency

The Value field specifies the SNR threshold. The Beacon SNR_LOW event is triggered when the average SNR in received beacons falls below this value.

The Frequency field specifies the reporting frequency for this event. If it is set to 0, the event is reported only once, and then automatically unsubscribed. If it is set to 1, the event is reported every time it occurs. If it is set to N (N > 1), an event is generated only when the condition happens N consecutive times.

10.4.6 MrvlIETypes_FailureCount_t

This TLV specifies the consecutive failure count threshold. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0106
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Failure count threshold
Frequency	UINT8	Reporting frequency

The Value field specifies the consecutive failure count threshold that triggers the generation of the MAX_FAIL event. Once this event is generated, the consecutive failure count is reset to 0.

The Frequency field specifies the reporting frequency for this event. If it is set to 0, the event is reported only once, and then automatically unsubscribed. If it is set to 1, the event is reported every time it occurs. If it is set to N, then the event is reported every Nth time it occurs. At initialization, the MAX_FAIL event is not subscribed by default.

10.4.7 MrvlIETypes_BeaconsMissed_t

This TLV specifies the number of consecutive missing beacons. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0107
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Number of missed beacons
Reserved	UINT8	Reserved (set to 0)

The Value field specifies the number of consecutive missing beacons which triggers the BEACON_LOSS/LINK_LOSS event. This event is generated only once, after which the firmware resets its state.

At initialization, the BEACON_LOSS/LINK_LOSS event is subscribed by default. The default value of MissedBeacons is 60.

At initialization, the RSSI_LOW, SNR_LOW and MAX_FAIL events are NOT subscribed by default.

10.4.8 MrvlIETypes_LedGpio_t

The fixed fields are followed by an optional TLV field to map the GPIO pin used for each LED. This TLV has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0108
Length	UINT16	Length of payload (equals twice the number of LEDs configured)
LedNumber	UINT8	LED number to be mapped to GPIO number below
GpioNumber	UINT8	GPIO pin number used to control LED number above
…		
LedNumber	UINT8	LED number to be mapped to GPIO number below
LedNGpio	UINT8	GPIO pin number used to control LED number above

The payload length is variable (depending on the number of LEDs the host wishes to configure). The configuration of LEDs that are not included in this TLV is left unchanged.

The GpioNumber field indicates the GPIO pin number (used to control LED indicated by the LedNumber field). The valid range for this field is [0,16]. If it is set to 16, the LED is disabled completely. By default, only LED 1 is enabled in the firmware GPIO[1] is used to control it.

The host must ensure that any GPIO pin it wants to use in the firmware to control the LEDs is available.

The LED on/off behavior can also apply to other GPIOs without problems since the on/off behavior does not need special LED IP.

10.4.9 MrvlIETypes_LedBehavior_t

This TLV specifies the status of the LED behavior. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0109
Length	UINT16	Length of payload (4 bytes)
FirmwareState	UINT8	Firmware state
LedNumber	UINT8	LED number
LedState	UINT8	LED state corresponding to the firmware state
LedArgs	UINT8	Arguments for LED state

Note: This TLV is implemented in v6.0 or later.

10.4.10 MrvlIETypes_Passthrough_t

This TLV adds IEEE IEs to the command. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x010A
Length	UINT16	Length of payload
Data	UINT8[n]	IEEE IE(s) to be added to the management frame generated by the command

The Data field of the pass through TLV is added to the management frame generated. If multiple IEEE IEs are in the Data field, they are parsed and inserted in the correct order. Multiple instances of the pass through TLV are accepted in an API.

10.4.11 MrvlIETypes_WmmQStatus_t

This TLV queues the status TLV. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0110
Length	UINT16	Length of payload
QueueIndex	UINT8	AC queue by priority 0x00 = AC_BK 0x01 = AC_BE 0x02 = AC_VI 0x03 = AC_VO
Disabled	UINT8	Set if data traffic is allowed on this AC Queue Flows are disabled when admission control (FlowRequired) is set by the AP and TSPEC negotiation (ADDTS messaging) has not yet been completed.
MediumTime	UINT16	Access time granted by the AP in units of 32 µs
FlowRequired	UINT8	Set if admission control is required for this AC
FlowCreated	UINT8	Set if a TSPEC has been admitted by the AP for this AC via Admission Control messaging
Reserved3	UINT32	Reserved (set to 0)

10.4.12 MrvlIETypes_Crypto_Data_t

This TLV determines the encryption data type. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0111
Length	UINT16	Length of payload (bytes)
Data	UINT8[Length]	Data

10.4.13 MrvlIETypes_WildCardSSIDParamSet_t

This TLV configures the Wildcard SSID parameters. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0112
Length	UINT16	Length of payload
maxLen	UINT8	Maximum length of wildcard SSID
Ssid	UINT8[Length - 1]	Wildcard SSID

Up to 10 instances combined of this TLV and MrvlIETypes_SSIDParamSet_t supported for each scan command.

Examples includes:

- To send a specific SSID probe and exactly match the Rx SSID:
  SSID = ap102-g or wc = ap102-g,0.
- To filter on ap102-g, ap103-g without generating specific probes (only broadcast probe generated):
  wc = ap102-g,7 wc = ap103-g,7
- To match the base of an SSID up to a maximum length:
  wc = ap10,7 (same as ap10* but with a length restriction)
- To specify a single matching character and only match the fixed length of the wildcard:
  wc = ap10*-g,* or wc = ap10$-g,$
- Standard UNIX pattern matching:
  wc = ap*-g,255 or wc = ??102*,255, and so on. (? = single character, * = string)

Use of probes argument (‘numprobes’) set by MrvlIETypes_NumProbes_t

- No wildcards or SSIDs specified
  ‘numprobes’ broadcast probes are sent and all received probe responses or beacons returns are considered a match.
- Only wildcards specified
  ‘numprobes’ broadcast probes sent and filtered against the wildcards.
- 1 specific SSID
  ‘numprobes’ with the SSID IE set to the specific SSID.
- 2 specific SSIDs
  ‘numprobes’ for each SSID IE.
- 2 specific SSIDs and 8 wildcard SSIDs
  ‘numprobes’ broadcast probes, ‘numprobes’ for each specific SSID = total of 3*’numprobes’ sent per channel.
- Match a set of APs, send a specific for an AP:
  (wc = ap103-g,7 wc = ap102-g,7 SSID = ap-findme):

‘numprobes’ broadcast probes and filtered on ap103-g, ap102-g, and ap-findme (no patterns), ‘numprobes’ with the SSID IE set to ap-findme = total of 2*’numprobes’ sent per channel.

Note:
SSID = ap_name
where ap_name is the SSID specified in MrvlIETypes_SSIDParamSet_t TLV.
wc = ap_name,maxLen
where ap_name and maxLen are specified in this TLV.
‘numprobes’ is specified in MrvlIETypes_NumProbes_t.

10.4.14 MrvlIETypes_TsfTimestamp_t

This TLV specifies the TSF timestamps. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0113
Len	UINT16	Length of payload
tsfTable	UINT64[]	TSF timestamps

10.4.15 MrvlIETypes_PowerAdapt_Group_t

This TLV configures the power adaptation group. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0114
Length	UINT16	Length of payload (8 * number of power group)
PowerAdapt_level	UINT16	Power level
RateBitMap	UINT16	Bitmap of rates
Reserved	UINT32	Reserved for future bitmap of rates (set to 0)
…	…	…
PowerAdapt_level	UINT16	Power level
RateBitMap	UINT16	Bitmap of rates
Reserved	UINT32	Reserved for future bitmap of rates (set to 0)

Note: Maximum of 5 power groups are supported.

10.4.16 MrvlIETypes_HostSleepFilterType1

This TLV specifies EthType packets that can wake up the host. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0115
Length	UINT16	Length of payload
Payload	UINT8[]	Payload

Payload contains 0 or more fixed size entries of EthTypeEntry. Each EthTypeEntry has the following format:

Field Name	Type	Description
AddrType	UINT16	Type of MAC address 0x0001 = bcast 0x0002 = unicast 0x0003 = multicast
EthType	UINT16	Ethernet type field in 802.2 header Example: 0x0806 = ARP
Ipv4Addr	UINT32	IP v4 address specific to EthType

AddrType specifies the type of MAC address. Each value of AddrType requires the corresponding Bcast/Unicast/Multicast bit in Criteria to be set. This field is not used if set to 0xFFFF.

EthType specifies the IEEE Ethernet type. For example, an EthType entry for matching ARP request has EthType 0x0806. For faster matching, EthType is in network byte order, that is, big endian. For the example of ARP, the lowest address byte of EthType is the most significant byte 0x08, and the higher address byte is 0x06.

Ipv4Addr is the IP v4 address specific to EthType. For example, ARP request matching specifies the IP v4 address of the host. This field is not used if set to 0xFFFFFFFF. For faster matching, Ipv4Addr is in network byte order. For the example of 192.168.0.2 (0xC0A80002), the lowest address byte of Ipv4Addr is 0xC0, followed by 0x80, 0x00, and 0x02 (the highest address byte).

Each EthTypeEntry specifies an *allow* filter, (matching packet wakes up the host). The AddrType (if not 0xFFFF), EthType, and Ipv4Addr (if not 0xFFFFFFFF) fields are AND'd together to form the filter. Filters are processed in the storage order and in an OR relationship.

As an example, if the host wishes to wake up on any multicast packet or an ARP packet (a broadcast packet) intended for the host, it should set bit 3 (multicast) and bit 0 (broadcast) in Criteria and include the following this TLV:

Field Name	Type	Description
Type	UINT16	Type ID (MrvlIETypes_HostSleepFilterType1)
Length	UINT16	16
AddrType	UINT16	3 (multicast)
EthType	UINT16	0x0800
Ipv4Addr	UINT32	0xFFFFFFFF
AddrType	UINT16	1 (broadcast)
EthType	UINT16	0x0806
Ipv4Addr	UINT32	192.168.0.88

10.4.17 MrvlIETypes_BeaconHighRssiThreshold_t

This TLV specifies the high beacon RSSI threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0116
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Absolute value of RSSI threshold value (dBm)
Frequency	UINT8	Reporting frequency

The Value field specifies the absolute value of the RSSI threshold, the actual value of RSSI threshold is always negative. The Beacon RSSI_HIGH event is triggered when the average RSSI in received beacons goes above the actual value.

The Frequency field specifies the reporting frequency for this event. If it is set to 0, the event is reported only once, and then automatically unsubscribed. If it is set to 1, the event is reported every time it occurs. If it is set to N (N > 1), an event is generated only when the condition happens N consecutive times.

10.4.18 MrvlIETypes_BeaconHighSnrThreshold_t

This TLV specifies the high beacon SNR threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0117
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	SNR threshold value (dB)
Frequency	UINT8	Reporting frequency

The Value field specifies the SNR threshold. The Beacon SNR_HIGH event is triggered when the average SNR in received beacons goes above this value.

The Frequency field specifies the reporting frequency for this event. If it is set to 0, the event is reported only once, and then automatically unsubscribed. If it is set to 1, the event is reported every time it occurs. If it is set to N (N > 1), an event is generated only when the condition happens N consecutive times.

10.4.19 MrvlIETypes_AutoTxMacFrame_t

This TLV configures the auto transmit settings. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0118
Length	UINT16	Length of MrvlIETypes_AutoTxMacFrame_t (excluding Type and Length fields) Set to (18 + n + p).
Interval	UINT16	Auto-transmission interval (seconds)
Priority	UINT8	User Priority (0 to 7)
Reserved	UINT8	Reserved (set to 0)
FrameLen	UINT16	Length of 802.3 MAC frame payload (octets) Set to (12 + n + p).
DestAddr	UINT8[6]	Destination station's IEEE 48-bit MAC address If it is 0, destination address will use AP's MAC address.
SrcAddr	UINT8[6]	Source station's IEEE 48-bit MAC address If it is 0, source address will use transmitter's address.
FrameData	UINT8[n]	802.3 MAC frame payload
FramePadding	UINT8[p]	802.3 MAC padding

10.4.20 MrvlIETypes_WPSEnrolleeProbeReqIE_t

This TLV includes WPS_IE contents in Enrollee Probe Request. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x011B
Length	UINT16	Length of payload
WPS_IE	Variable length	WPS_IE contents that are included in Enrollee Probe Request Frames

10.4.21 MrvlIETypes_WPSRegistrarBeaconIE_t

This TLV includes WPS_IE contents in Registrar Beacons. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x011C
Length	UINT16	Length of payload
WPS_IE	Variable length	WPS_IE contents that are included in Registrar Beacons

10.4.22 MrvlIETypes_WPSRegistrarProbeRespIE_t

This TLV includes WPS_IE contents in Registrar Probe Response. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x011D
Length	UINT16	Length of payload
WPS_IE	Variable length	WPS_IE contents that are included in Registrar Probe Response Frames

10.4.23 MrvlIETypes_AuthType_t

This TLV sets the authentication type for association. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x011F
Length	UINT16	Length of payload (always 2 bytes)
AuthType	UINT16	Authentication Type to be used to authenticate with AP 0x0000 = Auth_Mode_Open 0x0001 = Auth_Mode_Shared 0x0080 = Auth_Mode_Network_EAP

10.4.24 MrvlIETypes_StaMacAddr_t

This TLV sets/gets the station MAC address. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0120
Length	UINT16	Length of payload
MacAddress	UINT8[6]	MAC address

10.4.25 MrvlIETypes_BSSIDList_t

This TLV sets/gets the BSSID. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0123
Length	UINT16	Length of payload
MacAddress	UINT8[n][6]	MAC address

10.4.26 MrvlIETypes_LinkQualityThreshold_t

This TLV configures the Link Quality event threshold. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0124
Length	UINT16	Length of payload (always 2 bytes)
Value(LQ_SNR)	UINT16	Link SNR threshold (dB)
Frequency(LQ_SNR)	UINT16	Link SNR frequency
Value(LQ_RATE)	UINT16	Second minimum rate value as per the rate table below
Frequency(LQ_RATE)	UINT16	Second minimum rate frequency
Value(LQ_TX_LATENCY)	UINT16	Tx latency value (μs)
Frequency(LQ_TX_LATENCY)	UINT16	Tx latency threshold

To generate the Link Quality event, configure the 3 thresholds in this TLV.

LQ_SNR and LQ_RATE setting are used in conjunction for generating the link quality event. LQ_TX_LATENCY is used standalone for the generation of this event. The frequency parameter for all these parameters can be any positive number. 0 is not a valid number for the frequency parameter.

The event is generated if the SNR of the received beacon falls below the value set in the Value field for LQ_SNR for consecutive 'n' number of times (number 'n' is equal to the value in the frequency parameter for LQ_SNR). The event is also generated when the rate has fallen back to the value equal to or less than the value set in the Value field of the LQ_RATE for consecutive 'm' number of data frame transmission (number 'm' is equal to the value in the frequency parameter for LQ_RATE).

The event is also generated when the Tx latency goes below the value set in the value parameter of Tx Latency for 'n' consecutive number of times (number 'n' is equal to the value in the frequency parameter for LQ_TX_LATENCY).

The value for LQ_RATE is set as per following:.

Physical Rate	Value for LQ_RATE parameter
1 Mbps	10
2 Mbps	20
5.5 Mbps	55
11 Mbps	110
6 Mbps	60
9 Mbps	90
12 Mbps	120
18 Mbps	180
24 Mbps	240
36 Mbps	360
48 Mbps	480
54 Mbps	540

10.4.27 MrvlIETypes_DataLowRssiThreshold_t

This TLV specifies the value of low data RSSI threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0126
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Absolute value of RSSI threshold value (dBm)
Frequency	UINT8	Reporting frequency

Value field specifies the absolute value of the RSSI threshold. The actual value of RSSI threshold is always negative. The Data RSSI_LOW event is triggered when the average RSSI in received beacons falls below the actual value.

Frequency field specifies the reporting frequency for this event. If it is set to 0, then the event is reported only once, and then automatically unsubscribed. If it is set to 1, then the event is reported every time it occurs. If it is set to N (N>1), an event is generated only when the condition happens N consecutive times.

10.4.28 MrvlIETypes_DataLowSnrThreshold_t

This TLV specifies the value of low data SNR threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0127
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	SNR threshold value (dB)
Frequency	UINT8	Reporting frequency

Value field specifies the SNR threshold. The Data SNR_LOW event is triggered when the average SNR in received beacons falls below this value.

Frequency field specifies the reporting frequency for this event. If it is set to 0, then the event is reported only once, and then automatically unsubscribed. If it is set to 1, then the event is reported every time it occurs. If it is set to N (N>1), an event is generated only when the condition happens N consecutive times.

10.4.29 MrvlIETypes_DataHighRssiThreshold_t

This TLV specifies the value of high data RSSI threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0128
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Absolute value of RSSI threshold value (dBm)
Frequency	UINT8	Reporting frequency

Value field specifies the absolute value of the RSSI threshold. The actual value of RSSI threshold is always negative. The Data RSSI_HIGH event is triggered when the average RSSI in received beacons goes above the actual value.

Frequency field specifies the reporting frequency for this event. If it is set to 0, then the event is reported only once, and then automatically unsubscribed. If it is set to 1, then the event is reported every time it occurs. If it is set to N (N>1), an event is generated only when the condition happens N consecutive times.

10.4.30 MrvlIETypes_DataHighSnrThreshold_t

This TLV specifies the value of high data SNR threshold value. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0129
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	SNR threshold value (dB)
Frequency	UINT8	Reporting frequency

Value field specifies the SNR threshold. The Data SNR_HIGH event is triggered when the average SNR in received beacons goes above this value.

Frequency field specifies the reporting frequency for this event. If it is set to 0, then the event is reported only once, and then automatically unsubscribed. If it is set to 1, then the event is reported every time it occurs. If it is set to N (N>1), an event is generated only when the condition happens N consecutive times.

10.4.31 MrvlIETypes_ChanBandList_t

This TLV is returned in the response of commands:

- CMD_802_11_SCAN
- CMD_802_11_BG_SCAN_QUERY
- APCMD_SYS_CONFIGURE

This TLV is included in APCMD_SYS_CONFIGURE to configure and query uAP's operating channel and enable Automatic Channel Selection feature. This TLV has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x012A
Length	UINT16	Length of payload
BandConfigType[1]	UINT8	Band Configuration Type Bits[7:6]: channel selection mode 00 = manual 01 = ACS on BSS start Bits[5:4]: secondary channel offset 00 = no secondary channel 01 = secondary channel is above primary channel 10 = reserved 11 = secondary channel is below primary channel Bits[3:2]: channel width 00 = 20 MHz channel width 01 = 10 MHz channel width 1x = reserved Bits[1:0]: band information 00 = 2.4 GHz band 01 = 5 GHz band 10 = 4 GHz band 11 = reserved
ChanNumber[1]	UINT8	Channel number for the specified band

1. Multiple instances of these fields, 1 set for each channel/band information.

10.4.32 MrvlIETypes_EncryptionProtocol_t

This TLV configures the encryption protocol. It has the following format.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0140</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload (2 bytes)</td></tr>
<tr><td>Protocol</td><td>UINT16</td><td>Bitmap for protocols supported
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:7</td><td>Reserved</td></tr>
<tr><td>6</td><td>CCKM—Support CCKM protocol
By default the embedded supplicant will not support CCKM. CCKM support will have to be explicitly enabled.</td></tr>
<tr><td>5</td><td>WPA2—Support WPA2 protocol</td></tr>
<tr><td>4</td><td>WPA-NONE (Ad-Hoc only)
This option is provided for Ad-Hoc security. Protocol priority. If enabled and available, the embedded supplicant will select protocols in the following order:
• CCKM
• WPA2
• WPA
• No RSN</td></tr>
<tr><td>3</td><td>WPA</td></tr>
<tr><td>2</td><td>WEP Dynamic
This feature is currently not supported by the embedded supplicant.</td></tr>
<tr><td>1</td><td>WEP Static
The embedded supplicant is currently not involved in static WEP transactions and does not need to be configured for to use static WEP.</td></tr>
<tr><td>0</td><td>No RSN
This configuration allows association that do not support WPA. WPA2, or CCKM to proceed when the embedded supplicant is enabled. By default, this option will be enabled in the embedded supplicant.</td></tr>
</table></td></tr>
</table>

The bits in the bitmap correspond to the cipher suites. Bit[0] is the LSb. Setting the bit to 1 indicates the cipher type is available to the embedded supplicant.

10.4.33 MrvlIETypes_Cipher_t

This TLV configures the cipher keys. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0142
Length	UINT16	Length of payload (2 bytes)
PairwiseCipher	UINT8	Pairwise Cipher Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bit[1]: WPA_CIPHER_WEP104 Bit[0]: WPA_CIPHER_WEP40
GroupCipher	UINT8	Group Cipher Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bit[1]: WPA_CIPHER_WEP104 Bit[0]: WPA_CIPHER_WEP40

The bits in the bitmap correspond to the cipher suites. Bit[0] is the LSB. Setting the bit to 1 indicates the cipher type is available to the embedded supplicant.

Within the selected protocol, and if available and configured, the embedded supplicant chooses the cipher suites in the following priority order:

- AES CCMP
- TKIP
- WEP 104
- WEP 40

WEP 104 and WEP 40 are not available as Pairwise cipher suites.

10.4.34 MrvlIETypes_PMK_t

This TLV sets/gets the Pairwise Master Key (PMK). It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0144
Length	UINT16	Length of payload (always 32 bytes)
PMK	UINT8[32]	PMK array

10.4.35 MrvlIETypes_Passphrase_t

This TLV sets/gets the passphrase. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0145
Length	UINT16	Length of payload
Passphrase	UINT8[n]	ASCII array containing the passphrase n <= 64

10.4.36 MrvlIETypes_PreBeaconMissed_t

This TLV configures the Pre Beacon Lost event. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0149
Length	UINT16	Length of payload (always 2 bytes)
Value	UINT8	Number of pre missed beacons Specifies the number of consecutive missing beacons which triggers the PRE_BEACON_ LOST event. This event is generated only once and changed to disable after triggered. The host need to perform the enable if needed after the event is triggered. The default value is 50 and should be smaller than the link lost counts. If this value is larger than the link lost count, the event will not trigger.
Reserved	UINT8	Reserved

10.4.37 MrvlRateDropPattern_t

This TLV configures rate drop pattern. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0151
Length	UINT16	Length of payload (excluding Type and Length fields)
RateDropMode	UINT32	Rate drop mode 0 = firmware defined rate drop, no "Rate_Drop_Control" LV is needed 1 = no rate drop, no "Rate_Drop_Control" LV is needed

10.4.38 MrvlIE_RateScope_t

This TLV configures the rate scope. It has the following format.

Field Name	Type	Description
Type	UINT16	Type ID = 0x0153
Length	UINT16	Length of payload (excluding Type and Length fields)
HT/DSSS Rate Bitmap	UINT16	HT/DSSS rate bitmap Bits[15:4]: reserved (set to 0) Bit[3]: CCK 11 Mbps Bit[2]: CCK 5.5 Mbps Bit[1]: DSSS 2 Mbps Bit[0]: DSSS 1 Mbps
OFDM Rate Bitmap	UINT16	OFDM rate bitmap Bits[15:8]: reserved (set to 0) Bit[7]: OFDM 54 Mbps Bit[6]: OFDM 48 Mbps Bit[5]: OFDM 36 Mbps Bit[4]: OFDM 24 Mbps Bit[3]: OFDM 18 Mbps Bit[2]: OFDM 12 Mbps Bit[1]: OFDM 9 Mbps Bit[0]: OFDM 6 Mbps
HT-MCS Rate Bitmap	UINT16[8]	MCS0 to MCS127

Field Name	Type	Description
VHT-MCS Rate Bitmap	UINT16[8]	VHT_MCS <table><tr><th>VHT_MCS Bit</th><th>Description</th></tr><tr><td>7</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=8</td></tr><tr><td>6</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=7</td></tr><tr><td>5</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=6</td></tr><tr><td>4</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=5</td></tr><tr><td>3</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=4</td></tr><tr><td>2</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=3</td></tr><tr><td>1</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=2</td></tr><tr><td>0</td><td>Bits[15:10]: reserved Bits[9:0]: MCS 0-9 Nss=1</td></tr></table>**NOTE:** This field exists if HW_SPEC report 802.11ac is supported.
HE-MCS Rate Bitmap	UINT16[8]	HE_MCS <table><tr><th>HE_MCS Bit</th><th>Description</th></tr><tr><td>7</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=8</td></tr><tr><td>6</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=7</td></tr><tr><td>5</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=6</td></tr><tr><td>4</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=5</td></tr><tr><td>3</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=4</td></tr><tr><td>2</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=3</td></tr><tr><td>1</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=2</td></tr><tr><td>0</td><td>Bits[15:12]: reserved Bits[11:0]: MCS0-11 Nss=1</td></tr></table>**NOTE:** This field exists if HW_SPEC report 802.11ax is supported.

The initial rate of the first transmission for each frame (that is, not re-transmission) is fixed if there is only 1 bit set in TxRateScope; otherwise the rate adaptation module in firmware will decide which rate is to be used based on the link status, and so on. The parameters listed below:

- TxRateScope configured by user
- Rates supported by BSS
- Rates supported by hardware

Only the intersection of rates above could be selected by rate adaptation module in firmware. f there is no rate available, the minimum supported rate will be selected. User can issue the SET command before association. The user rate drop mode is no longer supported in v17.0.

10.4.39 MrvlIETypes_Power_Group_t

This TLV configures the power group. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0154
Length	UINT16	Length of payload (excluding Type and Length fields, 8 * number of rate groups)
PowerGroups[n]	Power_Group_t	Power groups (see Table 84, Power Groups, on page 339)

Table 84: Power Groups

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>ModulationClass</td><td>UINT8</td><td>Modulation class
0x03 = HR/DSSS
0x07 = OFDM
0x08 = HT</td></tr>
<tr><td>FirstRateCode</td><td>UINT8</td><td>MCS code or legacy RateID
<table>
<tr><th>ModulationClass</th><th>FirstRateCode</th></tr>
<tr><td>0x03 = HR/DSSS</td><td>0x00 = 1 Mbps
0x01 = 2 Mbps
0x02 = 5.5 Mbps
0x03 = 11 Mbps
Others = reserved</td></tr>
<tr><td>0x07 = OFDM</td><td>0x00 = 6 Mbps
0x01 = 9 Mbps
0x02 = 12 Mbps
0x03 = 18 Mbps
0x04 = 24 Mbps
0x05 = 36 Mbps
0x06 = 48 Mbps
0x07 = 54 Mbps
Others = reserved</td></tr>
<tr><td>0x08 = HT</td><td>0x00 = MCS-0
0x01 = MCS-1
0x02 = MCS-2
0x03 = MCS-3
0x04 = MCS-4
0x05 = MCS-5
0x06 = MCS-6
0x07 = MCS-7
:
0x0F = MSC-15
0x20 = MCS-32
Others = reserved</td></tr>
<tr><td>0x09 = VHT</td><td>0xNM
where N = Nss-1 (0/1). M = MCS 0-9</td></tr>
</table>
</td></tr>
<tr><td>LastRateCode</td><td>UINT8</td><td>MCS code or legacy RateID (see above)</td></tr>
<tr><td>Reserved</td><td>SINT8</td><td>Reserved (set to 0)</td></tr>
</table>

Table 84: Power Groups (Continued)

Field Name	Type	Description
Reserved	SINT8	Reserved (set to 0)
Power_Max	SINT8	Maximal Tx power level (dBm)
Bandwidth	UINT8	HT bandwidth 0x00 = 20 MHz 0x01 = 40 MHz 0x02 = 80 MHz 0x03 = 160 MHz
Reserved	UINT8	Reserved (set to 0)

10.4.40 MrvlIETypes_Bss_Scan_Rsp_t

This TLV configures the BSS scan response. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0156
Length	UINT16	Length of payload
BSSID	UINT8[6]	BSS ID
FraneBody	UINT8[6]	Beacon/probe Response Frame Content

10.4.41 MrvlIETypes_Bss_Scan_Info_t

This TLV configures the BSS scan information. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0157
Length	UINT16	Length of payload
RSSI	SINT16	RSSI value as received from peer
Rserved	SINT16	Reserved (set to 0)
Reserved	UINT8	Reserved (set to 0)
Band	UINT8	Band information where the peer is located
Channel	UINT8	Channel number where the peer is located
Reserved	UINT8	Reserved (set to 0)
TSF	UINt64	TSF value

10.4.42 MrvlIETypes_WAPI_IE_t

This TLV configures the WAPI IE. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x015E
Length	UINT16	Length of payload
WAPI IE Content	UINT8[n]	Contents will be added to associate request frame as is n is the length.

10.4.43 MrvlIETypes_RobustCoex_t

This TLV enables/disables robust coexistence mode. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0160
Length	UINT16	Length of payload (always 4 bytes)
Enable	UINT8	Robust coexistence mode enable 0x00 = disable 0x01 = enable
Reserved	UINT8[3]	Not used (for padding purposes)

10.4.44 MrvlIETypes_RobustCoex_Period_t

This TLV sets robust coexistence period. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0161
Length	UINT16	Length of payload (always 12 bytes)
Mode	UINT16	Mode 0x0000 = not used 0x0001 = time share mode 0x0002 = aggressive mode
Reserved	UINT16	Bits[15:1]: reserved (set to 0) Bit[0]: BTTime and Period 0 = BTTime and Period is in μs unit (default) 1 = BTTime and Period is in ms unit
BTTTime	UINT32	Length of time to give to Bluetooth (μs) Bits[31:16]: BTTime specific to A2DP coex Bits[15:0]: BTTime for generic ACL coex BTTime for generic ACL must be set, but BTTime for A2DP is optional. If BTTime for A2DP is 0, firmware uses the default value (which is 20% of Period). It is recommended to set BTTime for A2DP as 0, so that the default is to be used.
Period	UINT32	Length of period (μs) Default = 20 ms

10.4.45 MrvlIETypes_WapiInfoSet_t

This TLV configures the WAPI information. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0167
Length	UINT16	Length of payload
Mutlicase PN	UINT8[16]	PN of multicast packet used

10.4.46 MrvlIETypes_MgmtFrameSet_t

This TLV configures the management frame set. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0168</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td>FrameControl</td><td>UINT16</td><td>802.11 frame control field:
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15</td><td>Order</td></tr>
<tr><td>14</td><td>WEP</td></tr>
<tr><td>13</td><td>More data</td></tr>
<tr><td>12</td><td>Power management</td></tr>
<tr><td>11</td><td>Retry</td></tr>
<tr><td>10</td><td>More fragment</td></tr>
<tr><td>9</td><td>From DS</td></tr>
<tr><td>8</td><td>To DS</td></tr>
<tr><td>7:4</td><td>Subtype</td></tr>
<tr><td>3:2</td><td>Type</td></tr>
<tr><td>1:0</td><td>Protocol version</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>FrameContents</td><td>UINT8[]</td><td>
Association request (Type = 0, Subtype = 0)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Capability</td><td>UINT16</td><td>IEEEtypes_CapInfo_t</td></tr>
<tr><td>Listen Interval</td><td>UINT16</td><td>IEEEtypes_ListenInterval_t</td></tr>
<tr><td>IE Buffer</td><td>UINT8[]</td><td>Association request IEs</td></tr>
</table>
Association response (Type = 0, Subtype = 1)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Capability</td><td>UINT16</td><td>IEEEtypes_CapInfo_t</td></tr>
<tr><td>Status</td><td>UINT16</td><td>IEEEtypes_StatusCode_t</td></tr>
<tr><td>AID</td><td>UINT16</td><td>IEEEtypes_AId_t</td></tr>
</table>
Re-association request (Type = 0, Subtype = 2)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Capability</td><td>UINT16</td><td>IEEEtypes_CapInfo_t</td></tr>
<tr><td>Listen Interval</td><td>UINT16</td><td>IEEEtypes_ListenInterval_t</td></tr>
<tr><td>Current AP Address</td><td>UINT8[6]</td><td>IEEEtypes_MacAddr_t</td></tr>
<tr><td>IE Buffer</td><td>UINT8[]</td><td>Association request IEs</td></tr>
</table>
Re-association response (Type = 0, Subtype = 3)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Capability</td><td>UINT16</td><td>IEEEtypes_CapInfo_t</td></tr>
<tr><td>Status</td><td>UINT16</td><td>IEEEtypes_StatusCode_t</td></tr>
<tr><td>AID</td><td>UINT16</td><td>IEEEtypes_AId_t</td></tr>
</table>
Probe request (Type = 0, Subtype = 4)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>IE Buffer</td><td>UINT8[]</td><td>Start from SSID to end of probe request frame packet</td></tr>
</table>
Action frame (Type = 0, Subtype = 13)
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>FrameContent</td><td>UINT8[]</td><td>Start from Category to end of action frame packet</td></tr>
</table>
</td></tr>
</table>

10.4.47 MrvlIETypes_MgmtIEList_t

This TLV sets/gets the user specified IEs. It has the following format.

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0169</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload (excluding Type and Length fields)</td></tr>
<tr><td>IEIndex0</td><td>UINT16</td><td>Index of IE buffer</td></tr>
<tr><td>MgmtSubTypeMask</td><td>UINT16</td><td>Management subtype mask for buffer index ‘IEIndex0’
Management frame subtype mask to which the IE is to be attached:
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:9</td><td>Reserved</td></tr>
<tr><td>8</td><td>Beacon</td></tr>
<tr><td>6:7</td><td>Reserved</td></tr>
<tr><td>5</td><td>Probe Response</td></tr>
<tr><td>4</td><td>Probe Request</td></tr>
<tr><td>3</td><td>Reassociation Response</td></tr>
<tr><td>2</td><td>Reassociation Request</td></tr>
<tr><td>1</td><td>Association Response</td></tr>
<tr><td>0</td><td>Association Request</td></tr>
</table>
If more than 1 bits are set, it will indicate that they use the same IE buffer.</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of IE buffer for buffer index ‘IEIndex0’</td></tr>
<tr><td>IEBuffer</td><td>UINT8[]
(up to 256)</td><td>IE buffer list for buffer index ‘IEIndex0’</td></tr>
<tr><td>...</td><td>...</td><td>...</td></tr>
<tr><td>IEIndex3</td><td>UINT16</td><td>Index of IE buffer</td></tr>
<tr><td>MgmtSubTypeMask</td><td>UINT16</td><td>Management subtype mask for buffer index ‘IEIndex3’
Management frame subtype mask to which the IE is to be attached:
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:9</td><td>Reserved</td></tr>
<tr><td>8</td><td>Beacon</td></tr>
<tr><td>6:7</td><td>Reserved</td></tr>
<tr><td>5</td><td>Probe Response</td></tr>
<tr><td>4</td><td>Probe Request</td></tr>
<tr><td>3</td><td>Reassociation Response</td></tr>
<tr><td>2</td><td>Reassociation Request</td></tr>
<tr><td>1</td><td>Association Response</td></tr>
<tr><td>0</td><td>Association Request</td></tr>
</table>
If more than 1 bits are set, it will indicate that they use the same IE buffer.</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of IE buffer for buffer index ‘IEIndex3’</td></tr>
<tr><td>IEBuffer</td><td>UINT8[]
(up to 256)</td><td>IE buffer list for buffer index ‘IEIndex3’</td></tr>
</table>

- The host can set up to small IE (length <= 64 bytes), medium IE (length <= 192 bytes), and large IE (length <= 256 bytes). The number of these buffers is depended on the release. The smaller size IE can be stored in larger biffer if the buffer is available.
- The GET action command can be used to read the IE buffers and the subtype field.
- The GET command for this TLV can have no payload with 'Length' field in TLV set to 0 and response will have all (index, mask, length, IE buffers) in response with the individual length fields and tag length populated correctly.
- The GET command can also have an 'IEIndex' field in the payload and the response will have the (index, mask, length and IE buffer) fields associated with that index in the buffer.
- Selective removal of certain IEs from the IE buffer will not be supported. Host will need to perform add/delete in application and give the final IE list to firmware.
- If the IE buffer length is set to 0, the IE buffer will be set as empty and it would indicate that no user specified IEs are to be attached for that subtype.
- Host should not include IEs that are being populated by firmware using this API and should implement any necessary error checking. Firmware will not perform any checks, but simply copy these IEs to the specified frames.
- These IEs are added by the firmware at the end of the existing IEs generated by firmware. Firmware will not re-sort the IEs after inclusion of the user-specified IEs. As a result, if this user specified IEs include IEs other than Vendor Specific IE (221), there could be interop issues since the final IE list may not be sorted.

10.4.48 MrvlIETypes_ApSleepParams_t

This TLV configures sleep parameters. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016A
Length	UINT16	Length of payload
ControlBitmap	UINT32	Bits[31:1]: reserved Bit[0]: 0 = disable Null-data (NAV) protection frame Tx before PS 1 = enable Null-data (NAV) protection frame Tx before PS
MinSleep	UINT32	Minimum sleep duration (μs)
MaxSleep	UINT32	Maximum sleep duration (μs)

The minimum value for MinSleep and MaxSleep is chip dependent. The actual minimum MinSleep value could be slightly lower for specific reference clock value used on the board. If the user tries to configure a MinSleep value that is less than the minimum required MinSleep value of the board, then that configured value will be rejected.

10.4.49 MrvlIETypes_ApSleepParamsInact_t

This TLV configures sleep inactivity parameters. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016B
Length	UINT16	Length of payload
PSInactivtyTmo	UINT32	Inactivity timeout (µs) after which to start sleep-awake cycles
MinAwake	UINT32	Minimum awake duration (µs) Minimum 2000 µs.
MaxAwake	UINT32	Maximum awake duration (µs) Minimum 2000 µs.

10.4.50 MrvlIETypes_ApBTCoexCommonConfig_t

This TLV configures common Bluetooth coexistence. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016C
Length	UINT16	Length of payload
ControlBitmap	UINT32	Bits[31:1]: reserved Bit[0]: override CTS2Self protection
Reserved	UINT32[4]	Reserved

10.4.51 MrvlIETypes_ApBTCoexScoConfig_t

This TLV configures Synchronous Connection Oriented (SCO) coexistence related settings. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016D
Length	UINT16	Length of payload
ProtectionFrmQTime	UINT16[4]	Index 0 = ONLY_SCO Index 1 = ACL_BEFORE_SCO Index 2 = ACL_AFTER_SCO Index 3 = reserved
AclFrequency	UINT16	ACL frequency
ProtectionFrmRate	UINt16	Protection frame rate
Reserved	UINT32[4]	Reserved

10.4.52 MrvlIETypes_ApBTCoexAclConfig_t

This TLV configures Access Control List (ACL) coexistence related settings. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016E
Length	UINT16	Length of payload
Enabled	UINT16	ACL coexistence 0 = disable 1 = enable
BtTime	UINT16	Bluetooth time
WlanTime	UINT16	Wi-Fi time
ProtectionFrmRate	UINT16	Protection frame rate
Reserved	UINT32[4]	Reserved

10.4.53 MrvlIETypes_ApBTCoexStats_t

This TLV gets the coexistence related statistics. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x016F
Length	UINT16	Length of payload
NullNotSent	UINT32	Null not sent
NumOfNullQueue	UINT32	Number of null queued
NullNotQueued	UINT32	Null not queued
NumOfCfEndQueued	UINT32	Number of CF end queued
CfEndNotQueued	UINT32	CF end not queued
NullAllocaitonFail	UINT32	Null allocation fail
CfEndAllocationFail	UINT32	CF end allocation fail
Reserved	UINT32[8]	Reserved

10.4.54 MrvlIETypes_Auto_DS_Param_t

This TLV configures the auto DS parameters. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0171
Length	UINT16	Length of payload
IdleTime	UINT16	Idle time (ms) Default = 100 ms Minimum = 10 ms

10.4.55 MrvlIETypes_Enh_Sta_PS_t

This TLV configures the enhanced station power save mode. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0172
Length	UINT16	Length of payload
MullPktInterval	UINT16	Null packet interval(s) to keep alive in infrastructure power save mode For other modes, this field is ignored. 0x0000 = unchanged 0x001E = 30s (default) 0xFFFF = disable
NumDtims	UINT16	Number of DTIM intervals to wake up 0x0000 = unchanged 0x0001–0x0005 = valid setting 0xFFFD = closest DTIM to the listen interval period will be used 0xFFFE = DTIM will be ignored, listen interval will be used 0xFFFF = disable Default = 1
BCNMissTimeOut	UINT16	Sets beacon miss timeout 0x0000 = unchanged 0xFFFF = disable Default = 0x0005
LocalListenInterval	UINT16	Local listen intervals 0x0000 = unchanged 0x0001–0x0031 = value in beacon intervals > 0x0032 = value in TUs 0xFFFF = disable Default = 0x000A
AdhocAwakePeriod	UINT16	Sets ad-hoc awake period 0x0000 = unchanged 0x0001–0x0031 = beacon interval 0x00FF = sleep after beacon sent out Default = 0x0004
PS_Mode	UINT16	Power save mode 0x0000 = unchanged 0x0001 = auto mode 0x0002 = PS-Poll mode 0x0003 = PS null packet mode (default)
DelayToPS	UINT16	Delay to PS (ms) Default = 500 ms

10.4.56 MrvlIETypes_ChanTRPCConfig_t

This TLV configures the TRPC channel. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x0189
Length	UINT16	Length of payload
StartFreq	UINT16	Starting frequency of the band for this channel • 2407, 2414, or 2400 for 2.4 GHz • 4000 • 5000
ChanWidth	UINT8	Channel width Valid values: • 10 • 20
ChanNum	UINT8	Logical 5 MHz channel number (1–255)
ModulationGroup1	UINT8	Modulation group 1 0x00 = CCK (1, 2, 5.5, 11) 0x01 = OFDM (6, 9, 12, 18) 0x02 = OFDM (24, 36) 0x03 = OFDM (48, 54) 0x04 = HT20 (0, 1, 2), VHT20 (1SS, MCS0, 1, 2) 0x05 = HT20 (3, 4), VHT20 (1SS, MCS3, 4) 0x06 = HT20 (5, 6, 7), VHT20 (1SS, MCS5, 6, 7) 0x07 = HT40 (0, 1, 2), VHT40 (1SS, MCS0, 1, 2) 0x08 = HT40 (3, 4), VHT40 (1SS, MCS3, 4) 0x09 = HT40 (5, 6, 7), VHT40 (1SS, MCS5, 6, 7) 0x0A = HT2_20 (8, 9, 10), VHT2_20 (2SS, MCS0, 1, 2) 0x0B = HT2_20 (11, 12), VHT2_20 (2SS, MCS3, 4) 0x0C = HT2_20 (13, 14, 15), VHT2_20 (2SS, 5, 6, 7) 0x0D = HT2_40 (8, 9, 10), VHT2_40 (2SS, MCS0, 1, 2) 0x0E = HT2_40 (11, 12), VHT2_40 (2SS, MCS3, 4) 0x0F = HT2_40 (13, 14, 15), VHT2_40 (2SS, 5, 6, 7) 0x10 = VHT20 (1SS, MCS8) 0x11 = VHT40 (1SS, MCS8, 9) 0x12 = VHT80 (1SS, MCS0, 1, 2) 0x13 = VHT80 (1SS, MCS3, 4) 0x14 = VHT80 (1SS, MCS5, 6, 7) 0x15 = VHT80 (1SS, MCS8, 9) 0x16 = VHT2_20 (2SS, MCS8) 0x17 = VHT2_40 (2SS, MCS8, 9) 0x18 = VHT2_80 (2SS, MCS0, 1, 2) 0x19 = VHT2_80 (2SS, MCS3, 4) 0x1A = VHT2_80 (2SS, MCS5, 6, 7) 0x1B = VHT2_80 (2SS, MCS8, 9) 0x1C = HE20 (1SS, MCS8, 9) 0x1D = HE20 (1SS, MCS10,11) 0x1E = HE40 (1SS, MCS10,11) 0x1F = HE80 (1SS, MCS10,11) 0x20 = HE20 (2SS, MCS8, 9) 0x21 = HE20 (2SS, MCS10,11) 0x22 = HE40 (2SS, MCS10,11) 0x23 = HE80 (2SS, MCS10,11) **NOTE:** Depending on firmware implementation, some power groups will be merged.

Field Name	Type	Description
Power1	UINT8	Tx power 1 (dBm)
...	...	...
ModulationGroupn	UINT8	Modulation group n
Powern	UINT8	Tx power n (dBm)

10.4.57 MrvlIETypes_RxBASync_t

This TLV configures the Rx BA synchronization. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0199
Length	UINT16	Length of payload
MacAddress	UINT8[6]	MAC address
TID	UINT8	TID
Reserved	UINT8	Reserved
SSN	UINT16	Starting sequence number of bitmap stream
Len	UINT16	Length of bitmap
Bitmap	UINT8[0–16]	Bitmap to indicate whether packet #N is received and filtered out by firmware Example: N = SSN + X, where 0 <= X < 128 Then Bit[Z] in byte Y in the Bitmap defined as UINT8[0–16] will be set to indicate that packet with sequence number N is received and filtered, where Y is quotient of (X/8) and Z the remainder of (X/8).

10.4.58 MrvlCoalesceRule_t

This TLV configures the coalesce rule. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x019A
Length	UINT16	Length of payload
Number_Of_Fields	UINT8	Number of coalesce filter field parameters (maximum 4)
Pkt_Type	UINT8	Specific which type of receive packets need to coalesce 00 = reserved 01 = unicast 10 = multicast 11 = broadcast
Max_Coalescing_Delay	UINT16	Specific maximum coalesce packets delay (ms) allowed
Field_Params[]	Coalesce_filter_field_param_t	Coalesce filter field parameter array

10.4.58.1 Coalesce_filter_field_param_t

The coalesce filter field parameter format:

Field Name	Type	Description
Operation	UINT8	0x80 = equal 0x81 = not equal
Operand_len	UINT8	Filter pattern length (maximum 4 bytes)
Offset	UINT16	Offset from incoming MAC packet Figure 16 shows the MAC packet format with offset starting from Packet Length.
Operand_byte_stream[4]	UINT8[4]	Filter pattern (maximum 4 bytes)

Figure 16: MAC Packet Format

Bit[31]	Bit[0]
Frame Control[15:0]	Packet Length[15:0]
Address1[15:0]	Duration/ID[15:0]
Address1[47:16]	
Address2[31:0]	
Address3[15:0]	Address2[47:32]
Address3[47:16]	
Address4[15:0]	Sequence Control[15:0]
Address4[47:16]	
802.11 Frame Body	

10.4.59 MrvlIETypes_KeyParamSet_v2_t

This TLV configures the key parameters. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x019C</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td>MacAddress</td><td>UINT8[6]</td><td>MAC address</td></tr>
<tr><td>KeyIdx</td><td>UINT8</td><td>Key index</td></tr>
<tr><td>KeyType</td><td>UINT8</td><td>Type of key
0x0 = WEP
0x1 = TKIP
0x2 = AES
0x3 = WAPI
0x4 = AES CMAC</td></tr>
<tr><td>KeyInfo</td><td>UINT16</td><td>Key information
<table>
<tr><th>Bits</th><th>Field Name</th></tr>
<tr><td>15:11</td><td>Reserved</td></tr>
<tr><td>10</td><td>isIGTKKey
Used as IGTK</td></tr>
<tr><td>9:6</td><td>Reserved</td></tr>
<tr><td>5</td><td>Rx
Used for Rx packet</td></tr>
<tr><td>4</td><td>Tx
Used for Tx packet</td></tr>
<tr><td>3</td><td>isDefaultKey
Used as default key</td></tr>
<tr><td>2</td><td>isKeyEnabled</td></tr>
<tr><td>1</td><td>isUnicastKey
Used as PTK</td></tr>
<tr><td>0</td><td>isMulticastKey
Used as GTK</td></tr>
</table></td></tr>
<tr><td>KeyParam</td><td>KeyParam_t</td><td>Key parameters</td></tr>
</table>

10.4.59.1 WEP Key Type

The KeyParam field in WEP key type is:

Field Name	Type	Description
KeyLen	UINT16	0x5 = 40-bits WEP 0xD = 104-bits WEP
WepKey	UINT8[]	WEP encryption/decryption key

10.4.59.2 TKIP Key Type

The KeyParam field in TKIP key type is:

Field Name	Type	Description
PN	UINT8[8]	GTK peer side Rx packet number Bits[7:6]: Reserved (set to 0) Bits[5:0]: 5 octets
KeyLen	UINT16	Length of key (32)
TkipKey	UINT8[16]	TKIP encryption/decryption key
TkipTxMicKey	UINT8[8]	TKIP transmission MIC key
TkipRxMicKey	UINT8[8]	TKIP receive MIC key

10.4.59.3 AES Key Type

The KeyParam field in AES key type is:

Field Name	Type	Description
PN	UINT8[8]	GTK peer side Rx packet number Bits[7:6]: Reserved (set to 0)
KeyLen	UINT16	Length of key = 0x10
AesKey	UINT8[]	AES encryption/decryption key

10.4.59.4 WAPI Key Type

The KeyParam field in WAPI key type is:

Field Name	Type	Description
PN	UINT8[16]	GTK peer side Rx packet number
KeyLen	UINT16	Length of key = 0x20
WapiKey	UINT8[16]	WAPI encryption/decryption key
micKey	UINT8[16]	WAPI MIC key

10.4.59.5 AES CMAC Key Type

The KeyParam field in AES CMAC key type is:

Field Name	Type	Description
IPN	UINT8[8]	IGTK peer side Rx packet number [0 to 5] octets Bytes[7:6]: Reserved (set to 0)
KeyLen	UINT16	Length of key = 0x10
AesCmacKey	UINT8[]	AES CMAC encryption/decryption key

10.4.60 MrvlIETypes_Multi_Chan_Info

This TLV configures the multi-channel information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01B7
Length	UINT16	Length of payload
Status	UINT16	Multi-channel operation status 0 = inactive 1 = active
TLV List	TLV	0 or more MrvlIETypes_Chan_Group_Info TLV(s)

10.4.61 MrvlIETypes_Chan_Group_Info

This TLV configures the multi-channel group information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01B8
Length	UINT16	Length of payload
ChanGroupId	UINT8	Identifier for this group
ChanBufferWeight	UINT8	Sum of individual buffer weights of all active interfaces in this group
ChanBandInfo	ChanBandInfo_t	Channel number and band information
ChannelTime	UINT32	Maximum channel time (μs) among all active interfaces in this group
Reserved	UINT32	Reserved (set to 0)
IntfID	UINT8	Identifier for multi-channel operation In case of USB, it indicates the EP number used for data packet Tx for this group.
NumIntf	UINT8	Number of active interfaces (N) present in this group
BssTypeNum0	UINT8	Upper Nibble for BssType and Low Nibble for BssNum for 1st interface in this group
BssTypeNum1	UINT8	Upper Nibble for BssType and Low Nibble for BssNum for 2nd interface in this group
...	...	...
BssTypeNumN	UINT8	Upper Nibble for BssType and Low Nibble for BssNum for Nth interface in this group

10.4.61.1 ChanBandInfo_t

The channel band information field parameter format:

Field Name	Type	Description
BandConfig	UINT8	Bits[7:6]: reserved Bits[5:4]: secondary channel offset 00 = secondary channel none 01 = secondary channel above 10 = reserved 11 = secondary channel below Bits[3:2]: channel width 00 = 20 MHz channel width 01 = 10 MHz channel width 10 = 40 MHz 11 = reserved Bits[1:0]: band information 00 = 2.4 GHz 01 = 5 GHz 10 = 4 GHz 11 = reserved
ChanNum	UINT8	Channel number of specified band

10.4.62 MRVL_SCAN_CHAN_GAP_TLV_ID

This TLV configures the scan channel gap time. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01C5
Length	UINT16	Length of payload
uChanScanGap	UINT16	Gap time between scan channels set (µs)

10.4.63 MrvlIEChannelStats_t

This TLV specifies the channel statistics. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01C6
Length	UINT16	Length of payload
chanStat	UINT8[]	Channel statistics

10.4.64 MrvlIETypes_ApBTCoexScoWinSize_t

This TLV configures the Bluetooth coexistence aggregation windows size. It has the following format:

Field Name	Type	Description
Type	UINT16	(MRVL_BTCOEX_WLAGGER_WINSIZE_TLV_ID) Type ID = 0x01CA
Length	UINT16	Length of payload
customWinSize	UINT8	Window size 0 = restore default window size 1 = use below window size
txWinSize	UINT8	Tx windows size
rxWinSize	UINT8	Rx windows size
Reserved	UINT8	Reserved

10.4.65 MrvlIETypes_BTCoexScanTime_t

This TLV configures the Bluetooth coexistence scan time setting. It has the following format:

Field Name	Type	Description
Type	UINT16	(MRVL_BTCOEX_WLSCANTIME_TLV_ID) Type ID = 0x01CB
Length	UINT16	Length of payload
state	UINT8	State 0 = disable 1 = enable
Reserved	UINT8	Reserved
minScanTime	UINT8	Minimum scan time

10.4.66 MrvlIETypes_SMCAddrRange_t

This TLV configures the valid range of multicast packets which would be forwarded to the host/driver. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x01CC</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload (8 bytes)</td></tr>
<tr><td>StartAddr</td><td>UINT8[6]</td><td>Starting address of valid multicast address range that would be forwarded to host/driver</td></tr>
<tr><td>EndAddr</td><td>UINT8[6]</td><td>Ending address of valid multicast address range that would be forwarded to host/driver</td></tr>
<tr><td>FilterTpye</td><td>UINT16</td><td>Filter type bitmap
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:2</td><td>Reserved (set to 0)</td></tr>
<tr><td>1</td><td>Rx frames from STAs</td></tr>
<tr><td>0</td><td>Rx frames from APs</td></tr>
</table></td></tr>
</table>

10.4.67 MrvlIEBSSScanType_t

This TLV contains BSS mode to support 0x0107 ext_scan command. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01CE
Length	UINT16	Length of payload
Bss_mode	UINT8	BSS scan type 0x1 = BSS_INFRASTRUCTURE 0x2 = BSS_INDEPENDENT 0x3 = BSS_ANY

10.4.68 MrvlIETypes_Eapol_Pkt_t

This TLV contains GTK frame in buffer, used for set GTK in firmware with embedded supplicant. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01CF
Length	UINT16	Length of payload
pkt_buf	UINT8[]	EAPOL packet buffer

10.4.69 IEEEtypes_WMM_ParamElement_t

This TLV sets AP WMM parameters. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D0
Length	UINT16	Length of payload
OuiType	UINT8[4]	OUI type: 00:50:F2:02
OuiSubType	UINT8	OUI subtype: 01
Version	UINT8	Version
QosInfo	IEEEtypes_QOS_Info_t	AP QoS information and STA QoS information
Reserved	UINT8	Reserved
AcParams	IEEEtypes_WMM_AC_Parameters_t[4]	WMM AC parameters

10.4.70 MrvlIETypes_SMCFrameFilter_t

This TLV configures various frame filters to forward matching filter frames to the host/driver. Multiple FrameType and DestType fields are allowed in this TLV. Length will indicate how many FrameType and DestType have been embedded in this TLV. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x01D1</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td>DestType</td><td>UINT8</td><td>More than 1 destination types are allowed
For example, Unicast + Broadcast or Unicast + Broadcast + Multicast.
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:3</td><td>Reserved</td></tr>
<tr><td>2</td><td>Multicast</td></tr>
<tr><td>1</td><td>Broadcast</td></tr>
<tr><td>0</td><td>Unicast</td></tr>
</table></td></tr>
<tr><td>FrameType</td><td>UINT8</td><td>Bits[7:6]: Reserved
Bits[5:2]: Subtype
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><th colspan="2">These maps with subtype filed of IEEE 802.11 management frames</th></tr>
<tr><td>8</td><td>Beacon</td></tr>
<tr><td>5</td><td>Probe response</td></tr>
<tr><td>4</td><td>Probe request</td></tr>
<tr><td>3</td><td>Reassociation response</td></tr>
<tr><td>2</td><td>Reassociation request</td></tr>
<tr><td>1</td><td>Association response</td></tr>
<tr><td>0</td><td>Association request</td></tr>
</table>
Bits[1:0]: Type
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><th colspan="2">These maps with type filed of IEEE 802.11 frames</th></tr>
<tr><td>00</td><td>Management</td></tr>
<tr><td>01</td><td>Control</td></tr>
<tr><td>10</td><td>Data</td></tr>
</table>
Subtypes for Data and Control types are currently ignored.</td></tr>
</table>

10.4.71 NAN_MASTER_OPERATION_MODE_TLV_ID

This TLV is used to program NAN 2.4 GHz or 5 GHz operations. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D2
Length	UINT16	1
OperationMode	UINT8	Operational Mode 0 = 5 GHz disabled 1 = 5 GHz enabled

10.4.72 NAN_DISC_BEACON_PERIOD_TLV_ID

This TLV is used to program beacon discovery period. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D3
Length	UINT16	2
DiscoveryBcnPeriod	UINT16	Beacon discovery period

10.4.73 NAN_SCAN_DWELL_TIME_TLV_ID

This TLV is used to program NAN scan dwell time. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D4
Length	UINT16	2
ScanDwellTime	UINT16	Scan dwell time on each channel

10.4.74 NAN_MASTER_PREFERENCE_TLV_ID

This TLV is used to get and set NAN master preference attribute. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D5
Length	UINT16	2
MasterPreference	UINT8	Information that is used to indicate a NAN device's preference to serve as the role of Master, with a larger value indicating a higher preference.

10.4.75 NAN_SDFRAME_TRS_TLV_ID

This TLV is used to get and set delay parameter used in sending service discovery frames in the contention mitigation window. If contention delay is configured to 0 then contention mitigation is enabled, otherwise disabled. Delay parameter is used to delay sending service discovery frame with specified time. Delay is from the start of DWStart boundary. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D6
Length	UINT16	4
Delay	UINT32	Time (ms) used to delay sending service discovery frames

10.4.76 MrvlIETypes_NAN_FAM_ATTRIBUTE_ID_TLV_ID

This TLV indicates the set of attributes for availability of NAN device for NAN Data Cluster (NDL) Schedule, post NDP Handshake. The values set in this TLV will be applicable to 1 or multiple DW periods. Firmware will reset this TLV at the end of the DW if repeat entry count reaches 0. Otherwise, same availability is continued until it is reset from the host.It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D7
Length	UINT16	Length of buffer
Data	UINT8[]	Data[0] to Data[Length - 1]

There can be multiple NDL Schedule availability attributes in this TLV. Each attribute has a structure of *type* to *AvailabilityEntry* (see Table 85). Each further availability attribute is followed by another attribute of structure *IntervalLen* to *AvailabilityEntry*.

The *Data* field of this TLV contains 1 or further availability attributes. The number of further availability attributes present in the TLV is *Length*/12.

Table 85: Further Availability Time Slot Attribute Format

Field	Type	Description
type	UINT8	Type 00 = FAV 01 = NDL 10 = RANGE (NAN ranging)
IntervalLen	UINT8	Length of interval Valid values: 16 TUs, 32 TUs, 64 TUs
repeatEntry	UINT8	Number of repeat attributes 0 = further availability attribute applicable forever n = further availability attribute applicable for n DWs
period	UINT8	Periodicity of availability entry Integer multiple of FAV window size.
offset	UINT8	Offset from DW0 Integer multiple of FAV window size.
operatingClass	UINT8	Operating class attribute as per IEEE 802.11-2012[3]
operatingChannel	UINT8	Channel number when device is available
AvailabilityEntry	UINT32	Availability for map ID

10.4.77 NAN_OPERATING_CHANNEL_TLV_ID

This TLV is used to get and set operating channel parameter used in NAN operation. Possible values of this parameters are 6 and 149. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D8
Length	UINT16	2
social2_4GChan	UINT8	Social channel in 2.4 GHz
5GChan	UINT8	Social channel in 5 GHz

10.4.78 NAN_WARMUP_TIME_TLV_ID

This TLV is used to get and set NAN start warmup timer. If warmup time is not configured then default warmup time is taken which is 12 seconds. Host can program warmup start prior to NAN_start (in seconds). If programmed after NAN_start then it will not take effect. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01D9
Length	UINT16	2
WarmupTime	UINT16	Time (seconds) used to specify NAN warmup time

10.4.79 MrvlIETypes_NAN_UNALIGNED_SCHED_t

This TLV is used to indicate to firmware about the parameters of unaligned schedule to follow. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01DB
Length	UINT16	47
startTime	UINT32	Starting time of first indicated Unaligned Window (ULW) expressed in terms of lower 4-bytes of NAN TSF
duration	UINT32	Duration for each ULW (µs)
period	UINT32	Time between consecutive ULWs (µs)
countDown	UINT8	Number of indicated ULWs 0x00 = remaining ULWs are canceled 0xFF = ULWs does not end until next schedule update
overwrite	UINT8	Whether unaligned schedule should take precedence over all NAN Availability attributes
blacklistValid	UINT8	If blacklist channel list is being provided
channelBlacklist	UINT32[8]	List of blacklist channels

10.4.80 MrvlIETypes_NAN_PMF_AND_SECURE_DATA_PATH_t

This TLV is used to indicate to firmware whether to enable management frame protection or not. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01DD
Length	UINT16	1
isPmfRequired	UINT8	Indicates if PMF is supported If a value is set, management frame encryption, such as NAN Data Path (NDP) negotiation frames, NDP security negotiation frames, and so on is enabled.

10.4.81 NeighborRprtTlv_t

This TLV is used to report Neighbor information to host. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01DE
Length	UINT16	Bytes of payload
Bssid	IEEEtypes_MacAddr_t	BSS MAC address
BssInfo	IEEEtypes_BSSID_Info_t	BSS information
regulatoryClass	UINT8	Regulatory class
channelNumber	UINT8	Channel
phyType	IEEEtypes_PhyType_e	802.11 IEEE PHY types
subElem	UINT8	Sub element

10.4.82 MRVL_BSS_TRANSITION_REQ_TLV_ID

This TLV is used to send 802.11v Wireless Network Management Action dialogToken. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01DF
Length	UINT16	Bytes of payload
Value	UINT8[]	Value of dialogToken

10.4.83 MrvlIETypes_BridgeParamSet_t

This TLV configures the Bridge parameter. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01E0
Length	UINT16	Length of payload
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID of external AP
Passphrase	MrvlIETypes_Passphrase_t	Passphrase information of external AP
SSIDParamSet	MrvlIETypes_SSIDParamSet_t	SSID of Bridge in-AP
Passphrase	MrvlIETypes_Passphrase_t	Passphtase information of Bridge in-AP

10.4.84 MrvlIETypes_RSSI_NF_t

This TLV specifies the 802.11 received signal. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01E5
Length	UINT16	Length of payload
PathID	UINT16	Bits[7:2]: Reserved Bits[1:0]: 00 = reserved 01 = path A 10 = path B 11 = combined signal of path A and B
DataRSSIlast	SINT16	Last data RSSI (dBm)
DataNFlast	SINT16	Last data NF (dBm)
DataRSSIAvg	SINT16	Averaging data RSSI (dBm)
DataNFAvg	SINT16	Averaging data NF (dBm)
BcnRSSIlast	SINT16	Last beacon RSSI (dBm)
BcnNFlast	SINT16	Last beacon NF (dBm)
BcnRSSIAvg	SINT16	Averaging beacon RSSI (dBm)
BcnNFAvg	SINT16	Averaging beacon NF (dBm)

10.4.85 MrvlIETypes_IPv6AddrParamSet_t

This TLV takes IPv6 address for router advertisement. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01E6
Length	UINT16	Bytes of payload
ipv6AddrSet	UINT8[16]	IPv6 internet address

10.4.86 MRVL_WIFI_PROBERSP_ONLY_TLV_ID

This TLV is used in scan command for WPS feature. It will take probe response as scan result only (not beacon frame). It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01E9
Length	UINT16	Bytes of payload
Value	UINT8	Value It will be set true when driver detected the WPS-IE in probe-request.

10.4.87 MrvlIETypes_CHANATTRCFG_t

This TLV is used to indicate the channel attribute configuration. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x01ED</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td colspan="3">Multiple instances (up to 159) of the following fields, 1 for each channel.</td></tr>
<tr><td>ChanNum</td><td>UINT8</td><td>Channel number</td></tr>
<tr><td>ChanAttrs</td><td>UINT8</td><td>Band configuration type
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>7</td><td>Channel disabled
0 = enable
1 = disable</td></tr>
<tr><td>6:5</td><td>Reserved</td></tr>
<tr><td>4</td><td>160 MHz bandwidth disabled
0 = enabled
1 = disabled</td></tr>
<tr><td>3</td><td>80 MHz bandwidth disabled
0 = enabled
1 = disabled</td></tr>
<tr><td>2</td><td>40 MHz bandwidth disabled
0 = enabled
1 = disabled</td></tr>
<tr><td>1</td><td>Dynamic Frequency Select (DFS) channel
0 = not DFS channel
1 = DFS channel</td></tr>
<tr><td>0</td><td>Passive scan allowed
0 = active scan allowed
1 = only passive scan allowed</td></tr>
</table></td></tr>
</table>

10.4.88 MrvlIETypes_REGIONINFO_t

This TLV is used to set the region information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01EE
Length	UINT16	Length of payload
CountryCode	UINT16	ISO 3166-1 alpha-2 country code
regionCode	UINT8	Region code 0x00 = null region 0x01 = Federal Communications Commission (FCC) region 0x02 = European Telecommunication Standards Institute (ETSI) region 0x03 = Message Integrity Code (MIC) region 0xFF = other region
environment	UINT8	Environment 0x20 = all environments ASCII ‘I’ = indoor ASCII ‘O’ = outdoor
Reserved	UINT16	Reserved

10.4.89 MrvlIETypes_FAST_LINK_INTERFACE_BASIC_INFO_t

This TLV is used to set the fast link interface basic information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01F2
Length	UINT16	Length of payload
status	UINT8	Status If STA type: 00 = not connected 01 = connected If uAP type: 00 = not started 01 = idle 10 = active

10.4.90 MrvlIETypes_FAST_LINK_EXT_STATION_INFO_t

This TLV is used to set the fast link external station information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01F3
Length	UINT16	Length of payload
mac_addr	UINT8[6]	Connected STA MAC address
is_wmm_enabled	UINT8	WMM enabled flag
Ht_capability	IEEEtypes_HTCap_t	Connected STA HT capability
Vht_capability	IEEEtypes_VHTCap_t	Connected STA VHT capability

10.4.91 MrvlIETypes_FAST_LINK_DATA_PORT_INFO_t

This TLV is used to set the fast link data port information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01F4
Length	UINT16	Length of payload
Mp_rd_bitmap	UINT32	Read bitmap
Mp_wr_bitmap	UINT32	Write bitmap
Curr_rd_bitmap	UINT8	Current read port
Curr_wr_bitmap	UINT8	Current write port

10.4.92 MrvlIETypes_EMBEDDED_DHCPD_CFG_t

This TLV is used to set the embedded DHCPD configuration. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0201
Length	UINT16	Length of payload
Host IP address	UINT32	Host IP address, network order Example: For 192.168.1.1: Addr[3] = 1 Addr[2] = 1 Addr[1] = 168 Addr[0] = 192
Subnet mask	UINT32	Network order
DHCP start IP address	UINT32	First address in IP pool, network order
Limit count	UINT8	Size of address pool Range: 0 to 6, default = 5.
Lease time	UINT32	Lease time Default = 43200s (12 hours)

10.4.93 MrvlIETypes_POWERTABLE_t

This TLV is used to set the power table. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0206
Length	UINT16	Length of payload
TlvBuffer	UINT8	TLV buffer First byte of variable length buffer (length indicated by Length field) containing 1 or more compressed Tx/specific absorption rate power tables.

10.4.94 MrvlIETypes_MULTI_CHAN_TIME_SLICE_t

This TLV is used to set the multi channel time slice configuration. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0207
Length	UINT16	Length of payload
ChanIdx	UINT16	Channel index of the setting Bits[15:2]: reserved Bit[1]: channel index 1 Bit[0]: channel index 0
ChanTime	UINT8	Channel time (TU) for ChanIdx for both Tx/Rx
SwitchTime	UINT8	Channel switch time (TU) for ChanIdx
UndozeTime	UINT8	Un-doze time (TU) for ChanIdx which should be less than SwitchTime
Mode	UINT8	Mode Rx traffic control scheme when channel switch for ChanIdx. Valid only for Group Client (GC)/STA interface. 0 = PM1 (default) 1 = Null2Self

10.4.95 MrvlIETypes_FTM_SESSION_CFG_LCI_t

This TLV is used to set FTM session configuration LCI. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x020E
Length	UINT16	Length of payload
Longitude	Double	Known longitude
Latitude	Double	Known latitude
Altitude	Double	Known altitude
Latitude Uncertainty	UINT8	Known latitude uncertainty
Longitude Uncertainty	UINT8	Known longitude uncertainty
Altitude Uncertainty	UINT8	Known altitude uncertainty
Z-Info	UINT8	1 byte of additional Z information

10.4.96 MrvlIETypes_FTM_SESSION_CFG_LOCATION_CIVIC_t

This TLV is used to set the FTM session configuration for location civic. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x020F
Length	UINT16	Length of payload
Country_code	UINT16	Country code
Address_type	UINT8	Civic address type
Address_length	UINT8	Civic address length
Address	Variable	Civic address

10.4.97 MrvlIETypes_FTM_SESSION_CFG_RESPONDER_t

This TLV is used to set the FTM session configuration for responder. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0210
Length	UINT16	Length of payload
Burst_exponent	UINT8	Number of burst instance requested for FTM session
Burst_duration	UINT8	Duration of a burst instance
Min_delta_FTM	UINT8	Minimum time between consecutive FTM frames
Is_ASAP	UINT8	ASAP/non-ASAP case
Per_burst_FTM	UINT8	Number of FTMs per burst
Channel_spacing	UINT8	FTM channel spacing (HT20/40)
Burst_period	UINT16	Interval between 2 consecutive burst instances

10.4.98 MrvlIETypes_FTM_SESSION_CFG_INITIATOR_t

This TLV is used to set the FTM session configuration for initiator. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0211
Length	UINT16	Length of payload
Burst_exponent	UINT8	Indicates how many burst instances are requested for FTM session
Burst_duration	UINT8	Duration of a burst instances
Min_delta_FTM	UINT8	Minimum time between consecutive FTM frames
Is_ASAP	UINT8	ASAP/non-ASAP case
Per_Burst_FTM	UINT8	Number of FTMs per burst
Channel_spacing	UINT8	FTM channel spacing Example: HT20, HT40, and so on.
Burst_period	UINT16	Interval between 2 consecutive burst instances
Civic_Req	UINT8	Indicates if civic location is present
Lci_Req	UINT8	Indicates if LCI is present

10.4.99 MrvlIETypes_11P_MODE_CONFIG_t

This TLV is used to set the 802.11p mode configuration. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0212
Length	UINT16	Length of payload
Operation	UINT8	802.11p operation 0 = disable 1 = enable
Channel	UINT8	802.11p operation starting channel Will be overwritten by the channel provided along with Tx.
Bandwidth	UINT8	802.11p operation starting bandwidth (10/20 MHz) Will be overwritten by the non-0 bandwidth provided along with Tx.
Tx_datarate	UINT8	802.11 interface default rate (in 0.5 Mbps unit) for all transmission on 802.11p interface Will be overwritten by the non-0 rate provided along with Tx,
Power	SINT8	Default power (dBm) for all transmission on 802.11p interface Will be overwritten by the non-0 power provided along with Tx.
Monitor	UINT8	Enable monitor mode to capture all packets in a particular channel 0 = disable 1 = enable
UseLPD	UINT8	User LLC Protocol Discrimination (LPD)/EtherType Protocol Discrimination (EPD) Ethernet packet descriptor for all packets send on OCB interface 0 = EPD packet header is attached to all packets sent on OCB interface 1 = LPD packet header is attached to all packets sent on OCB interface
Protocol_Ctrl_len	UINT8	Number of EtherType in Protocol_Ctrl field
Protocol_Ctrl	UINT16[]	Array of EtherType for which per-packet control information in Tx/Rx is desired Driver attaches and consumes the per-packet control information for non-0 EtherTypes in this array. Example: If the array contains {0x88DC, 0x8947}, the driver will attach and consume per-packet control information for all WSMP and Geonet packets.

10.4.100 MrvlIETypes_11P_TDM_CONFIG_t

This TLV is used to set the 802.11p TDM configuration. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0213</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td>Operation</td><td>UINT8</td><td>Time Division Multiplexing (TDM) operation
0 = cancel any ongoing TDM schedule
1 = enable TDM operation</td></tr>
<tr><td>Repeat_count</td><td>UINT16</td><td>Number of times to repeat following schedule
0xFFFF = repeat until canceled</td></tr>
<tr><td>Num_tdm</td><td>UINT8</td><td>Number of channel-time tuples in body</td></tr>
<tr><td>Chan_time tuples</td><td>chan_time</td><td>Channel time tuples
It has the following format::
<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Chan</td><td>UINT8</td><td>Channel number</td></tr>
<tr><td>Bandwidth</td><td>UINT8</td><td>Bandwidth (10 MHz/20 MHz)
0 = default bandwidth configured during initialization as part of MrvlIETypes_11P_MODE_CONFIG_t TLV</td></tr>
<tr><td>Time_in_ms</td><td>UINT16</td><td>Total time to be present on this channel (ms)</td></tr>
</table></td></tr>
</table>

10.4.101 MrvlIETypes_11P_RADIO_COUNTERS_t

This TLV is used to set the 802.11p radio counters. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0214
Length	UINT16	Length of payload
Tx_frame	UINT32	Number of Tx frames
Mcast_tx_frame	UINT32	Number of Tx multicast frames
Failed	UINT32	Failed Tx frames
Retry	UINT32	Retry count
Multi_retry	UINT32	Multiple retry count
Frame_dup	UINT32	Duplicate frame count
Ack_failure	UINT32	Acknowledgment failure count
Rx_frag	UINT32	Rx fragmentation count
Mcast_rx_frame	UINT32	Multicast Rx frames
Fcs_error	UINT32	Number of Frame Check Sequence (FCS) errors
Tx_frag	UINT32	Number of Tx fragments

10.4.102 MrvlIETypes_RTT_RESPONDER_INFO_ID_t

This TLV is used to set the RTT responder information ID. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0228
Length	UINT16	Length of payload
channel	UINT8	Channel number
bandcfg	Band_Config_t	Band configuration
preamble	UINT8	Preamble 0x01 = WIFI_RTT_PREAMBLE_LEGACY 0x02 = WIFI_RTT_PREAMBLE_HT 0x04 = WIFI_RTT_PREAMBLE_VHT

10.4.103 MrvlIETypes_UAP_STA_FLAGS_t

This TLV is used to set the uAP/STA flags. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0239</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload</td></tr>
<tr><td>Sta_flag</td><td>UINT32</td><td>STA flag
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>31:7</td><td>Reserved</td></tr>
<tr><td>6</td><td>STA_FLAG_ASSOCIATED
STA is associated</td></tr>
<tr><td>5</td><td>STA_FLAG_TDLS_PEER
STA is a TDLS peer</td></tr>
<tr><td>4</td><td>STA_FLAG_AUTHENTICATED
STA is authenticated</td></tr>
<tr><td>3</td><td>STA_FLAG_MFP
STA uses management frame protection</td></tr>
<tr><td>2</td><td>STA_FLAG_WME
STA is WME/QoS capable</td></tr>
<tr><td>1</td><td>STA_FLAG_SHORT_PREAMBLE
STA is capable of receiving frames with short barker preamble</td></tr>
<tr><td>0</td><td>STA_FLAG_AUTHORIZED
STA is authorized (802.1x)</td></tr>
</table></td></tr>
</table>

10.4.104 MrvlIETypes_DMCS_STATUS_t

This TLV is used to set the DMCS status. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x023A
Length	UINT16	Length of payload
MappingPolicy	UINT8	Mapping policy 0 = direct mapping 1 = dynamic mapping
radioStatus[MAX_NUM_MAC]	dmcsStatus_t	DMCS radio status 1 = single MAC 2 = dual MAC

Table 86: dmcsStatus_t

Field Name	Type	Description
RadioID	UINT8	Radio ID 0 = radio 0 1 = radio 1
runningMode	UINT8	Running mode 0x0 = idle 0x1 = Dual Band Concurrent (DBC) 0x2 = DRCS
chanStatus[2]	dmcsChanStatus_t	Radio channel status Currently maximum channel number on 1 MAC is 2 for DRCS purpose. If only 1 channel is active on this radio, chanStatus[1] will be set to 0.

Table 87: dmcsChanStatus_t

Field Name	Type	Description
channel	UINT8	Channel number
apCount	UINT8	Number of AP on this channel
staCount	UINT8	Number of STA on this channel

10.4.105 MrvlIETypes_ZERO_DFS_t

This TLV is used to set the 0DFS interface. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x023B
Length	UINT16	Length of payload
Enable/Disable	UINT8	0DFS enable/disable 0 = disable 1 = enable

10.4.106 MrvlIETypes_ChanRuPwrCfg _t

This TLV configures the Channel RU Power. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0244
Length	UINT16	Length of payload
StartFreq	UINT8	Starting frequency of the band for this channel • 2407, 2414, or 2400 for 2.4 GHz • 4000 • 5000
ChanWidth	UINT8	Channel width. Valid values: • 20 • 40 • 80
ChanNum	UINT8	Channel number
RU power	SINT8*6	Power limit in dBm. Range from -64 dBm to +63 dBm in steps of 1 dBm Array of 6 values (RU=26,52,106,242,484,996)

10.4.107 MRVL_DOT11AX_OBSS_PD_OFFSET_TLV_ID

This TLV is used to set the OBSS PD offset value. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0243
Length	UINT16	Length of payload
SRG offset	UINT8	SRG offset value
NON-SRG offset	UINT8	Non-SRG offset value

10.4.108 MRVL_DOT11AX_ENABLE_SR_TLV_ID

This TLV is used to enable/disable SR for 802.11ax. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0242
Length	UINT16	Length of payload
Enable/disable	UINT8	SR enable/disable 0 = disable 1 = enable

10.4.109 MRVL_TX_RATE_CFG_TLV_ID

This TLV is used to indicate the Tx packet parameters like preamble type, STBC, GI duration and HE-LTF sizes for 802.11ax packets. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x023F
Length	UINT16	Length of payload

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>TxRateCfg</td><td>UINT16</td><td>
<table>
<tr><th>Bit</th><th>Description</th></tr>
<tr><td>15:14</td><td>Reserved</td></tr>
<tr><td>13:12</td><td>Max PE (Packet Extension)
0 - 0 us
1 - 8 us
2 - 16 us</td></tr>
<tr><td>11:10</td><td>Doppler
Bit[11:10]
00: Doppler0
01: Doppler 1 with Mma =10
10: Doppler 1 with Mma =20</td></tr>
<tr><td>9</td><td>AdvCoding
0: BCC
1: LDPC</td></tr>
<tr><td>8</td><td>DCM
0: no DCM
1: DCM used</td></tr>
<tr><td>7</td><td>STBC
0: no STBC
1: STBC</td></tr>
<tr><td>6:5</td><td>Short Guard Interval
For legacy: not used
For 802.11n: 00 = normal, 01 =shortGI, 10/11 = reserved
For 802.11ac: SGI map to VHT-SIG-A2[0]
VHT-SIG-A2[1] is set to 1 if short guard interval is used and NSYM mod 10 = 9, otherwise set to 0.
For 802.11ax:
00 = 1xHELTF+GI0.8usec
01 = 2xHELTF+GI0.8usec
10 = 2xHELTF+GI1.6usec
11 = 4xHELTF+GI0.8 usec if both DCM and STBC are 1
11 = 4xHELTF+GI3.2 usec otherwise
Continues...</td></tr>
</table>
</td></tr>
</table>

Field Name	Type	Description
TxRateCfg	UINT16	Continued

Bit	Description
4:2	Bandwidth **For 802.11n and 802.11ac** traffic: bandwidth 0 = 20 MHz 1 = 40 MHz 2 = 80 MHz 3 = 160 MHz 4-7 = reserved **For legacy rate**: BW>0 implies non-HT duplicates. **For HE SU PPDU**: 0 = 20 MHz 1 = 40 MHz 2 = 80 MHz 3 = 160 MHz 4-7 = reserved **For HE ER SU PPDU**: 0 = 242-tone RU 1 = upper frequency 106 tone RU within the primary 20 MHz **For HE MU PPDU**: 0 = 20 MHz 1 = 40 MHz 2 = 80 MHz non-preamble puncturing mode 3 = 160 MHz and 80+80 MHz non-preamble. 4 = for preamble puncturing in 80 MHz, where in the preamble only the secondary 20 MHz is punctured. 5 = for preamble puncturing in 80 MHz, where in the preamble only one of the two 20 MHz sub channels in the secondary 40 MHz is punctured. 6 = for preamble puncturing in 160 MHz or 80 MHz + 80 MHz, where in the primary 80 MHz of the preamble only the secondary 20 MHz is punctured. 7 = for preamble puncturing in 160 MHz or 80 MHz + 80 MHz, where in the primary 80 MHz of the preamble the primary 40 MHz is present.
1:0	Preamble type **For legacy 802.11b**: preamble type 00 = long 01 = short 10/11 = reserved **For legacy 802.11g:** reserved **For 802.11n**: Green field PPDU indicator 00 = HT-mix 01 = HT-GF 10/11 = reserved. **For 802.11ac**: reserved. **For 802.11ax**: 00 = HE-SU 01 = HE-EXT-SU 10 = HE-MU 11 = HE trigger based

10.5 Micro AP TLV

Table 88: Micro AP Formats

Table 88: Micro AP Formats (Continued)

Format	Description	Page
Section 10.5.33, MrvlIETypes_Mgmt_Passthru_Mask_t	Allows management frame pass-through	page 392
Section 10.5.34, MrvlIETypes_PTK_Timeout_t	Specifies pairwise key handshake timeout value for WPA/WPA2	page 392
Section 10.5.35, MrvlIETypes_PTK_Retry_t	Specifies pairwise key handshake retry count	page 393
Section 10.5.36, MrvlIETypes_GTK_Timeout_t	Specifies group key handshake timeout value	page 393
Section 10.5.37, MrvlIETypes_GTK_Retry_t	Specifies group key handshake retry count	page 393
Section 10.5.38, MrvlIETypes_PS_STA_Ageout_t	Specifies age-out value for STA in PS state	page 394
Section 10.5.39, MrvlIETypes_PTK_Cipher_t	Specifies pairwise cipher keys for specified protocol	page 394
Section 10.5.40, MrvlIETypes_GTK_Cipher_t	Specifies groupwise cipher key type	page 394
Section 10.5.41, MrvlIETypes_ApBSS_Status_t	Queries firmware BSS status	page 395
Section 10.5.42, MrvlIETypes_Tx_Data_Pause_t	Specifies pause Tx state	page 395
Section 10.5.43, MrvlIETypes_Sticky_Tim_Config_t	Gets/sets sticky TIM feature parameters	page 395
Section 10.5.44, MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t	Communicates MAC address of associated STA for which to activate/deactivate sticky TIM	page 396
Section 10.5.45, MrvlIETypes_2040_BSS_Coex_Ctrl_t	Enable/disable 20/40 MHz BSS coexistence schemes	page 396
Section 10.5.46, MrvlIETypes_Api_Ver_Info_t	Populates the version info corresponding to a specific API version ID	page 397
Section 10.5.47, MrvlIETypes_Max_Supported_STA_Count_t	Stores information on the maximum number of connections supported in the P2P and AP modes respectively	page 397
Section 10.5.48, MrvlIETypes_FW_Cap_Info_t	Points to the information on the firmware capabilities and firmware extended capabilities	page 397

10.5.1 MrvlIETypes_HT_Capability_t

This TLV is used to configure the 802.11n parameters in firmware. Because of hardware limitation, not all parameters are configurable. If all the bits of SupportedMCSSet are cleared, then 802.11n will be disabled; otherwise, 802.11n will be enabled. The interpretation of the bits is a per IEEE specification. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x002D</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Number of bytes in payload</td></tr>
<tr><td>HTCapInfo</td><td>UINT16</td><td>HT capability information
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15</td><td>L Sig TxOP protection (reserved, set to 0)</td></tr>
<tr><td>14</td><td>40 MHz intolerant (reserved, set to 0)</td></tr>
<tr><td>13</td><td>PSMP (reserved, set to 0)</td></tr>
<tr><td>12</td><td>DSSS Cck40MHz mode</td></tr>
<tr><td>11</td><td>Maximal AMSDU size (reserved, set to 0)</td></tr>
<tr><td>10</td><td>Delayed BA (reserved, set to 0)</td></tr>
<tr><td>9:8</td><td>Rx STBC (reserved, set to 0)</td></tr>
<tr><td>7</td><td>Tx STBC (reserved, set to 0)</td></tr>
<tr><td>6</td><td>Short GI 40 MHz</td></tr>
<tr><td>5</td><td>Long GI 20 MHz</td></tr>
<tr><td>4</td><td>GF preamble</td></tr>
<tr><td>3:2</td><td>MIMO power save (reserved, set to 0)</td></tr>
<tr><td>1</td><td>SuppChanWidth</td></tr>
<tr><td>0</td><td>LDPC coding (reserved, set to 0)</td></tr>
</table></td></tr>
<tr><td>AMPDUParam</td><td>UINT8</td><td>AMPDU parameter
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:5</td><td>Reserved</td></tr>
<tr><td>4:2</td><td>MPDU density</td></tr>
<tr><td>1:0</td><td>Maximum Rx AMPDU factor</td></tr>
</table></td></tr>
<tr><td>SupportedMCSSet</td><td>UINT8[16]</td><td>Supported MCS bitmap</td></tr>
<tr><td>HTExtCap</td><td>UINT16</td><td>All fields are reserved and set to 0
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>15:12</td><td>Reserved</td></tr>
<tr><td>11</td><td>RdResponder</td></tr>
<tr><td>10</td><td>HtcSupport</td></tr>
<tr><td>9:8</td><td>McsFeedback</td></tr>
<tr><td>7:3</td><td>Reserved</td></tr>
<tr><td>2:1</td><td>TransitionTime</td></tr>
<tr><td>0</td><td>PCO</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>TxBFCap</td><td>UINT32</td><td>All fields are reserved and set to 0
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>31:29</td><td>Reserved</td></tr>
<tr><td>28:27</td><td>ChanEstCapability</td></tr>
<tr><td>26:25</td><td>CSIMaxMaxBeamSupported</td></tr>
<tr><td>24:23</td><td>CompSteerMatrixBFAntennae</td></tr>
<tr><td>22:21</td><td>UcompSteerMatrixBFAntennae</td></tr>
<tr><td>20:19</td><td>CSINumMFAntennae</td></tr>
<tr><td>18:17</td><td>MinimalGrouping</td></tr>
<tr><td>16:15</td><td>ExplCompSteerMatrixFeedback</td></tr>
<tr><td>14:13</td><td>ExplUcompSteerFeedback</td></tr>
<tr><td>12:11</td><td>ExplBFCSIFeedback</td></tr>
<tr><td>10</td><td>ExplCompSteerMatrix</td></tr>
<tr><td>9</td><td>ExplUcompSteerMatrix</td></tr>
<tr><td>8</td><td>ExplCSITxBF</td></tr>
<tr><td>7:6</td><td>Calibration</td></tr>
<tr><td>5</td><td>ImplicitTxBF</td></tr>
<tr><td>4</td><td>TxNDPCap</td></tr>
<tr><td>3</td><td>RxNDPCap</td></tr>
<tr><td>2</td><td>TxStaggeredSounding</td></tr>
<tr><td>1</td><td>RxStaggeredSounding</td></tr>
<tr><td>0</td><td>TxBFCapable</td></tr>
</table></td></tr>
<tr><td>ASEL</td><td>UINT8</td><td>All fields are reserved and set to 0
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7</td><td>Reserved</td></tr>
<tr><td>6</td><td>TxSoundingPPDUs</td></tr>
<tr><td>5</td><td>RxASCapable</td></tr>
<tr><td>4</td><td>AntIndicesFeedback</td></tr>
<tr><td>3</td><td>ExplicitCSIFeedback</td></tr>
<tr><td>2</td><td>IndicesFeedbackTxAS</td></tr>
<tr><td>1</td><td>ExplCSITxASCapable</td></tr>
<tr><td>0</td><td>ASCapable</td></tr>
</table></td></tr>
</table>

10.5.2 MrvlIETypes_VHT_Capability_IE_t

This TLV describes the VHT capability information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x00BF
Length	UINT16	Length of payload (6 bytes)
VHTCapInfo	UINT32	As described in 802.11ac specifications
VHTSupportedMCSSet	UINT8[8]	As described in 802.11ac specifications

10.5.3 MrvlIETypes_VendorSpecific_t

This TLV describes the vendor specific parameters (WWM parameters). It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x00DD
Length	UINT16	Number of bytes in payload
OuiType	UINT8[4]	WMM OUI type 00:50:F2:02
OuiSubType	UINT8	OUI subtype = 01
Version	UINT8	Version = 01
QosInfo	IEEEtypes_QAP_QOS_Info_t	As defined in IEEE specification
Reserved	UINT8	Reserved, set to 0
AcParams	Parameter records for BE, BK, VI and VO	As defined in IEEE specification

10.5.4 MrvlIETypes_ApMacAddr_t

This TLV describes the MAC address for the AP, which will become the BSSID of the infrastructure network created by the AP. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x012B
Length	UINT16	Length of payload (6 bytes)
ApMacAddr	UINT8[]	AP MAC address

10.5.5 MrvlIETypes_Beacon_Period_t

This TLV describes the beacon period. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x012C
Length	UINT16	Length of payload (2 bytes)
BecaonPeriod	UINT16	Beacon period in TU (1 TU = 1024 μs)

10.5.6 MrvlIETypes_DTIM_Period_t

This TLV describes the DTIM period. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x012D
Length	UINT16	Length of payload (1 byte)
DtimPeriod	UINT8	DTIM period in beacon periods

10.5.7 MrvlIETypes_ApTx_Power_t

This TLV specifies the power level that the AP must use for transmitting packets to client stations in the BSS. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x012F
Length	UINT16	Length of payload (1 byte)
TxPower_dBm	UINT8	Transmit power level (dBm)

10.5.8 MrvlIETypes_ApBCast_SSID_Ctrl_t

This TLV enables/disables the SSID broadcast feature (also known as the hidden SSID feature). When broadcast SSID is enabled, the AP responds to probe requests from client stations that contain null SSID. When broadcast SSID is disabled, the AP

- Does not respond to probe requests that contain null SSID
- Generates beacons that contain null SSID

It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0130
Length	UINT16	Length of payload (1 byte)
BcastSsidCtl	UINT8	Broadcast SSID feature 0 = disable 1 = enable

10.5.9 MrvlIETypes_Preamble_Ctrl_t

This TLV specifies the type of preamble in use, that is, short preamble or long preamble. This is a read-only attribute and will only support GET operation for uAP. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0131
Length	UINT16	Length of payload (1 byte)
PreambleType	UINT8	Preamble type 0x01 = short preamble 0x02 = long preamble

10.5.10 MrvlIETypes_Antenna_Ctrl_t

This TLV selects the antenna. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0132
Length	UINT16	Length of payload (2 bytes)
WhichAntenna	UINT8	Antenna 0 = receive antenna 1 = transmit antenna
AntennaMode	UINT8	Antenna Mode 0x01 = antenna A 0x02 = antenna B

10.5.11 MrvlIETypes_RTS_Threshold_t

This TLV specifies the RTS threshold to be set in the AP. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0133
Length	UINT16	Length of payload (2 bytes)
RtsThreshold	UINT16	RTS threshold value

10.5.12 MrvlIETypes_TxData_Rate_t

This TLV specifies the data rate to use for unicast packet transmission. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0135
Length	UINT16	Length of payload (2 bytes)
TxDataRate	UINT16	Data rate to use for packet transmission 0 = auto rate Others = fix Tx data rate at this value (in 500 Kbps) Example: Value of 11 specifies a data rate of 5.5 Mbps.

10.5.13 MrvlIETypes_PKT_Fwd_Ctrl_t

This TLV specifies the data packet forwarding mode.

Bit[0] defines whether the firmware must forward packets from a station to another station to the host system, or just bridge it inside the Wi-Fi module. Firmware overrides this configuration to “forward all packets to host” if 802.11n is enabled in firmware.

Bits[3:1] controls BSS privacy settings which decide whether (intra/inter)-BSS unicast/broadcast packets are allowed. Depending on the value of Bit[0] and what kind of packet received, the rules for BSS privacy need to be enforced by either driver or firmware.

It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x0136</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload (1 byte)</td></tr>
<tr><td>PktFwdCtl</td><td>UINT8</td><td>Packet forwarding control
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:4</td><td>Reserved (set to 0)</td></tr>
<tr><td>3</td><td>Allow/deny inter-BSS unicast packets
0 = allow
1 = deny</td></tr>
<tr><td>2</td><td>Allow/deny intra-BSS unicast packets
0 = allow
1 = deny</td></tr>
<tr><td>1</td><td>Allow/deny intra-BSS broadcast packets
0 = allow
1 = deny</td></tr>
<tr><td>0</td><td>Intra-BSS forwarding done by (used along with Bit[1] and Bit[2] to take decision)
0 = host
1 = firmware (default)</td></tr>
</table>
If WMM is enabled in the BSS, Bit[0] setting is overridden by firmware automatically to route all received packets through host (Bit[0] = 0), at the time of BSS start.</td></tr>
</table>

Packets received with broadcast destination address:

Broadcast Intra-BSS Packet Forward		
Master Packet Forward Control	**0 = allow**	**1 = deny**
0 = host	Send to host	Send to host
1 = firmware	Send to host and also send to DS task for Tx	Send to host

Packets received with intra-BSS unicast destination address:

Unicast Intra-BSS Packet Forward		
Master Packet Forward Control	**0 = allow**	**1 = deny**
0 = host	Send to host	Send to host if packet is Rx on RxBA-link, or is an AMSDU, else drop packet in firmware
1 = firmware	Send to DS task for Tx	Drop packet in firmware

Packets received with inter-BSS unicast destination address:

Unicast Inter-BSS Packet Forward		
Master Packet Forward Control	**0 = allow**	**1 = deny**
0 = host	Send to host	Send to host if packet is Rx on RxBA-link, or is an AMSDU, else drop packet in firmware
1 = firmware	Send to host	Drop packet in firmware

10.5.14 MrvlIETypes_STA_Info_t

This TLV describes the information of a client station. Only the GET action is supported. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0137
Length	UINT16	8
MacAddress	UINT8[6]	Station MAC address
PowerMgtStatus	UINT8	Power management status 0 = active (not in power save) 1 = in power save
Rssi	SINT8	Received signal strength indication (RSSI) value (dBm)

10.5.15 MrvlIETypes_STA_MAC_Filter_t

This TLV specifies the client station filter table based on MAC address. A filter table SET operation will always overwrite the previous filter table configuration (for all parameters, that is, mode as well as table entries). It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0138
Length	UINT16	2 + (6 * count)
FilterMode	UINT8	Mode of operation 0x0 = filter table is disabled, allow any MAC 0x1 = allow MAC addresses specified in the table to associate, and block everything else (allowed list) 0x2 = block MAC addresses specified in the table from associating, and allow everything else (banned list)
Count	UINT9	Number of MAC address entries that follow (up to 16)
MacAddress	UINT8[6]	List of MAC addresses, 6 bytes per address

10.5.16 MrvlIETypes_STA_Ageout_t

This TLV specifies the timer value for aging out inactive client stations. It has the following format:

Field Name	Type	Description
Tag	UINT16	Type ID = 0x0139
Length	UINT16	Length of payload (4 bytes)
StaAgeoutTimer_100ms	UINT32	Inactive client station age out timer value (100 ms) 0 = no age out Minimum = 300 (30 s) maximum = 864000 (86400s = 1 day)

10.5.17 MrvlIETypes_WEP_Key_t

This TLV specifies the static WEP keys. Four WEP keys can be specified. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x013B
Length	UINT16	Length of payload
KeyIndex	UINT8	Index of WEP key to use (0 to 3)
IsDefault	UINT8	Transmit key index 0 = KeyIndex is not the default 1 = KeyIndex is the default transmit key
Key	UINT8[]	40 bit or 104 bit key

10.5.18 MrvlIETypes_WPA_Passphrase_t

This TLV specifies the WPA or WPA2 passphrase. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x013C
Length	UINT16	Up to 64
Passphrase	UINT8[]	Passphrase The ASCII array contains the passphrase. If length is 64, then the passphrase is treated as the actual 256-bit key value and the hash operation is not performed to derive the key value from the passphrase. If this (host computes and programs 256-bit passphrase value) is used, the embedded CPU will not have to perform the CPU-intensive computation before starting BSS and hence the BSS start process will complete faster.

10.5.19 MrvlIETypes_Encryption_Protocol_t

This TLV selects the encryption protocol. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0140
Length	UINT16	Length of payload
Protocol	UINT16	Bitmap for supported protocols Bits[15:8]: reserved Bit[7]: WAPI Bit[6]: CCKM–Support CCKM protocol Bit[5]: WPA2–Support WPA2 protocol Bit[4]: WPA-None (Ad-Hoc only) Bit[3]: WPA Bit[2]: WEP dynamic Bit[1]: WEP static Bit[0]: no RSN

10.5.20 MrvlIETypes_AKMP_t

This TLV is used to set the authentication and key management type when using WPA or WPA2 for encryption. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0141
Length	UINT16	Length of payload
KeyMgmt	UINT16	Key management bitmap Bits[7:3]: reserved Bit[2]: KEY_MGMT_NONE Bit[1]: KEY_MGMT_PSK Bit[0]: KEY_MGMT_EAP
Operation	UINT16	Operation Bits[15:2]: reserved Bit[1]: Authenticator Bit[0]: KeyExchange

10.5.21 MrvlIETypes_Cipher_t

This TLV is to be used for backward compatibility.

This TLV is used to set the key types for pairwise and groupwise keys to be used for WPA and WPA2. There cannot be more than 1 groupwise cipher even through it is defined as a bitmap. There can be more than 1 unicast ciphers. uAP firmware does not support WEP40 and WEP104 as cipher types for WPA/WPA2. Only TKIP and CCMP are valid ciphers.

It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0142
Length	UINT16	Length of payload (2 bytes)
Pairwise Cipher	UINT8	Pairwise cipher bitmap Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bit[1]: WPA_CIPHER_WEP104 Bit[0]: WPA_CIPHER_WEP40
Group Cipher	UINT8	Group cipher bitmap Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bit[1]: WPA_CIPHER_WEP104 Bit[0]: WPA_CIPHER_WEP40

10.5.22 MrvlIETypes_Frag_Threshold_t

This TLV specifies the fragmentation threshold to be set in the AP. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0146
Length	UINT16	Length of payload (2 bytes)
FragThrehold	UINT16	Fragmentation threshold value

10.5.23 MrvlIETypes_GTK_Rekey_t

This TLV is used to set the group re-key time interval for WPA/WPA2. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0147
Length	UINT16	Length of payload (4 bytes)
Group Re-key Time	UINT32	Group key re-key interval (seconds) Only applicable if protocol is WPA or WPA2. The value 0 means never re-key.

10.5.24 MrvlIETypes_Max_STA_Counts_t

This TLV specifies the maximum number of stations that will be allowed to connect to the uAP. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0155
Length	UINT16	Length of payload (2 bytes)
MaxStaCount	UINT16	Maximum number of stations that are allowed to connect to uAP From 1 to N. N varies for different chip/feature-pack. See the product release notes. Default = maximum

10.5.25 MrvlIETypes_Chanrpt_Chan_Load_t

This TLV provides the channel utilization measurement (as a fraction of total measurement time for which CCA is busy) as observed by the measuring STA. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0159
Length	UINT16	Length of payload (1 byte)
CcaBusyFraction	UINT8	Fraction of measurement time for which CCA indicates the medium is busy Value varies from 0 to 255.

10.5.26 MrvlIETypes_Chanrpt_Noise_Hist_t

This TLV provides the power histogram measurement of non-IEEE 802.11 noise power by sampling the channel when virtual carrier sense indicates idle and the STA is neither transmitting nor receiving a frame. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x015A
Length	UINT16	Length of payload (13 bytes)
Anpi	SINT16	Average Noise Pulse Interference (ANPI) power observed during the measurement
RpiDensities	UINT8[11]	RPI densities observed during the measurement

10.5.27 MrvlIETypes_Chanrpt_11H_Basic_t

This TLV defines 802.11h basic report. It has the following format:

<table>
<tr><th>Field Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Type</td><td>UINT16</td><td>Type ID = 0x015B</td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of payload (1 byte)</td></tr>
<tr><td>Map</td><td>UINT8</td><td>Map
<table>
<tr><th>Bits</th><th>Description</th></tr>
<tr><td>7:5</td><td>Reserved (set to 0)</td></tr>
<tr><td>4</td><td>Unmeasured
0 = channel has been measured
1 = channel has not been measured</td></tr>
<tr><td>3</td><td>Radar
0 = radar not detected
1 = radar detected</td></tr>
<tr><td>2</td><td>Unidentified
0 = significant power not detected
1 = significant power is detected which cannot be characterized as radar, OFDM preamble, valid MPDU</td></tr>
<tr><td>1</td><td>OFDM preamble
0 = no sequence detected
1 = at least 1 sequence of short training symbols is detected without valid signal field</td></tr>
<tr><td>0</td><td>BSS
0 = no valid MPDU
1 = at least 1 valid MPDU was received from other BSS or IBSS</td></tr>
</table>
</td></tr>
</table>

10.5.28 MrvlIETypes_Retry_Limit_t

This TLV specifies the retry limit to use for packet transmissions. This will set both the long and short retry limit to the specified value. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x015D
Length	UINT16	Length of payload (1 byte)
Retry Limit	UINT8	Tx retry limit (short and long) 0 to 14 (default = 7)

10.5.29 MrvlIETypes_WAPI_t

This TLV specifies the WAPI IE Content. It has the following format

Field Name	Type	Description
Type	UINT16	Type ID = 0x015E
Length	UINT16	Length of payload
WAPI IE Content	UINT8[n]	Contents will be added to associate request frame as is n is the length.

10.5.30 MrvlIETypes_MCBC_Data_Rate_t

This TLV specifies the data rate to use for multicast and broadcast packet transmission. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0162
Length	UINT16	Length of payload (2 bytes)
MCBCDataRate	UINT16	Data rate to use for packet transmission 0 = auto rate Others = fix MCBC data rate at this value (in 500 Kbps) Example: Value of 11 specifies a data rate of 5.5 Mbps.

10.5.31 MrvlIETypes_RSN_Replay_t

This TLV controls whether the firmware will disable or enable RSN replay attack protection. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0164
Length	UINT16	Length of payload (1 byte)
RSNReplayProtection	UINT8	RSN replay protection 0 = protection disabled (default) 1 = protection enabled (frames detected as replay frames will be dropped)

10.5.32 MrvlIETypes_WapiInfoSet_t

This TLV specifies the WAPI information. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0167
Length	UINT16	Length of payload (1 byte)
Multicase_PN	UINT8[16]	PN of multicase packet used

10.5.33 MrvlIETypes_Mgmt_Passthru_Mask_t

This TLV is defined to allow management frame pass-through which can be used with APCMD_SYS_CONFIGURE. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0170
Length	UINT16	Length of payload
MgmtSubTypeMask	UINT32	If Bit X is set in mask, IEEE Management Frame SubType X is to be filtered and passed through to host

SET action can be used to set the mask when the BSS is active or inactive. A GET action for APCMD_SYS_CONFIGURE can be used to retrieve the mask settings. Default value for mask will be 0, that is, no pass through. The RxPD format for the forwarded management frames will be as follows:

Field Name	Type	Description
RxBSSTType	UINT8	Packet description type (should be set to 1) 0 = reserved 1 = uAP
RxBSSNum	UINT8	BssNum on which the management packet was received
RxPacketLength	UINT16	Number of bytes in the receive data packet
RxPacketOffset	UINT16	Offset to the beginning of the receive data packet
RxPacketType	UINT16	Packet type 0xE5 = 802.11 packet
Reserved	UINT8[10]	Set these to 0 by firmware; ignored by the driver

10.5.34 MrvlIETypes_PTK_Timeout_t

This TLV specifies the pairwise key handshake timeout value used for EAPOL frame exchange by internal PSK/1X authenticator for WPA/WPA2. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0175
Length	UINT16	Length of payload (4 bytes)
PwkHskTimeout_ms	UINT32	EAPOL pairwise key handshake timeout (ms) Default = 100 ms This setting should be modified according to the DTIM frequency in the system. For example, if DTIM occurs every 500 TU (BI = 100 TU and DTIM period = 5), then the handshake timeout should be preferably x1.2 to x1.5 (600 ms to 750 ms).

10.5.35 MrvlIETypes_PTK_Retry_t

This TLV specifies the pairwise key handshake retry count value used for EAPOL frame exchange by internal PSK/1X authenticator for WPA/WPA2. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0176
Length	UINT16	Length of payload (4 bytes)
PwkHskRetries	UINT32	EAPOL pairwise key handshake retry count Default = 3 If a message transmission from uAP to STA for pairwise key exchange does not get any response from STA, the message will be retried by uAP these number of times after configured timeout. Handshake failure is triggered if there is no successful response received from STA after all retries are used up.

10.5.36 MrvlIETypes_GTK_Timeout_t

This TLV specifies the groupwise key handshake timeout value used for EAPOL frame exchange by internal PSK/1X authenticator for WPA/WPA2. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0177
Length	UINT16	Length of payload (4 bytes)
GwkHskTimeout_ms	UINT32	EAPOL groupwise key handshake timeout (ms) Default = 100 ms This setting should be modified according to the DTIM frequency in the system. For example, if DTIM occurs every 500 TU (BI = 100 TU and DTIM period = 5), then the handshake timeout should be preferably x1.2 to x1.5 (600 ms to 750 ms).

10.5.37 MrvlIETypes_GTK_Retry_t

This TLV specifies the groupwise key handshake retry count value used for EAPOL frame exchange by internal PSK/1X authenticator for WPA/WPA2. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0178
Length	UINT16	Length of payload (4 bytes)
GwkHskRetries	UINT32	EAPOL groupwise key handshake retry count Default = 3 If a message transmission from uAP to STA for groupwise key exchange does not get any response from STA, the message will be retried by uAP these number of times after configured timeout. Handshake failure is triggered if there is no successful response received from STA after all retries are used up.

10.5.38 MrvlIETypes_PS_STA_Ageout_t

This TLV specifies the age-out value to be used for STA that is in PS state. If the station is in or transitions to awake state, it will use the age-out setting configured using MrvlIETypes_STA_Ageout_t. However, if the station transitions to PS state, the age-out value for this will be set to this TLV. PS age-out would be by default be lower than standard age-out. Default PS age-out is 400 (that is, 40s). It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x017B
Length	UINT16	Length of payload (4 bytes)
PsStaAgeout_100ms	UINT32	PS station age-out (in 100 ms)

10.5.39 MrvlIETypes_PTK_Cipher_t

This TLV specifies the pairwise cipher keys type to be used for specified protocol. There can be more than 1 unicast ciphers. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0191
Length	UINT16	Length of payload (4 bytes)
Protocol	UINT16	Protocol bitmap for which to update cipher Bits[15:6]: reserved Bit[5]: WPA2 Bit[4]: reserved Bit[3]: WPA Bit[2:0]: reserved
Pairwise Cipher	UINT8	Pairwise cipher Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bit[1]: WPA_CIPHER_WEP104 (not supported) Bit[0]: WPA_CIPHER_WEP40 (not supported)
Reserved	UINT8	Reserved

10.5.40 MrvlIETypes_GTK_Cipher_t

This TLV specifies the groupwise cipher key type to be used for specified protocol. There cannot be more than 1 groupwise cipher even through it is defined as a bitmap. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0192
Length	UINT16	Length of payload (2 bytes)
Groupwise Cipher	UINT8	Groupwise cipher Bits[7:4]: reserved Bit[3]: WPA_CIPHER_CCMP Bit[2]: WPA_CIPHER_TKIP Bits[1:0]: reserved
Reserved	UINT8	Reserved

10.5.41 MrvlIETypes_ApBSS_Status_t

This TLV queries the firmware BSS status (active/inactive). Only GET action is supported on this TLV. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0193
Length	UINT16	Length of payload (2 bytes)
BSS Status	UINT16	BSS status 0 = inactive (BSS is stopped) 1 = active (BSS has been started)

10.5.42 MrvlIETypes_Tx_Data_Pause_t

This TLV specifies the pause Tx state. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0194
Length	UINT16	Length of payload (2 bytes)
PeerMACAddr	UINT8[6]	Peer MAC address
TxPauseState	UINT8	0 = Tx data free flowing 1 = Tx data paused
TotalQueued	UINT8	Total packets queued for the client

10.5.43 MrvlIETypes_Sticky_Tim_Config_t

This TLV specifies the GET/SET of the sticky Traffic Indication Map (TIM) feature parameters. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0196
Length	UINT16	Length of payload
Enable	UINT16	Enable or disable control for this feature in firmware 0x0000 = disable sticky TIM bit feature (duration/StickyBitmask in this command are ignored for this action) 0x0001 = enable sticky TIM bit feature with new duration/StickyBitmask (duration/StickyBitmask in this command are ignored for this action) 0x0002 = enable sticky TIM bit feature with previous duration/StickyBitmask
Duration	UINT16	Number of beacons TIM bit should ‘stick’ to SET after activity 0 = sticky the TIM bit once SET due to activity, will permanently remain ‘stuck’ 1 to 65535 = number of beacon intervals the TIM bit should ‘stick’ to SET after activity
StickyBitmask	UINT16	Bit[1]: • When SET, MoreData bit is made sticky • When CLEAR, normal MoreData bit updates resume Bit[0]: • When SET, TIM bit is made sticky • When CLEAR, normal TIM bit updates resume StickyBitmask = 0 is NOT a valid configuration value.

10.5.44 MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t

This TLV is defined to communicate the MAC address of an associated STA for which to activate/deactivate this feature. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0197
Length	UINT16	Length of payload
Control	UINT16	Control 0 = deactivate sticky TIM bit for that STA 1 = activate sticky TIM bit for that STA **NOTE:** TIM bit for an associated STA is made sticky only if both conditions are satisfied: • Enable = 1 or 2 in most recently received MrvlIETypes_Sticky_Tim_Config_t TLV • Control = 1 in most recently received MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t TLV which has the corresponding STA's MAC address
StaMacAddr	UINT8[6]	MAC address of an associated STA

This TLV can only be sent after the STA with MAC address specified in StaMacAddr field is associated to the uAP. For the host ACTION_GET request, host should fill in the STA MAC address it wants this information for, and the firmware would fill in the control field for that peer. For host ACTION_GET request with TLV length set to 0, firmware would return the information for all the STAs as sequence of MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t TLVs (1 per peer).

Example use cases:

- Configure the sticky TIM parameter and enable the feature—MrvlIETypes_Sticky_Tim_Config_t TLV is sent with Enable = 1 and valid values for other parameters.
- Activate the feature for a particular STA (after the STA associates)—MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t TLV is issued by host containing the desired STA's MAC address, and Control = 1.
- Deactivate the feature for a particular STA—MrvlIETypes_Sticky_Tim_STA_MAC_Addr_t TLV is issued by host containing the desired STA's MAC address, and Control = 0. Also note that firmware does not 'remember' the feature status for any STA across associations, that is, once the client STA re-connects, this feature will be reset to its default value of 'deactivated' after the association regardless of whether the feature was activated for this STA during earlier associations.
- Temporarily suspend the feature—MrvlIETypes_Sticky_Tim_Config_t TLV with Enable = 2 to re-enable the feature. Note that after the second command is issued, firmware will activate the feature only on previously configured STA links.

10.5.45 MrvlIETypes_2040_BSS_Coex_Ctrl_t

This TLV is used to control whether the uAP should enable the use of 20/40 MHz BSS coexistence schemes or not. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0198
Length	UINT16	Length of payload (1 byte)
2020CoexControl	UINT8	20/40 MHz BSS coexistence 0 = disable 1 = enable (default)

10.5.46 MrvlIETypes_Api_Ver_Info_t

This TLV is used to populate the version info corresponding to a specific API version ID. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x01C7
Length	UINT16	Length of payload
apiId	UINT16	API ID 1 = KEY_API_VER_ID 2 = HOST_API_VER_ID 3 = BSS_START_API_VER_ID 4 = CHANRPT_API_VER_ID 5 = FW_HOTFIX_VER_ID
major	UINT8	API major version for the corresponding API ID
minor	UINT8	API minor version for the corresponding API ID

10.5.47 MrvlIETypes_Max_Supported_STA_Count_t

This TLV stores information on the maximum number of connections supported in the P2P and AP modes respectively. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x0217
Length	UINT16	Length of payload
MaxSupportedSta Count	UINT8*2	MaxSupportedStaCount[0]: P2P_GO_MAX_STATION MaxSupportedStaCount[1]: AP_MAX_STATION

10.5.48 MrvlIETypes_FW_Cap_Info_t

This TLV points to the information on the firmware capabilities and firmware extended capabilities. It has the following format:

Field Name	Type	Description
Type	UINT16	Type ID = 0x023E
Length	UINT16	Length of payload
fwCapBitmap	UINT32	Firmware capability bitmap for a specified radio
fwCap_ExtBitmap	UINT32	Firmware capability extended bitmap for a specified radio

11 Command and Event List

Table 89: Host Command List

Symbolic Name	Value	Description
Initialization		
CMD_802_11_SNMP_MIB	0x0016	Gets/sets SNMP MIB
CMD_802_11_MAC_ADDR	0x004D	GEts/sets the Wi-Fi MAC address
CMD_MAC_MULTICAST_ADR	0x0010	Gets/sets MAC multicast filter table
CMD_FUNC_INIT	0x00A9	Initializes data structures and hardware blocks and activate threads
CMD_FUNC_SHUTDOWN	0x00AA	Releases resources, deactivates hardware blocks and threads, and re-initializes SDIO driver
CMD_802_11_MGMT_IE_LIST	0x00F2	Gets/sets user-specified IEs
CMD_SDIO_SINGLE_PORT_RX_AGGR	0x0223	Enables/disables SDIO single port Rx aggregation feature
CMD_IPV6_MSG_OFFLOAD	0x0238	Prepares and enables IPv6 address offload
CMD_STACMD_SYS_CONFIGURE	0x023F	Gets AP configuration parameters
CMD_EMBEDDED_DHCPD_CONFIG	0x0248	Gets/sets DHCP server configuration
CMD_ADD_NEW_STATION	0x025F	Adds STA information to firmware
MAC/Baseband/RF Control		
CMD_MAC_CONTROL	0x0028	Controls hardware MAC
CMD_802_11_RF_ANTENNA	0x0020	Gets/sets Tx and Rx antenna mode
CMD_802_11_RF_CHANNEL	0x001D	Gets/sets RF channel
CMD_CHANNEL_TRPC_CONFIG	0x00FB	Gets/sets TRPC channel configuration
CMD_802_11_REMAIN_ON_CHANNEL	0x010D	Keeps channel on specific channel until abort or expired
CMD_CW_TONE_CTRL	0x0239	Controls CW level on Tx antenna
CMD_CHAN_REGION_CFG	0x0242	Configures channel attributes and/or region information
CMD_DYN_BW	0x0252	Gets/sets dynamic bandwidth
Register and Memory Access		
CMD_BBP_REG_ACCESS	0x001A	Peeks and pokes baseband processor hardware register
CMD_RF_REG_ACCESS	0x001B	Peeks and pokes RF hardware register
CMD_MAC_REG_ACCESS	0x0019	Peeks and pokes MAC hardware register
CMD_EEPROM_ACCESS	0x0059	Retrieves the EEPROM data
CMD_PMIC_REG_ACCESS	0x00AD	Pokes NXP power management device registers
CMD_CAU_REG_ACCESS	0x00ED	Pokes NXP CAU registers
RF Calibration Data		

Table 89: Host Command List (Continued)

Symbolic Name	Value	Description
CMD_802_11_CAL_DATA_EXT	0x006D	Gets/sets RF calibration data
CMD_CFG_DATA	0x008F	Downloads new optimized register settings or calibrating data
Status Information		
CMD_GET_HW_SPEC	0x0003	Gets hardware specifications
CMD_802_11_GET_LOG	0x000B	Gets Wi-Fi log
CMD_GET_TSF	0x0080	Retrieves current TSF value
CMD_TX_PKT_STATS	0x008D	Reports current Tx packet status
CMD_802_11_AVG_RATE_INFO	0x00A3	Gets Tx or Rx last successful rate and average rate
CMD_802_11_RSSI_INFO	0x00A4	Gets received signal strength
CMD_802_11_RSSI_INFO_EXT	0x0237	Gets received signal strength for each path
LED Control		
CMD_802_11_LED_CONTROL	0x004E	LED control
Scan		
CMD_802_11_SCAN	0x0006	Starts scan process
CMD_802_11_BG_SCAN_CONFIG	0x006B	Gets/sets background scan configuration
CMD_802_11_BG_SCAN_QUERY	0x006C	Gets background scan results
CMD_802_11_SCAN_EXT	0x0107	Sends scan results to host by SCAN_REPORT_BY_EVENTS event
Network Start/Stop/Join		
CMD_802_11_ASSOCIATE	0x0012	Initiates an association with the AP For backwards compatibility, the CMD_802_11_ASSOCIATE can be defined as either 0x0050 or 0x0012. However, the CMD_802_11_ASSOCIATE return CmdCode is always defined as (0x0012 | 0x8000).
CMD_802_11_AD_HOC_START	0x002B	Starts an Ad-Hoc network
CMD_802_11_AD_HOC_JOIN	0x002C	Joins an Ad-Hoc network
CMD_802_11_AD_HOC_STOP	0x0040	Stops Ad-Hoc network
CMD_802_11_IBSS_COALESCING_STATUS	0x0083	Configures IBSS coalescing settings
CMD_802_11_DEAUTHENTICATE	0x0024	Starts de-authentication process with the AP
CMD_SET_BSS_MODE	0x00F7	Sets BSS mode
CMD_802_11_NET_MONITOR	0x0102	Sets net monitor (promiscuous) configuration
CMD_802_11_RX_MGMT_INDICATION	0x010C	Starts/stops management frame forwards to host through datapath
CMD_FAST_LINK_SYNC_QUERY	0x0244	Gets fast link synchronization operation information/status
Security		

Table 89: Host Command List (Continued)

Symbolic Name	Value	Description
CMD_802_11_KEY_MATERIAL	0x005E	Gets/sets key material used to perform Tx encryption or Rx decryption
CMD_802_11_CRYPTO	0x0078	Provides AES cipher engine API between firmware and driver
CMD_SUPPLICANT_PMK	0x00C4	Configures embedded supplicant information
CMD_SUPPLICANT_PROFILE	0x00C5	Configures supplicant profile
CMD_CRYPTO	0x025E	Sets crypto format
Rate Adaptation		
CMD_TX_RATE_QUERY	0x007F	Reports current Tx rate
CMD_TX_RATE_CFG	0x00D6	Gets/sets 802.11n transmission data rate
CMD_PACKET_AGGR_CTRL	0x0251	Gets/sets software Tx/Rx aggregation configuration parameters in firmware
CMD_REGION_POWER_CFG	0x0249	Configures region information and/or power table
Event Subscription		
CMD_802_11_SUBSCRIBE_EVENT	0x0075	Subscribes to events and sets thresholds
Power Management		
CMD_802_11_SLEEP_PARAMS	0x0066	Gets/sets sleep parameters
CMD_802_11_INACTIVITY_TIMEOUT_EXT	0x008A	Gets/sets inactivity timeout enhancement value
CMD_802_11_FW_WAKE_METHOD	0x0074	Configures firmware wake method
CMD_802_11_AUTO_TX	0x0082	Configures auto transmit settings
CMD_SDIO_GPIO_INT_CONFIG	0x0088	Configures SDIO/GPIO host interface interrupt
CMD_802_11_PS_MODE_ENH	0x00E4	Enables/disables power save
CMD_802_11_HS_ENH	0x00E5	Configures wakeup semantics for host driver
CMD_HS_WAKEUP_REASON	0x0116	Gets host wakeup reason from firmware
CMD_802_11_SLEEP_PERIOD	0x0068	Defines sleep period command
Bluetooth Coexistence		
CMD_802_11_BCA_CONFIG_TIMESHARE	0x0069	Configures BCA timeshare interval and percentage of time in this timeshare interval
CMD_ROBUST_COEX	0x00E0	Sets infrastructure power save mode and period
Wi-Fi Multimedia		
CMD_WMM_GET_STATUS	0x0071	Retrieves current WMM state
CMD_WMM_ADDTS_REQ	0x006E	Sends an ADDTS frame to the associated AP
CMD_WMM_DELTS_REQ	0x006F	Sends a DELTS to the associated AP
CMD_WMM_QUEUE_CONFIG	0x0070	Gets, sets, or defaults the configuration of a WMM queue
CMD_WMM_QUEUE_STATS	0x0081	Manages the WMM AC queues statistics collection

Table 89: Host Command List (Continued)

Symbolic Name	Value	Description
CMD_WMM_TS_STATS_CMD_802_11	0x005D	Retrieves current state of a given Traffic Stream (TSID) from the firmware
CMD_WMM_PARAM_CONFIG	0x023A	Configures WMM parameters from host
802.11d		
CMD_802_11_D_DOMAIN	0x005B	Gets/sets 802.11d domain information
802.11h		
CMD_802_11_MEASUREMENT_REQUEST	0x0062	Sends measurement request
CMD_802_11_MEASUREMENT_REPORT	0x0063	Gets measurement response report frame
CMD_802_11_CHAN_SW_ANN	0x0061	Broadcasts a channel switch announcement
CMD_DFS_REPEATER_MODE	0x012B	Enables/disables DFS repeater mode
802.11n		
CMD_11N_CFG	0x00CD	Gets/sets 802.11n configuration
CMD_11N_ADDBA_REQ	0x00CE	Sends ADDBA request to recipient node
CMD_11N_ADDBA_RSP	0x00CF	Sends ADDBA response to originator node
CMD_11N_DELBA	0x00D0	Deletes Block Ack agreement
CMD_RECONFIGURE_TX_BUFF	0x00D9	Gets/sets Tx buffer size
CMD_AMSDU_AGGR_CTRL	0x00DF	Enables/disables AMSDU aggregation
CMD_11N_2040COEX	0x00E9	Sends 20/40 coexistence management action to AP
CMD_RX_PKT_COALESC_CFG	0x012C	Gets/sets Rx packet coalescing
CMD_802_11_EAPOL_AMSDU_INDICATION	0x012E	Sends GTK frame to firmware
802.11ac		
CMD_11AC_CFG	0x0112	Gets/sets 802.11ac VHT Tx/Rx capability and supported MCS set
Memory Efficient Filtering		
CMD_MEF_CFG	0x009A	Sets up firmware filtering options
Small Debug Print		
CMD_DBGS_CFG	0x008B	Configures small debug print facility
CMD_GET_MEM	0x008C	Retrieves firmware vitals or trace memory
Wi-Fi Direct		
CMD_WIFI_DIRECT_PARAMS_CONFIG	0x00EA	Sets/modifies/deletes Wi-Fi Direct protocol related parameters
CMD_WIFI_DIRECT_MODE_CONFIG	0x00EB	Sets device mode externally
CMD_802_11_ACTION_FRAME	0x00F4	Sends different types of Wi-Fi Direct action frames
CMD_WIFI_DIRECT_SERVICE_DISCOVERY_PACKET	0x00EC	Sends GAS packet sent by driver to Wi-Fi

Table 89: Host Command List (Continued)

Symbolic Name	Value	Description
Data Flow Control		
CMD_CFG_TX_DATA_PAUSE	0x0103	Controls Tx data packet buffering by firmware
CMD_COALESCE_CFG	0x010A	Sets rules to filter and buffer certain type of packet
Multi-Channel Operation		
CMD_MULTI_CHAN_TIMESLICE_STATISTICS	0x011D	Gets timeslice and channel switching statistics
CMD_MULTI_CHAN_CONFIG	0x011E	Configures multi-channel operation
CMD_MULTI_CHAN_POLICY	0x0121	Gets/sets multi-channel policy
CMD_MULTI_CHAN_TIMESLICE_CONFIG	0x024A	Configures timeslice of each channel for DRCS
CMD_DMCS_CONFIG	0x0260	Gets/sets status of DMCS agent
TDLS		
CMD_802_11_TDLS	0x0100	Supports embedded supplicant TDLS features
CMD_TDLS_OPERATIONS	0x0122	Setup or teardown TDLS link with each peer
Neighbor Awareness Network		
CMD_NAN_PARAM_CONFIG	0x0228	Gets/sets NAN configuration parameters used by NAN MAC layer
CMD_NAN_MODE_CONFIG	0x0229	Gets/sets NAN mode
CMD_NAN_TRANSMIT_SDFRAME	0x022A	Sends GAS packet formed by NAN discovery engine to publish or subscribe NAN services
CMD_NAN_UPDATE_SERVICE_HASH_DB	0x022B	Adds/removes service hashes
CMD_NAN_STATE_CONFIG	0x022C	Sets runtime parameters of NAN state
CMD_NAN_FLUSH_SD_QUEUE	0x022D	Flushes all SD frames queued in SD queue to stop advertising of SD frames
802.11k		
CMD_GET_NEIGHBORLIST	0x0231	Gets Neighbor list
Smart Mode Configuration		
CMD_SMART_MODE_CFG	0x012D	Configures smart configuration mode
ED-MAC Control		
CMD_ED_CTRL	0x0130	Supports ED_MAC control feature
Temperature Sensor		
CMD_GET_SENSOR_TEMPERATURE	0x0227	Gets temperature from temperature sensor
Bridge Mode Control		
CMD_BRIDGE_MODE_CTRL	0x022E	Controls bridge mode for bridge device
802.1AS Timesync		
CMD_TIMESYNC_CFG	0x0246	Synchronizes host (system) and firmware (device) clocks

Table 89: Host Command List (Continued)

Symbolic Name	Value	Description
Wi-Fi Location		
CMD_FTM_CONFIG_SESSION_PARAMS	0x024D	Configures Wi-Fi location session configuration parameters for Initiator and Responder
CMD_FTM_SESSION_CTRL	0x024E	Triggers/stops FTM procedure by requesting firmware to initiate/stop FTM session
CMD_FTM_FEATURE_CTRL	0x024F	Configures Wi-Fi location feature responsibilities between host application and firmware
CMD_FTM_CONFIG_RESPONDER	0x0255	Gets RTT Responder information
Channel State Information		
CMD_CSI	0x025B	Informs firmware to start/stop CSI collection
802.11P		
CMD_802_11P_CONFIG	0x024B	Gets/sets 802.11p specific parameter configuration
CMD_802_11P_STATS	0x024C	Gets/sets 802.11p status
Independent Reset		
CMD_INDEPENDENT_RESET	0x0243	Configures Wi-Fi OOB independent reset mode and GPIO pin
mac80211		
CMD_MAC80211_STA_ADD	0x0253	Stores peer information during connection through mac80211 feature
SoC Spectral Analysis Unit		
CMD_SSU	0x0259	Passes buffer address, length, and other baseband parameters to device for SSU feature
Virtual Dynamic Library Loading		
CMD_VDLL	0x0240	Downloads VDLL block as requested by firmware through event
Micro Access Point		

Table 90: uAP Command List

Symbolic Name	Value	Description
APCMD_SYS_INFO	0x00AE	Returns system information
APCMD_SYS_RESET	0x00AF	Resets the uAP to its initial state
APCMD_SYS_CONFIGURE	0x00B0	Gets/sets AP configuration parameters
APCMD_BSS_START	0x00B1	Starts BSS
APCMD_BSS_STOP	0x00B2	Stops active BSS
APCMD_STA_LIST	0x00B3	Gets list of client stations currently associated with the AP
APCMD_STA_DEAUTH	0x00B5	De-authenticates a client station
CMD_CFG_DATA	0x008F	Facilitates downloading of calibration data
CMD_CHAN_REPORT_REQUEST	0x00DD	Performs channel load, channel noise histrogram, basic measurement on a given channel
CMD_PMF_PARAMS	0x0131	Gets/sets PMR support on uAP with embedded authenticator
CMD_APCMD_ACS_SCAN	0x0224	Supports ACS feature on DFS channels for uAP
CMD_UAP_OPERATION_CTRL	0x0233	Gets/sets uAP operation control parameters in non-DRCS case
802.11ax		
CMD_802_11AX_CFG	0x0266	Gets/sets 802.11ax HE configuration
CMD_802_11AX_CMD	0x026D	Gets/sets 802.11ax HE configuration using sub commands

Table 91: Command Result Codes

Symbolic Name	Value	Description
CMD_STATUS_SUCCESS	0x0000	No error
CMD_STATUS_ERROR	0x0001	Command failed
CMD_STATUS_UNSUPPORTED	0x0002	Command is not supported

Table 92: Event Support List

Event Name	Event ID	Used by		Description
		Host	uAP	
Dummy Host Wakeup Signal	0x0001	X	--	Dummy event used to generate an interrupt to wakeup the host through the host interface if configured
Beacon Loss/Link Loss	0x0003	X	--	Beacon loss/link loss
Link Sense	0x0004	X	--	In the IBSS network, if the total number of joined stations (including self) changes from one station to more than one station, Link Sense is issued
Deauthenticate	0x0008	X	--	Generated when the firmware receives a 802.11 deauthenticate management frame from the AP (see Section 8.2, SDIO Event Extended Format, on page 279)
PS_AWAKE	0x000A	X	X	Generated when the firmware first takes the Wi-Fi SoC device out of Sleep mode
PS_ASLEEP	0x000B	X	X	Generated when the firmware is about to put the Wi-Fi SoC device in Sleep mode
Group MIC Error	0x000C	X	--	MIC error generated for Broadcast packets
UNICAST MIC Error	0x000D	X	--	MIC error generated for Unicast packets
Ad-Hoc Beacon Lost	0x0011	X	--	In the IBSS network, if the total number of joined stations (including self) changed from more than one station to only one station, Ad-Hoc Beacon Lost is issued
Stop Transmit	0x0013	X	--	Generated to indicate the firmware is not ready to receive data packets. Generation of this event can occur during measurements, quiet periods, or channel switches.
Start Transmit	0x0014	X	--	Generated to indicate the firmware is ready to receive data packets
Measurement Report Ready	0x0016	X	--	Generated when the firmware has a measurement report waiting for driver retrieval using the Section 6.17.1, CMD_802_11_MEASUREMENT_REQUEST, on page 154.
WMM Status Change	0x0017	X	--	Generated when the firmware monitors a change in the WMM state. This is a result of an AP change in the WMM Parameter IE or a change in the operational state of one of the AC queues (see Section 6.15.1, CMD_WMM_GET_STATUS, on page 145).
Background Scan Report	0x0018	X	--	Generated as a result of a completed background scan report (see Section 6.7.2, CMD_802_11_BG_SCAN_CONFIG, on page 70)
BeaconRSSI Low	0x0019	X	--	Subscribed event (see Section 6.12, Event Subscription, on page 115)
BeaconSNR Low	0x001A	X	--	
MAX Failures	0x001B	X	--	
BeaconRSSI High	0x001C	X	--	
BeaconSNR High	0x001D	X	--	

Table 92: Event Support List (Continued)

Event Name	Event ID	Used by		Description
		Host	uAP	
IBSS Coalesced	0x001E	X	--	Generated when the IBSS Coalescing process is finished and the BSSID has changed (see Section 6.8.5, CMD_802_11_IBSS_COALESCING_STATUS, on page 86)
BG Scan Timeout	0x001F	X	--	Generated when the firmware is unable to find a WPS registrar within the timeout duration specified by (see Section 10.4.20, MrvlIETypes_WPSEnrolleeProbeReqIE_t, on page 329)
ADHOC_STATION_CONNECT	0x0020	X	--	Generated when a station is ad-hoc connected
ADHOC_STATION_DISCONNECT	0x0021	X	--	Generated when a station is ad-hoc disconnected
Data RSSI Low	0x0024	X	--	Subscribe event (see Section 6.12, Event Subscription, on page 115)
Data SNR Low	0x0025	X	--	
Data RSSI High	0x0026	X	--	
Data SNR High	0x0027	X	--	
Link Quality Indication	0x0028	X	--	
PORT_RELEASE	0x002B	X	--	Triggered when controlled port are released
MICRO_AP_EV_ID_STA_DEAUTH	0x002C	--	X	Generated when a client station leaves the BSS
FORWARD_MGMT_FRAME	0x002D	X	--	Generated when a management frame is forwarded
MICRO_AP_EV_ID_STA_ASSOC	0x002D	--	X	Generated when a client station joins in the BSS
MICRO_AP_EV_ID_BSS_START	0x002E	--	X	Generated when BSS is started
Pre Beacon Lost	0x0031	X	--	Subscribe event (see Section 6.12, Event Subscription, on page 115)
DOT11N_ADDBA	0x0033	X	X	Generated when ADDBA request action frame is received
DOT11N_DELBA	0x0034	X	X	Generated when DELBA request action frame is received
BA_STREAM_TIMEOUT	0x0037	X	X	Block Ack inactivity timeout
AMSDU_AGGR_CTRL	0x0042	X	--	Reports aggregation size available for host
MICRO_AP_EV_ID_BSS_IDLE	0x0043	--	X	Generated when BSS is idle
MICRO_AP_EV_ID_BSS_ACTIVE	0x0044	--	X	Generated when BSS is active
EVT_WEP_ICV_ERR	0x0046	X	--	Triggered by WEP decryption error
HS_ACT_REQ	0x0047	X	X	For enhanced power management
DOT11N_BWCHANGE_EVENT	0x0048	X	--	Reports bandwidth change
MICRO_AP_EV_ID_MIC_COUNTERMEASURES	0x004C	--	X	Generated when firmware starts or stops MIC countermeasure
MEF_HOST_WAKEUP	0x004F	X	--	Triggered to wakeup host when MEF condition matched
CHANNEL_SWITCH_ANNOUNCEMENT	0x0050	X	--	Reports channel switch

Table 92: Event Support List (Continued)

Event Name	Event ID	Used by		Description
		Host	uAP	
MICRO_AP_EV_ID_RSN_CONNECT	0x0051	--	X	Generated when STA successfully completed 4-way handshake
TDLS_GENERIC	0x0052	X	--	Triggered in firmware when it detects TDLS link does not exist and informs driver to perform a teardown
RADAR_DETECTED	0x0053	--	X	Generated if master mode radar detection mode is enabled in firmware and if firmware detects radar on the current operating channel or firmware receives a measurement report/autonomous measurement report which indicate that radar is detected on the current operating channel
CHANNEL_REPORT_RDY	0x0054	--	X	Generated when channel report measurement is completed
TX_DATA_PAUSE	0x0055	X	X	Sent when a uAP or GO client in PS exceeds the allowed PS buffering, or when the client has buffering available
SCAN_REPORT_BY_EVENT	0x0058	X	--	Used to scan results for CMD_802_11_SCAN_EXT
EV_RXBA_SYNC	0x0059	X	--	Triggered by firmware to synchronize Rx BA window bitmap
REMAIN_ON_CHANNEL_EXPIRED	0x005C	X	--	Informs host about CMD_802_11_REMAIN_ON_CHANNEL command expired
MULTI_CHAN_INFO	0x006A	X	--	Triggered whenever any interface becomes active or inactive
EVENT_FW_DUMP_INFO	0x0073	X	--	Triggered when firmware dumps memory for debug usage
EVT_TX_STATUS_REPORT	0x0074	X	--	Triggered by firmware to return tx_status_report for requested host Tx management frame fro mdriver
EVENT_CLASS_NAN	0x0075	X	--	Used to send all type of NAN events to the host
EVT_BTCOEX_WLAN_PARA_CHANGE	0x0076	X	--	Sent to driver in 2 cases: • GO/MMH+SCO • GC+OPP_TX
EV_SMC_GENERIC	0x0077	X	--	Generated by firmware while Smart mode configuration is under progress
EV_NEIGHBOR_REPORT	0x0079	X	--	Triggered when driver issues CMD_GET_NEIGHBORLIST command
EV_BSS_TRANSITION_REQUEST	0x007A	X	--	Used for DOT11V, voice enterprise certification
MICRO_AP_EV_ID_BSS_STOP	0x007B	--	X	Triggered to indicate to driver that in-STA disconnects from ext-AP and uAP will stop automatically depending on setting.
EVT_VDLL_IND	0x0081	X	--	Sent to host by firmware in 2 cases: • Case 1 – Inform driver the VDLL image offset in firmware during initialization • Case 2 – Request VDLL block at any time

Table 92: Event Support List (Continued)

Event Name	Event ID	Used by		Description
		Host	uAP	
EVT_CHAN_REG_INFO	0x0082	X	--	Generated by firmware when channel attributes and/or region information has been changed
EVT_SSU_DUMP_DMA	0x008C	X	--	Sent to driver by firmware when SSU DMA complete or error detected
WIFI_DIRECT_EVENT	--	X	--	Generated by firmware to send data received from peer to driver and application
WIFI_DIRECT_SERVICE_DISCOVERY_EVENT	--	X	--	Passes received GAS service discovery packet to host

12 Acronyms and Abbreviations

Table 93: Acronyms and Abbreviations

Acronym	Definition
AC	Access Category
ACK	Acknowledgement
ACL	Access Control List
ACS	Automatic Channel Selection
AES	Advanced Encryption Standard
AIFS	Arbitration Inter Frame Space
AMSDU	Aggregated MAC Service Data Unit
ANPI	Average Noise Pulse Interference
AP	Access Point
API	Application Programming Interface
APSD	Automatic Power Save Delivery
ATIM	Announced Traffic Indication Message
BA	Block Acknowledgment
BCA	Bluetooth Coexistence Arbitration
BSS	Basic Service Set
BW	Bandwidth
CAU	Common Analog Unit
CCA	Clear Channel Assessment
CCKM	Cisco Centralized Key Management
CCMP	Counter mode with Cipher Block Chaining Message Protocol
CF	CompactFlash
CMAC	Cipher-based Message Authentication Code
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
CSI	Channel State Information
CW	Contention Window
DCC	Decentralized Congestion Control
DMA	Direct Memory Access
DSSS	Direct Sequence Spread Spectrum
DTIM	Delivery Traffic Indication Message
DW	Discovery Window
EAP	Extensible Authentication Protocol
EAPOL	EAP Over Lan

Table 93: Acronyms and Abbreviations (Continued)

Acronym	Definition
EDCA	Enhanced Distributed Channel Access The prioritized CSMA/CA access mechanism used by QSTAs in a QBSS. This access mechanism is also used by the QAP and operates concurrently with a controlled channel access mechanism based on polling.
EEPROM	Electrically Erasable Programmable Read-Only Memory
FTM	Fine Time Measurement
GAS	Generic Advertisement Service
GF	Greenfield
GI	Guard Interval
GPIO	General Purpose Input/Output
G-SPI	Generic Serial Peripheral Interface
GTK	Group Transient Key
GWK	Groupwise Key
HAL	Hardware Abstraction Layer
HE	High Efficiency
HT	High Throughput
IBSS	Independent Basic Service Set
IE	Information Element
IFS	Inter Frame Space
IGTK	Integrity Group Transient Key
IP	Internet Protocol
LAN	Local Area Network
LED	Light Emitting Diode
LSb	Least Significant Bit
MAC	Media/Medium Access Controller
MCBC	Multicast Broadcast
MCS	Modulation and Coding Scheme
MEF	Memory Efficient Filtering
MIB	Management Information Base
MIC	Message Integrity Code
MIMO	Multiple Input Multiple Output
MLME	MAC Sublayer Management Entity
MPDU	MAC Protocol Data Unit
MSDU	MAC Service Data Unit
MU	Multiple User
NAN	Neighbor Awareness Network
NF	Noise Floor
OBSS	Overlapping Basic Service Set

Table 93: Acronyms and Abbreviations (Continued)

Acronym	Definition
OFDM	Orthogonal Frequency Division Multiplexing
OID	Object ID
OOB	Out-of-Band
P2P	Point-to-Point
PA	Power Amplifier
PCI	Peripheral Component Interconnect
PER	Packet Error Rate
PMF	Protected Management Frame
PMK	Pairwise Master Key
PPS	Proprietary Periodic Sleep
PS	Power Save
PSK	Pre-Shared Key
PTK	Pairwise Transient Key
PWK	Pairwise Key
QAP	QoS Enhanced Access Point
QBSS	QoS Enhanced Basic Service Set
QoS	Quality of Service
QSTA	QoS Enhanced Station A label for those MSDUs for which a common set of EDCA parameters are used for contending the channel. Each QSTA has four ACs.
RAM	Read Access Memory
RIFS	Reduced Interframe Space
RF	Radio Frequency
RFI	Radio Frequency Interference
RSN	Robust Secure Network
RSSI	Receive Signal Strength Indicator
RTS	Request to Send
RTT	Round Trip Time
SCO	Synchronous Connection Oriented
SD	Service Discovery
SDIO	Secure Digital Input/Output
SISO	Single Input Single Output
SNMP	Simple Network Management Protocol
SNR	Signal to Noise Ratio
SoC	System-on-Chip
SQU	Internal SRAM Unit
SR	Spatial Reuse in 802.11ax

Table 93: Acronyms and Abbreviations (Continued)

Acronym	Definition
SSID	Service Set ID A 32-character unique identifier attached to the header of packets sent over a Wi-Fi that acts as a password when a mobile device tries to connect to the BSS.
SSU	SoC Spectral Analysis Unit
STA	Station (Client)
SU	Single User
TBTT	Target Beacon Transmission Time
TC	Traffic Class; For example, admission class
TCP	Transmission Control Protocol
TDLS	Tunneled Direct Link Setup
TID	Traffic Stream Identifier
TIM	Traffic Indication Map
TKIP	Temporal Key Integrity Protocol
TLV	Tag Length Value
TPC	Transmit Power Control (802.11h)
TRPC	Transmit Rate-based Power Control
TSC	TKIP Sequence Counter
TSF	Time Synchronization Function
TSID	Time Stream ID
TSPEC	Traffic Specification
TU	Time Unit
uAP	Micro Access Point
UAPSD	Unscheduled Automatic Power Save Delivery
UDP	User Datagram Protocol
UINT16	16-bit unsigned Integer
UINT32	32-bit unsigned Integer
UINT8	8-bit unsigned Integer
USB	Universal Serial Bus
VDLL	Virtual Dynamic Library Loading
WAPI	Wi-Fi Authentication and Privacy Infrastructure
WCB	Wireless Control Block
WEP	Wired Equivalent Privacy
WFA	Wi-Fi Alliance
Wi-Fi	Wireless Local Area Network
WMM	Wi-Fi Multimedia
WPA	Wi-Fi Protected Access
WPS	Wi-Fi Protected Setup

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